The Regeneration of Transport Infrastructure as Green Infrastructure:
A Nature-Based Solution Approach for the Territory of Odivelas

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Scientific Dissertation specially prepared for the award of the degree of
Mestre in Territorial Management and Urban Studies

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June 2026

Acknowledgments

First and foremost, I would like to thank God for guiding me along this path that I am still discovering.

I am deeply grateful to my family - my father, my mother, and my brother - for their unwavering support and for giving me every condition to develop this work.

I would like to express my sincere gratitude to my supervisors, Prof. Dr. João Rafael Santos and Prof. Dr. Isabel Loupa Ramos, for their guidance, contributions, and continuous support throughout this research. I also wish to thank the Universidade de Lisboa, and in particular the faculty and coordinators of the Master's Programme in Spatial Planning and Urbanism (MOTU), for the rich learning environment and the discussions that shaped this work.

A special acknowledgement goes to my undergraduate professor, Dr. Cláudio Manetti, for helping me develop a critical way of thinking, and for his encouragement and support in my decision to pursue this Master's degree.

I would also like to thank the all@cityscape group and all the faculty involved for their contributions to the discussions, methodologies, and data that were essential to the development of this research, as well as the Câmara Municipal de Odivelas for making available the various datasets that were fundamental to understanding the territory.

Finally, I wish to thank Spacesyntax Brasil for the workshops and discussions that directly contributed to the conclusions and research presented here.

Summary

Throughout the twentieth century, high-capacity road infrastructure played a central role in shaping metropolitan territories, enabling urban expansion while simultaneously generating fragmentation, ecological discontinuity, and vast quantities of residual, inaccessible, and functionally indeterminate space. These interstitial spaces, produced by the spatial logic of highways, ring roads, and radial corridors, occupy significant portions of the metropolitan fabric yet remain largely absent from planning instruments and ecological management frameworks.

This research asks whether transport infrastructure, historically conceived as a barrier to ecological and spatial quality, can be reimagined as a carrier of it, through the systematic transformation of its interstitial spaces and the infrastructure itself into green infrastructure. Drawing on the EU Nature Restoration Law (Regulation EU 2024/1991) as a regulatory framework and on the growing body of international practice in grey-green integration, the research proposes that Green Infrastructure and Nature-Based Solutions (NBS) represent a concrete operative tool for this transformation.

The research develops a multi-scale territorial diagnosis of Odivelas, a municipality in the Lisbon Metropolitan Area where three major highway corridors - the IC17, IC22, and IC18 - have profoundly conditioned the spatial, ecological, and morphological development of the territory. Operating across metropolitan, municipal, and corridor scales, the diagnosis identifies and characterises the interstitial spaces produced by each corridor through complementary analytical layers, including Space Syntax, Fragstats landscape metrics, NDVI analysis, and field observation.

From this diagnosis, a typological classification of interstitial spaces is developed - Understructure and Node Spaces, Primary Marginal Open Spaces, and Embankments and Marginal Linear Spaces - and five representative cases are selected for the application of a NBS framework. Each case is evaluated through three analytical conditions - Barrier Condition, Latent Connectivity, and Functional Indetermination - and three operative capacities - Mediation, Requalification, and Articulation - producing a transferable methodology for the identification, classification, and programmatic transformation of infrastructural interstices through nature-based strategies.

Keywords: Infrastructural Interstitial Spaces; Nature-Based Solutions; Highway Infrastructure; Green Infrastructure; Territorial Regeneration;

Resumo

Ao longo do século XX, as infraestruturas rodoviárias de alta capacidade desempenharam um papel determinante na configuração das áreas metropolitanas europeias. Concebidas para assegurar mobilidade regional e suportar processos de expansão urbana, estas infraestruturas contribuíram simultaneamente para a fragmentação do território, a descontinuidade dos sistemas ecológicos e a produção de extensas áreas residuais situadas entre faixas de circulação, nós viários, taludes e espaços sobrantes de processos de urbanização associados à lógica automóvel. Embora frequentemente tratadas como subprodutos inevitáveis da infraestrutura, estas áreas representam uma componente significativa da paisagem metropolitana contemporânea e permanecem, em grande medida, ausentes dos instrumentos convencionais de planeamento urbano e ecológico.

A relevância desta problemática tornou-se particularmente evidente perante os desafios ambientais e climáticos que atualmente afetam os territórios urbanos europeus. A intensificação de fenómenos extremos, como ondas de calor, eventos de precipitação intensa e inundações urbanas, têm evidenciado os impactos acumulados de décadas de impermeabilização do solo, artificialização dos cursos de água e fragmentação ecológica. Paralelamente, a entrada em vigor do Regulamento (UE) 2024/1991 relativo ao Restauro da Natureza estabelece metas vinculativas para a recuperação dos ecossistemas urbanos e para o aumento da conectividade ecológica, reforçando a necessidade de identificar novas oportunidades territoriais para a implementação de estratégias de restauro ecológico e adaptação climática. Neste contexto, os espaços intersticiais associados às infraestruturas rodoviárias emergem como um recurso territorial amplamente disponível, mas ainda insuficientemente explorado.

A presente investigação parte da hipótese de que estes espaços podem ser compreendidos não como resíduos territoriais, mas como componentes estratégicos de uma infraestrutura verde metropolitana. A questão central que orienta o trabalho consiste em compreender de que forma as infraestruturas rodoviárias, historicamente associadas à fragmentação espacial e ecológica, podem ser reimaginadas como suportes de processos de regeneração territorial através da transformação sistemática dos seus espaços intersticiais e da sua integração em estruturas verdes multifuncionais.

O enquadramento teórico da investigação organiza-se em torno de três eixos complementares. O primeiro analisa o papel das infraestruturas de transporte na produção do território contemporâneo, abordando simultaneamente a sua capacidade estruturadora e os impactos ecológicos, morfológicos e sociais que lhes estão associados. Particular atenção é dada à emergência dos espaços intersticiais enquanto consequência da lógica infraestrutural moderna e à forma como diferentes conceitos contribuem para a sua interpretação.

O segundo eixo desenvolve uma proposta conceptual para a transformação destes espaços através do espaço público. Em vez de compreender o espaço público apenas como elemento de qualificação urbana, a investigação propõe uma lógica operativa assente em três capacidades complementares: mediação, requalificação e articulação. A mediação corresponde à capacidade de negociar conflitos e descontinuidades produzidas pela infraestrutura entre diferentes sistemas, escalas e usos do território. A requalificação refere-se à transformação dos espaços residuais em ativos multifuncionais capazes de

responder simultaneamente a necessidades ecológicas, sociais e urbanas. A articulação corresponde à integração destas intervenções em redes territoriais mais amplas, permitindo que contribuam para a conectividade ecológica, a mobilidade suave e a coesão espacial à escala metropolitana.

O terceiro eixo centra-se na Infraestrutura Verde e nas Soluções Baseadas na Natureza enquanto paradigma operativo para a concretização desta transformação. São analisadas diferentes definições, abordagens e tipologias de intervenção, bem como exemplos internacionais e nacionais que demonstram o potencial das estratégias ecológicas aplicadas a contextos infraestruturais. Esta análise permite estabelecer um conjunto de princípios orientadores para a adaptação das Soluções Baseadas na Natureza às condições específicas dos espaços intersticiais rodoviários.

A investigação adota o município de Odivelas como caso de estudo. Localizado na Área Metropolitana de Lisboa, o município constitui um exemplo representativo de urbanização periférica fortemente condicionada por grandes infraestruturas rodoviárias. Os corredores do IC17 (CRIL), IC22 (Radial de Odivelas) e IC18 (CREL) desempenharam um papel central na estruturação do território, influenciando os seus padrões de ocupação, acessibilidade, fragmentação ecológica e configuração morfológica. Odivelas é assim entendido não como uma exceção, mas como um território representativo de uma condição partilhada por numerosas periferias metropolitanas europeias.

A metodologia desenvolve-se através de uma abordagem multiescalar organizada em três níveis de análise. À escala metropolitana, o município é enquadrado nas dinâmicas territoriais da Área Metropolitana de Lisboa, identificando os principais processos de urbanização, infraestruturação e estruturação ecológica que influenciam o contexto de estudo. À escala municipal é realizada uma caracterização integrada do território, incorporando informação biofísica, usos do solo, estrutura urbana, vulnerabilidades ambientais, métricas de ecologia da paisagem obtidas através do software Fragstats e indicadores de acessibilidade espacial produzidos através do Place Syntax Toolkit. Esta análise permite identificar padrões de fragmentação, conectividade e potencial ecológico relevantes para a investigação.

À escala dos corredores rodoviários, cada infraestrutura é analisada através de três dimensões complementares: Condição Infraestrutural, Condição Urbana e Porosidade, e Estrutura Verde e Azul. A partir desta leitura é desenvolvido um sistema de interpretação assente em três condições analíticas: Condição de Barreira, Conectividade Latente e Indeterminação Funcional. Estas condições constituem um dos principais contributos conceptuais da investigação, permitindo compreender simultaneamente os problemas existentes e os potenciais de transformação associados a cada espaço.

Os espaços intersticiais identificados são posteriormente organizados em três tipologias principais, sendo: (1) Espaços sob Estrutura e Nós, (2) Espaços Abertos Adjacentes e Taludes e (3) Margens Lineares. Esta classificação evidencia que os interstícios não constituem uma categoria homogénea, mas um conjunto diversificado de condições espaciais que exigem estratégias diferenciadas de intervenção. A tipificação proposta constitui um contributo metodológico adicional, fornecendo uma base sistemática para a análise e comparação de situações semelhantes noutros territórios.

Com base nesta classificação são selecionados cinco casos de estudo representativos. Cada caso é avaliado através das três condições analíticas e das três capacidades operativas do quadro conceptual desenvolvido. Esta avaliação é sintetizada através de fichas analíticas que funcionam simultaneamente como instrumento de diagnóstico e como ferramenta de apoio à definição de estratégias de intervenção. Para cada caso é desenvolvido um programa específico de Soluções Baseadas na Natureza ajustado às suas características ecológicas, urbanas e infraestruturais.

Os resultados demonstram que os espaços intersticiais associados às infraestruturas rodoviárias podem ser identificados, classificados e avaliados através de uma metodologia replicável, permitindo a definição de estratégias de transformação ajustadas às suas condições específicas. Demonstram igualmente que estes espaços possuem capacidade para desempenhar funções ecológicas, sociais e urbanas relevantes, contribuindo para o aumento da conectividade ecológica, a mitigação dos efeitos das alterações climáticas, a melhoria da acessibilidade pedonal e ciclável e a qualificação do espaço público.

A investigação conclui que a infraestrutura rodoviária não deve ser entendida exclusivamente como uma barreira territorial, mas também como um potencial suporte para novas formas de infraestrutura verde. Os espaços intersticiais que produz constituem uma reserva estratégica de solo capaz de contribuir para os objetivos contemporâneos de restauro ecológico, adaptação climática e coesão territorial. Através da integração entre espaço público, infraestrutura verde e Soluções Baseadas na Natureza, torna-se possível transformar territórios historicamente associados à fragmentação em sistemas capazes de promover conectividade, resiliência e qualidade espacial.

Entre as principais limitações identificadas destacam-se a ausência de uma análise aprofundada dos mecanismos de governança necessários à implementação das propostas, a delimitação do estudo ao município de Odivelas e a natureza ainda evolutiva dos métodos de avaliação do desempenho das Soluções Baseadas na Natureza. Nomeadamente, como exemplo, as fichas analíticas foram produzidas externamente, sem a participação das comunidades que habitam estes espaços, cujo conhecimento situado seria fundamental para uma programação verdadeiramente contextualizada das intervenções propostas. Estas limitações apontam para futuras linhas de investigação relacionadas com modelos de gestão intermunicipal, processos participativos de planeamento e aplicação do quadro metodológico desenvolvido a outros contextos metropolitanos nacionais e internacionais.

Palavras-Chave: Espaços Intersticiais Infraestruturais; Soluções Baseadas na Natureza; Infraestruturas Rodoviárias; Infraestrutura Verde; Regeneração Territorial;

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1. Introduction

1.1. The Regeneration of Infrastructural Interstitial Spaces relevance and Contemporary Urgency

Throughout the twentieth century, transport infrastructures - roads, highways, viaducts, and other linear systems - played a central role in shaping modern cities. Driven by a technicist and expansionist logic, these structures were designed to optimise mobility, integrate territories, and support urban growth (Graham & Marvin, 2001; Hauck et al., 2011). Understanding these impacts requires looking beyond the road itself, towards the broader spatial system it produces. As argued by Hauck et al. (2011) and Santos (2018), infrastructures should not be understood as single, simple spaces, but as sets of three-dimensional spatial conditions - composed of the vehicle circulation structure and its structural components, such as easements, viaducts, embankments, and channel spaces - that together constitute what can be defined as infrastructural space. Despite their evident benefits in terms of connectivity and economic dynamics, their implementation has frequently occurred without integration into broader urban planning frameworks, generating significant and persistent spatial, ecological, and morphological impacts on the city and its territory

While transport infrastructures operate across a wide range of territorial contexts - from rural corridors to dense urban centres - their spatial consequences are particularly acute in the consolidated and semi-consolidated peripheries of metropolitan areas, where road-led urbanisation occurred rapidly and without integrated planning, producing a specific condition of urban fragmentation and ecological discontinuity. It is in this context that the present research is situated (Hauck et al., 2011; Santos, 2018).

From an environmental perspective, high-capacity road infrastructures - such as highways and ringroads - have created ecological barriers, fragmented habitats, intensified soil impermeabilisation processes, and suppressed the hydrological systems that regulate water flows (Forman, 2003). Morphologically, the implementation of high-capacity road infrastructure introduced abrupt discontinuities into the urban territory - whether by cutting through existing fabric or by structuring new development in ways that left residual, disarticulated, and often neglected spaces alongside and between the infrastructural domain. These spaces, broadly defined as interstitial spaces (Hauck et al., 2011), belong neither to the road itself, nor to traditional public space, nor to productive private use; they exist in a condition of functional indeterminacy, systematically overlooked by planning instruments and absent from land-use maps, yet occupying vast areas of the metropolitan territory (Solà-Morales, 1995; Hauck et al., 2011; Santos, 2018).

These interstitial spaces are an integral part of urban development while carrying a structural duality: they are simultaneously remnants of an expansionist urban logic and spaces of latent transformative potential. Meaning, their condition of abandonment and functional indeterminacy is, in paradox, also what makes them available for new forms of intervention, fields of potential capable of mediating urban conditions, articulating fragmented fabrics, reintroducing ecologies, and morphologically requalifying the city (Berger, 2006; Vancells Guerin, 2015; Santos, 2018; Silva, 2024). Understanding interstitial spaces not merely as voids but as dormant connectors to be activated has therefore become one of the central challenges of contemporary urban planning, particularly in densely urbanised metropolitan territories where conventional spaces for ecological and public space intervention are increasingly scarce (Carmona, 2010; Silva, 2024).

Even more so in Europe, this challenge has acquired a new and urgent regulatory dimension. Urban ecosystems, to the majority of European citizens, play a fundamental role in regulating water cycles, mitigating urban heat islands, purifying water and air, and supporting biodiversity, functions that are increasingly compromised by the impermeabilisation and ecological fragmentation produced by decades of infrastructure-led urbanisation (EU Nature Restoration Law - Regulamento (UE) 2024/1991). As a consequence of climate change and extreme weather events across Europe, including the severe flooding and heat waves that affected large parts of Portugal in early 2026, directly exposing the consequences of suppressed ecological systems and impermeabilised urban territories, demonstrated that the restoration of these functions is not merely desirable but necessary.

The EU Nature Restoration Law (Regulation EU 2024/1991), which entered into force in 2024, translates this necessity into legally binding obligations. States Members are required to ensure no (net) loss of urban green space and tree canopy cover by 2030, and to achieve an increasing trend in urban green space from 2031, also considering the integration of green space into buildings and infrastructure (Regulamento UE 2024/1991). Portugal, in this case, has committed to presenting the Plano Nacional de Restauro da Natureza (National Restoration Plan - Art.15: Regulamento UE 2024/1991) to the European Commission by the second semester of 2026, a document that must translate these European obligations into concrete territorial measures, which must include restoration measures for urban ecosystems alongside other ecosystem types covered by the law. (Regulamento UE 2024/1991; ICNF, 2024). These frameworks represent a significant reinforcement of existing ecological planning obligations, moving from nationally defined instruments such as: the Reserva Ecológica Nacional (REN) (D. L. nº 166, 2008), a public utility restriction governed by a special territorial regime that establishes a set of constraints on land occupation, use, and development, while defining the uses and activities that are compatible with the objectives of this regime, binding European-level restoration targets with explicit quantitative commitments.

It is in this context that the concepts of Green Infrastructure (GI) and Nature-Based Solutions (NBS) emerge as operative tools capable of transforming infrastructural spaces into vectors of territorial regeneration (Hansen et al., 2017; EU Commission, 2015; Cohen, 2016; UNEP, 2025). Operating within the constraints imposed by road safety legislation and infrastructural easements, NBS proposes working with natural processes and ecological dynamics as the basis for transforming urban space, integrating green and grey, and configuring a more cohesive and ecologically functional urban system (Cohen, 2016; Jones, 2022). Emerging practices in different parts of the world already demonstrate the viability of this approach, showing how infrastructural interstices can be transformed into public space systems at multiple scales.

The regeneration of interstitial spaces associated with transport infrastructure through NBS therefore represents a concrete opportunity to respond to the ecological restoration obligations, transforming zones of discontinuity into vectors of ecological, morphological, and social integration, and aligning urban development with contemporary demands for sustainability, resilience, and quality of life (IUCN, 2016; Browder et al., 2019). It is from this idea, and from a central critique of the technical separation between mobility infrastructure and other urban dimensions, that the present research departs.

The central question that motivates this research is therefore: can transport infrastructure - historically conceived as a barrier to ecological and spatial quality - be reimagined as a carrier of it, through the systematic transformation of its interstitial spaces into green infrastructure?

1.2. Objectives

The research is structured around answering this question and has a primary objective and four specific objectives. The **primary objective** is to discuss how the regeneration of infrastructural interstitial spaces - associated with major highway corridors - through its integration in Green Infrastructure and of Nature-Based Solutions, can contribute to systematic ecological and morphological connectivity in densely urbanised territories. Also in this research - as specific objectives, it is important to: **[1] To understand** how road infrastructural spaces have historically conditioned and continue to affect the spatial, ecological, and morphological configuration of urban territories, examining the mechanisms through which infrastructure generates fragmentation, impermeability, and interstitial conditions at multiple scales. Based on this, it becomes necessary **[2] to examine** how Green Infrastructure and Nature-Based Solutions have been operationalised in international practice in the context of transport infrastructure regeneration, identifying the typologies, scales, and ecosystem services that ground the analytical and propositional framework of this research. This way, it is possible **[3] to develop** an empirical multi-scale territorial diagnosis that documents the spatial, ecological, and urban conditions generated by highway infrastructure in a representative metropolitan context, producing an evidence base that grounds the identification and characterisation of interstitial spaces. **[4] To propose**, then, a method for the application of Nature-Based Solutions to identified interstitial spaces, calibrated to their specific conditions of barrier, ecological fragmentation, and transformative potential, aiming to contribute to a transferable planning approach applicable beyond the specific case studied.

1.3. Methodology

The research follows a qualitative and applied methodology, structured around four sequential phases. The first phase consists of a literature review, examining the theoretical dimensions of transport infrastructure and its spatial, morphological and ecological, consequences; the transformative potential of public space and interstitial conditions; also the operative frameworks of Green Infrastructure and Nature-Based Solutions, establishing the conceptual foundations and analytical ideas that guide the empirical work.

The second phase involves the selection and justification of a case study, whose territorial condition reflects, in a particularly legible and representative way, the spatial contradictions produced by infrastructure-led urbanisation.

The third phase develops a multi-scale territorial diagnosis of Odivelas, reading the territory across three scales - from its position within the Lisbon Metropolitan Area, to the municipal scale, to the individual highway corridor - through complementary analytical layers that document the spatial, ecological, and urban conditions generated by the highway network, and produce an evidence base for the identification and characterisation of interstitial spaces.

The fourth phase develops an application of Nature-Based Solutions - discussed in the first phase - to a representative selection of identified interstitial spaces, tailored to their specific conditions and aligned with the operative dimensions developed in the theoretical framework.

1.4. Structure of the Document

The dissertation is organised in two main parts, comprising seven chapters. The first part - Chapters 2 and 3 - establishes the conceptual and operative foundations of the research: Chapter 2 examines the contradictions of transport infrastructure, the transformative logic of public space, and the operative paradigm of GI and NBS, concluding with a comparative synthesis of international references and typologies; Chapter 3 presents the methodological framework that guides the empirical work. The second part - Chapters 4 and 5 - applies this theoretical basis to a concrete territorial context: Chapter 4 develops the territorial diagnosis of Odivelas across three scales - metropolitan, municipal, and corridor - identifying and characterising the interstitial spaces produced by the highway network; Chapter 5 proposes NBS applied to representative cases selected from the identified typologies, structured around an analytical and operative reading of each space. The conclusions, presented in Chapter 7, synthesise the main findings, discuss the limitations of the research, and outline directions for future investigation.

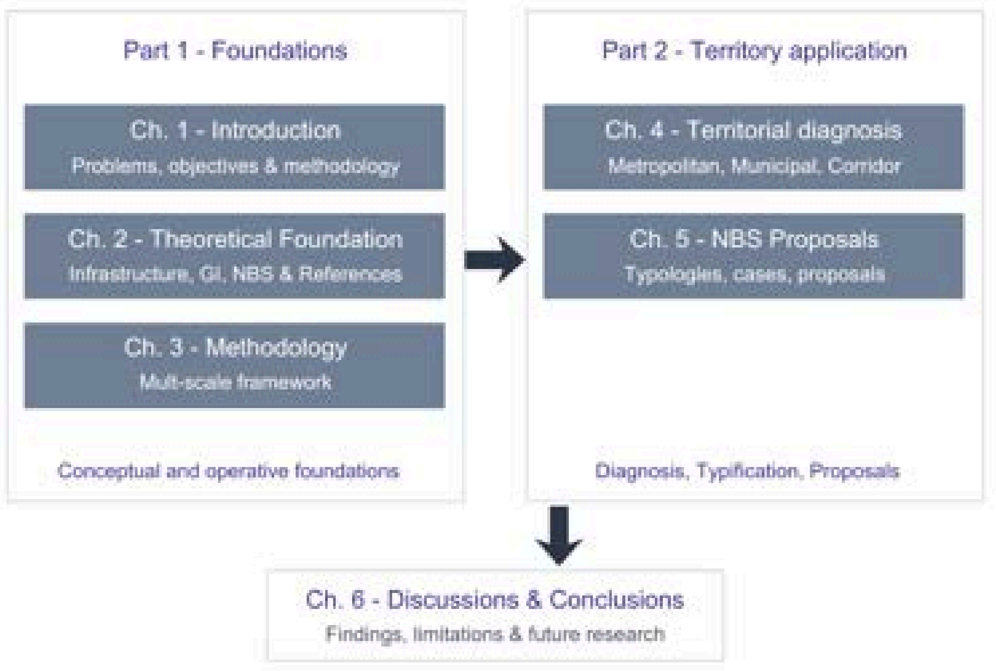


Figure 1. Document Structure Scheme. Source: Own Elaboration.

2. Theoretical Foundation

2.1. Critical dimensions of transport infrastructures - between urban expansion and residual spaces production.

The contemporary urban space, particularly as it developed through the twentieth century, was conceived- an idealization of a continuous and cohesive space - in which spatial relations are expressed by free and functional design of the territory (Solà-Morales, 1997; Graham & Marvin, 2001; Hauck et al., 2011).

However, this “utopian” vision was never fully realized. On the contrary, the contemporary city is increasingly marked by discontinuities, spatial inequalities, and parallel systems of access, a phenomenon that Graham & Marvin (2001) termed as *Splintering Urbanism*. Under these conditions, this means that infrastructure becomes - far from only a neutral support - an active agent of territorial differentiation and morphology, producing and perpetuating zones of inclusion and exclusion, connectivity and isolation, visibility and invisibility (Hauck et al., 2011; Santos & Silva Leite, 2021; Silva, 2024).

Thus, when analyzing the city, the infrastructural systems - such as utilities systems (telecommunications, energy sanitation and others), and specially transport systems - emerges as fundamental objects of analysis, as they reveal a dynamic, complex and technical perspective on the process of spatial occupation. These interactions must be examined across different scales and lenses (Graham & Marvin, 2001; Hauck et al., 2011).

2.1.1. Transport infrastructure essential role for the modern urban development

Linear transport infrastructures like highways, railways, viaducts and mobility corridors in general, are a fundamental part of the spatial configuration of the city, both with positives and negatives aspects. Cities historically developed around infrastructural capacity, with their territorial configuration being a direct reflection of their historical evolution (Newman & Kenworthy, 1999; Neuman e Smith, 2010; Hauck et al., 2011).

Mobility infrastructures, in particular, have long acted as major structuring forces of development. When transport was based on horses and carts, urban spaces were generally more compact and dense, oriented towards pedestrian access (Newman & Kenworthy, 1999; Neuman e Smith, 2010). With the advent of railway lines, urban space significantly expanded, although walking remained a central aspect, development occurred along and around stations, creating types of “sub-cities” (Newman & Kenworthy, 1999), allowing for greater territorial expansion and the emergence of new dynamics (Solà-Morales, 1997; Hauck et al., 2011).

Yet, a fundamental shift occurred largely with the advent of the automobile, especially during the 20th century. Transport infrastructure expanded rapidly in scale and speed of implementation, materialising through high capacity roads that prioritised vehicular flow over urban integration. In this context, mobility structures no longer merely played a structural role of development, but instead embodied a technical logic of flows that responded not only to local demands, but defined: how and where they grew; how they were and are organized; and how they should relate with the surrounding territories. Infrastructure thus became the guiding skeleton of cities' land use occupation, residential density, services disposition, and even - to some extent - the cultural identity/imagery (Lynch, 1964) of neighborhoods. This logic is still being perpetuated until current days (Hauck et al., 2011; Newman & Kenworthy, 1999; Neuman e Smith, 2010; Bernsteiner, 2025).

Moreover, modern cities of the last century were (re)designed for speed, regional scale and uninterrupted continuity. The construction of highways not only connected centers to peripheries, but also opportunities for new centralities that became hubs of commerce, housing and services, rivaling so-called “historic cores” (Newman & Kenworthy, 1999; Neuman e Smith, 2010). From the (by then) new paradigm, physical distance became secondary to travel time as the measure of urban proximity. In other words, a suburb, located 30km from a city center could be more “central” than a congested historical neighbourhood, subverting the traditional centre-periphery relationship, as accessibility by car redefined what it meant to be central (Newman & Kenworthy, 1999; Hauck et al., 2011).

Based on the elemental role of these infrastructures in the development of the urban environment, it is also necessary to reflect on the consequences brought by this logic. The same logic that, grounded in technical efficiency and continuous flow, enabled the expansion and modernization of the city, also generated - as side effects - fragmentation, impermeability and others (Forman, 2003; Neuman & Smith, 2010) . While infrastructure connected vast regions, it fragmented locally, introducing discontinuities into existing urban fabric where it cut through consolidated areas, and conditioning the fragmented pattern of new development where it preceded urbanisation (Graham & Marvin, 2001; Bernsteiner, 2025).

Road infrastructure thus became a true urbanisation agent, determining not only where investments concentrated and which areas were valorised, but also which territories would fall outside the jurisdiction of municipal planning, managed by road authorities according to mobility and safety criteria without necessary consideration for the urban context in which they are inscribed. Yet, in many cases, it was these infrastructures - rather than legal planning instruments - that effectively shaped the city, imposing a self-reinforcing logic of horizontal and fragmented expansion that generates its own demand for further infrastructural investment.

This logic was particularly determinant in the formation of suburban and peri-urban territories, the scale at which road infrastructure operated most freely, without the constraints of existing urban fabric, and where its consequences on spatial organisation, ecological continuity, and social cohesion were most acute.

2.1.2. Infrastructural Spaces: morphological and ecological consequences

Even within this infrastructural logic, transport infrastructures should not be understood solely as technical systems of movement, but as spatial agents that actively produce urban space, even if not the most usual one (Santos & Silva Leite, 2021). As argued by Hauck et al. (2011), infrastructure generates its own spatial condition, giving rise to what can be defined as *infrastructural spaces*: spaces which are shaped, conditioned and/or dominated by the presence and logic of infrastructural systems. These spaces are not limited to the physical footprint of the infrastructure itself, but also include the areas beneath, alongside and between these linear elements, where technical requirements intersect with urban life.

The concentration of development around these infrastructural corridors fundamentally reconfigured the capacity and morphology of suburban and peri-urban territories. Unlike compact urban forms, suburban development patterns became characterized by low-density, fragmented occupation dependent on high-capacity road access (Newman & Kenworthy, 1999; Neuman e Smith, 2010). The urban growth, in these areas, is not continuous anymore, instead it became splintered, that is, a set of uneven discontinuous islands, forming a type of urban archipelago (Graham & Marvin, 2001; Silva & Ma, 2022).

From an ecological perspective, highways operate as hard boundaries that subdivide continuous habitats into smaller, isolated patches, triggering a cascade of interrelated primary effects that collectively degrade the functioning of ecosystems (Forman, 2003; COST 341, 2003). Adapting from Forman (2003), COST 341 (2003) and Van der Ree et al. (2015), these primary effects, strongly interconnected and best understood in conjunction, include five distinct dimensions:

First, **direct habitat loss** occurs due to inherent physical occupation and sealing of soil within the right-of-way, permanently removing land from ecological processes. Second, **road-killing or wildlife mortality**, this being a main research topic when talking about road infrastructure impacts (Forman, 2003; Seiler & Helldin, 2006; Hallegate et al., 2019), as traffic represents a continuous source of lethal risk for species attempting to cross active corridors, whether through direct collision or through increased exposure to predation in the open and uncovered conditions of the road corridor. A third dimension is the generation of **barrier effects**, which operates both physically and behaviourally. This means infrastructure interrupts the movement of species across the landscape, reducing gene flow, limiting access to food and mates, and compromising the functional connectivity between habitat patches (COST 341, 2003; Hallengate et al., 2019). Fourth - and being alongside the third effect the ones more perceptible (Forman, 2003; Hauck et al., 2011) - **edge effects and pollution**, where the margins of infrastructure corridors are exposed to elevated levels of noise, light, chemical contamination, and microclimatic disturbance, conditions that degrade the ecological quality of adjacent habitats well beyond the physical footprint of the road itself. Fifth, **habitat transformation and corridor effects**, whereby the infrastructure corridor itself can paradoxically function as a vector for the spread of invasive alien species and for the transformation of habitat types in ways that alter species composition along its length.

The combination of these primary effects produces a secondary and cumulative outcome, like for example, a barrier effect stimulating the decline of a specific animal population, due to less reproduction, which could lead to biodiversity loss (Klopatek, 1999; COST 341, 2003; Forman, 2003; Van der Ree et al., 2015).

High-capacity road infrastructures impose geometric constraints that exceed their physical footprint. The combination of lane widths, safety easements, embankments, access restrictions, and buffer zones creates effective barrier widths, meaning zones of impermeability, for people and for nature (Forman, 2003). Furthermore, infrastructural spaces are characterized by high levels of soil impermeabilization, which alters natural drainage patterns, concentrates runoff, and increases flood risks in adjacent areas (Lu et al., 2024; Zhou et al. 2022; Xiong et al., 2023). Simultaneously, the accumulation of impervious surfaces (asphalt, concrete) combined with reduced vegetation cover intensifies the Urban Heat Island (UHI) effect along infrastructure corridors (Oke, 1973; Xi et al., 2023; Mirabi e Davis, 2024).

More than simply organizing the space, linear transport infrastructures also exert a systemic pressure on urban and peri-urban ecosystems. Their presence is not limited to the ground level or the immediate landscape; rather, it extends across multiple dimensions - climatic, hydrological, biological, and social - profoundly altering the functioning of the territories they traverse (Seidu, 2024; Van der Ree et al., 2015; Forman, 2003).

Once again, designed to maximize the efficiency of a single flow - whether of cars, cables, or water - these systems are conceived in a rigid and closed manner, excluding any other function or secondary use (Santos, 2018; Bernsteiner, 2025). They can therefore be understood as barriers, that is, spaces impermeable to the contexts in which they are embedded, fragmenting ecosystems, cutting through social networks, and creating both physical and symbolic barriers (Forman, 2003; Berger, 2006; Hauck et al., 2011).

2.1.3. The generation of Interstitial as residuals conditions

Rigid alignments, curves designed for high speeds, elevated or underground structures, and their wide rights-of-way (buffer areas) have contributed - and continue to contribute - to these ruptures: spaces that belong neither to the lane itself, nor to traditional public space, nor even cost effective to the private domain development (Hauck et al., 2011; Santos, 2018). These are the so-called interstitial spaces in some contexts; in others they are referred to as *terrain vague* (Solà-Morales, 1995), “en margen” (Delbaere, 2020), Fuzzy Boundaries (Dal Cin et al., 2025) or infrastructural mediation spaces (Santos, 2018). Despite the different terms, the underlying idea remains the same: neglected spaces with no clear definition, often empty and “inaccessible,” yet presenting themselves as spaces of potential transformation (Hauck et al., 2011)

> *"Dictionary definitions of interstices refer to small, narrow intervening spaces. Although diverse in form, when interstitial spaces are initially captured by urban infrastructures they share common conditions of enclosure, emptiness and abandonment"*
>
> **Wall, 2011, p.146**

The statement discussed by Ed Wall reinforces this notion of interstitial spaces generated by infrastructure. A highway typically uses only a part of its designated surface: the functional part is just the road itself, while the other part of the space is occupied by slopes, verges, embankments, and similar features (Delbaere, 2020). Described as empty, abandoned, or marginal, they are not accidents or failures of planning. Rather, they are direct products of the infrastructural logic that produced them, emerging in different forms and defined by geometries imposed by infrastructural necessity. The result is an urban landscape marked by spatial dualities: on one hand, the “direct and longitudinal” fluidity of the transport network; on the other, the concrete stagnation of the surrounding territories (Vancells Guerin, 2015). These interstitial spaces often remain invisible in official representations of the city - absent from land-use maps, neglected in planning instruments, and lacking clear management frameworks. They constitute “non-places” that escape traditional typologies of public space (Augé, 1995; Berger, 2006).

It is nonetheless important to nuance this understanding by recognising that this interstitial condition is not limited to the spaces immediately and physically generated by infrastructural geometry, be it the embankments, the margins, the voids beneath viaducts, or the technical easements (Forman, 2003; Vancells Guerin, 2015; Santos, 2018). What it means is that the influence of high-capacity infrastructure extends well beyond its physical footprint, producing a broader zone of territorial disruption in which adjacent spaces gradually lose their urban legibility, functional coherence, and spatial integration (Hauck et al., 2011; Dal Cin et al., 2025). Understanding these spaces requires looking into the nature of the boundaries that produce them. With the notion of boundary carrying multiple simultaneous meanings - entry, exclusion, contact, porosity, marginality, and transformation - suggesting that boundaries are never purely physical constructs but complex spatial conditions that operate across material and social dimensions (Dal Cin et al., 2025).

In this sense, a vacant plot, an underused edge, or a residual open space that was never developed - not because of any intrinsic condition, but because the presence of a nearby corridor rendered it inaccessible, undesirable, or simply overlooked by planning instruments - must equally be understood as an infrastructural interstice. Not one produced by geometry, but by consequential effect. This distinction is relevant because it expands the scope of what can and should be considered a space candidate for transformation. Thus the interstitial condition is also a question of the territorial obsolescence that infrastructure, as a barrier, actively produces in the spaces around it (Santos, 2018; Silva, 2024; Dal Cin et al., 2025). This is particularly relevant in the context of infrastructural interstices, where the boundary between the road corridor and the surrounding territory is not a clean line but a zone of friction, ambiguity, and latent exchange. The concept of edge effect further reinforces this reading - liminal spaces and/or spaces of in-betweenness, tend to attract rather than repel human activity precisely because of their threshold condition (Dal Cin et al., 2025).

This difficulty in determining a scale and/or boundary to these spaces is even more truthful if looking from a landscape ecology perspective, in which the infrastructure corridor itself can be understood as a 'macro-interstice' - a continuous, linear space of flow that simultaneously functions as a zone of interruption (Forman, 2003; Van der Ree et al., 2015). Challenging, while also complementing the definition of interstice previously discussed. Just as smaller interstitial spaces emerge as residuals of infrastructural geometry, the highway right-of-way operates at the metropolitan scale as an extensive and continuous in-between (Forman, 2003).

Beyond their shared condition of functional indeterminacy and abandonment, interstitial spaces are not uniform, they manifest in distinct territorial configurations that reflect the specific logic through which each space was produced and marginalized. From this reading, interstitial spaces, in infrastructural spaces, can be understood as residual spatial units produced by infrastructural logic, characterised by functional indeterminacy, spatial marginalisation, and latent transformative potential. Rather than a uniform condition, however, they manifest in three distinct analytical configurations: Barrier Condition, where the infrastructural logic has produced a space of near-total impermeability and disconnection from both the urban fabric and the ecological system; Latent Connectivity, where ecological or morphological potential exists but remains structurally interrupted, present in the territory but unable to function as part of a coherent system; and Functional Indetermination, where the space exists in an unresolved condition between systems, designations, and uses, neither truly urban nor truly natural, carrying transformative potential precisely because of its undefined character. These three conditions are not mutually exclusive - most interstitial spaces present elements of all three simultaneously - but they allow for a more precise reading of what each space is, what it lacks, and what type of intervention it demands.

Paradoxically, however, it is precisely this condition of abandonment, functional indeterminacy, and lack of strict control that opens space for alternative forms of appropriation. In other words, this interstitial condition - of being “in-between” systems, scales, and functions - reveals these spaces, although marginalized, as non-inert: they are fields of potential that should be recognized for their capacity to mediate urban conditions, articulate fragmented fabrics, reintroduce ecologies, and morphologically requalify the city (Berger, 2006; Vancells Guerin, 2015; Santos, 2018; Silva Leite, 2024).

2.2. A potential public space transformation logic: mediation, requalification and articulation

Interstitial spaces, therefore, supported by public space emerge not as a residual category, but as a strategic anchor for systematic territorial transformation (Coelho, 2017; Beja et al., 2024). Central to understanding this transformative potential is the concept of urban porosity - the capacity of an urban environment to offer transitions between systems, spaces, and scales (Wolfrum, 2018; Castel’Branco & Ricardo da Costa, 2024). Porosity, as developed by Secchi and Viganò in their strategic vision for Greater Paris, operates not as an abstract ideal but as an analytical instrument: it is precisely its absence that reveals where the urban fabric has failed: areas where flows of people, water, air, and ecology are interrupted, and where collective life has no material support (Castel'Branco & Ricardo da Costa, 2024). In this sense, high-capacity infrastructure does not merely fragment the city physically, but it systematically erodes its porosity, producing zones of impermeability - relational, ecological, and social - in which interstitial spaces accumulate as the direct expression of that erosion.

This transformation can therefore not be reduced to "filling" voids or "painting them green." Rather, public space must actively reconfigure the relations between mobility, ecology, and collective life, restoring, where and when possible, the porosity that infrastructure has suppressed (Wolfrum, 2018).

In fragmented metropolitan peripheries, especially, where infrastructures have severed local fabrics, this role becomes more critical. Areas in which public space should not merely be an amenity, but an infrastructure for urban interaction (Jacobs, 1992; Carmona, 2019). It can and must operate within fragmented territories, acting as a counterbalance to the 'splintering' effects of high-capacity corridors (Graham & Marvin, 2001). This means understanding public space through three interdependent foundational capacities. First, as a support for *well-being* - the availability of well-designed and accessible urban spaces directly shapes opportunities for leisure, sociability, contact with nature, and emotional balance, acting as tangible indicators of quality of life (MEA, 2005; Carmona, 2019; OECD, 2023). Urban design and land use influence access to employment, health, and education, while public green spaces reduce pollution, strengthen social cohesion, and improve both physical and mental health of residents - transforming highway margins from technical residues into territories of negotiation and collective life (Santos, 2018; Silva & Ma, 2021). Second is understanding public space as a domain defined by *collective ownership*, understood not through land ownership, but through access, use, and shared benefit. Meaning, highway margins, embankments and residual corridors frequently fall under the jurisdiction of national infrastructure agencies rather than municipal planning frameworks - for example, in case of Portugal, such as Infraestruturas de Portugal (IP) and concessions to Brisa and Ascendi - creating legal and administrative tensions for integrated intervention (Hauck et al., 2011). However, public space in fragmented territories should be understood as a domain that enables access, multifunctionality and connection, qualities derived - not from legal status - but from design, governance, and social use (Gaffikin et al., 2010; Coelho, 2017).

To fulfill its transformative potential, public space must equally operate as a *multiscalar* system, not as isolated pockets, but as a connected network that enables ecological continuity, supports active mobility corridors, and fosters cohesion across fragmented urban fabrics (Beja et al., 2024; Santos, 2018). A park, a square, or a linear corridor only achieves its desired impacts when conceived as part of a broader pattern of morphological and ecological connectivity (Coelho, 2017). Connectivity here is not merely physical continuity, but functional integration, meaning the capacity of a space to facilitate flows of people, water, air, and biodiversity, while accommodating collective use. This requires a fundamental shift in planning logic, in which infrastructure is not solely a matter of transport, and the park is not solely an issue of environment.

Building on these ideas, this transformation must seek to prioritize neighbourhoods' connection and cohesion, serving as a goal for multi-functional projects aimed at providing urban structure (Santos, 2018; Vancells Guerin, 2015), promoting urban vitality and social interaction, especially in economically disadvantaged areas and spaces surrounding new local facilities (Gaffikin et al., 2010; Beja et al., 2024). Ultimately, these three foundational capacities - well-being, ownership, and multiscalarity - require operational mechanisms to be realized within the specific condition of infrastructural interstices. Therefore, the following sections propose a public space transformation through three interdependent operational dimensions of Mediation, Requalification, and Articulation, not as prescriptive categories, but as analytical lenses through which spaces can be read, understood, and ultimately transformed.

2.2.1. Mediation: Public space as urban interface.

Mediation here is proposed the capacity of public space to negotiate conflicts, transitions and discontinuities produced by infrastructure (Santos, 2018; Silva & Leite, 2024). In a context of infrastructural interstices, mediation is not merely about physical adjacency, but about integrating the space with its own utilization and the systems it affects. A mediated interstitial space functions as critical interfaces between neighbourhoods, ecosystems, and mobility networks. They can be understood not as “voids,” but rather as zones of friction and potential, precisely because of their condition of “non-belonging,” where the city can be “stitched together” (Hauck et al., 2011; Vancells Guerin, 2015).

In practice, mediation implies navigating tensions between competing systems and scales. A space beneath a viaduct, for instance, must simultaneously negotiate the regional flow logic of the highway above, the local pedestrian needs of the neighbourhood beside it, and the ecological dynamics of the drainage system running through it. These are not merely physical conflicts, but conflicts of jurisdiction, function, and meaning, and it is precisely the capacity to hold and negotiate these tensions, rather than resolve them through a single dominant logic, that defines a truly mediating space (Santos, 2018; Hauck et al., 2011; Dal Cin et al., 2025).

Essentially, a mediation space must address discontinuities generated by infrastructure, be it social, neighbourhoods, services, employment hubs, ecological activities. That is, for example, the spaces between, beneath, and around high-speed roads constitute zones of friction between scales (regional/metropolitan/local), between functions (residential/ transport/ ecological), and between flows (vehicles/ pedestrians/ water/ heat). The transformation of interstitial spaces into public space is not merely a question of physical requalification, but of creating conditions for shared use in contexts of spatial and social fragmentation, ensuring it integrates not just with other structures, but with the space itself and its utilization.

Mediation, in this sense, is not one dimension among three equals, but the analytical starting point from which the subsequent operative logics of requalification and articulation derive their direction and purpose (Beja et al., 2024; Vancells Guerin, 2015).

2.2.2. Requalification: from technical residues to an active multifunctionality

If mediation defines the relational role of space, its requalification defines the possible operational content of public space. It is not about “beautifying” the void with benches, new pavements, and trees, but rather about converting the void into a multifunctional asset. It is important to distinguish from regeneration, as here, requalification stands as an operative framework, while regeneration is established as a more broad theoretical aim (UN-Habitat, 2022). This means Requalification addresses the capacity of the public space to transform, adapt and function in different uses, directly affecting the quality of the space, and consequently, the quality of life of itself and of its surroundings (Carmona, 2010; Vancells Guerin, 2015).

Thus, the possibilities for intervention within these structures manifest in different forms and typologies. These interstitial spaces can and should act as catalysts, being understood in planning as spaces of transformation - parks, ecological corridors, collective facilities, and soft mobility infrastructures - where public space becomes a site of material and functional transformation, turning what was once a residual condition into something active and responsive (Santos & Silva Leite, 2024). What defines requalification is not the specific typology chosen, but the operative logic behind it: transforming a space conceived for a single, closed function into one that is open, adaptive, and capable of responding to multiple simultaneous demands.

This is where multifunctionality becomes the defining quality standard of requalification. A requalified interstitial space should deliberately integrate multiple layers of function into a single spatial solution, a vegetated embankment is simultaneously a noise buffer and a stormwater retention surface, a shaded pedestrian path is simultaneously a mobility corridor and a heat mitigation strategy. An interstitial space below a highway used merely as parking, for instance, fails precisely because it reduces a space of potential multiplicity to a single technical use, lacking accessibility, legibility, and the capacity to accommodate diverse needs.

This capacity to perform across ecological, social, and morphological dimensions simultaneously is not a bonus, but the essential characteristic against which any intervention should be measured (Hansen et al., 2017; Browder et al., 2019; Jones et al., 2022). Requalification, then, is intrinsically linked to mediation and articulation. How the space should be requalified depends on what systems and conflicts it mediates, but also on how and with what structures it connects.

2.2.3. Articulation: the public space as a metropolitan network.

Finally, interconnected with the other two aspects and following its multi-scalar foundation, articulation defines the capacity of public space to position local interventions within wider territorial networks. This means an interstitial public space only fulfills its potential when it moves beyond the local scale and becomes connected in a bigger scale (Beja et al., 2024).

Interstitial spaces, from a morphological point of view, exist as fragments, as isolated remnants of infrastructural logic that, by itself, hold transformative power (Vancells Guerin, 2015; Santos, 2018; Santos & Silva Leite, 2021). However, when understood as potential nodes within a broader network, these spaces gain a strategic significance. Articulation, in this case, is the process of connecting these fragments into a coherent system that transcends their residual conditions. Imagining a metropolitan network of public space means envisioning territorial integration capable of overcoming fragmentation, reorganizing urban design through a connected framework.

Articulation operates primarily at the question of scale and infrastructural connectivity, not in an administrative metropolitan sense, but in the sense of establishing meaningful connections with other structures, whether a river, an ecological corridor, a railway, or another mobility network. Where mediation negotiates conflicts between adjacent systems, articulation extends the reach of interstitial space beyond its immediate context, transforming each fragment into a potential link within a broader chain of connectivity (Vancells Guerin, 2015; Beja et al., 2024). In this sense, articulation is not merely about connecting parks, but about using interstitial public space to restore pedestrian continuity and ecological coherence across a fractured landscape, ensuring that local interventions contribute to a systemic territorial logic rather than remaining isolated gestures.

Hence, interstices must function as “nodes”, corridors and matrix of intervention. This requires recognizing that, when articulated, interstices stop being “non-places” (Augé, 1995), but the basis for a new metropolitan configuration, where grey infrastructure and green infrastructure are complementary (Forman, 2003; Benedict, 2006; Browder et al., 2022). A potential linear park along a highway shoulder, for example, is not just a local amenity, it can simultaneously manage stormwater, mitigate noise, provide shade and serve as pedestrian/cycling corridors in a broader framework, being both multifunctional and multiscalar and working as a living infrastructure (Forman, 2003; Hauck et al., 2011; Beja et al., 2024)

Together, these three dimensions - mediation, requalification, and articulation - do not operate as isolated categories but as an integrated and mutually dependent system. A space that mediates without being requalified remains a zone of friction without resolution; one that is requalified without articulation risks becoming an isolated intervention disconnected from the broader territorial logic it should serve. Only when conceived and implemented in conjunction do these dimensions fulfill the transformative potential of interstitial spaces, converting what were once barriers into connective tissue of a more cohesive, resilient, and ecologically sensitive metropolitan fabric. This redefinition of public space - not as mere backdrop, but as collective infrastructure - nevertheless requires a framework capable of materializing these ideals, leading to the integration of Green Infrastructures and Nature-Based Solutions.

2.3: A new paradigm: The integration of transport in Green Infrastructures and Nature-Based Solutions

As previously noted, contemporary urban transformations, associated with the expansion of infrastructural spaces, result in significant impacts on land use and environmental dynamics, including the exacerbation of climate change, increased noise pollution, soil sealing, and biodiversity loss (Cohen et al., 2016; EU Urban Agenda, 2018; Jones et al., 2022).

Recognizing the interstitial spaces of highways as potential public spaces that go beyond mere "green rehabilitation," the restoration of linear transport infrastructures and their integration with Nature-Based Solutions constitute a strategic approach for more resilient and sustainable cities (Newman & Kenworthy, 1999; Hansen et al., 2017; Browder et al., 2019, Silva & Ma, 2022). From this perspective, public space ceases to be a passive "setting" and is understood as living infrastructure - a multifunctional support operating simultaneously across three dimensions: ecological, social, and morphological (Cohen et al., 2019). Thus, the integration of green and grey infrastructures not only optimizes technical performance but also enhances social and ecological benefits (Seidu, 2025; Browder et al., 2019).

2.3.1. The concepts of GI and NBS

In current debates on resilience and sustainability, alternatives such as Nature-Based Solutions (NBS) and Green Infrastructure (GI) have emerged, proposing integrated and multifunctional systems capable of combining environmental management, climate mitigation, and social benefits, thus overcoming the limitations of traditional grey infrastructures (Jones, 2022; Kabisch et al., 2016; Hansen, 2017).

The term Green Infrastructure, in specific, has gained increasing prominence in recent years, both in spatial planning, public policy, and scientific research (Naylor, 2017; EU Urban Agenda, 2018).

Aligning with Europe Green Infrastructure Strategy (EU Commission, 2013) and Magalhães (2013), green infrastructure can be understood as a broader ecological structure composed of natural and semi-natural areas, as well as artificial elements, functioning as a multifunctional network across urban, rural, coastal, and marine contexts. This network includes elements such as parks, forest reserves, wetlands, hedgerows, as well as infrastructures like ecoducts and cycle paths, collectively contributing to ecosystem health and resilience while enhancing ecosystem services and human well-being (Magalhães, 2013). In urban contexts - more specifically - green infrastructure is understood as an urban planning strategy focused on developing an interconnected system that includes green and blue spaces - such as parks, street trees, rivers, green roofs, and green walls - designed and managed to provide ecosystem services through Nature-Based Solutions (EU Urban Agenda, 2018; Jones, 2022; Hansen et al., 2017).

Aligned with this idea, Hansen et al. (2017) define four principles for green infrastructure: [1] a connected cohesive green system, [2] multifunctionality, [3] integration of green and grey infrastructure, and [4] the promotion of community participation with social inclusion. These interventions aim to mitigate climate change, protect biodiversity, promote a green economy, and enhance social cohesion (Hansen et al., 2017).

Expanding on these objectives, central to understanding the multiple benefits of green infrastructure is the concept of ecosystem services, the benefits that functioning ecosystems provide to human societies, broadly organized into four interconnected categories (Contanza et al., 1997, 2017; Haines-Young, 2023; EU Commission, 2015): provisioning services, related to the direct supply of goods such as food, fresh water, and raw materials; regulating services, which include climate regulation, flood control, air purification, and carbon sequestration; cultural services, encompassing recreation, mental well-being, aesthetic experience, and sense of place; and supporting services, such as nutrient cycling, soil formation, and habitat provision, which constitute the ecological foundation upon which all other services depend. When properly planned and connected, these systems generate multiple benefits across all these domains simultaneously (Magalhães, 2013; Van der Ree et al., 2015). In terms of biodiversity and species protection, green infrastructure supports habitat creation, ecological connectivity, and species permeability. Regarding climate change, it plays a dual role in both mitigation and adaptation by reducing urban heat island effects through evapotranspiration, shading, reinforcing ecosystem resilience, improving stormwater management, and enabling carbon sequestration. It further supports food production and security by preserving agricultural soils, promoting nutrient cycles, and preventing soil erosion. At the social level, it provides spaces for recreation, improves air quality, and strengthens the relationship between communities and natural environments. Economically, it can increase property values and highlight the value of ecosystem services, while also contributing to local identity, education, social interaction, and tourism (Magalhães, 2013; Haines-Young, 2023).

In this sense, a crucial aspect for GI implementation in practice is through the use of Nature-Based Solutions (NBS) (EU Commission, 2013). The concept of Nature-Based Solutions is relatively recent and seeks to encompass the possibility of using ecosystem services instead of conventional engineering solutions (Cohen, 2016; IISD Report, 2022). This concept emerges as an integrated response to the ecological and social crises brought by the increasing number of grey infrastructure in the 21st century (see section 2.1), proposing the active use of natural processes and structures as part of planning and infrastructural solutions (Cohen et al., 2016). According to the IUCN (2016), NBS are actions aimed at protecting, managing, and sustainably restoring natural or modified ecosystems. At the same time, they should provide benefits for both human well-being and biodiversity (EU Urban Agenda, 2018; Jones, 2022; Cohen et al., 2016, UNEP, 2025).

It is also worth pointing out that the IUCN (Cohen et al., 2016) defines several principles for Nature-Based Solutions (NBS). These are: [1] Nature conservation; [2] Integration with other solutions (they can be combined with grey infrastructures); [3] Adaptation to the local context; [4] Equity and participation; [5] Respect for ecosystem diversity and dynamics; [6] Implementation at a territorial scale; [7] Management of trade-offs; and [8] Integration into public policies: to achieve scale and lasting impact, NBS must be anchored in institutional plans, regulations, and strategies.

Furthermore, the operationalization of NBS has been supported by the development of practical catalogues and typologies that classify interventions according to their functions, benefits, and sustainability considerations. The most widely referenced is the World Bank (2021) catalogue, which organizes NBS into typologies based on the urban challenges they address: from stormwater management and heat mitigation to biodiversity support and social cohesion. Still, other websites such as Urban Nature Atlas and Urbinat Catalogue also heavily contribute in demonstrating the types of interventions around the world. These catalogues are particularly relevant for planning contexts, as they provide a structured framework for matching territorial diagnostic conditions with appropriate nature-based interventions, translating broad ecological principles into concrete and measurable spatial solutions.

Therefore, these new strategic tools, NBS and green infrastructures, emphasize not “isolated green solutions,” but integrated responses to contemporary urban challenges (Loupa-Ramos, 2026). However, it is necessary to understand how this can be effectively implemented, aligning directly with the proposed logic of a mediated, requalified, and interconnected public space.

2.3.2. A systematic integration of green infrastructure and interstitial spaces

As a starting point for understanding how green infrastructures and Nature-Based Solutions can serve as potential agents for occupying and transforming interstitial spaces, filling the “in-between,” the term grey-green infrastructure becomes increasingly relevant. It refers to the process of “greening” an infrastructure that cannot be removed, which can only be transformed (Naylor et al., 2017; Browder et al., 2019; Seidu et al., 2024)

The integration of grey and green infrastructure is imposed not as an aesthetic choice, but as a systemic necessity. This perspective is reinforced by studies that show how these interstitial suburban spaces support ecological functions, often ignored in traditional planning practices (Silva, 2024; Silva & Ma, 2022; Forman, 2003; Klopatek, 1999). Therefore, multifunctionality - a central characteristic of Nature-Based Solutions (NBS) - allows these voids (and consequently the infrastructures themselves) to be reinterpreted as living infrastructures, capable of regulating climate and mitigating heat islands (Seidu et al., 2024; Xi et al., 2023), managing stormwater (Xiong et al., 2021; Yang e Zhang, 2024), promoting biodiversity, and simultaneously accommodating collective uses: pedestrian paths, bike lanes, social spaces, and connections between neighbourhoods (Browder et al., 2019). However, this integration is only effective when guided by a logic of territorial Connectivity, one of the principles of GI planning (Hansen et al., 2017). That is, interstitial spaces, due to their intrinsic position between technical networks and urban fabrics, can cease to function as barriers and instead act as ecological and morphological stitching, especially when embedded within metropolitan networks of public space, green corridors, and soft mobility systems.

This means that a bike path shaded by native trees is not simply a “path plus trees”, but a grey-green system where mobility infrastructure and ecological elements mutually reinforce each other, creating simultaneously a corridor for cooling, biodiversity, and active transport. It means that a strip of native vegetation along a highway margin is not simply a visual buffer, but a living system that gradually restores ecological connectivity across fragmented urban fabric. It means that a green corridor connecting two disconnected neighbourhoods is not merely a “green linear park”, but a sustainable infrastructure that continues to perform - filtering air, managing water, supporting biodiversity - independently of active human management. Most importantly, it means that the quality of metropolitan public space should be measured not only by social usability, but also by its ecological capacity and climate resilience. In this way, NBS are not a 'complementary theme' to public space, they represent the new operative foundation of public space in the contemporary metropolis (Kabisch et al., 2016; Coppola et al., 2024; UNEP, 2024).

This systemic integration proves particularly significant in territories peri-urban areas, where the coexistence of dense urban fabric, high-capacity infrastructure corridors, and ecological vulnerability creates both the urgency and the opportunity for a grey-green approach. Here, the transformation of interstitial spaces through NBS is not merely a planning option, but a strategic necessity - one that will be further explored in the following section through the specific contradictions and potentials of a selected study area (Odivelas, Portugal) territorial condition.

For doing this, this study adapts from the World Bank catalogue of Nature-Based Solutions (2021) and Naylor et al. (2017), which provides a structured framework for translating this systemic logic into concrete spatial interventions. Organizing these interventions by scale of application and physical location within the infrastructural space, with the following typologies illustrating a range of NBS families applicable to highway, railway and canal contexts - from metropolitan-scale forest corridors to local bioretention systems - demonstrating that grey-green integration operates not as a single gesture, but as responses calibrated to specific territorial conditions.

NBS family	Scale	Where it applies	Key functions
Metropolitan / territorial			
Urban Forests	Metropolitan	Highway embankments, slopes alongside active corridors	Heat stress reduction. Biodiversity support. Air quality improvement. Carbon sequestration.
Green Corridors	Metropolitan	Linear margins alongside roads, railways and canals	Biodiversity support. Heat stress reduction. Social cohesion
River and Stream Renaturation	Metropolitan	Valley floors where highway crosses watercourse	Flood risk reduction. Biodiversity support. Water quality improvement
River Floodplains	Metropolitan	Low lying margins adjacent to highway / watercourse	Flood risk reduction. Biodiversity support. Water quality improvement
Municipal / district			
Terraces and Slopes	Municipal	Embankments, cut slopes and retaining walls alongside highways and railways	Flood risk reduction. Biodiversity support. Heat stress reduction.
Open Green Spaces	Municipal	Residual interchange areas, covered highway platforms	Heat stress reduction. Social cohesion. Biodiversity support.
Natural Inland Wetlands	Municipal	Low-lying valley margins where highway intersects drainage basin	Flood risk reduction. Water quality improvement. Biodiversity support
Constructed Inland Wetlands	Municipal	Residual node areas, low ground inaccessible interchange surfaces	Flood risk reduction. Water quality improvement. Social cohesion.
Urban Farming	Municipal	Accessible residual surfaces, rooftops of highway structures	Social cohesion. Heat stress reduction. Carbon sequestration
Green Crossings	Municipal	Existing Crossings and disconnected areas	Biodiversity support. Social Cohesion. Air quality improvement
Local / site			
Bioretention Areas	Local	Highway margins, drainage channels alongside road platforms	Flood risk reduction. Water quality improvement. Heat stress reduction.
Building Solutions *	Local	Green Viaduct columns, elevated structure green walls, highway green rooftops	Heat stress reduction. Air quality improvement. Biodiversity support

Source: Adapted from World Bank (2021) – A Catalogue of Nature-Based Solutions for Urban Resilience and Naylor et al. (2017)

[illegible]

Frame 1. NBS Families. (Source: Own Elaboration)

Which NBS families are applicable - whether urban forests and green corridors along active margins, river renaturation in valley corridors, or open green spaces on residual spaces- depends, however, on the type of action taken over the infrastructure itself: whether it can be removed, reconverted, or only activated at its margins. It is this distinction that structures the following section.

2.4. From Barriers to Living Infrastructure: An Analytical and Instrumental Overview of Grey-Green Integration

2.4.1. A spectrum of Projectual Approaches: Removal, Reconversion and Activation

The occupation and regeneration of interstitial spaces associated with transport infrastructure is not a novel proposition, projects such as the Management of Abandoned Areas in Montpellier [Figure 02], the Sistema de Parques de Sant Cugat del Vallès [Figure 03], Stalled Spaces in Glasgow [Figure 04] and Esto no es un Solar in Zaragoza [Figure 05] demonstrate this across different scales and planning cultures. However, when these spaces are specifically associated with major transport infrastructure, their transformation demands a more complex and systemic approach, one that the following three categories seek to organise. A critical reading of international practices reveals a spectrum of approaches that differ fundamentally in their actions over the infrastructure, its scale of intervention, and the type of NBS mobilised. From this, three broad categories of transformation can be identified

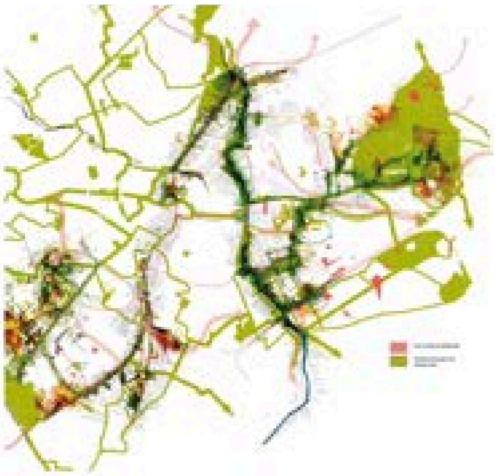


Figure 02. Montpellier Strategie
Source: Clément e Coloco (2010)

Figure 03. Sant Cugat del Vallès park system.
Source: Batlleiroig Arquitectura (1993 - 1998)

Figure 04. ParkHead Community Garden. Stalled Places
Source: Glascow City Council Local Spaces

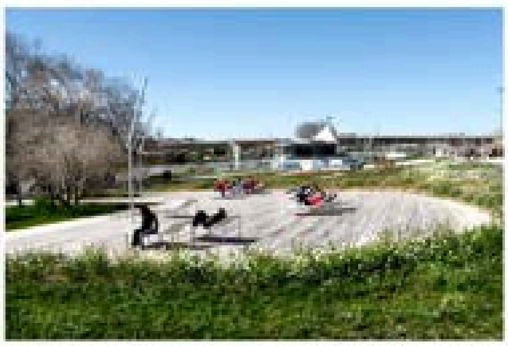


Figure 05. Park Vandorey. Esto no Es un Solar.
Source: Gravalos & Di Monte (2009)

The first one is removal, which refers to the partial or total demolition of existing infrastructure, liberating urban land for ecological and public space regeneration. The Cheonggyecheon stream restoration in Seou [Figures 06 & 07] is the paradigmatic example, a complete demolition of an elevated expressway that restored a buried watercourse as a linear ecological corridor, mobilising river renaturation, wetlands, and open green spaces. Removal represents the most radical form of grey-green transformation, but presupposes a pre-established change in the metropolitan mobility system, meaning it is only feasible when the infrastructure's function can be redistributed or is no longer needed, conditions that are difficult to achieve in extensive and polycentric metropolitan areas where infrastructures remain essential for daily use.

Figure 06. Cheonggyecheon before intervention.
Source: Harvard Graduate School of Design (2023).

Figure 07. Cheonggyecheon after intervention.
Source: Harvard Graduate School of Design (2023).

A second approach category is reconversion, this the transformation of infrastructure function while retaining its physical structure, a road reprogrammed, an elevated corridor becoming a pedestrian green spine, a disused viaduct becoming a linear park. The High Line in New York, the Goods Line in Australia [Figures 08 & 09], and the Tokyo KK Line proposal are the exemplary references, demonstrating how reconversion mobilises green corridors and open green spaces integrated into the retained structure. In active metropolitan contexts, these actions may be more feasible than total removal, but still depends on prior infrastructural decisions.

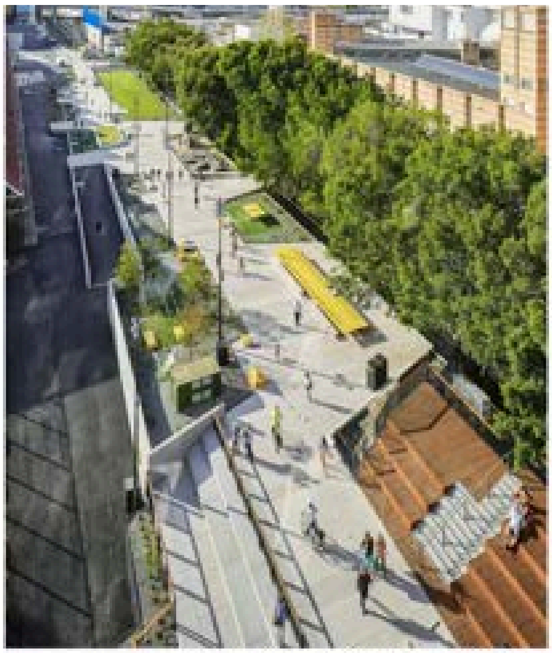


Figure 08. The Goods Line, Sydney.
Source: ASPECT Studio (2015)

Figure 09. High Line, New York.
Source: @Timothy Schenck

Some projects extend this logic through downgrading [Figures 10-11], by transforming active highways into urban boulevards, as proposed for the Bonaventure Expressway in Montreal and the A35 in Strasbourg, reducing speed, lane capacity, and barrier effect while introducing green corridors and ecological continuity. As with removal, reconversion depends on prior infrastructural decisions and significant institutional commitment.

Figure 10. Bonaventure Expressway, Montreal.
Source: L'Institut Paris Region (2021)

Figure 11. A35, Strasbourg.
Source: Agence Ter (2019)

The third category - and the focus of this study - is marginal/interstitial activation. This means the transformation of the interstitial spaces generated by active, retained infrastructure, without altering the road platform themselves. Rather than transforming infrastructure, these strategies operate through spatial, ecological and functional augmentation of its margins, voids and intersections, revealing the latent capacity of linear systems to host Nature-Based Solutions.

This means that it is within this third category that Nature-Based Solutions find their most immediate operative actions. More than constituting a single type of intervention, NBS applicable to highway margins must be organised by the type of infrastructural situation they address, that is, by the spatial typologies, scale, type of intervention and ecological effects condition.

2.4.2. Emerging Practices and References

Highway Green Systems
Likoto Linear Forest - Lille-Kortrijk-Tournai, Belgium and France

Figure 12. LIKOTO Linear Forestry.
Source: Likoto Research Project. Hypotheses

The Likoto Linear Forest [Figure 12], developed by Denis Delbaere, is a transnational project between the cities of Lille, Kortrijk, Tournai (Lille-Kortijk-Tournai Eurometropolis) - hence the name Likoto - is one of the most methodologically relevant references for this research - not because it is an interventive project, but by being a project focused on conservation and maintenance. Field research demonstrated that highway margins simultaneously function as biodiversity reservoirs, informal social spaces, and latent climate regulation systems. Likoto proposes to consolidate and manage these spontaneous linear woodlands as a metropolitan ecological corridor, not through construction, but through a shift in maintenance strategies. Delbaere terms this a métaprojet: a landscape project that nobody designed, but that infrastructure itself produced (Delbaere, 2020)

A16 Highway - Rotterdam, Netherlands

Figure 13. A16 Groene Boog.
Source: H+N+S Landscape Architects (2018–2024)

A contrasting position is occupied by the A16 Highway in Rotterdam [Figure 13] among these references, because it does not address an existing highway but a new one. Rather than transformation, it represents an attempt to integrate green and ecological thinking into new grey infrastructure from the outset. The proposal works with its margins and crossing points - introducing a metropolitan green and blue network that extends ecological and active mobility corridors from existing parks into disconnected neighbourhoods, minimising barrier effects through new crossings and reactivating residual land alongside the highway. But it also simultaneously acknowledges a contradiction: new highway construction generates new interstitial spaces and fragmentation, however carefully mitigated. For this reason, the 2024 ateliers of the International Architecture Biennale Rotterdam further explore its interstitial spaces and integration potential. In the sense of this research, the A16 serves as a case to understand the limits of green-grey integration - demonstrating both the potential and the boundaries of it when applied to infrastructure expansion rather than regeneration.

De Groene Singel - Antwerp, Netherlands

Where the A16 represents a limit case of new construction, the Groene Singel (Green Singel) [Figures 14 - 16] in Antwerp addresses a huge infrastructural corridor surrounding the city’s ring road - an ecologically and spatially fragmented space between the inner and outer city, at the time considered not particularly attractive, yet already a habitat for fauna and flora. Rather than a single transformative intervention, the project operates as a masterplan in multiple incremental sub-plans, with different segments of the areas with different contexts and typologies - like South Junction and RUP Ringpark Het Schijn. By focusing on unifying the fragmented space through green, ecological and affordable guidelines for the imagery, public space, ecological features, infrastructure and buildings, the project aims to transform unused barred space surrounding the ring into a new centrality or attraction point, while also improving air quality, reducing noise, enhancing biodiversity and improving water management. The Groene Singel demonstrates exactly that a metropolitan transport infrastructure can be systematically reinterpreted as a green infrastructure.

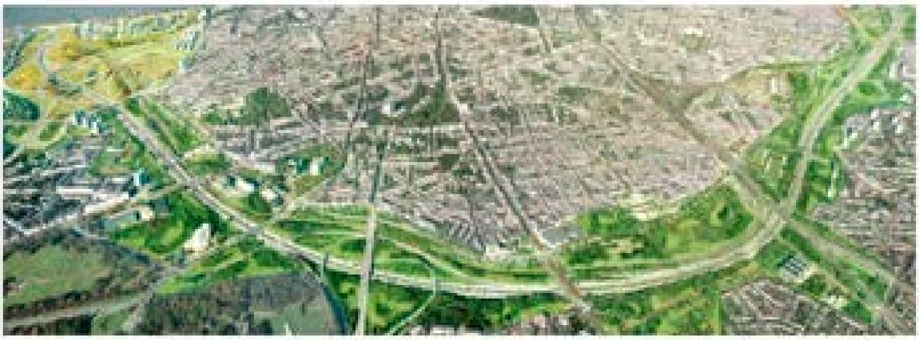


Figure 14. Green Singel. Antwerp.
Source: Karres en Brands et al. (2011–2017)

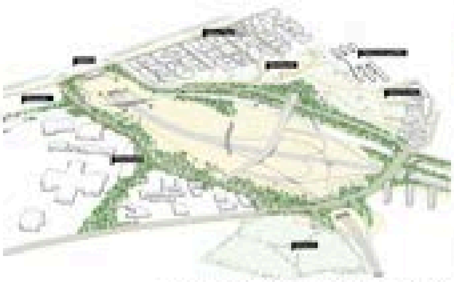


Figure 15. Antwerp South Junction.
Source: Karres en Brands et al. (2018)

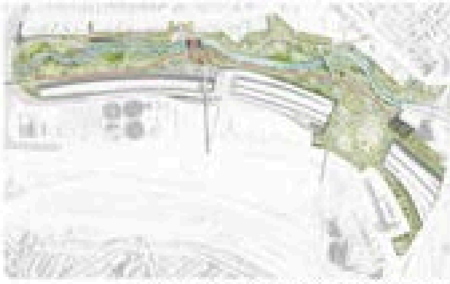


Figure 16. RUP Ringpark Het Schijn.
Source: Stad Antwerpen (2024)

Buffalo Bayou Park & the Greening Houston's Freeway Proposal - Houston, USA

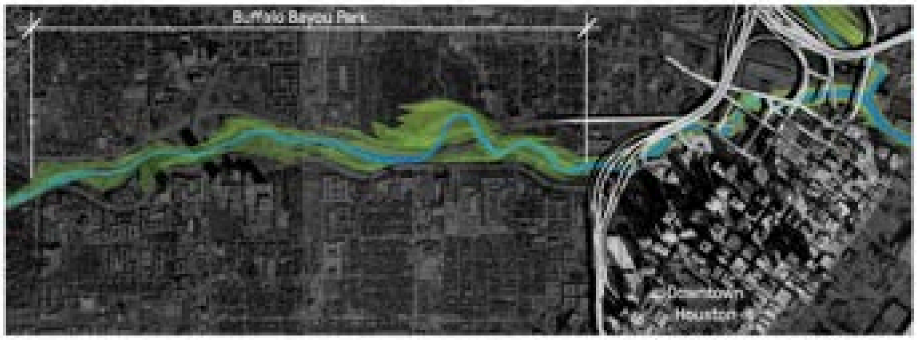


Figure 17. Buffalo Bayou Park, Houston.
Source: Urban Land Institute (2015)

Figure 18. Before. Elevated Freeway.
Source: SWA Group (2015)

Figure 19. After. Elevated Freeway.
Source: SWA Group (2015)

A different dimension emerges where highway infrastructure directly intersects with water and other biophysical aspects. Buffalo Bayou [Figures 17 - 19] regenerates what was once an abandoned and degenerated property space, along and beneath a tangled junction of the city's elevated freeways and bridges. Transforming a flood-prone waterway, neglected beneath the highway urban open space, simultaneously functioning as critical flood infrastructure and public park. This means the park was designed to flood, integrating a previous risk to the area into a feature of green-grey integration, and the previous interstitial condition now carries ecological, climate and promotes neighbourhoods connections. As an example of this, areas for bat colony research and other wildlife monitoring protection and conservation tools were established in the park. Other projects around the world have similar approaches as the Buffalo Bayou Park.

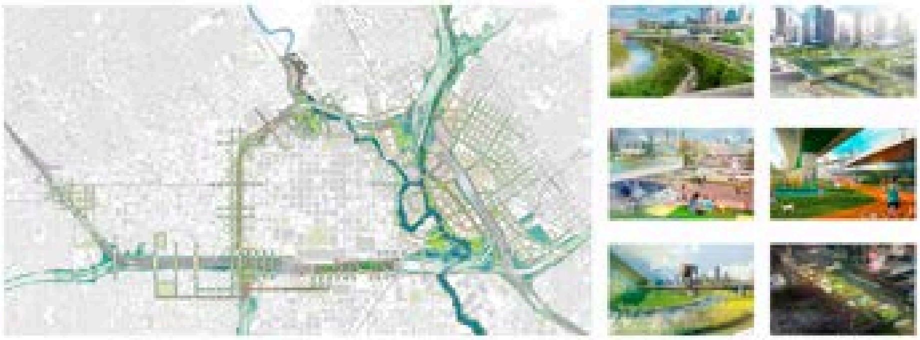


Figure 20. Greening Houston's Highway.
Source: SWA Group

Still in Houston, at a broader scale, the proposal of Greening Houston's Freeways [Figure 20] extends this logic across the entire highway ring of downtown Houston, identifying multiple intervention points distributed across the network, from neighbourhood parks beneath elevated sections to ecological corridors alongside embankments. Together, both projects represent two complementary scales of the same ambition: one demonstrating what is possible at a single infrastructural intersection, the other proposing what becomes possible when that logic is applied systematically across an entire urban highway system.

Local, Punctual and Transversal Interventions

At the same time, the projects discussed so far tend to operate at a more general and continuous scale, taking the highway itself as the main vector of intervention, either by working along it or by attempting to reframe its overall condition. However, there are also projects that engage with infrastructure in a more fragmented or/ and localized way. Instead of addressing the entire corridor, these interventions focus on specific sections of the system - acting on adjacent, residual nodes, underpasses, or crossing points - where targeted actions can reduce impacts and improve ecological and spatial connections.

At the scale of specific nodes and residual spaces, documented interventions are comparatively scarce, a gap that could by itself reflect the underexplored character of these conditions noted earlier. The projects that were looked into this research demonstrate, however, a range of possibilities.

Nu de la Trinitat & Meguro Sky Garden - Barcelona / Tokyo

The Parc del Nu Trinitat, in Barcelona [Figure 21], occupies a central interstice of a complex highway junction, transforming its geometry into a connected park with native vegetation, sports facilities and urban farming spaces for local residents, demonstrating that even the most "technical" residual space carries ecological and social potentials. Also working around these residual spaces - and sort of a more particular project - part of a list of junctions interventions in Ohashi Highways, in Tokyo (Bernsteiner, 2025) - the Meguro Sky Garden [Figure 22] works as a suspended garden inside a junction without much connection with surrounding ecological system, specially due to its urban location. Yet, the project promotes a new social green space with communal spaces and facilities, also native trees and shrubs of different species, which promote biodiversity to a highly urbanized area.

Figure 21. Parc del Nus Trinitat.
Source: @ Antonio Navarro Wijkmark

Figure 22. Meguro Sky Garden. Ohashi Junction.
Source: Meguro City Government (2024)

Ricardo Lara Park - California, USA

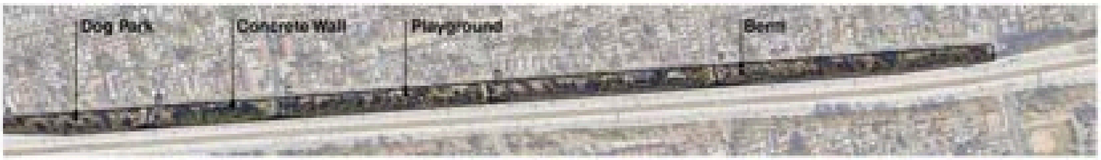


Figure 23. Before. Park Intervention.
Source: Landscape Performance Series

Figure 24. After. Park Intervention.
Source: Landscape Performance Series

The Ricardo Lara Linear Park in Lynwood, California transforms five blocks of residual land alongside an Interstate Highway into a mile-long community park, organized into programmatic segments including recreation, fitness, and community gardening, while integrating stormwater capture from the adjacent elevated freeway [Figure 23 & 24]. Beyond its ecological performance, the park exemplifies how highway-adjacent spaces can become active urban edges that support everyday life in underserved communities, connecting to regional active mobility networks and reducing surface temperatures significantly.

TERRABANK

At a more speculative register, the Terrabank proposal [Figure 25], by Fletcher Studio imagines these same highway embankments as productive live-work landscapes - operative landscapes and programmatically flexible units that metabolize contaminated water, organic waste, and recyclable materials while generating food, shelter, and sustainable revenue. It is nonetheless important to acknowledge that the activation of highway margins and residual spaces through vegetation and food production is not without risk. Current researches points to the dangers of planting and cultivation in close proximity to active road infrastructure, due to the accumulation of heavy metals, particulate matter, and other pollutants in soils and plant tissues adjacent to high-traffic corridors (Petit et al., 2011; Kibblewhite, 2018). This raises questions of the viability of urban farming and edible planting around highways and the need to combine these interventions with processes such as phytoremediation (Guo et al., 2023; Lewandowski et al., 2025). Despite its speculative nature, Terrabank is nonetheless analytically relevant, precisely because it pushes boundaries of marginal activation can mean - not as a build proposal to be replicated, but as a critical instrument for expanding the conceptual horizon of grey-green integration.

Figure 25. TERRABANK
Source: Fletcher Studio

Bicentenário Park & The Bentway

A different set of projects operate focusing more on the transversal aspect, not along the infrastructure but across it, reconnecting what it has severed. These interventions range from the activation of space beneath an existing structure to the most radical act, almost reaching an act of removal: covering the road entirely.

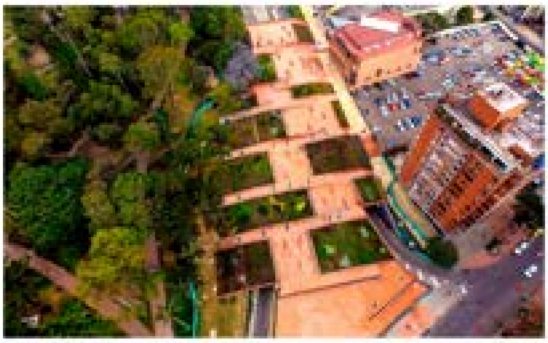


Figure 26. Bicenternario Park, Bogotá.
Source: Grupo Mazzanti. Green Roofs.

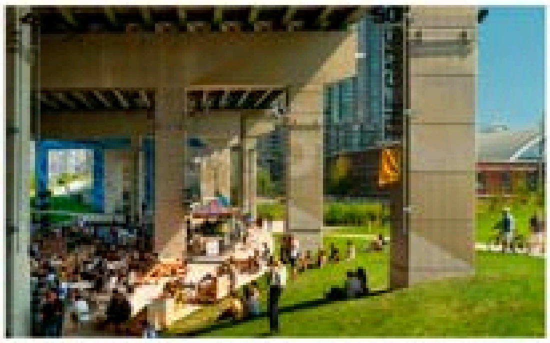


Figure 27. The Bentway, Toronto.
Source: PUBLIC WORK (2018)

At a more contained scale, the Parque Bicentenario in Bogotá [Figure 26] covers the Calle 26 with a public platform integrating native vegetation, pedestrian paths, and cycling connections, reconnecting cultural institutions and public spaces severed by a major urban artery, while also incentivizing the use of native vegetation.

The Bentway in Toronto [Figure 27] follows a different but related logic: rather than covering the road - similar to what was done in Buffalo Bayou Park - it activates the void beneath the expressway as a linear public corridor, integrating pedestrian paths, community programming, and bioretention planters that filter stormwater from the highway above. This approach is not uncommon, with different projects like The Canopy Project in Orlando, USA and Underpass Park in Toronto, Canada.

Wildlife Crossings

Both the Bentway and Parque Bicentenario demonstrate that infrastructure crossings and voids can become active public spaces with ecological character, though their primary dimension remains social and cultural rather than strictly ecological. Another spectrum of interventions is focusing more on ecological performance, like wildlife crossing structures. These crossing structures increase permeability of roads and other linear infrastructure for wildlife by allowing safe crossing under or over roads , reducing risks of wildlife vehicle collisions and restoring ecological activities between fragmented populations, ranging from large ecoducts - wide overspasses usually greater than 50 meters - to small animals crossings [Figure 29] and modified drainage ducts [Figures] (Smith et al., 2015). At the same time, these structures can be multifunctional, with spaces of convergence of human and wildlife crossings depending on the ecosystem. The Natuurbrug Zanderij near Hilversum [Figure 28], in the Netherlands, is one of the iconic interventions when talking about these wildlife crossing interventions, but a lot of different typologies exist internationally. They represent a form of transversal intervention invisible to the urban dweller, but fundamental to ecological function.

Figure 28. Natuurbrug Zanderij Crailoo
Source: Atlas Obscura.

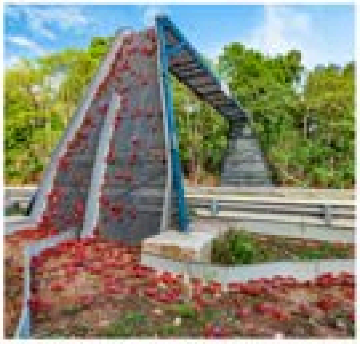


Figure 29. Crab Bridge, Christmas Island
Source: Wondrous World Images

Figure 30. Loop Bridge, Yellowstone.
Source:RtR Report (2026)

A7 Highway - Hamburg, Germany

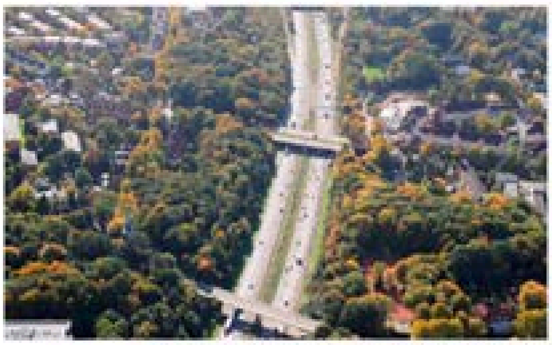


Figure 31. Before. A7 Hamburg Covering
Source: Inhabitat.

Figure 32. After. A7 Hamburg Covering
Source: Inhabitat.

Still on highway transversality and representing a more radical proposition is the Covering of A7 in Hamburg [Figures 31 - 32]. Here, the highway remains fully active below, while its surface becomes continuous urban green space. Around 3,5 kilometers of the highway is covered, and different parks are built in sections of the city, promoting city connectivity, urban farming, and development of native vegetation.

National References: Grey-Green References from Portugal

As a matter of, within the Lisbon Metropolitan Area (AML), a set of recent interventions demonstrates that green-grey integration is not an exclusively northern european practice, but an emerging reality in the Portuguese context as well.

Alta do Lumiar Green Corridor - Lisbon

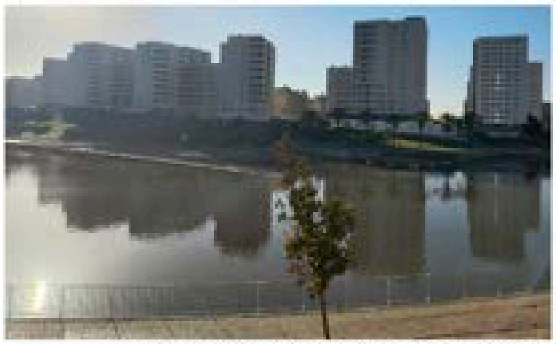


Figure 33. Retention Basin Alta do Lumiar.
Source: Original.

Figure 34. Mixed Green Crossing. Alta do Lumiar.
Source: Original.

The Alta do Lumiar Green Corridor is perhaps the most complete example of this logic currently built in the AML. Developed as a floodable park [Figure 33]with new leisure facilities and retention basins right next to a highway (Eixo Norte-Sul), the project highlights distinct crossing structures over road infrastructure: one seamless connected to the park [Figure 34], designed for shared ecological and pedestrian use, and another pre-existing, primarily focused on wildlife and ecological connectivity.Together, these interventions demonstrate how a single project can balance ecological connectivity, flood risk, and public space activation.

Parque da Bela Vista - Lisbon

Figure 35-37. Mixed Green Crossing. Bela Vista. Lisbon.
Source: Original.

In Parque da Bela Vista [Figures 35 - 37], a park in the middle of Lisbon, integrates community participation through native vegetation planting and the creation of shared leisure spaces, reinforcing social dimension. Beyond this dimension, a green crossing over the avenue allows for the ecological connection and pedestrian access between two green areas.

Ponte Verde - Queluz

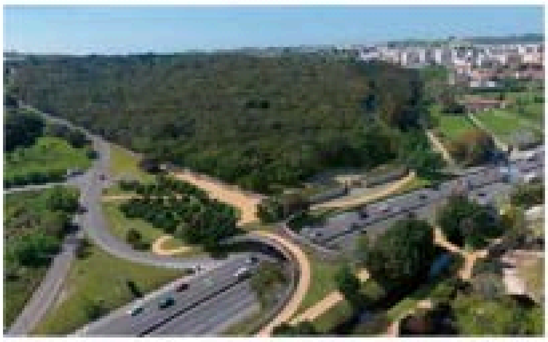


Figure 38. Ponte Verde de Queluz.
Source: União das Freguesias de Queluz e Belas.

Figure 39. Queluz Project. Green Park.
Source: Documentation. 1st place. OASRS

Though still in planning, the Ponte Verde in Queluz addresses the fragmenting condition of the IC19 through a marginal and transversal intervention [Figure 38]. A green and pedestrian crossing that reconnects historically linked spaces on either side of the highway, integrating into the broader Eixo Verde e Azul, a green-blue corridor system of the AML. Aside from the crossing itself, the proposal incorporates retention basins, urban farming areas, natural infrastructure noise barriers, and native species planting in the marginal spaces [Figure 39], demonstrating an ambition that goes well beyond infrastructure mitigation and positions the intervention as a node within a larger ecological and mobility network.

Urban Agriculture and Spontaneous Occupation - Vale de Chelas and Amadora

Figure 40. Vale de Chelas Urban Farm.
Source: Original.

A different and particularly revealing dimension of interstitial occupation emerges when looking at the agricultural use of highway margins and residual spaces within the AML.

In Vale de Chelas [Figure 40], what began as informal occupation, now has been recognized and integrated into a more planned framework, with urban agriculture now operating at a relatively organized scale in the residual land along Av. Santo Condestável, that - even though not a highway - is an arterial road in Lisbon’s context. This transformation - from spontaneous appropriation to planned intervention - is itself an argument: demonstrating that communities had already identified and activated the potential of these spaces long before formal planning instruments caught up (Luz e Pirez, 2015).

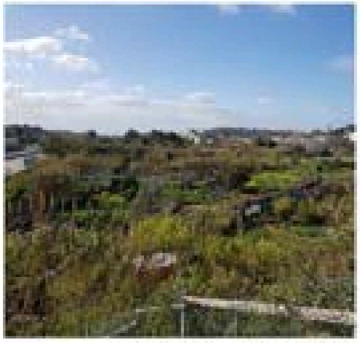


Figure 41. Zona de Hortas. Source: Maia (2018).

Figure 42. Urban Farm Along Highway. Source: SOL Journal.

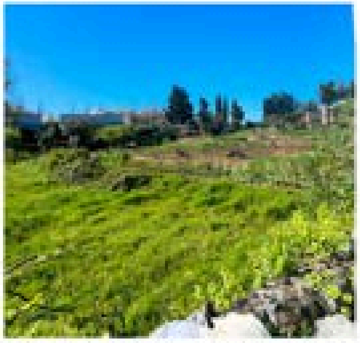


Figure 43. Spontaneous Farm, Buraca. Source:Original.

At a smaller and more fragmented scale, similar dynamics can still be observed in their more spontaneous form in Amadora [Figures 41 - 43], where marginal and residual spaces alongside the IC19 have been informally occupied with vegetable plots and small cultivation areas, occurring mainly by unemployed residents, immigrants, and retired individuals for whom these margins represent both economic resource and social space. The institutional response to this reality is removal rather than integration, with Estradas de Portugal classifying as illegal and citing risks to slope stability and driver safety, while the Câmara Municipal de Amadora acknowledged its social value, being very liberal towards it, but equally defending its careful assessment and communicating with Estradas de Portugal.

This reality is significant precisely because it demonstrates how communities recognize and occupy these interstices, while planning must navigate together with these different needs and concerns. These examples raise, though, the critical discussion earlier in relation to Terrabank: the proximity of cultivation to active road infrastructure carries risks of soil and plant contamination, suggesting that any planned integration of productive uses in these margins must be accompanied by strategies such as phytoremediation strategies and careful ecological assessment. Meaning the potential could be real, but so are the conditions to do it.

2.4.3. A comparative Synthesis: Typologies, Ecosystem Services and Scales of Intervention

Taken together, these references reveal that the transformation of infrastructural spaces into green structures is not utopian, but factual and already happening across different scales, with different degrees of intention, and with different levels of ecological, morphological and social ambition. From the spontaneous forestry management of Likoto to the radical highway covering of Hamburg's A7, from community gardens to ecoducts of the Netherlands, each project demonstrates a distinct operative logic in relation to the infrastructure it addresses.

Reading the projects comparatively, it is possible to identify a range of typologies within the activation category: interventions that work longitudinally along the structure as urban green corridors, such as Ricardo Lara Park and the Groene Singel; those that cap or deck active roads to create new surfaces, such as Parque Bicentenário and the A7 Highway; those that activate underneath elevated structures, such as The Bentway; those that work with hydrological dynamics through wildlife crossings; and those that introduce productive uses through urban farming. Each mobilises different NBS families simultaneously and generates different ecosystem service contributions. A table was established to organise this analysis [APPENDIX 01], allowing typologies, ecosystem services, and scales of intervention to be read simultaneously and comparatively.

It should be noted that the classifications proposed in this table reflect analytical judgements based on the available documentation of each project, and should be read as interpretive instruments rather than definitive attributions.

3. Methodology

In pursuit of the primary objective - understanding how the regeneration of infrastructural interstitial spaces through Green Infrastructure and NBS can contribute to ecological and morphological connectivity - the methodology is organised around three sequential and mutually reinforcing scales of analysis, followed by a propositional phase, contextualized to the territory of Odivelas [Figure 44]. The selection of Odivelas as case study responds to a set of converging criteria: its peripheral position within a metropolitan context as a mid-ring municipality shaped by successive waves of infrastructural investment, the legibility of its fragmentation patterns, the presence of interstitial spaces along high-capacity corridors, and the existence of ecological networks with connectivity potential, criterias that will be further explored in the following sections. What makes Odivelas particularly relevant as a case study is therefore not the exceptionality of its condition, but its representativeness, a territory where the mechanisms of infrastructural urbanism are not hidden beneath layers of subsequent planning, but remain legible and spatially explicit, offering a direct window into the processes that this research seeks to understand and ultimately transform. At a first moment, a metropolitan scale reading contextualises Odivelas within the Lisbon Metropolitan Area, establishing its historical, ecological, and planning position in relation to the broader territorial system.

This is followed by a municipal scale multi-dimensional diagnosis, which reads the territory across three complementary layers - biophysical characterisation, urban structure and land use, and environmental vulnerabilities - using data from multiple sources including DGT LiDAR, Landsat 9, Copernicus Land Monitoring Service, DGADR, and data provided by the Municipality of Odivelas (Câmara Municipal de Odivelas). This dimensional reading is complemented by a landscape metrics (Klopatek, 1999; Leitão et al., 2012) and spatial syntax analysis, combining Fragstats landscape metrics - Edge Density, Cohesion, SHDI - and Space Syntax Analysis through the Place Syntax Toolkit (Hillier, 2007; Ståhle et al., 2004; Stavroulaki et al., 2023), to capture both the ecological fragmentation and the circulatory marginalisation produced by the highway network.

At a third and more granular scale, each major highway corridor is read individually, identifying the specific spatial, ecological, and morphological conditions that characterise its interstitial spaces and establishing the basis for their classification and transformation.

Building on this diagnostic foundation, the propositional phase applies Nature-Based Solutions to a representative selection of interstitial fragments, organised according to their transformative potential and their alignment with the three operative dimensions of mediation, requalification, and articulation developed in the theoretical framework. The final step synthesises these applications into a broader discussion of their implications for planning practice and metropolitan ecological strategy.

Each step builds directly on the outputs of the previous one, ensuring that the identification of interstitial spaces and the subsequent NBS proposals remain grounded in territorial evidence and aligned with the operative framework developed throughout this dissertation. The goal is to identify, classify, and propose a new systematic planning approach for the interstitial spaces along the major highways of Odivelas, through the integration of Nature-Based Solutions.

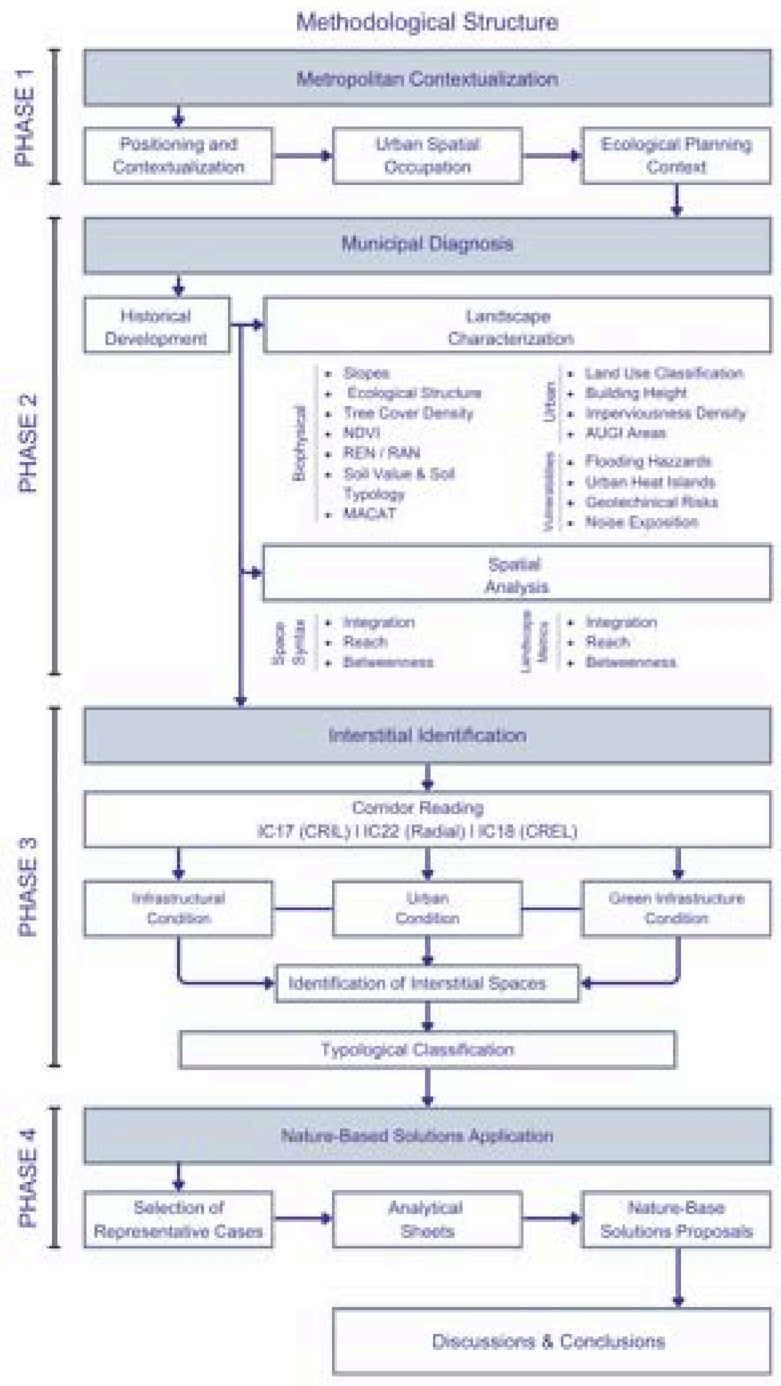


Figure 44. Methodological Structure Proposal. Source: Own Elaboration

4. Case Study: Odivelas

Building on this methodology, the research operates through three sequential scales of analysis - metropolitan, municipal, and corridor - each contributing a distinct layer of evidence that feeds into the subsequent identification and transformation of interstitial spaces.

4.1. Metropolitan Scale

4.1.1. Metropolitan Contextualization and Development

As one of the municipalities Lisbon Metropolitan Area (AML) Located northwest of Lisbon and bounded by Loures (northeast), Amadora (southwest), and Sintra (northwest) [APPENDIX 02], Odivelas it is established at the intersection of almost all of Lisbon's most important infrastructure corridors [Figure 45]: the Cintura Regional Interior de Lisboa (CRIL / IC17), Cintura Regional Exterior de Lisboa (CREL), IC16, and A40. This positioning is not incidental but comes from Portugal's accelerated road-centric urbanisation between the 1970s and 2000s, which prioritised motorway density over cohesive planning (Pereira e Nunes da Silva, 2008; Padeiro, 2018; Afonso et al., 2025).

Figure 45. Major Transport Network. Source: adapted from DGT "Redes de Transporte"

Odivelas is a recent administrative entity, established as an autonomous municipality only in 1998 (Lei n.º 84/1998) following its separation from Loures (Câmara Municipal de Odivelas). This timing is structurally significant: its administrative formation coincides closely with Portugal's motorway expansion and the consolidation of the Lisbon metropolitan periphery (Padeiro, 2018). Previously characterized by a dispersed, agrarian-monastic landscape, from the 1970s onward the territory experienced rapid urban expansion driven by housing pressure due to migration process occurring around the 50-60 decades, but without integrated regional planning, and largely subordinate to the high-capacity corridors that were simultaneously being constructed around and through it (Câmara Municipal de Odivelas).

4.1.2. Lisbon Metropolitan Area and Odivelas Historical Expansion

The spatial configuration of Odivelas cannot be understood without tracing the history of Portugal's motorway expansion, a process that was not merely technical, but fundamentally territorial, reshaping the metropolitan periphery of Lisbon through successive waves of infrastructural development that prioritized regional connectivity over local coherence (Padeiro, 2018; Afonso et al., 2025).

Between the 1970s and 2000s, Portugal's motorway network grew from virtually non-existent to one of Europe's highest densities, with the AML representing approximately 450 km of a national network that exceeded 2,800 km by 2011 (Padeiro, 2018; Afonso et al., 2025). This expansion - organised around concentric rings and radial corridors - occurred without integrated regional governance: Lisbon maintained structured planning, while surrounding municipalities like Odivelas accommodated fragmented decisions from the capital, enabling dispersed urbanisation models where residential, commercial, and logistical functions scattered across the territory without forming cohesive urban tissue (Padeiro, 2018; Pereira e Nunes da Silva, 2008).

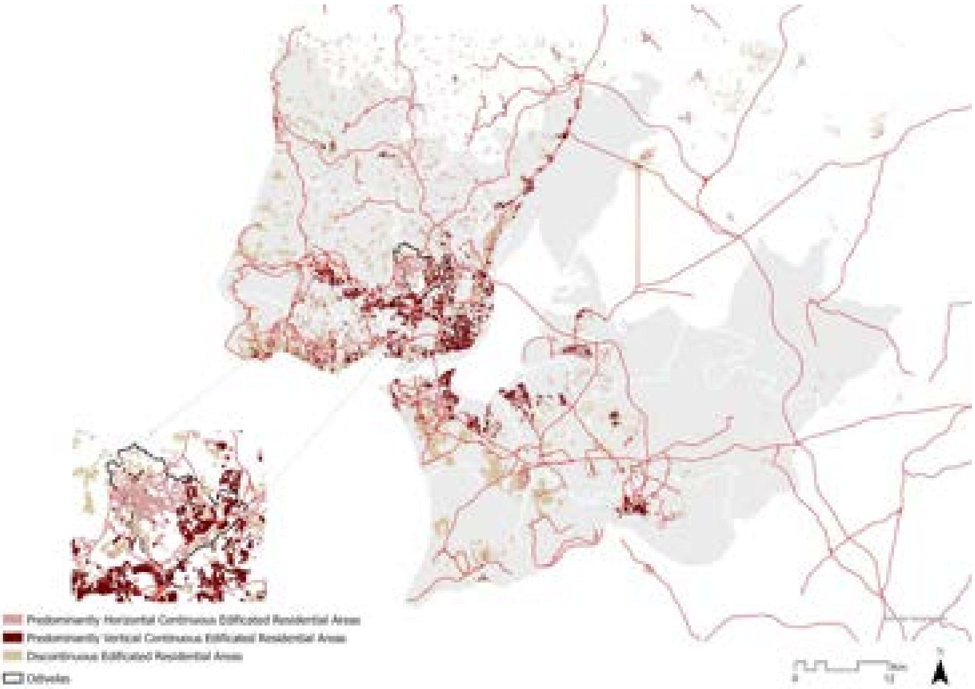


Figure 46. Territorial Occupation. Source: adapted from COS 2018 and DGT.

This resulted in a "fragmented metropolitanization" [Figure 46], in which suburbs grew not as integrated extensions of the city, but as archipelagos of development connected by highways yet disconnected from each other at the local scale around the same structure (Pereira e Nunes da Silva, 2008; Santos, 2018). Odivelas is, of course, not excluded from this trend. Its infrastructural capacity thus determined not only where growth occurred, but also what type of urban form emerged - one inherently dependent on automotive mobility and structurally prone to generating interstitial voids between fragmented parcels.

This infrastructural legacy directly shapes contemporary mobility patterns across the AML. The region exhibits a pronounced pendular movement, this is, daily commuting flows between peripheral municipalities like Odivelas and Lisbon's employment hubs, not merely a consequence of distance, but of a spatial organisation structured around automotive dependency (CCDR LVT, 2022; Padeiro, 2018). In Odivelas, this car-centric reality perpetuates the very fragmentation that generates interstitial spaces, creating a self-reinforcing cycle in which infrastructure promotes isolation, and isolation consolidates infrastructural dependency, confirming that the interstitial conditions of Odivelas are not accidental, but structural expressions of Lisbon's road-centric metropolitan development.

4.1.3. The ecological condition of Odivelas in a Metropolitan context

The Lisbon Metropolitan Area is a highly diverse territory in terms of biophysical characteristics and land uses, where densely populated urban areas coexist with agricultural land, forests, and wetlands (Loupa Ramos et al., 2026).

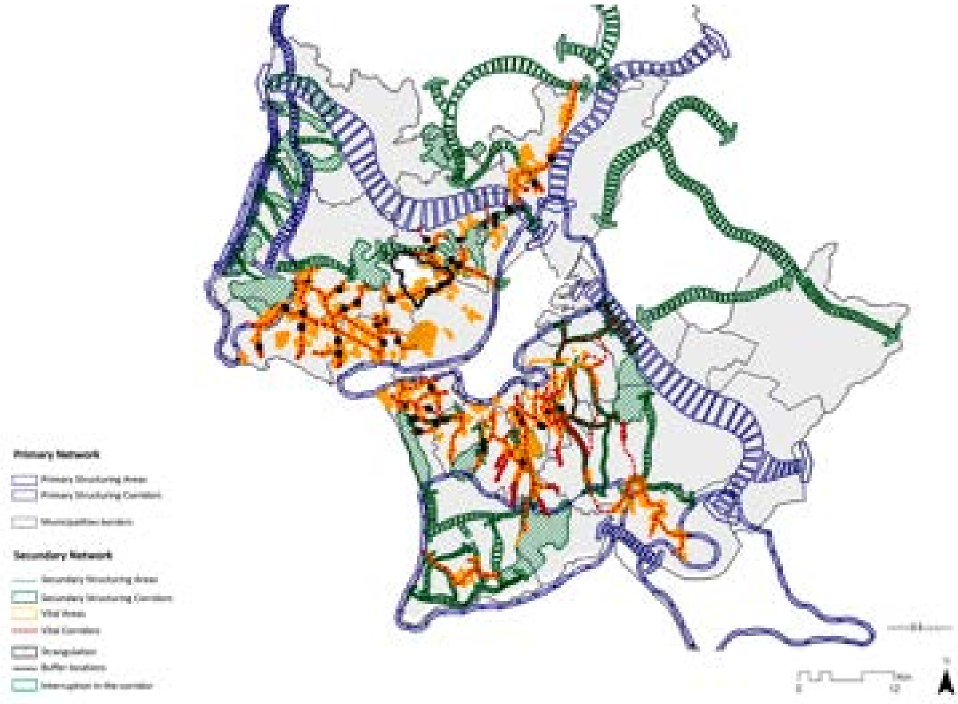


Figure 47. Metropolitan Ecological Network. Source: adapted from Loupa Ramos (2026)

Within the Portuguese planning system, Regional Strategic Plans must be adopted by lower-level instruments such as Municipal Master Plans, and since 1999, green infrastructure planning has been formally integrated into spatial planning, following the strategic core components of the Regional Ecological Network established under the Strategic Plan for the Lisbon Metropolitan Area (PROT-AML, 2002). However, the PROT-AML lacks clear policy guidance for implementing ecological functions at the local level, leading each municipality to apply its own interpretation, resulting in an incoherent patchwork of disconnected green infrastructures across administrative boundaries (Loupa Ramos et al., 2026).

At the same time, when examining climate adaptation and ecological connectivity - at the metropolitan scale - Odivelas is identified as a priority area for urban flooding and slope instability risks in the Plano Metropolitano de Adaptação às Alterações Climáticas (PMAAC, 2019), a document orientating coordinated transmunicipal action toward climate resilience.

Even more so as Odivelas is embedded within one of the densest and most continuous urban zones of the metropolitan area (PROT-AML, 2002; Nicolau et al., 2026), while also being vital for the ecological network. These vital areas, defined in the metropolitan ecological structure (PROT-AML Vol IV, 2002) as interstitial open spaces within consolidated urban fabric that are essential for ecological function, green continuity, and climate resilience in densely built territories, have been gradually occupied by built development, reducing connectivity and concentrating environmental vulnerabilities in the remaining open areas (Nicolau et al., 2026).

At the municipal scale, these broader ecological objectives are translated through the Municipal Ecological Structure (Estrutura Ecológica Municipal - EEM), incorporated within the Odivelas Municipal Master Plan (PDM). The EEM identifies the municipality's principal ecological support systems, including watercourses, flood-prone valleys, areas of relevant vegetation cover, and other spaces considered essential for ecological continuity, environmental regulation, and landscape functioning. However, the scope is largely confined to the municipal territory itself. While they provide an important basis for local ecological planning, they do not fully address the metropolitan-scale ecological continuities and infrastructural systems that extend beyond administrative boundaries, particularly those associated with the major highway corridors analysed in this research. Still, in counter to this, closely associated with this framework is the National Ecological Reserve (Reserva Ecológica Nacional - REN), a statutory land-use designation that protects environmentally sensitive areas and restricts forms of development incompatible with their ecological functions. Together, the EEM and REN establish the primary ecological planning framework within Odivelas.

This means that highways such as IC17 and IC22 traverse the municipality serving metropolitan flows, while consolidating an extensive and fragmented urban layer. At the same time, ecological and climate adaptation strategies - though present in metropolitan planning documents and partially addressed in local instruments such as the PDM and the Municipal Climate Adaptation Plan (PAC. Câmara Municipal de Odivelas, 2024) - rarely translate into concrete spatial interventions at the scale where they are most needed, remaining operative at levels that effectively bypass the municipality's most vulnerable territories. The result is a territory that bears significant environmental vulnerabilities - flood-prone valleys, thermally stressed urban fabrics, fragmented habitats, making it one area where the conditions that make ecological integration most difficult are also the conditions that make it most necessary [Figure 47].

Recognizing this paradox is essential for the present research. Odivelas, a territory whose urban form was produced by infrastructural logic from its inception, acknowledged but insufficiently addressed in metropolitan planning frameworks (Padeiro, 2018), and whose spatial configuration remains structurally dependent on the very highways that fragment it - highlights, more than most, the contradictions this research seeks to address. The interstitial spaces generated by its major corridors are not incidental leftovers but foundational elements of its metropolitan development, carrying within them the latent potential for a new territorial logic. Their transformation through grey-green integration and nature-based intervention may therefore become not merely a local regeneration exercise, but a strategic mechanism for operationalizing ecological and morphological cohesion.

4.2. Municipal Scale: Territorial Characterization and Multi-Diagnosis.

The municipal diagnosis establishes the evidence base that documents how transportation infrastructures have structurally conditioned the spatial, ecological, and functional configuration of Odivelas. It departs from a historical reading of urban form, followed by a characterisation across biophysical, urban, and vulnerability layers, complemented by landscape metrics and space syntax analysis. These layers feed into a transversal synthesis organised around three analytical conditions - Barrier Condition, Latent Connectivity, and Functional Indetermination - that structure both the identification of interstitial spaces and the subsequent NBS proposals.

4.2.1. The historical development and the Road Network as Structuring Logic

Figure 48. Calçada do Carriche in 1961.
Source: AJ. Goulart in Odivelas Slideshow.

Figure 49. Agriculture School in Paiã, 1962.
Source: AJ. Goulart in Odivelas Slideshow.

While the metropolitan context and historical development of Odivelas have been previously addressed at a broader scale, it is also important to understand the relationship between the implantation of each major highway corridor and the progressive transformation of the local urban fabric.

Odivelas was, historically, an agricultural and pastoral landscape on the outskirts of Lisbon, characterized by dispersed rural settlements and monastic structures embedded predominantly within the valleys and watercourses that structured the territory [Figures 48 & 49]. Rather than developing in harmony with these fluvial systems, subsequent urban expansion frequently occurred through their occupation and transformation. Constrained by the municipality's topography and guided by the accessibility offered by valley floors, urban growth progressively encroached upon river corridors and flood-prone areas, reducing the spatial prominence of these ecological systems within the emerging urban fabric (CMO. Plano de Mobilidade e Transportes, 2018; PDM Odivelas. Vol IV, 2018). Its connection to the capital city was made primarily through the Calçada de Carriche, a historical road axis that linked the southeast part of the territory to Lisbon, and which for a long period constituted the main and only structuring element of movement and settlement in the area (CMO. Plano de Mobilidade e Transportes, 2018; PDM Odivelas. Vol IV, 2018). Early urban development consolidated gradually along this axis and along the national road network - particularly the EN250, EN250-2 and EN8, and Estrada da Paiã - which together functioned as the primary structuring axes around which built fabric slowly emerged. Historical areas such as the centre of Odivelas, Caneças, Póvoa de Santo Adrião, and other smaller settlements developed linearly along these roads, establishing a pattern of growth that was fundamentally tied to road proximity rather than to any planned urban logic (CMO. Plano de Mobilidade e Transportes, 2018).

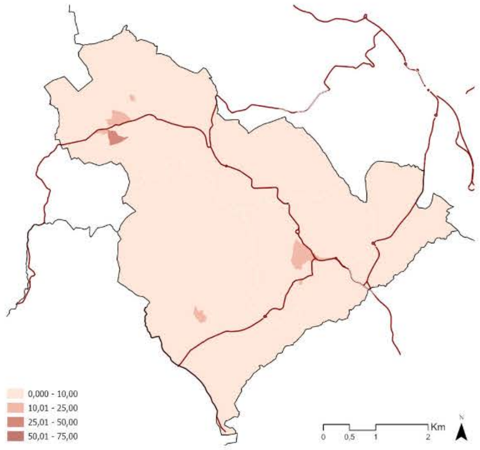


Figure 50. Nº of Buildings per Statistics Subsection until 1945.
Source: adapted from BGRI Census INE (2021)

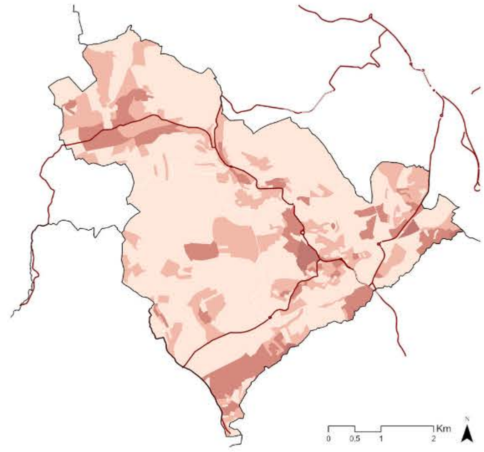


Figure 51. Nº of Buildings per Statistics Subsection until 1980.
Source: adapted from BGRI Census INE (2021)

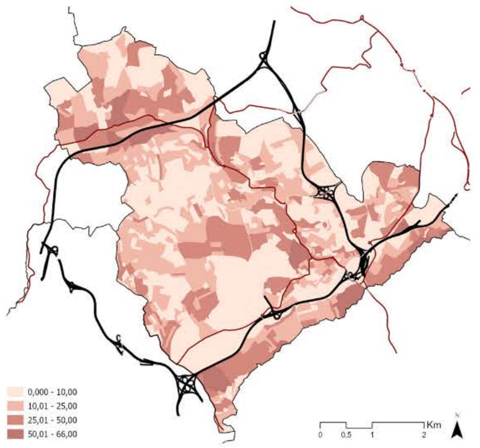


Figure 52. Nº of Buildings per Statistics Subsection until 2000.
Source: adapted from BGRI Census INE (2021)

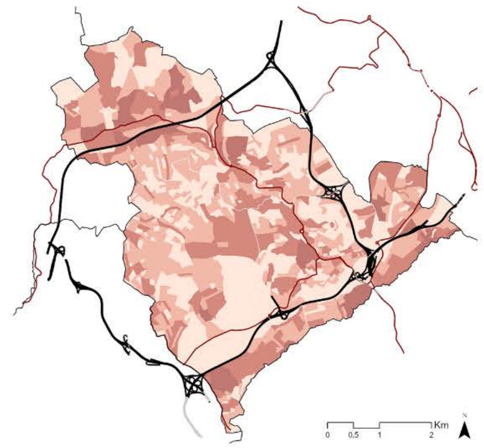


Figure 53. Nº of Buildings per Statistics Subsection until 2020.
Source: adapted from BGRI Census INE (2021)

Note: Spatial distribution of buildings according to their age, based on 2021 census substatical units. Does not reflect building typology and/or population density

As the sequence of maps illustrate [Figures 50 - 53], this alignment between road infrastructure and urban growth intensified progressively throughout the later parts of the twentieth century. The CRIL (IC17), the CREL (IC18) development both connected and actively shaped urban development as well. The Radial de Odivelas (IC22) and Radial de Pontinha (IC16) - especially after the completion of the Pontinha Node in the 2000's - contributed to the development of more centrally located areas, such as Casal de Cambra.

Still, while this development in alignment with the National Roads enabled expansion, the highways also defined the "limits" of said expansion in the municipality. As the Câmara Municipal de Odivelas acknowledges, these roads created an urban mesh based on proximity to arterial routes, while simultaneously generating a fragmented territory divided by the very infrastructures that enabled its growth. The current road hierarchy [Figure 54] reflects this layered condition: the national roads forming the historical "skeleton", and the high-capacity corridors over it as a later and more disruptive layer.

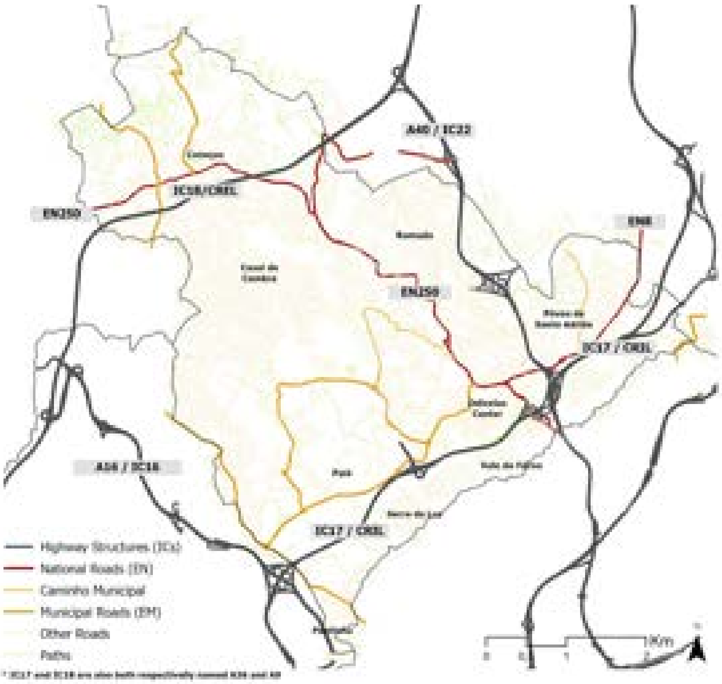


Figure 54. Road Hierarchy. Source: adapted from Odivelas Municipality

The spatial configuration of Odivelas today remains a direct reflection of the successive waves of infrastructural investment that shaped it, with the major corridors continuing to define not only movement patterns but the very boundaries of neighbourhoods, green areas, and residual spaces across the municipality.

This distinction is definitive for understanding the analytical focus of the present research. While the national roads - the EN250 and EN8 - could be considered structural barriers within the municipal territory, the urban fabric grew alongside and around them over decades, establishing a relationship of coexistence and legibility. The high-capacity ring roads and radials, in contrast, arrived over an existing urban tissue, imposing their geometry, their wide junctions, embankments, and access restrictions - onto a fabric that had not been planned around them. It is this condition of imposition over consolidation, rather than structuring from the beginning, that produces the most severe forms of fragmentation, and that makes these corridors the primary object of analysis in the following sections.

4.2.2. Landscape Characterization

The following characterisation reads the territory of Odivelas across three complementary analytical layers - its biophysical structure, its urban form and land use, and its environmental vulnerabilities - each contributing a distinct dimension to the understanding of how infrastructural logic has shaped and conditioned the municipal territory. Together, these layers aim to build an evidence base that is not merely descriptive but also interpretive, revealing the spatial, ecological, and functional consequences of the road network in the territory.

4.2.2.1. Biophysical Characterization

This biophysical structure in Odivelas reveals, from the outset, a landscape marked by a fundamental tension between its natural ecological potential and the infrastructural logic that has progressively constrained, as seen on the existence on Vital Areas on the Ecological

Network on the PROT-AML (2002). The topographic structure [APPENDIX 03] - characterized by a moderate but varied relief, with steeper slopes and ridges to the north and southeast, and a system of valleys in the center of the territory following the hydrographic network - constitutes the primary physical organizing force of the territory. This means the municipality presents a clear topographic gradient, descending from the higher elevations of the northern sectors towards the valley systems and the pronounced southern escarpment that marks the transition to Lisbon (PDM Odivelas - Caracterização Biofísica, 2001). This southeastern slope constitutes a major geomorphological feature of the territory, functioning as a natural barrier that predates, and partly conditioned, the infrastructural barriers that followed. It is along these valley corridors that water, air, fauna and ecological processes naturally concentrate. This reading is supported by the National Ecological Network framework developed by Magalhães (2013) [APPENDIX 04], which identifies the ecological systems considered fundamental to the functioning of natural processes at a national scale. Although not a statutory planning instrument, the framework provides an analytical basis for recognising these valleys as part of a latent ecological structure within the municipality and, unlike the Municipal Ecological Structure, allows these ecological relationships to be interpreted consistently across administrative boundaries.

This happens because most of the major highway structures (ICs) and the national roads (ENs) relate to it. Rather than adapting to the terrain, these infrastructure imposed themselves across valleys and slopes, creating areas in the municipality marked by slopes, embankments, cuttings and viaducts, occupying both ridge positions and valley floors simultaneously. This tension becomes even more prominent and evident when reading the vegetation cover in Odivelas. Both Tree Cover Density [APPENDIX 05] and NDVI [APPENDIX 06] confirms that the municipality of Odivelas is predominantly low on green structure, with meaningful arboreal “patches” and vegetation mainly on central and northern areas of the valleys system, and minor patches alongside the highways. This could mean spontaneous vegetation establishment of green cover on embankments and slopes, which could be ecologically significant, but also misleading as a green infrastructure, as these areas are usually technically inaccessible, unmanaged, and isolated, existing as a byproduct of infrastructural geometry rather than a functional green structure. The presence of vegetation should therefore not be interpreted automatically as an indicator of ecological functionality or public accessibility, but as representation of its potential for intervention. This ecological backbone is actually identified in the current planning tools of Odivelas though the National Ecological Reserve (Decree Law. 166/2008 - REN), which delimits the most sensitive areas of the territory, which usually - as will be seen further into the reading - with the very spaces most exposed to infrastructural pressure [APPENDIX 07].

Analysing the soil dimension, a significant, if not major, portion of the territory is characterized by “non-soil” according to the Portuguese Soil Chart [APPENDIX 08]. The remaining soil types are distributed in smaller sections and patterns that are themselves ecologically and spatially revealing. Along the valley floors and lower areas (IC17), alluvial soils predominate - fertile, with good water infiltration capacity, and theoretically well-suited for productive and ecological uses, which are confirmed a bit by where the temporary crop that occurs [APPENDIX 09]. This productive landscape is partially protected by the National Agricultural Reserve (RAN) [APPENDIX 10], a planning instrument designed not to reflect current agricultural land use, but to safeguard soils with strategic agricultural potential. As such, the presence or absence of active cultivation does not necessarily correspond to the boundaries of the reserve, which are instead defined according to the intrinsic characteristics and productive capacity of the soil.

The distribution of soil types across the municipality reveals significant differences between the major infrastructural corridors. The CRIL corridor intersects areas of alluvial soils associated with valley systems and watercourses. In contrast, substantial portions of the CREL and IC22 corridors are located on clay-rich and litholic soils, whose physical characteristics impose different limitations regarding cultivation, infiltration, and ecological performance. These distinctions are reflected in the Soil Value Chart [APPENDIX 11], which highlights the concentration of higher-value agricultural soils within the valley landscapes crossed by the CRIL.

Together, these conditions provide more than a description of the municipality's biophysical substrate. They reveal how different infrastructural corridors have been imposed upon territories with distinct ecological and productive capacities, establishing important constraints and opportunities for future ecological restoration and Nature-Based Solutions intervention

4.2.2.2. Urban Structure and Use

The territorial reading established in the previous section - a landscape shaped by valleys, slopes, and a fragmented ecological structure - has a direct foundation and impact on its urban development. Odivelas is, in its overwhelming majority, an urban territory [APPENDIX 12]. The land use maps read together - from the Urban Atlas and the PDM - confirm that the municipality is predominantly occupied by built fabric, with rural and natural areas reduced to mainly residual patches concentrated mainly in the northern area and along the valley corridors previously identified [APPENDIX 13 & APPENDIX 14].

Within this predominantly urban condition, the built fabric itself is far from homogeneous. The building footprint and height data [APPENDIX 15] reveal a territory marked by significant morphological diversity, from consolidated medium and high-density residential blocks concentrated along the major arterial corridors and previously identified arterial road (such as EN-250), to lower-density and more fragmented occupation in peripheral, far from Lisbon, transitional areas.

The relative absence of green structure in the municipality reinforces this reading. The existing parks and green spaces are few and spatially dispersed, with the most significant concentrations marked in the PDM as proposed or protected green areas. However, it is necessary to make a critical distinction here, many of these spaces, though formally designated, do not function as genuinely qualified green areas either from an ecological or from a social perspective. Frequently, they are residual, poorly maintained, and disconnected from each other and from the broader ecological network identified in the biophysical reading, failing to constitute a coherent green system at the municipal scale. The cumulative effect of this urban occupation is most directly expressed in the imperviousness map [APPENDIX 16]. Surface impermeability is extensive across the municipality, significantly exceeding the building footprint alone, reflecting the cumulative contribution of roads, parking areas, and paved surfaces associated with the dispersed urban model.

A further layer of complexity is added by the presence of AUGIs (Áreas Urbanas de Génese Ilegal) and active execution units across the municipality [APPENDIX 17]. Their concentration in the more central areas of the territory, alongside several highway corridors, reinforces the historical reading already established, where infrastructure created conditions for informal occupation in its margins, producing a fabric whose reconversion remains an ongoing planning challenge and whose spatial coincidence with the corridors analysed in this research adds a further dimension of urban complexity to the ecological and environmental vulnerabilities already identified.

Together, these readings confirm what was said in section 3 that Odivelas is a territory where urban consolidation has occurred largely at the expense of its natural structure - compressing ecological potential into residual spaces, reducing green continuity to formally designated but functionally insufficient areas, and sealing a significant portion of its surface in ways that directly amplify the environmental vulnerabilities that follow.

4.2.2.3. Urban and environmental vulnerabilities

Therefore, the environmental risks of Odivelas concentrate systematically along and around the same infrastructural corridors that structured the urban development, along the natural features suppressed by this infrastructures which changed its biophysical dimensions.

One of the most common and popularly legible of these risks is noise. The major highway corridors are the primary source of acoustic impact in the municipality [APPENDIX 18], generating high levels of sounds in areas where residential fabric and equipment are present. Acoustic barriers of significant height are installed along these corridors to mitigate it, but in actuality signals for the problem instead of the solution, as they generate visual and physical weight that highlights the barrier condition that is already produced by the infrastructure. The same happens with slope instability risk [APPENDIX 19] - mapped in the PDM of Odivelas across numerous areas of the municipality, specially in the northwestern part. These soles and embankments characterize margins of highway corridors exposed to risk, meaning that the infrastructural space itself is not only a source of environmental impact, but also a zone of physical fragility, which carries maintenance cost and safety application measures are required to protect those affected by it.

As for other vulnerabilities, flood risks mainly occur along the Ribeira da Costa and IC17 [APPENDIX 20], which is, once again, characterized by its alluvial soil, low-leveled valley floors and hydrographic network associated with it. The imposition of the highway over the valley creates a particular critical condition - points in which infrastructural impermeability meets natural water flow, concentrates runoff and amplifies inundation risks in ecologically valuable areas, with predisposition for drainage. But rather than managing this hydrological condition, the infrastructural logic has actually just literally gone over it, treating natural drainage areas as obstacles to be crossed rather than to be integrated.

The urban heat island patterns [APPENDIX 21] reinforce this reading from a climatic perspective. The most intense thermal concentrations coincide with the areas of highest imperviousness, while the coolest areas correspond almost exclusively to the green corridors and valley systems identified in the biophysical dimension. This further confirms the role of the ecological network as a thermal regulation system for the municipality, and demonstrates how its fragmentation and inaccessibility carry both climatic and ecological consequences. While the present research does not undertake a detailed ventilation analysis, the municipality's topographic configuration suggests that valley systems may also play an important role as potential air circulation corridors. . The occupation of valley systems by urban development and transport infrastructure further compromises the natural ventilation corridors through which cooler air would traditionally circulate, which could result in heat accumulation.

4.2.3. Landscape Metrics and Spatial Syntax

Through the readings already established - from the biophysical structure and ecological potential, to the urban structure diversity, to the environmental vulnerabilities concentrated along the infrastructural corridors - a consistent territorial portrait has begun to emerge. However, these layered readings, while individually revealing, do not yet capture the structural logic that denotes the fragmentation condition of Odivelas as a whole. To address this, the following section aims to achieve a complementary analytical reading that connects landscape structure and spatial configuration through two distinct but mutually reinforcing methodological frameworks: Landscape Metrics Analysis (Klopatek, 1999; Leitão et al., 2012) and Space Syntax Analysis through the Place Syntax Toolkit (Hillier, 2007; Ståhle et al., 2004; Stavroulaki et al., 2023).

This approach is based on the understanding that the interstitial condition in Odivelas is not solely a product of physical fragmentation, nor of spatial inaccessibility, but rather the result of both operating simultaneously. As argued by Klopatek (1999), Forman (2003) and Leitão (2012), landscape fragmentation should be read through different lenses - patch geometry, ecological connectivity, and the structures that produce it, in the case of this thesis, the infrastructural transport corridors. Equally, Space Syntax theory establishes that the configuration of a street network - its integration and accessibility patterns - directly conditions movement and, consequently, the spatial marginalisation of adjacent areas (Hillier, 2007; Ståhle et al., 2004). Together with the previous sections, these two frameworks contribute to an ecological and morphological reading of Odivelas, capturing the structural conditions of fragmentation caused by the highway network simultaneously from a landscape and a circulatory perspective.

4.2.3.1. Space Syntax

Integration (800m)

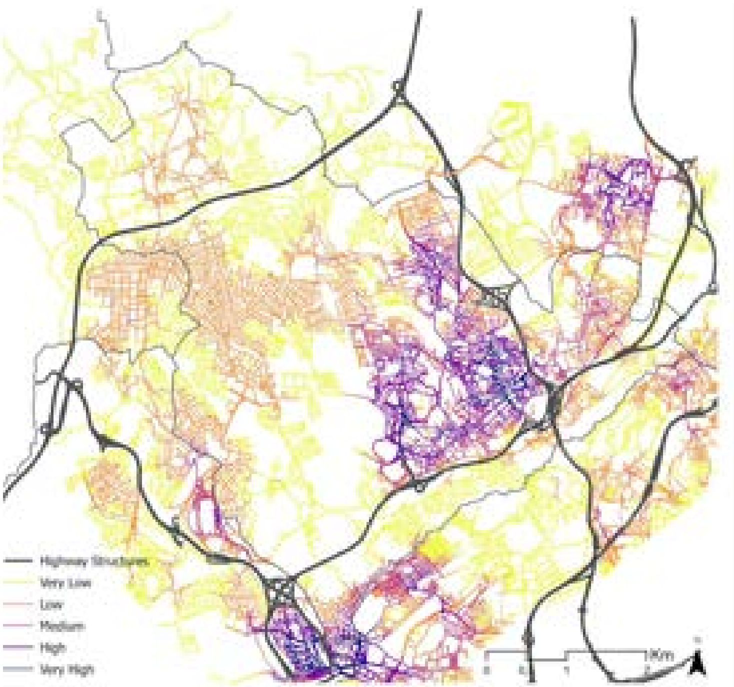


Figure 55. PST Angular Integration (800m radius). Source: Own Elaboration

The Space Syntax Integration analysis [Figure 55], calculated at an 800m normalised radius of Hillier, measures the degree to which each street segment is spatially close - in topological terms - to all other segments within the defined radius. In practical terms, higher integration values indicate spaces that are more likely to be naturally frequented and movement-rich, while lower values signal spaces that are more segregated, harder to reach, and therefore more likely to be marginalised in terms of urban life and activity. The analysis was produced excluding the major highway corridors from the network, a methodological choice that aims to attribute their condition as barriers rather than connectors, and that allows the reading to reveal the integrative capacity of the urban fabric independently of infrastructural accessibility.

The resulting map reveals a spatially coherent but internally fragmented pattern. The highest integration values concentrate around the historic centre of Odivelas and the more consolidated urban nucleus, confirming that the areas of greatest network legibility correspond precisely to the territories that developed earliest and most “organically”, a condition that is both a cause and a consequence of their historical consolidation. These integrated cores are surrounded by progressively lower integration values, but the transition is not gradual. Rather than forming a smooth gradient outward, the map presents abrupt discontinuities - zones of high integration immediately adjacent to zones of very low integration - a spatial signature that reflects the presence of physical barriers interrupting network continuity rather than a natural decay of urban density.

These abrupt transitions coincide consistently with the major highway corridors, particularly along the CREL (IC18), where the barrier condition is most spatially legible, and in the southern areas of the CRIL (IC17), where the AUGI zones previously identified also present markedly low integration values. This double condition - informal urban fabric combined with low spatial integration - reinforces the argument that these areas have developed outside both the planned urban logic and the connected network structure of the municipality, producing spaces that are simultaneously physically present and spatially marginalised.

Reach (300m)

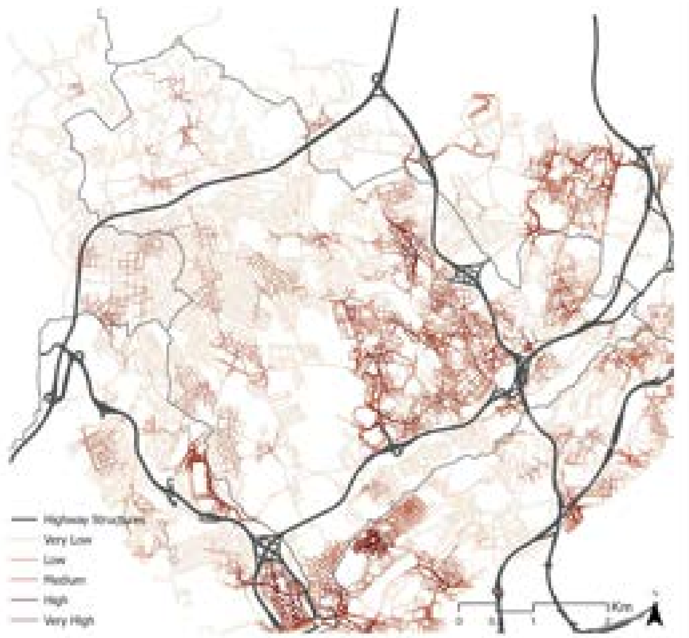


Figure 56. PST Angular Reach (300m radius). Source: Own Elaboration

The Reach analysis [Figure 56], calculated at a 300m normalised radius, measures the number of street segments reachable from each point within a defined walking distance, offering a reading of local accessibility and network density rather than global connectivity. Read alongside the integration map, it produces a complementary and more nuanced portrait of the spatial structure of Odivelas. Rather than contradicting the historical growth pattern of Odivelas, this polycentric distribution reinforces it, the multiple centralities of reach mirror the dispersed development that historically consolidated along the national road network, suggesting a territory that never organised around a single dominant core. However, it is important to clarify that higher reach values indicate topological proximity within the network, not necessarily actual occupation or urban activity. A space can present high reach - meaning, many segments are reachable within 300m - while remaining functionally marginal, underused, or physically inaccessible in practice.

This distinction becomes particularly relevant when the reach and integration maps are read together. Areas with high reach but low integration reveal a recurring pattern: local network density that remains disconnected from the broader circulatory structure, particularly evident adjacent to highway infrastructure. The juxtaposition of these two measures thus exposes a structural fragmentation, pockets of local accessibility that do not translate into systemic connectivity, pointing to the ruptures that green infrastructure interventions could potentially bridge

Betweenness (800m)

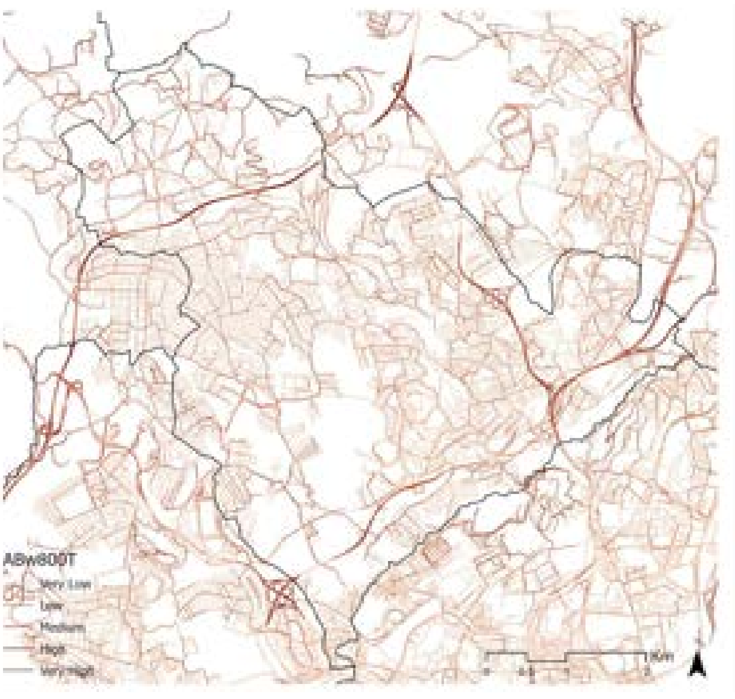


Figure 57. PST Angular Betweenness (800m radius). Source: Own Elaboration

The Betweenness analysis [Figure 57], in this case, calculated including the major highway corridors, measures the frequency with which each street segment lies on the shortest path between all other segments in the network, in other words, it identifies which routes carry the most movement of passage, or through-movement, across the territory. Unlike integration, which measures how accessible a space is from its surroundings, betweenness reveals which spaces are structurally used as connectors between different parts of the network. High betweenness values therefore indicate corridors of intense through-movement, while low values signal spaces that are bypassed by movement flows - present in the network but not used as connectors.

The map confirms, with particular clarity, the dominant role of the highway corridors and the major national roads as the primary through-movement structures of the municipality. The IC17, IC18, IC22 and IC16, together with the EN250 and EN8, concentrate the highest betweenness values, meaning that the vast majority of movement crossing Odivelas does so through these infrastructures, reinforcing their condition as metropolitan and regional flow corridors rather than as urban connectors. The local street network, by contrast, presents predominantly low betweenness values, functioning almost exclusively for local access rather than passage.

Even more particularly relevant is reading this map alongside the integration and reach analyses considering the spaces immediately adjacent to the highway corridors. These margins present simultaneously low integration, low reach, and low betweenness, in other words, they are not accessible, not locally dense, and not used as connectors. These three conditions of spatial marginalisation are not merely a consequence of low urban density, but a structural effect of the highway barriers themselves, which trap these spaces between the metropolitan flow logic of the infrastructure and the local network logic of the urban fabric.

4.2.3.2. Landscape Metrics.

Already understanding the connections and movement patterns of the territory through the Space Syntax analysis, it becomes equally important to examine how the current land use configuration shapes the landscape structure of Odivelas. While Space Syntax reveals the circulatory logic of the urban fabric, Landscape Metrics - calculated using FRAGSTATS (McGarigal et al., 2023) based on a land use classification of Urban Atlas (2018) divided into five classes: Natural, Urban, Agriculture, Identified Vacant Spaces, and Transport - offer a complementary reading of the spatial texture, fragmentation, and ecological coherence of the territory as a whole.

Edge Density

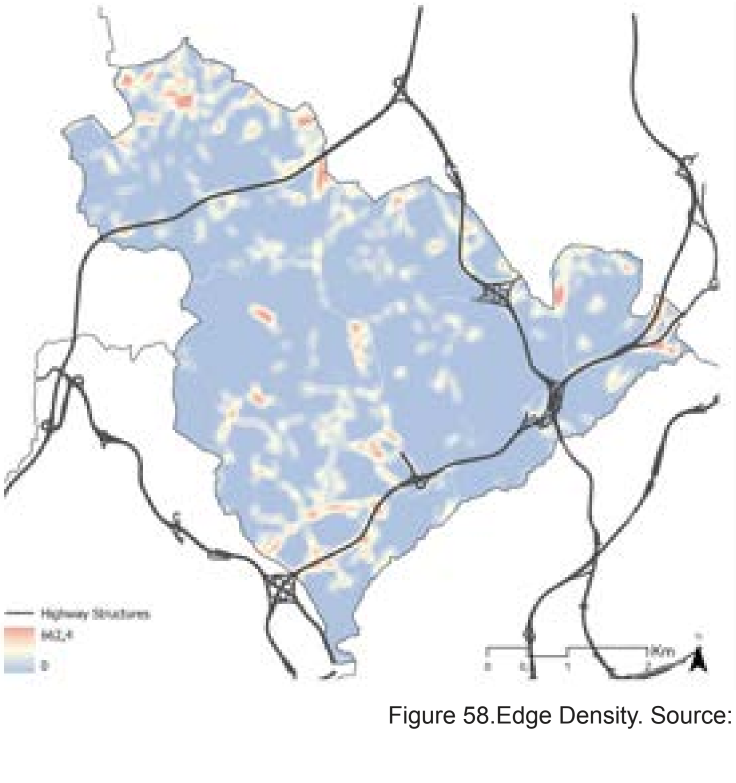


Figure 58.Edge Density. Source: Own Elaboration

The Edge Density map [Figure 58] measures the total length of edges - boundaries between different patches of land use classes - per unit area, offering a quantitative reading of spatial transition and fragmentation across the territory. Higher values indicate areas where different land use classes meet frequently, signalling zones of greater landscape heterogeneity and transition, while lower values reflect more homogeneous and continuous spatial conditions.

The resulting map presents a predominantly low edge density across the municipality, reflecting the largely continuous and homogeneous character of the urban fabric already established in previous readings. The higher values - concentrated in dispersed patches and linear occurrences - correspond primarily to the transition zones between urban and natural land use classes, confirming that the most significant landscape boundaries in Odivelas are precisely those where the built fabric meets residual natural or agricultural spaces. This pattern is spatially consistent with the valley corridors, ecological margins, and infrastructural edges already identified as the primary zones of ecological potential and vulnerability in the territory.

SHDI

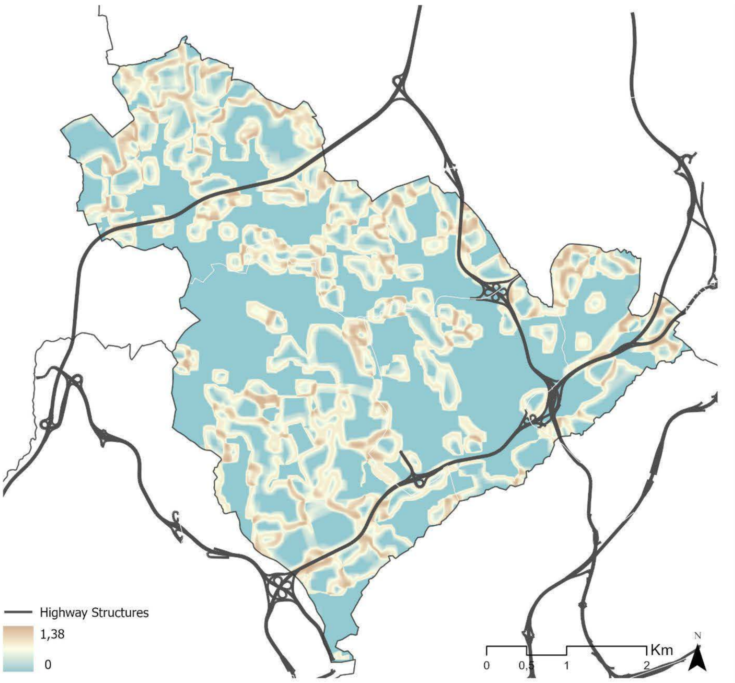


Figure 59. SHDI. Source: Own Elaboration

The Shannon's Diversity Index (SHDI) [Figure 59] measures the diversity of land use classes present within a given spatial unit, the higher the value, the greater the variety of different land use types coexisting in that area. It is therefore not a measure of ecological quality in itself, but of landscape heterogeneity: spaces with high SHDI present a mixture of different classes in close proximity, while spaces with low SHDI are dominated by a single land use type.

The map reveals a predominantly low diversity landscape across the majority of the municipality, aligning with the homogeneous urban fabric already documented in the Edge Density. The areas of higher diversity, appearing as dispersed white and beige patches concentrated primarily in the northern and central areas of the municipality, correspond to the zones where the urban, natural, agricultural, and transport classes coexist in closer proximity.

The higher SHDI values could possibly indicate zones of greater landscape complexity, where different territorial logics - ecological, productive, infrastructural, and urban - meet and/or overlap without a clear spatial resolution. The areas of higher diversity, appearing as dispersed white and beige patches, concentrate primarily in the northern and central areas of the municipality, and are spatially consistent with the locations already identified in previous readings, the valley corridors, the ecological margins, and the spaces alongside the major highway corridors. Rather than ecological richness, these zones of diversity may reflect conditions of spatial indetermination, where the absence of a dominant land use logic has allowed multiple classes to persist in an unresolved coexistence. This reading enforces an idea that the most heterogeneous areas of Odivelas are not necessarily more ecologically determinant, but rather its most ambiguous spaces where intervention potential and landscape fragmentation converge simultaneously.

Cohesion

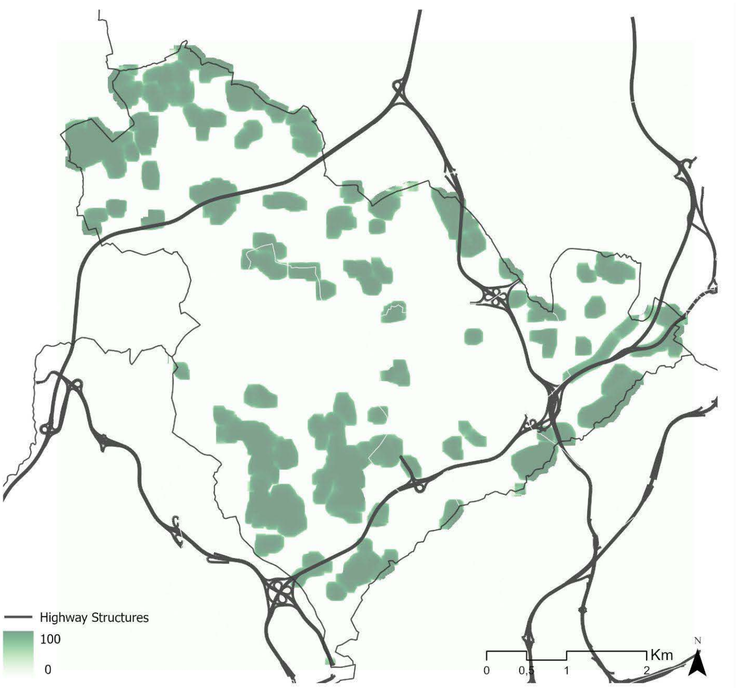


Figure 60. Natural Cohesion. Source: Own Elaboration

The Cohesion index map [Figure 60] maps the physical connectedness of patches belonging to the same land use class - in this case, natural elements - indicating the degree to which these patches are spatially coherent and proximate to one another. Higher cohesion values reflect larger and more internally continuous natural patches, while lower values signal smaller, more isolated, and physically disconnected occurrences. It is therefore a direct metric of ecological connectivity potential at the landscape scale.

The map reveals a pattern that is both spatially clear and analytically consistent with all previous readings. Natural patches are present across the municipality but are distinctly fragmented, distributed as dispersed and isolated occurrences without forming a continuous or connected system. The largest and most cohesive patches concentrate in the northern area, consistent with the higher ecological value, vegetation cover, and lower urban density already identified in that zone. Towards the centre and south of the municipality, the patches become progressively smaller and more isolated, reflecting the increasing intensity of urban consolidation and infrastructural pressure in these areas.

There are a lot of patches alongside the major highway corridors that do not really reach into the highways, a condition that reinforces, now through a quantitative landscape metric, what had already been observed qualitatively in the NDVI, tree cover density, and ecological soil value analyses, while also contradicting what is said in the PDM. Their physical isolation means that despite their presence, they contribute little to landscape connectivity in their current condition. This reading does not diminish their relevance, on the contrary, it precisely identifies them as the spaces where connectivity restoration is most needed and where strategic intervention could produce the greatest structural impact on the overall ecological coherence of the municipality.

In this way, by reading Space Syntax through - low integration, low reach, low betweenness - the Landscape Metrics confirm through landscape structure: isolated natural patches, abrupt land use transitions, and heterogeneous but ecologically incoherent conditions. These are not coincidental overlaps, but structural ones, produced by the same infrastructural barriers that simultaneously fragment the urban network and the ecological system.

Odivelas emerges from these readings as a territory of profound internal contradictions. It is a dense, consolidated, and historically legible municipality, yet simultaneously one of the most fragmented in the Lisbon Metropolitan Area, not disconsidering to its urban consolidation, but partly because of it. The highway network that enabled its rapid growth over the second half of the twentieth century is the same infrastructure that now divides its neighbourhoods, isolates its natural patches, and marginalises the spaces between them. The integrated reading reveals a municipality where ecological potential and spatial marginalisation consistently coincide: the valley corridors that carry the greatest biodiversity value are also the most physically inaccessible; the spaces with the highest landscape heterogeneity are also the most territorially unresolved; and the margins of the highway corridors remain structurally disconnected from both the urban fabric and the ecological network they border.

It is this condition - of a territory rich in latent potential yet structurally prevented from realising it - that defines the propositional focus of the following sections.

4.3. Corridor Scale: Infrastructural - Spatial Analysis of Odivelas Highways

4.3.1. The infrastructural condition of Odivelas' municipality

The infrastructural-spatial dimension focuses on the physical matrix generated by the highway network in Odivelas. Understanding the geometric reality of these corridors, their profiles [Figure 61 - 63] their domain boundaries, barrier typologies, elevated and submerged sections [Figures 64 - 66], and the density of infrastructure nodes such as viaducts and junctions, is fundamental to characterise how these structures occupy and condition the surrounding territory. Each of these elements contributes directly to a cohesive territorial reading, defining the zones of impermeability and rupture that infrastructure imposes on the urban fabric, and establishing the primary spatial generator against which residual conditions can be evaluated.

Taken together, these corridors position Odivelas not as a peripheral municipality bypassed by regional infrastructure, but as a node at the intersection of some of the most heavily trafficked axes of the metropolitan area. Yet, as the road hierarchy map reveals, the major highway corridors - IC17, IC22, IC18, and IC16 - are mainly located at or near the boundaries of the municipality, structuring Odivelas not from within but from its edges. Notably, the IC16, despite not running through the municipality itself, demarcates its boundary as decisively as the corridors inside it. As previously addressed in the historical development of Odivelas, this peripheral positioning played a decisive role in shaping the territory's urban consolidation, concentrating development in the central areas enclosed by these corridors.


Figure 61. IC17 - CRIL Profile.
Source: Original.

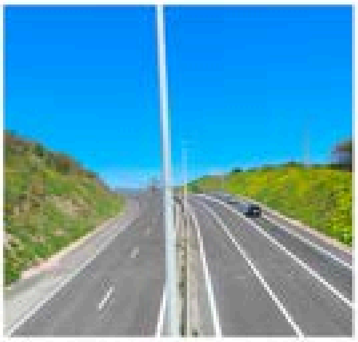


Figure 62. IC22 Profile..
Source: Original.

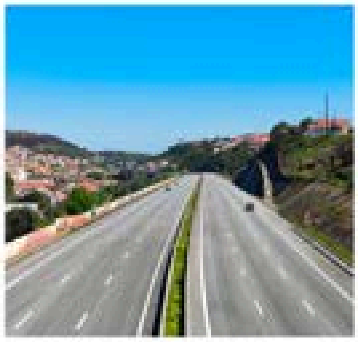


Figure 63. IC18 - CREL Profile.
Source: Original.

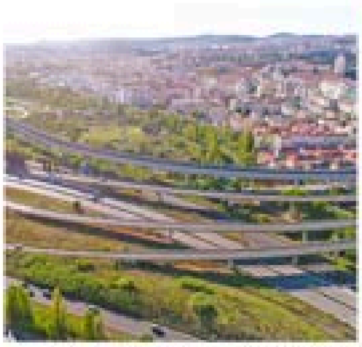


Figure 64. Interchange IC17-IC22
Source: CM Odivelas (2024)

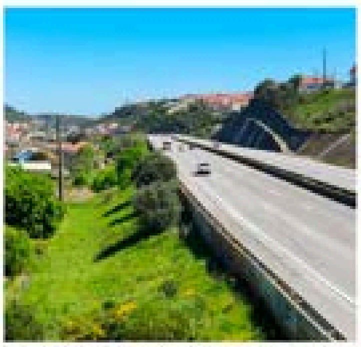


Figure 65. Example. Barriers Structures
Source: Original.


Figure 66. Example. Crossings.
Source: Original.

Understanding this positioning, however, is only the first layer of the analysis. Each corridor must also be read as a three-dimensional spatial structure that conforms to its own infrastructural space, not merely as a line on a map. The domain widths, the relationship between the road platform and the terrain, and the presence of elevated or submerged sections all determine the nature and intensity of the barrier condition each highway produces.

From a first observation, it becomes possible to establish a systemic inventory of the physical elements that together compose the highway network's spatial impact across the municipality. These are organised into three categories: Main Structures, meaning the major interchanges and tunnels that concentrate infrastructural intensity at specific nodes; Physical Barriers, which manifest across the corridors as embankments, retaining walls, acoustic walls, and slopes; and Crossings, which define the points of negotiation between the highway and its surrounding territory [Figure 67] [APPENDIX 22 & 23]. While each corridor shares this general structural logic, the way these elements combine, overlap, and condition the surrounding territory differs significantly between them - differences that the following sections read individually and in depth.

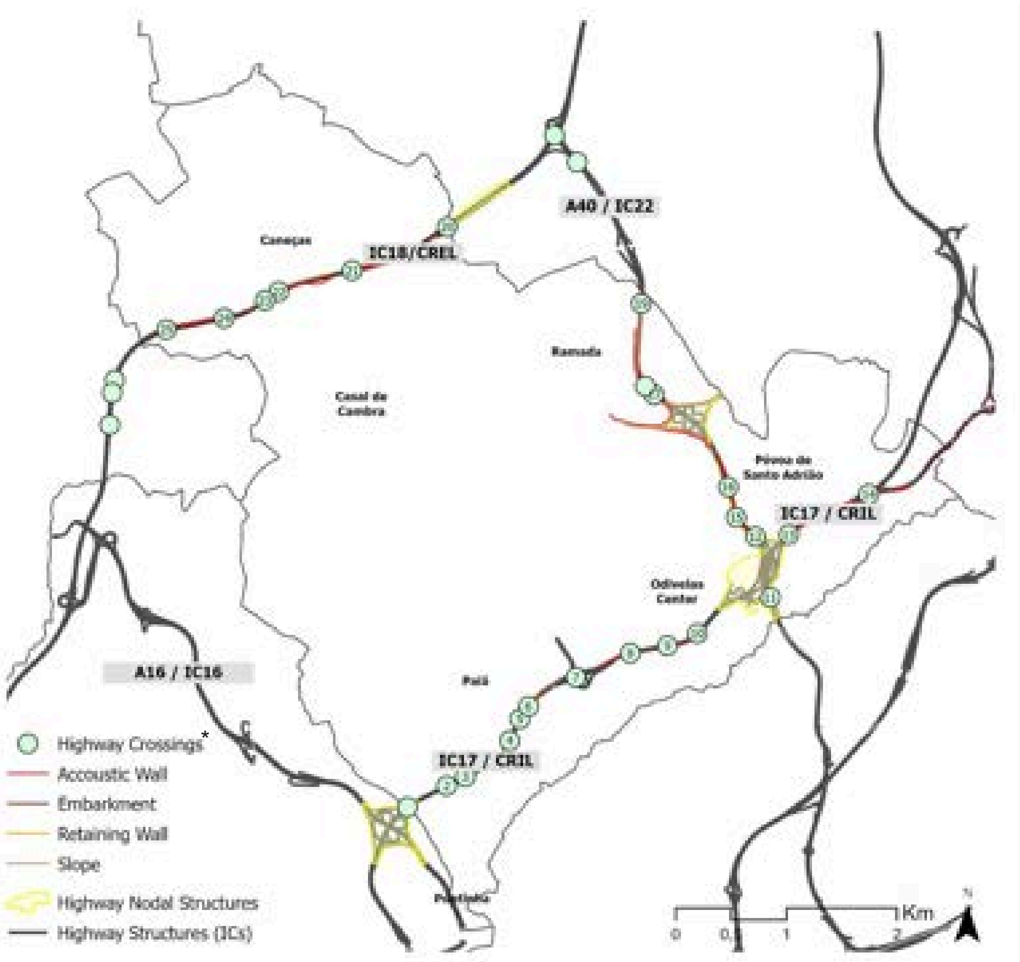


Figure 67: Odivelas Highways Structure. Source: Own Elaboration
** Each number corresponds to a crossing further discussed at Appendix 22*

4.3.2. IC17 (CRIL) - The convergence of territorial systems

Of the three major highway corridors that structure Odivelas, the IC17 (CRIL) represents the most complex and analytically rich case. Because it is the corridor most deeply inserted into the consolidated urban fabric, concentrating the highest number of mixed-use crossings, the most significant flood risk zones associated with its valley position and alluvial soils, and some of the highest ecological soil values in the municipality (Magalhães, 2013). The significance of the IC17 lies less in any individual condition than in the way multiple territorial systems overlap within the same corridor.

The corridor analysis is conducted within a 300-metre buffer on each side of the highway axis, following Forman's (2003) documented range of road ecological effects, and consistent with the local reach radius applied in the Space Syntax analysis. This boundary is understood as an analytical reference rather than a rigid limit.

Infrastructural Condition

The IC17 operates with a heavy cross-section, typically 3 lanes in each direction, plus the shoulder lanes along most of its extent through Odivelas. This means that at any given crossing, the corridor occupies at least 30 meters width (or more) of an inaccessible, non-porous surface in ecological and pedestrian aspects. This is before even considering additional offsets generated by embankments, slopes and other retaining structures. The barriers and crossing maps [Figure 75] reveals the spatial consequence of this infrastructural weight with more clarity by highlighting its immediate barriers - retaining walls, acoustic walls, embankments and slopes - which form an almost continuous line along both sides of the corridor, with very few interruptions. This defines an almost total physical closure that is also not limited to the infrastructural domain itself, but extends into the adjacent urban fabric, where the general barriers layers confirm how the surrounding territory organized itself in a closed manner and with little permeability towards and across the corridor.

Figure 68. Interchange IC17-IC16
Source: Google Earth.

Figure 69. Intersection IC17 - Strada
Source: Google Earth.

Figure 70. Interchange IC17 - IC22
Source: Google Earth.

Three major junctions mark the corridor within or at the boundary of the municipal territory: the western node connecting to Lisbon and the IC16 [Figure 68]; a smaller central interchange providing local access to the consolidated residential fabric of Odivelas [Figure 69]; and the eastern node shared with the IC22, where the convergence of the ring road and the radial produces a condition of extraordinary spatial complexity [Figure 70], with a layered accumulation of ramps, elevated sections that occupies a vast footprint and generates one of the most impermeable and indeterminate zones in the municipality.

Against this general situation of closure, the crossings appear as rare and punctual exceptions [APPENDIX 24]. The IC17 concentrates the highest number of mixed-use crossings of the three corridors - this means - shared by vehicles and pedestrians. These crossings materialise in very different spatial forms, from underpasses and tunnels that pass beneath the carriageway, to elevated walkways and viaducts that cross above it, to at-grade mixed crossings that negotiate the transition at road level. Each typology of crossings produces a distinct spatial experience and a distinct relationship with the surrounding urban fabric [Figures 71 - 74] However, through field readings, it is possible to confirm that these crossings are frequently characterized by reduced sidewalks widths, poor lighting (if none) and limited spatial connectivity, revealing a contradiction: the corridor that most demands pedestrian permeability is also the one where this permeability is most compromised.

The barrier map reveals not only the physical configuration of the corridor itself, but also the first indication of its surrounding urban condition.More importantly, these infrastructural conditions do not operate in isolation. Their significance becomes evident when read alongside the urban, ecological and hydrological systems that converge within the corridor. Further in the reading this will be explored in greater detail.

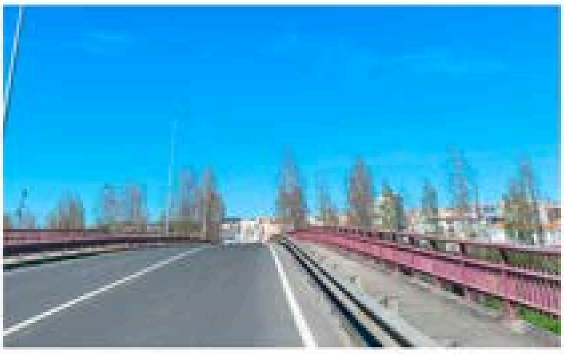


Figure 71. Mixed used Crossing.
Source: Original.

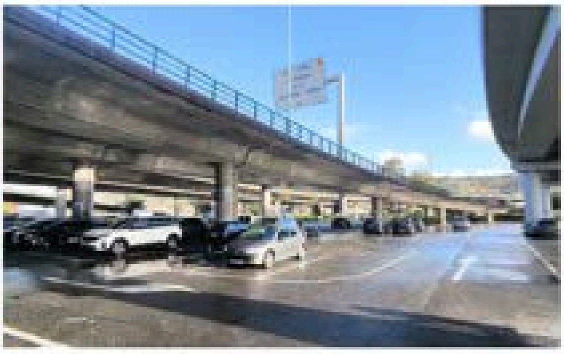


Figure 72. Parking Lot in Interchange IC17 - IC22
Source: Original.

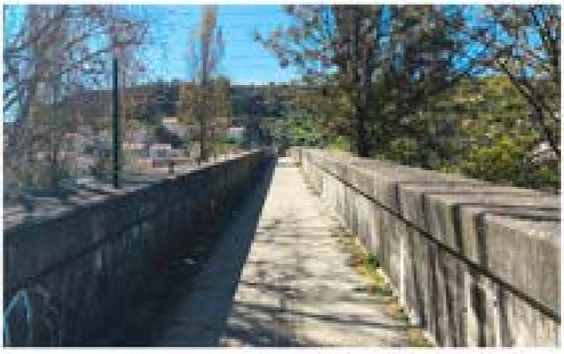


Figure 73. Pedestrian Crossing;
Source: Original.

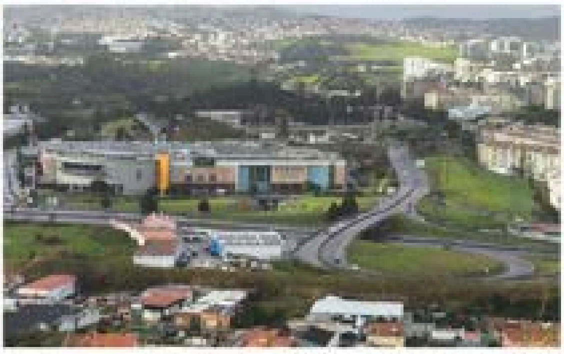


Figure 74. Minor Interchange IC17.
Source: Original.

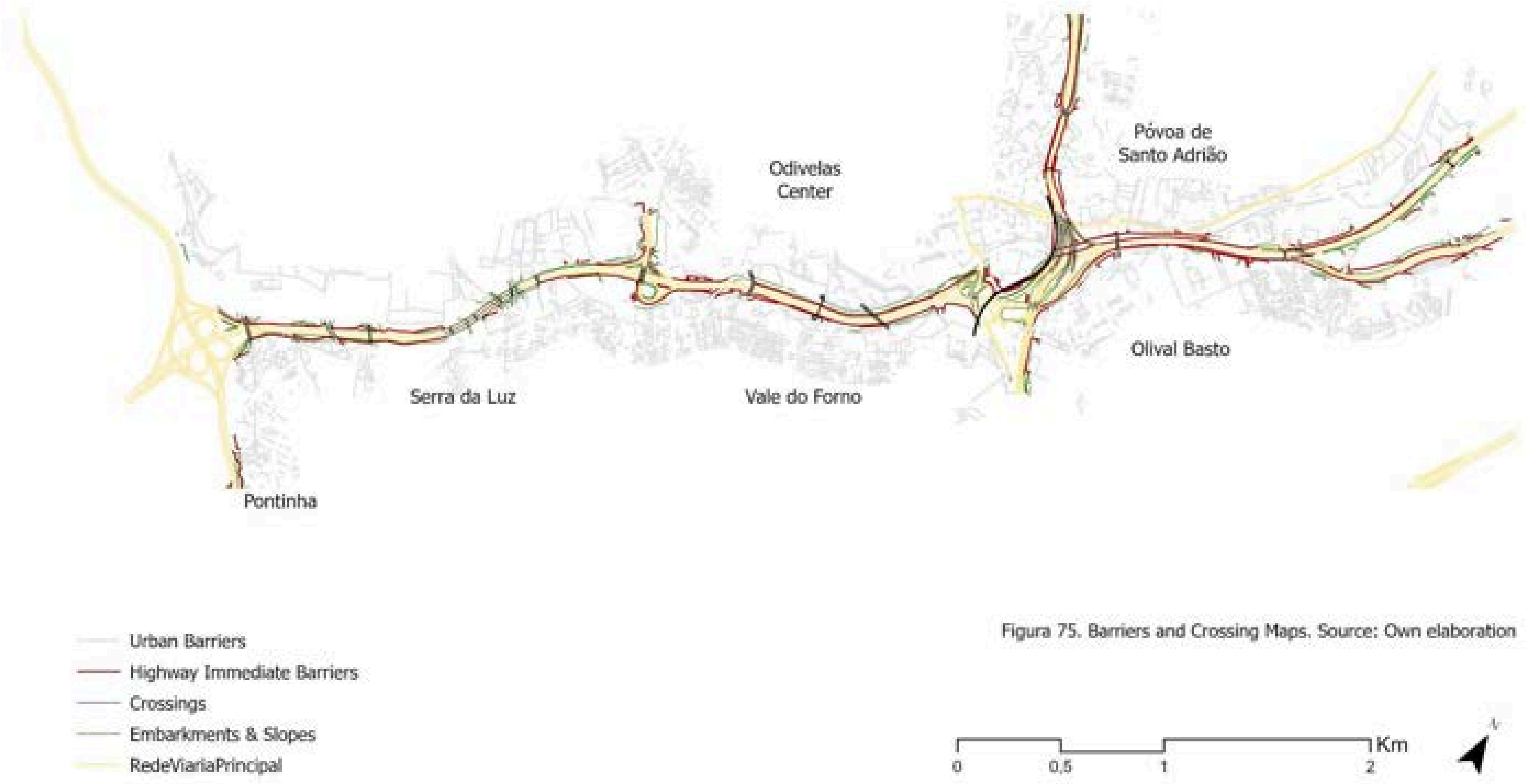


Figura 75. Barriers and Crossing Maps. Source: Own elaboration

Urban Condition and Porosity.

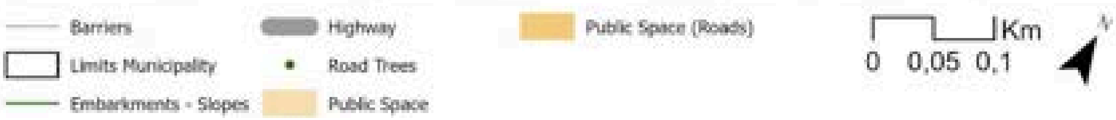
The IC17 corridor runs in close proximity to some of the consolidated urban fabric in the municipality of Odivelas [Figure 82]. The urban area is marked by the PDM in its majority [APPENDIX 25], even if the direct surroundings are marked as transitional green areas, a designation that, as the built fabric reflects, does not always reflect the actual occupation of the territory. This urban proximity to the corridor generates a clear spatial asymmetry between the two sides of the corridor. Based on previous readings, to the north of the highway, the urban fabric expands in a relatively continuous manner - in areas such as Odivelas center and Póvoa de Santo Adrião - with a denser and more consolidated tissue expanded from the national road already described. To the south, in areas such as Serra da Luz and Vale do Forno, this expansion is limited and interrupted by the via itself, and it is precisely on this rupture that the informal occupation previously identified emerges, with the AUGI zones concentrating along the southern margin of the corridor in a pattern completely different from the ones on the other side, being areas in which the urban tissue pressures the corridor edge without being able to negotiate with it [APPENDIX 26].

Figure 76 - Public Space Porosity. Vale do Forno.
Source: adapted from Câmara Munciipal de Odivelas

Figure 77 - Public Space Porosity. Odivelas Center.
Source: adapted from Câmara Munciipal de Odivelas

Figure 78 - Public Space Porosity. Industrial Park.
Source: adapted from Câmara Munciipal de Odivelas

Figure 79 - Public Space Porosity. Pv. de St. Adrião.
Source: adapted from Câmara Munciipal de Odivelas



In more consolidated central areas of Odivelas, buildings tend to be longer, taller and multifamily [Figure 81], generating greater pedestrian activity and a more active public realm [Figures 76 - 79]. Yet this urban intensity rarely translates into a meaningful relationship with the corridor itself. Along much of the IC17, buildings, public spaces and local streets are oriented towards the interior of neighbourhoods rather than towards the highway. Active frontages become increasingly rare near the corridor, while walls, fences and residual spaces dominate its edges, reinforcing the highway as a limit condition rather than an urban frontage.

In the more industrial stretches, buildings are predominantly closed and oriented toward vehicle access, while in the AUGI zones the fabric becomes more monofunctional and introverted, composed predominantly of low-rise dwellings [Figure 80] with limited and fragmented public space. None of these areas establishes a relationship with the highway. Odivelas Center represents the one moment along the corridor where collective spaces and a denser street network create some degree of negotiation with the infrastructure, yet even here this relationship remains limited. Still, most of the space that does exist is occupied by cars [Figure 81], a reflection of the car dependency in the area. By contrast, stretches of lower barrier density, generally associated with rural, less consolidated or planned areas, present greater spatial porosity and latent potential for intervention.

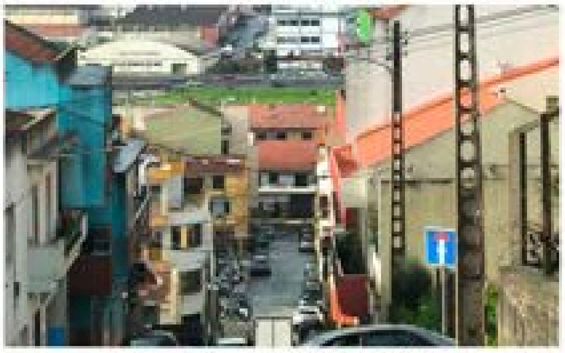


Figure 80. Serra da Luz AUGI. Low Porosity.
Source: Original.

Figure 81. Odivelas Central Area. Mid Porosity.
Source: Original.

The Reach Analysis [APPENDIX 27] reveals that spaces immediately adjacent to the highway - despite the presence of crossings - do not necessarily present higher accessibility values than would be expected if those crossings were functioning as genuine connectors. Read alongside field observations, the Reach analysis suggests that existing crossings fail to generate the level of spatial integration their number would imply. As a result, the corridor remains a significant rupture in the local movement network despite being crossed at multiple locations.

Looking into the facilities and services mapping [APPENDIX 28], this reading is further reinforced. Daily-use activities such as schools, markets and pharmacies, together with health and civic services, tend to concentrate within the consolidated northern fabric of Odivelas Center and Póvoa de Santo Adrião, while the southern side presents a sparser and less coherent distribution of services. Green spaces are few and isolated, rarely contributing to a connected public or ecological system. The projects documented by the MetroPublic Net project indicate a growing planning interest in spaces adjacent to the highway and the AUGI areas, yet these interventions remain fragmented and disconnected from the broader ecological and morphological logic of the corridor.

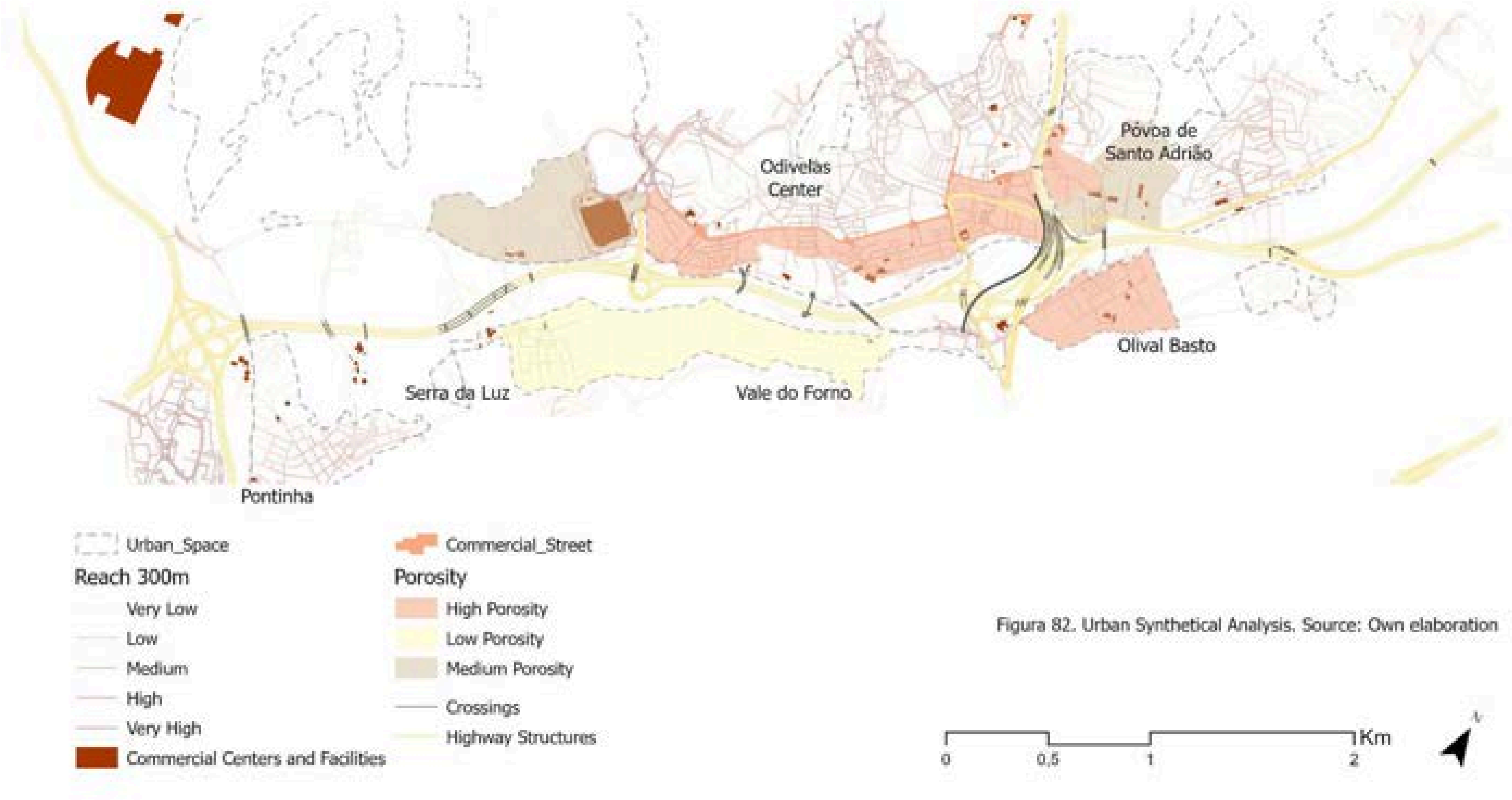


Figura 82. Urban Synthetical Analysis. Source: Own elaboration

Green Infrastructure Condition.

The ecological condition of the IC17 cannot be understood independently from its urban and infrastructural realities. It is precisely the overlap between valley systems, urban occupation and metropolitan infrastructure that produces the corridor's most significant ecological opportunities and constraints. The reading of this condition requires going back to thematics already discussed in municipal scale. From reading different data is possible to elaborate a map [Figure 89]: the NDVI above a threshold of 0,2 (LandSat 9, 2023), which is the standard deviation, as measure of active green occupation; the Natural Areas derived from Fragstats Cohesion analysis; the National Ecological Structure proposed by Magalhães (2013) as a strategic reference for identifying ecological continuities and territorial support systems; the hydrographic network and its water crossing points with the corridor, identified as the precise locations where water flow intersects the infrastructural barrier; and the existing parks and green spaces mapped from the PDM and MetroPublic Net.

The western sector concentrates the most expressive ecological condition, with denser and more continuous vegetation, a more active hydrographic network - even if not directly connected to the highway - coinciding with the less consolidated and/or more rural areas of the municipality [APPENDIX 29]. Here, one of the largest crossings of the corridor occurs beneath a viaduct structure, where more water crossings occur and green presence is mapped. However, field observation reveals a different reality, one in which the space is largely abandoned [Figure 83 & 84], neither ecologically managed nor publicly activated, existing as a clear residual condition beneath the infrastructure rather than as a functioning ecological and/or social asset.

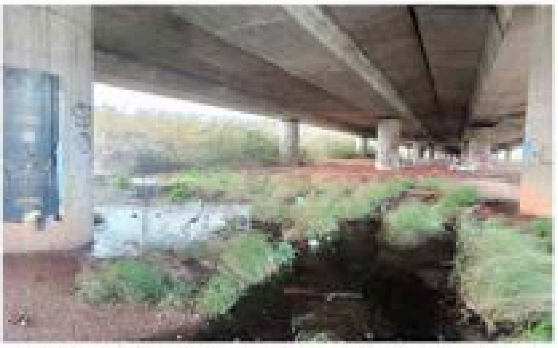


Figure 83. Ponding water in space below viaduct. IC17.
Source: Original.

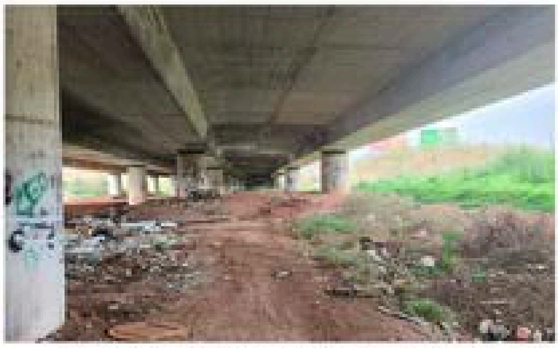


Figure 84. Bare Ground and trash disposal. IC17
Source: Original.

The hydrographic crossing points represent a location where water flows through infrastructural barriers - simultaneously a point of hydrological constraint - as confirmed in flood risk mapping [APPENDIX 30]. Unlike the ecological crossings that could allow natural hydrological continuity, the watercourses that intersect the IC17 do so almost entirely through culverts and buried passages beneath the carriageway, which suppresses their natural flow dynamics and eliminates the riparian vegetation that would otherwise develop alongside them. Rather than negotiating with the hydrographic system, the infrastructure has effectively buried it. At the same time, the watercourses along the highway are mainly accompanied by narrow and discontinuous lines of spontaneous vegetation - which are, a lot of the times, invasive species with low ecological value - working as thin ecological threads that signal where connectivity should exist, but does not [Figure 85 & 86]. This condition is not limited to the riparian margins, extending to the broader infrastructural domain, where the geometry of the corridor consistently generates technical spaces that fall outside both ecological management and public use logic.

These spaces carry a real and demonstrable potential, as represented by Parque Urbano do Rio da Costa, one of the only green parks in Odivelas, occupying the highway margin while also contributing to flood management and thermal regulation [Figures 87 & 88]. It stands as a prime example of what these spaces could become. Yet even here, a critical gap persists, there is no connection to this park from the other side of the highway, confirming that even where activation has occurred, the barrier condition of the corridor remains unresolved. Field observation also identified informal agricultural uses along the corridor, in which spontaneous cultivation occupies flat and vacant spaces adjacent to the highway that communities have appropriated in the absence of formal planning, signalling a latent productive potential that formal planning has neither protected nor strategically integrated

Following this idea, the formally designated green spaces along the corridor are few, spatially dispersed and critically disconnected from both the hydrographic crossing points and areas of higher NDVI and tree cover. This is confirmed by the Natural Cohesion analysis [APPENDIX 31], the natural patches are present on both sides of the corridor but never form a continuous longitudinal system, systematically interrupted at precisely the points where the valley and hydrographic network would support ecological continuity.

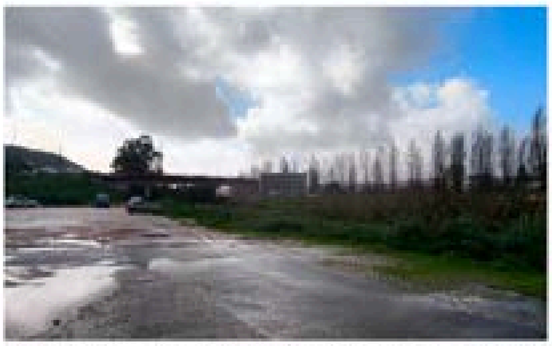


Figure 85. Illegal Parking space and vegetation thread. Source: Original.

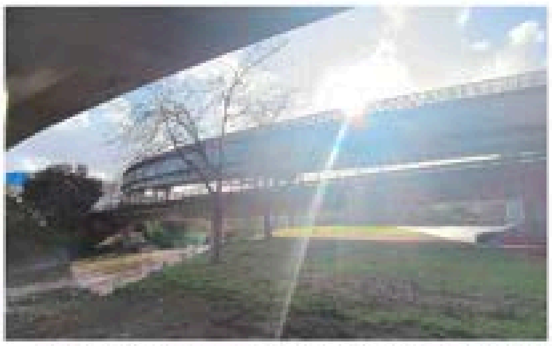


Figure 86. Underpass Parking beside Rio da Costa. Source: Original.

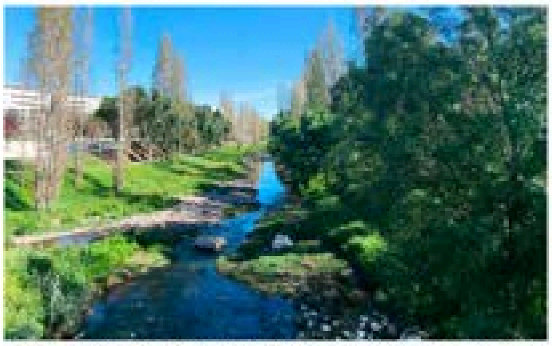


Figure 87. Rio da Costa Urban Park right next to IC17. Source: Original.

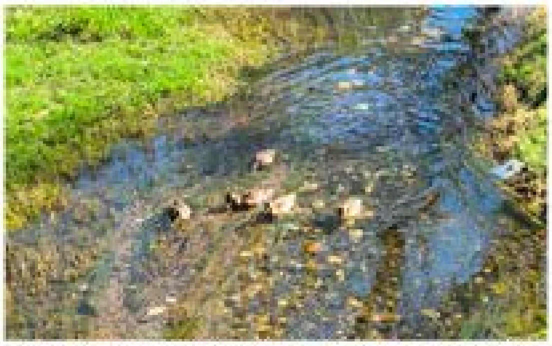


Figure 88. Wildlife recovery in urban area. Rio da Costa Source: Original.

This condition has direct climatic consequences. The absence of functional vegetation cover along the corridor - combined with the extensive impervious surfaces of the infrastructural domain and the sealed fabric of the adjacent urban tissue - contributes directly to the thermal stress conditions [APPENDIX 32]. The heat island pattern is not an abstract metropolitan phenomenon here, but a localised and experienced reality, concentrated precisely in the spaces where ecological and social activation has been most systematically absent. The marginal condition of these spaces is, therefore, not only a planning failure, but a climatic one.

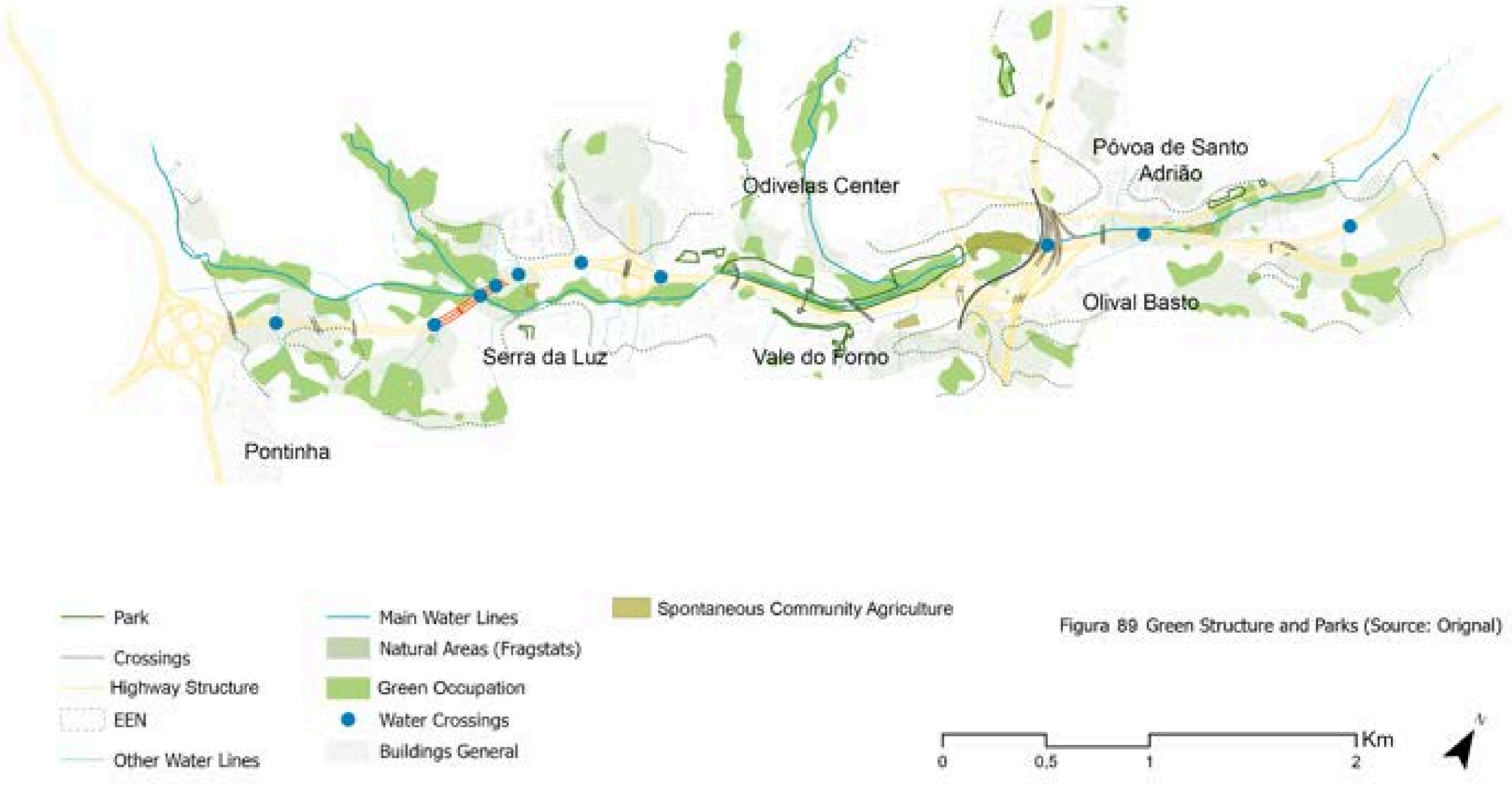


Figura 89 Green Structure and Parks (Source: Orignal)

A synthetical view of IC17 and its latent potential condition.

More than any other corridor analysed, the IC17 is defined by the convergence of multiple territorial systems within the same spatial framework. Rather than operating as a simple transport infrastructure, the corridor concentrates hydrological processes, ecological structures, informal urbanisation, and metropolitan mobility within a shared valley landscape. As a result, environmental, social and spatial pressures do not occur independently, but accumulate and interact throughout the corridor. This convergence becomes particularly evident in the valley structure itself. The same topographic system that concentrates water flows and flood-prone areas also accommodates ecological remnants, informal occupations and major infrastructural interventions. Consequently, spaces that might otherwise be understood as isolated environmental, urban or infrastructural issues emerge here as interconnected territorial conditions.

This condition is visible across the corridor. The AUGI areas of Serra da Luz and Vale do Forno occupy spaces that are simultaneously informal, flood-prone and ecologically significant, while the western sector retains the municipality's most continuous ecological presence. Between these extremes, industrial, residential and infrastructural landscapes overlap without forming a coherent territorial structure.

Ecological fragmentation is one of the clearest manifestations of this overlap. As throughout the corridor, natural vegetation persists in disconnected patches - on embankments, along watercourses and its riparian margins - never forming a continuous system capable of sustaining ecological processes or movement. The crossings that exist, despite being the most numerous of the three corridors analysed, only partially address this discontinuity. Morphologically, they provide limited pedestrian and vehicular negotiation across the barrier; ecologically, they are almost entirely absent, no crossing along the IC17 is designed, positioned or presents the conditions to support the movement of fauna, the continuity of riparian vegetation, or promote the natural restoration of the hydrological connections that the infrastructure has severed.

Following the criteria established in Chapter 2, the interstitial spaces identified along the IC17 concentrate precisely where these overlapping conditions become most evident: along the valley floor, around hydrographic systems, at the margins of the AUGI areas, and within the large infrastructural nodes generated by the corridor [Figure 90].

The interstitial spaces identified along this corridor are therefore zones of overlapping and simultaneous conditions, spaces where the valley floor concentrates flood risk, ecological potential, and informal occupation; where the AUGI margins of Serra da Luz and Vale do Forno signal both vulnerability and latent activation; and where the patched vegetation along embankments and watercourses reveals a green system that exists in fragments but has never been allowed to function as one. The crossings therefore reveal one of the corridor's central contradictions: while multiple systems converge within the same territory, the mechanisms capable of connecting them remain partial and highly selective.

Their boundaries, like the territory itself, resist clean definition. The transformative potential of the IC17 lies precisely in this condition of overlap, where interventions directed at a single issue - ecological restoration, flood management, public space or accessibility - can simultaneously affect multiple territorial systems. In this sense, the corridor's greatest challenge and its greatest opportunity derive from the same condition: the convergence of systems that are currently fragmented, yet remain spatially intertwined.

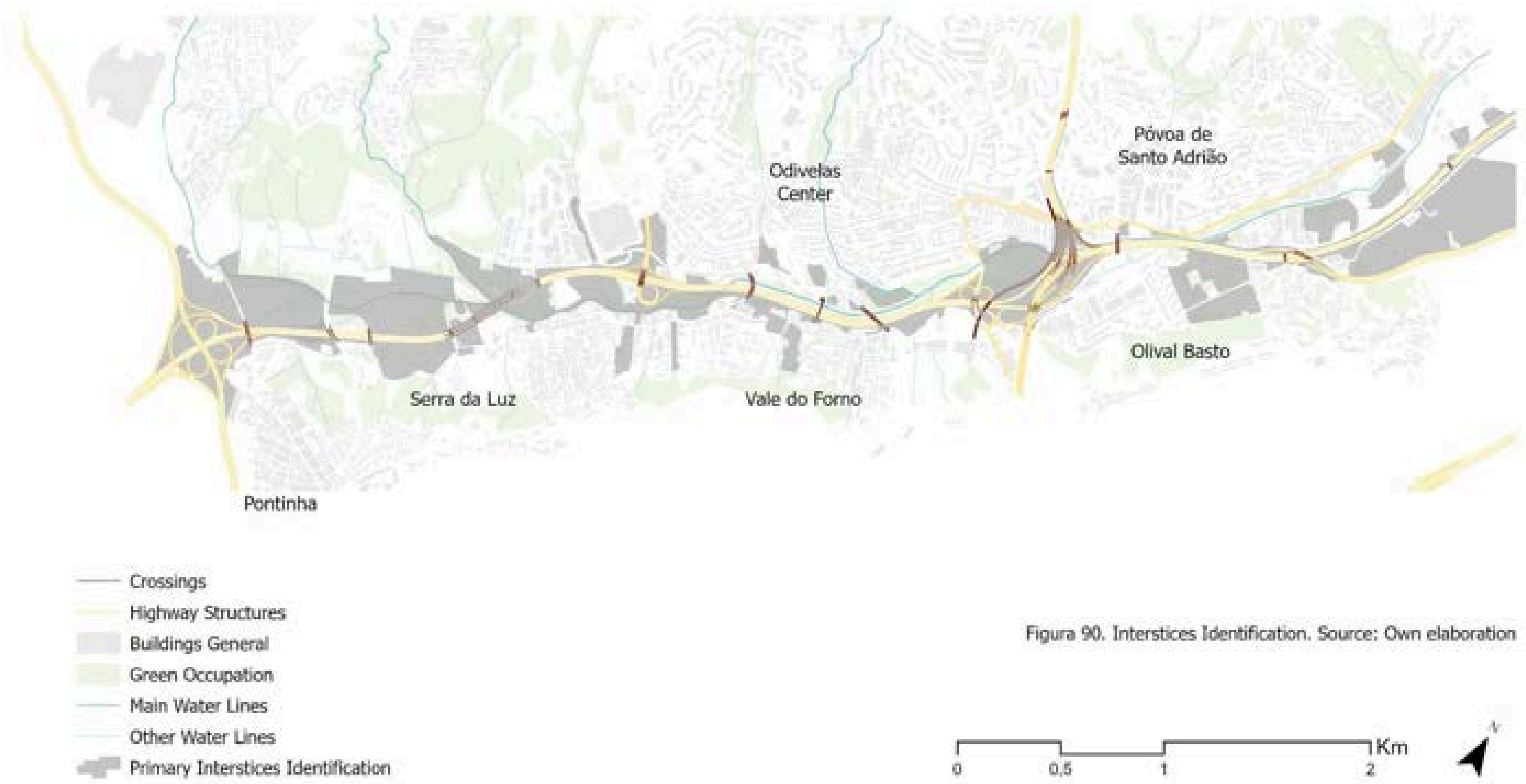


Figura 90. Interstices Identification. Source: Own elaboration

4.3.3. IC22. The urban compression of Odivelas Radial Highway

Unlike the IC17, where multiple territorial systems converge within a valley structure, the IC22 traverses a predominantly urbanised territory where the principal condition is one of compression. Urban development, mobility infrastructure and limited ecological space compete for the same surface, producing a corridor characterised by spatial discontinuity and ecological scarcity. As a radial corridor, it directly connects Odivelas to Lisbon - linking southward towards the Calçada de Carrichel - while simultaneously connecting the IC17 (CRIL) to the IC18 (CREL), making it a structurally significant axis within the metropolitan network while also being a major local structuring element.

The corridor extends northward beyond the Odivelas municipal boundary into Loures, where the territorial condition transitions progressively towards a less consolidated and more rural landscape. The present analysis focuses on the extent within Odivelas, where the urban and ecological conditions overlap, concentrates and where the data collected for this research is most complete. As with the previous corridor, the analytical boundary is understood as a reference rather than a rigid limit, incorporating relevant spatial relationships that extend beyond it where necessary.

Infrastructural Condition

The IC22 operates with a comparatively lighter cross-section than the ring roads, with 2 lanes towards Lisbon and 3 lanes towards Loures, complemented by hard shoulders on both sides. Four major junctions mark the corridor along its extent, of which two fall within the Odivelas municipal boundary. The first and most complex [Figure 91] is the interchange shared with the IC17 (CRIL) at the southern end - already discussed in the previous section - where the convergence of the ring road and the radial produces a zone of maximum infrastructural intensity, separating the Centro and Santo Adrião areas. The second junction [Figure 92], located in the central sector, between Odivelas Center, Póvoa de Santo Adrião, Ramada and Quintinha, marks a significant point of flow distribution within the municipality, marking a transition between the more consolidated western fabric and the progressively less dense eastern areas. Beyond the municipal boundary, a third interchange is visible [Figure 93], connecting with the National Road (EN250) at the edge of the analysed territory in Loures, before the corridor reaches a fourth junction connecting to the IC18 (CREL), closing the infrastructural ring around the municipality.

Figure 91. Interchange IC17-IC22
Source: Google Earth

Figure 92. Central Interchange - IC22
Source: Google Earth

Figure 93.Interchange IC22 - EN250
Source: Google Earth

The corridor is predominantly dominated by embankments and slopes as the primary form of physical closure, directly associated with the topographic adjustments required by the radial alignment across the valley system. Acoustic barriers are also extensively present along the stretches adjacent to Centro and Santo Adrião, reflecting the proximity of consolidated residential fabric and the noise exposure it generates. Against this condition of predominantly topographic and acoustic closure, the crossings concentrate primarily at the southern node and in the central sector, with the IC22 presenting the greatest proportion of pedestrian-oriented crossings of the three corridors. This combination of the corridor's radial alignment, its elevated and embanked sections, and the acoustic barriers installed along much of its urban extent still generates a significant condition of physical closure and impermeability in relation to the surrounding fabric [Figure 94] [Figure 98].

Against this condition, crossings are scarce distributed along the corridor [APPENDIX 33]. The mixed-use crossings that exist materialise mainly as underpasses - passing beneath the highway rather than above it - producing darker and more enclosed spatial conditions than the elevated crossings observed along the IC17 [Figure 95]. The pedestrian crossings present a comparatively more qualified spatial condition, with wider footpaths and greater legibility, especially between the center and Santo Adrião [Figure 96]. Further from this central concentration, towards and the northern stretches, the pedestrian infrastructure becomes progressively narrower and more recent [Figure 97], a condition consistent with the more peripheral character of these areas. In all cases, however, these crossings are oriented exclusively towards human movement, with no consideration for ecological permeability or the natural systems that the corridor intersects.

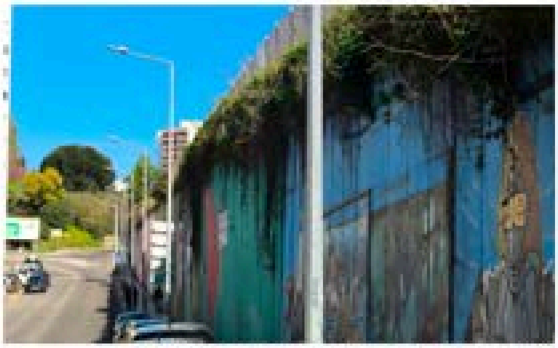


Figure 94. Barred space of the Highway. IC22.
Source: Original.

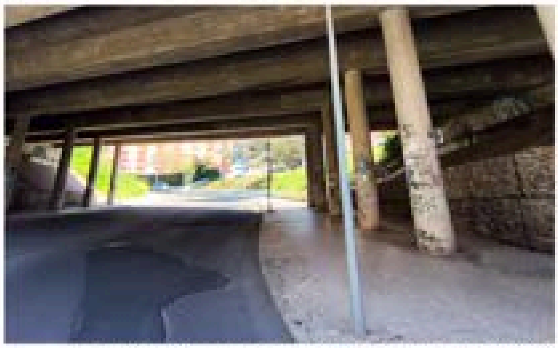


Figure 95. Crossing enclosed Space under IC22.
Source: Original.

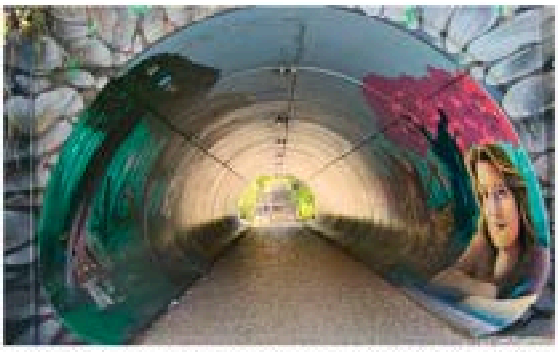


Figure 96. .Tunel Pais das Maravilhas - Center - Póvoa.
Source: Original.

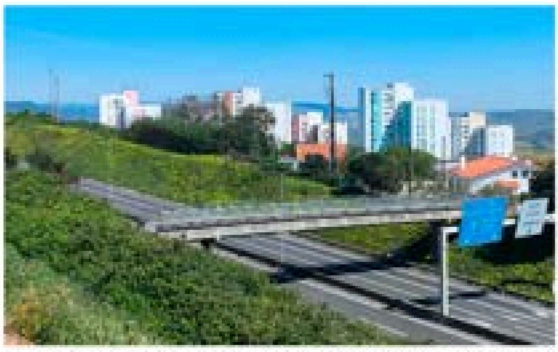


Figure 97. Recent Crossing Quitinha - Ramada in IC22
Source: Original.

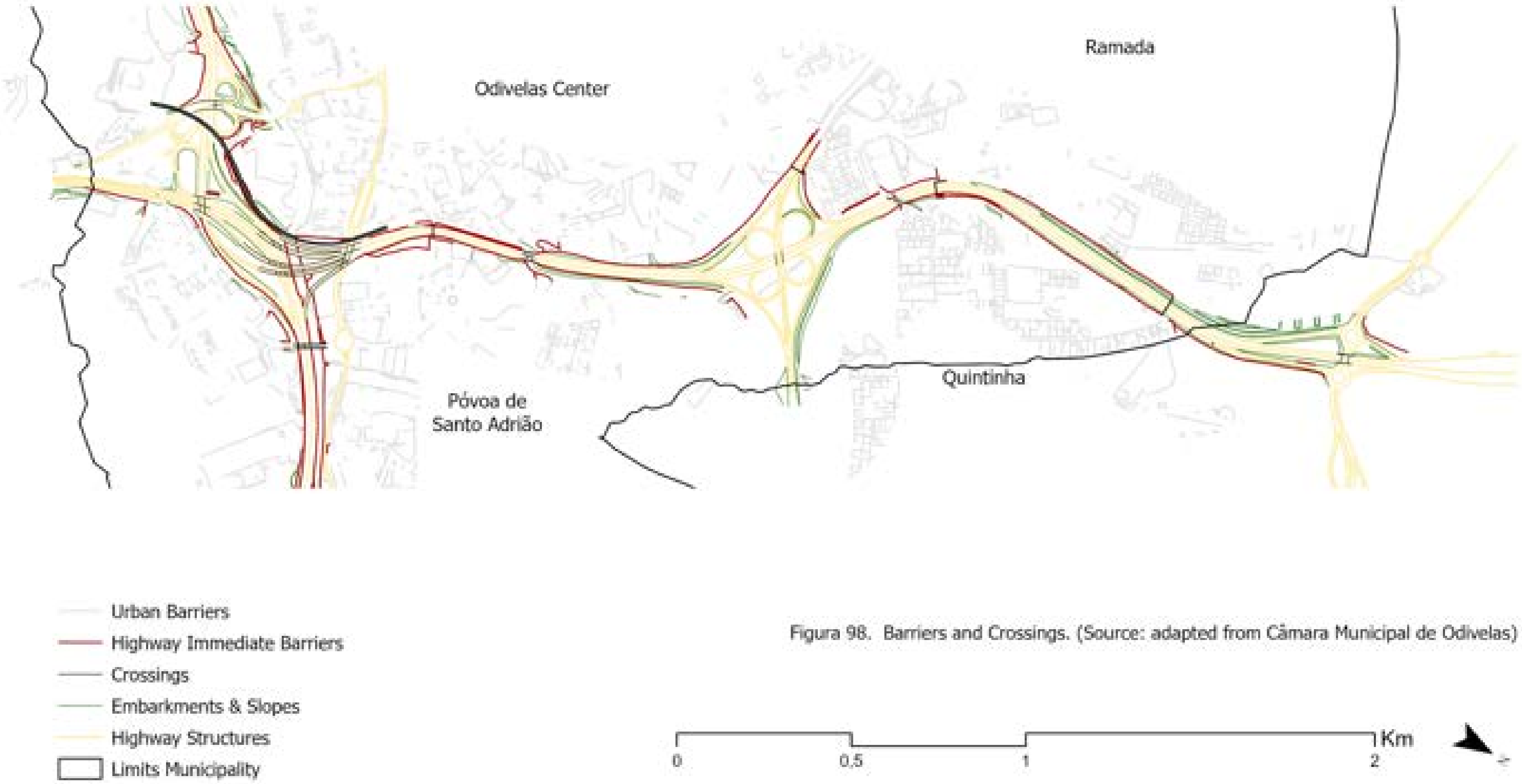


Figura 98. Barriers and Crossings. (Source: adapted from Câmara Municipal de Odivelas)

Urban Condition and Porosity.

The IC22, in contrast with the IC17, is mostly surrounded by urban fabric, except when it arrives in Loures Municipality, which then is mostly a rural area. Similar to IC17, the corridor divides the territory into the central area, and its informal AUGIS in the periphery. This contrast changes even more when merging with density, centrality, and accessibility. At a first reading, a territory organised around four distinct spatial moments - Centro, Póvoa de Santo Adrião, Ramada, and Quintinha - .each presenting a different urban condition in direct relation to the infrastructural logic of the IC22 and the barriers it creates across the territory [Figure 105]. As already suggested in the infrastructural reading, slope variation along the corridor is significant, with the terrain rising considerably from the southern valley, a condition that has also directly shaped where and how the crossing typologies occur in the corridor.

Figure 99. Póvoa de Santo Adrião Neighbour - IC22
Source: Original.

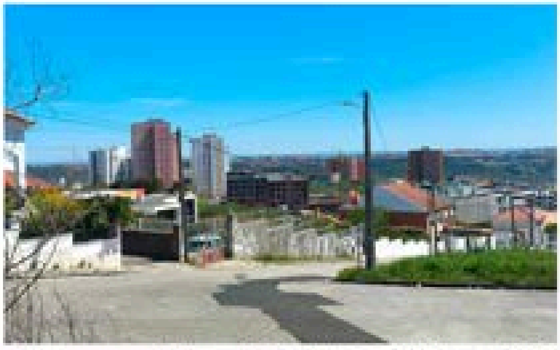


Figure 100. Quintinha Neighbour - IC22
Source: Original.

The central sector - around Odivelas Center and Póvoa de Santo Adrião [Figure 99] - presents the most consolidated condition along the corridor [APPENDIX 34]. Odivelas Center concentrates the highest building densities, commercial activity and variety of daily-use equipment, reflecting its historical role as the municipal core. Further east, Ramada and Quintinha represent two very different responses to the same infrastructural pressure. Ramada is a more recent and mixed area, with taller buildings and greater morphological diversity, a condition partly explained by its development alongside the EN250, which runs parallel to the IC22, producing a fabric that is compressed between them, with different development spurs emerging in the spaces in between. Quintinha, on the other hand, is an AUGI [APPENDIX 35] predominantly composed of low-rise single-family dwellings and remains largely informal in character [Figure 100]. Yet, unlike the AUGI zones of the IC17, this informal fabric is comparatively more open, with wider sidewalks and a greater degree of spatial legibility at street level. Equipment is nonetheless almost entirely absent on this side of the corridor, with daily use facilities and commercial activity concentrated almost exclusively on the western central side, in Centro and Ramada [APPENDIX 36]. At the same time, there are almost no crossings connecting Quintinha and Póvoa to these services or to the green spaces that exist on the other side, making it a spatially isolated area not only in infrastructural terms but also in terms of its access to the basic urban resources that the corridor separates it from. Despite these morphological differences, all four areas reveal a similar relationship with the corridor. The IC22 separates areas with distinct levels of urban consolidation and access to services, while the opportunities for movement between them remain concentrated in a limited number of crossings. As a result, proximity to the highway does not necessarily correspond to greater accessibility, but often to greater spatial fragmentation.

This condition is directly reflected in the movement capacity of the territory. The Reach analysis [APPENDIX 37] reveals that accessibility does not distribute continuously across the corridor. Higher values concentrate on the western side, while significant reductions occur across the highway, particularly towards Quintinha. This discontinuity suggests that the crossings provided by the IC22 are insufficient to overcome its barrier effect. Rather than supporting transversal movement between adjacent urban areas, accessibility remains largely organised on each side of the corridor independently.

Figure 101 - Public Space Porosity Odivelas Center.
Source: adapted from Câmara Munciipal de Odivelas

Figure 102 - Public Space Porosity. Ramada.
Source: adapted from Câmara Munciipal de Odivelas

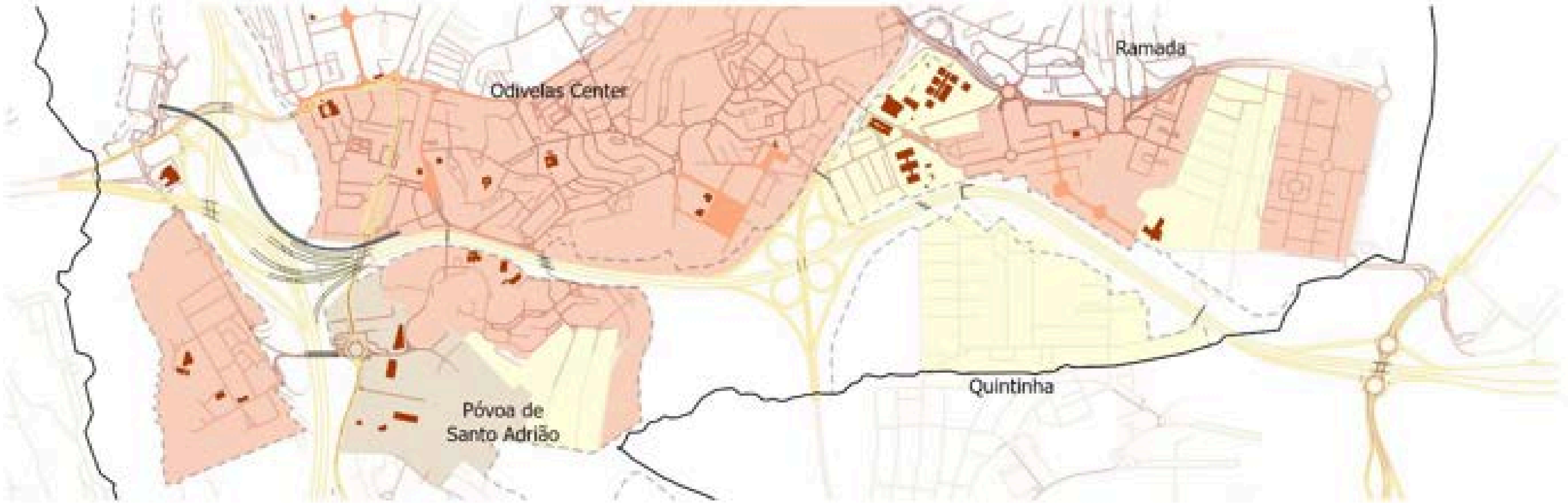
Figure 103 - Public Space Porosity. Pv. de Santo Adrião
Source: adapted from Câmara Munciipal de Odivelas

Figure 104 - Public Space Porosity. Quintinha.
Source: adapted from Câmara Munciipal de Odivelas

Reading the fabric at a closer scale [Figures 101 - 104] confirms and deepens this reading across the four distinct spatial moments. In the center, the fabric presents comparatively greater openness towards the public - buildings are mostly taller, more active, and the relationship with the street more legible - even if physical barriers remain present and the corridor itself is still unresolved. Póvoa de Santo Adrião, in its direct relationship with the IC22, presents a more heterogeneous condition. As, while predominantly residential in character, with taller buildings that generate a comparatively more open condition at street level, the area also contains industrial and warehouse uses that introduce a different spatial logic, one of larger footprints, closed facades, and vehicle-oriented access. This mix of residential and industrial occupation, compressed between the IC22 and the consolidated urban core, produces a fabric that is simultaneously denser and less legible than the Centre, with limited public space and few connections across the highway Ramada is more diverse and heterogeneous. Some spaces are completely closed and introverted, while others - particularly those associated with commercial and mixed uses - present a greater degree of openness and activity. It is also worth noting that in the areas of Ramada and Quintinha - being more recent - when closer to the corridor, more distant from the CRIL, and further from the EN250, spaces without urban occupation are more frequently encountered. Some of these are formally planned for future development, but others simply exist as residual and undefined conditions, accumulating in the spaces that the infrastructural and planning logic has not yet claimed.

Another dimension of this condition becomes evident in the distribution and continuity of public space. Along the IC22, public spaces rarely extend towards or across the corridor, becoming progressively smaller, more fragmented, or entirely absent in its immediate proximity. Much of the space between buildings - which could, instead, function as gathering spaces, social interaction, or pedestrian activity, is actually predominantly occupied by parked vehicles. Meaning, as a municipality shaped by road infrastructure from its inception, Odivelas developed around commuting, rather than local urban life, and this is spatially expressed in the appropriation of public space by the car. Along the IC22, where the urban fabric is denser and increasingly compressed by grey infrastructure, the demand for public space is higher. However, the dominance of road infrastructure has appropriated much of the available urban realm, reducing the capacity for social, recreational, and ecological functions. The contradiction lies in the fact that the areas most in need of public space are also those where infrastructure most severely limits its provision.

The elaboration of an urban synthesis map [Figure 105] consolidates this reading into a spatial pattern that reveals - beyond the individual conditions of each area - a broader territorial logic. Centro and Quintinha, even though having their significant differences in building height, typology, and history, converge towards two opposite but equally consistent conditions, one of relative openness and centrality, the other of closure and marginalisation. Ramada and Póvoa de Santo Adrião, by contrast, present more heterogeneous conditions, where different morphologies, uses, and degrees of porosity coexist without resolving into a coherent spatial logic. What is consistent across all four areas, however, is the unresolved condition of the corridor itself - much as in IC17 - the IC22 remains a barrier that fragments the continuity between these fabrics. Their internal urban conditions differ substantially, yet the interruption of transversal movement, public space continuity, and ecological connections remains a consistent characteristic along the corridor.



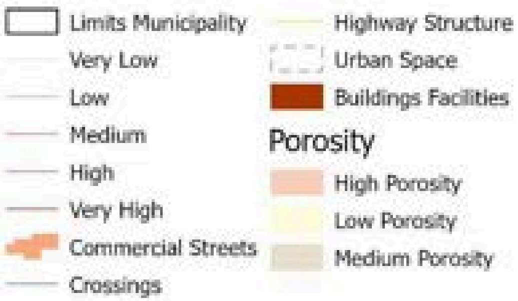


Figura 105. Urban Synthetical Analysis. (Source: Own Elaboration)

Green Infrastructure condition

The green infrastructure reading of the IC22 is markedly different from the IC17, and in some ways, could be more severe. The CRIL corridor, despite its fragmentation, remained a meaningful ecological backbone, especially due to its partial rural land use, valley position and hydrographic network. The IC22 transverses a predominantly urban and impermeable territory in which both vegetation and water are almost entirely absent.

The NDVI maps, aligned with the the Tree Cover Density (Coppernicus, 2018), the National Ecological Structure (Magalhães, 2013), the hydrographic network [APPENDIX 38] reveals a space that confirms that the immediate surroundings of the IC22 - at least within Odivelas - are overwhelmingly dominated by dense urban fabric, with vegetation concentrating at the central interchange - where the geometry of the node left residual spaces that allowed vegetation to partially occupy the space - or at the few points where the hydrographic network, though extremely limited, is still present. The crossings, as highlighted in the infrastructural reading, the crossings are mainly pedestrian and/or car focused, with little to none green/ecological crossings.This creates a space that, supposedly with great ecological potential becoming one with little to none.

Perhaps, one of the most significant readings emerges from the relationship between the built fabric and the National Ecological Structure (Magalhães, 2013). A substantial portion of the territory along the IC22 has been urbanised within areas identified by this framework as ecologically relevant for the continuity of natural processes and landscape functions. This overlap suggests not merely a reduction of ecological space, but a progressive occupation of areas that, from an ecological planning perspective, would ideally contribute to the municipality's structural ecological support system. Their location within an urbanised corridor highlights the potential role of restoration and Green Infrastructure interventions in recovering ecological continuity and environmental performance. This is not simply about ecological absence, but of ecological indebtedness. In other words, spaces that the planning framework recognizes as ecologically significant have been urbanised, and their restoration is now both a planning obligation and, with the EU Nature Restoration Law, a legal one.

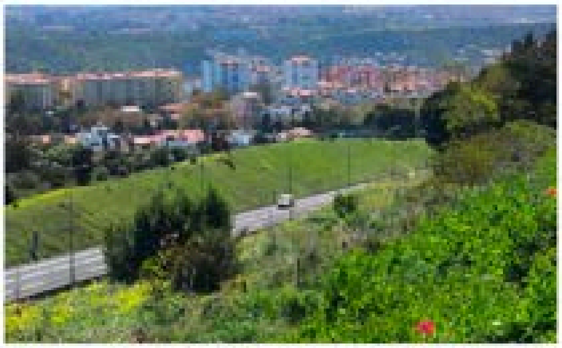


Figure 106. Embankment in Ramada - IC22
Source: Original.

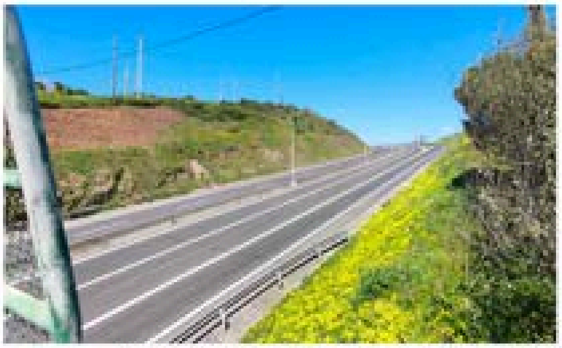


Figure 107. Embankment in Quintinha - IC22
Source: Original.

Like previously said, the IC22 highway is very much outlined by embankments [Figures 106 & 107], occupying varying stretches and having different dimensions. While these spaces carry no ecological concern in their current condition, they however, represent the most significant contributors for a possible scenario of transformation, allowing for a different intervention than the ones present along in CRIL, precisely due to their scale and relative openness.

At the same time, the formally designated green spaces along the corridor are few, spatially disconnected and, as said in the urban condition reading, focused on the western central area of the corridor. Parks appear as isolated contours without much of an ecological coherence or meaningful connection to each other or the broader municipal green structure. In some areas, the morphology and typology of buildings allows for internal squares that contribute marginally to green infrastructure, but these are not common. Some new park developments are also visible in the eastern sector, in the AUGI area of Ramada [Figures 108 & 109]. Yet these remain pedestrian-focused rather than ecologically oriented, confirming the pattern already identified along the IC17.

The Fragstats Natural Cohesion analysis [APPENDIX 39] demonstrates larger patches on the eastern side of the corridor, mainly due to its less urbanised and partially undeveloped character, a condition that makes precise classification difficult, but simultaneously signals another form of transformative potential. More specifically, coming from Póvoa de Santo Adrião, natural areas can be seen pressing towards the corridor before being interrupted by it, with a similar dynamic occurring between Quintinha and Ramada, with the exception of the park previously addressed.

Figure 108. New Park Development in Ramada - IC22
Source: Original.

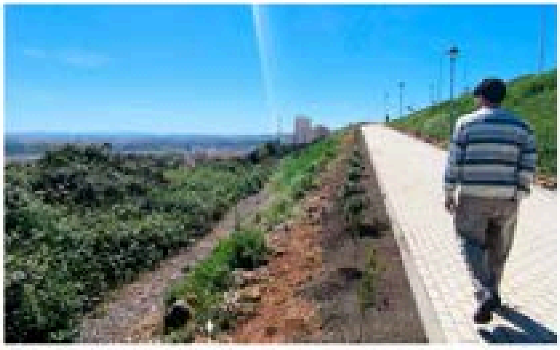


Figure 109. Embankment occupation, Ramada - IC22
Source: Original.

This condition has direct climatic consequences, with low vegetation cover, high imperviousness, dense urban fabric, and thermal mass of the infrastructural domain itself creates conditions of heat islands concentration along the corridor [APPENDIX 40], a condition that the absence of any meaningful green infrastructure makes structurally difficult to mitigate.

The green and blue synthesis map [Figure 110] consolidates this reading into a spatial portrait that is, at its core, one of absence - the absence of a coherent ecological system, of functional green connectivity, and of any meaningful relationship between the corridor and the natural structure that surrounds it. What green presence exists is fragmented, residual, and disconnected - concentrated at the edges of the analysed territory and in the spontaneous occupation of infrastructural margins, rather than forming any system capable of delivering ecological or climatic services at the scale the corridor demands. The IC22, is, in ecological terms, the most exposed, and maybe least resourced of the three corridors, which is a condition that makes the identification and activation of its interstitial spaces not merely an opportunity, but an ecological necessity.

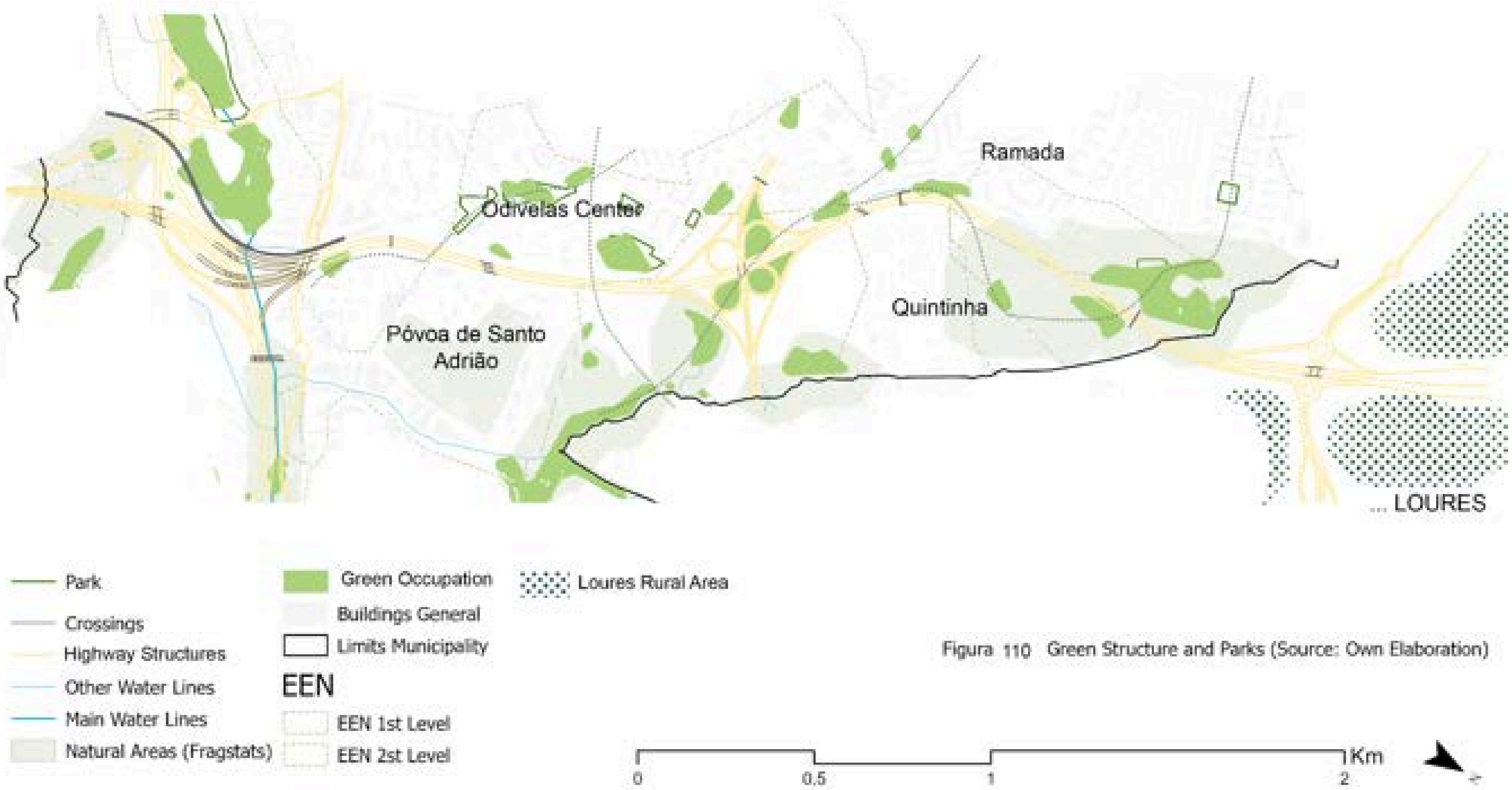


Figura 110 Green Structure and Parks (Source: Own Elaboration)

A synthetical view of IC22 and its latent potential condition.

The IC22 presents a territorial condition fundamentally different from both the IC17 and the IC18. While the former is characterised by the convergence of multiple territorial systems and the latter by its territorial detachment, the defining condition of the IC22 is one of urban compression. Traversing one of the most urbanised and impermeabilised sectors of Odivelas, the corridor concentrates competing demands for mobility, urban development, public space, and ecological function within a limited territorial framework. As a result, the corridor operates less as an isolated infrastructural barrier and more as a mechanism that intensifies existing spatial inequalities and discontinuities between adjacent urban areas.

This condition is reflected directly in the distribution of its interstitial spaces. Unlike the IC17, where residual spaces emerge from the interaction between infrastructure, hydrology, and ecological systems, the interstitial spaces of the IC22 are primarily generated by urban incompleteness, infrastructural geometry, and topographic adjustment. They concentrate around the central interchange, along embankments, and within areas where urbanisation remains unfinished or spatially unresolved. Their significance lies not in their ecological quality, which is generally limited, but in their capacity to accommodate functions that the surrounding urban fabric has struggled to provide.

The corridor's ecological condition reinforces this reading. Ecological systems are not merely fragmented, but largely absent. Large portions of the territory identified within the National Ecological Structure have been progressively urbanised, producing a condition of ecological indebtedness in which restoration depends not on reconnecting existing systems, but on re-establishing ecological functions that have been displaced. The few green spaces that exist remain isolated and disconnected, while the embankments and undeveloped margins represent some of the only continuous surfaces available for ecological intervention. Following the criteria established in Chapter 2, the interstitial spaces identified along the IC22 concentrate precisely where these conditions overlap: at infrastructural nodes, within undeveloped or partially urbanised areas, and in the spaces generated by the corridor's topographic and geometric requirements [Figure 111]. Their boundaries remain ambiguous, reflecting the broader condition of a corridor suspended between urban consolidation and unresolved territorial transformation.

The transformative potential of the IC22 lies precisely in this condition. In a corridor where urbanisation, mobility infrastructure, and ecological systems compete for limited space, the interstitial areas represent some of the few opportunities capable of simultaneously expanding public space, restoring ecological functions, and reducing the corridor's persistent condition of urban compression. Their value resides not in what they currently are, but in their capacity to accommodate the territorial functions that the surrounding urban fabric has progressively lost.

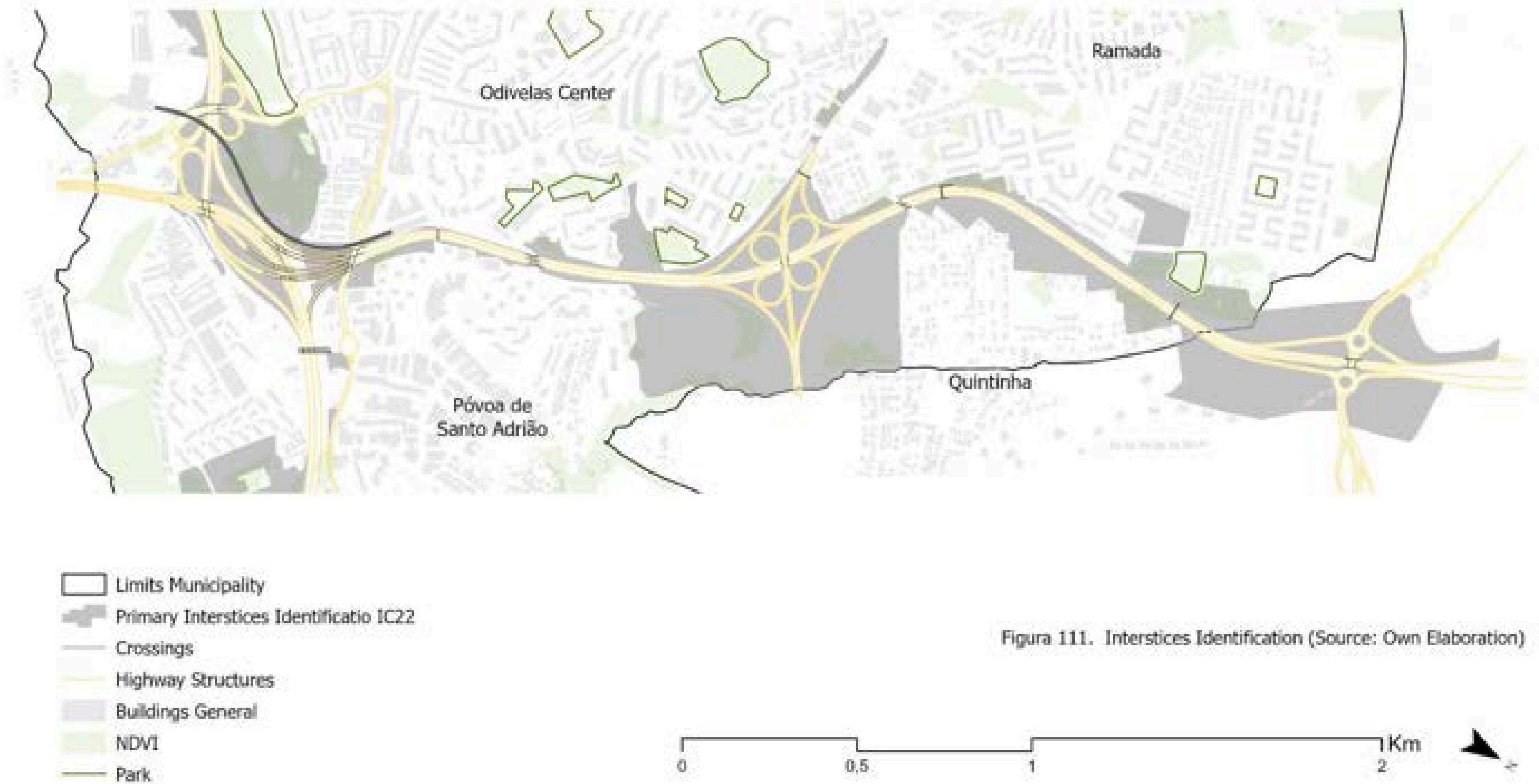


Figura 111. Interstices Identification (Source: Own Elaboration)

4.3.4. IC18 - Territorial marginality and infrastructural isolation

The IC18 (CREL) represents the most territorially peripheral detached corridor, traversing Odivelas without establishing direct urban relationships with it. Due to it being the outer ring road of the Lisbon Metropolitan Area, it defines the western and northern boundary of Odivelas, operating at the edge of the municipal territory rather than through its consolidated fabric. And unlike the IC17, which divides the urban interior, or the IC22, which connects the municipality radially to Lisbon, the IC18 passes through Odivelas without establishing connections to it, its junctions located outside the municipal boundary, in Amadora to the west and Loures to the east, leaving the corridor as a largely uninterrupted line that traverses the territory without belonging to it.

Infrastructural Condition

The IC18 operates with a heavy cross-section - typically 3 lanes in each direction plus hard shoulders - generating, as with the IC17, a barrier width of 30 or more metres of inaccessible surface before even considering the additional offsets produced by the corridor's highly variable topographic relationship with the terrain. This relationship is one of the most defining characteristics of the IC18, and is what distinguishes it clearly from the other two corridors. Also, rather than a uniform condition of embankments or slopes, the CREL alternates between deep terrain cuts [Figure 112] - where the road is excavated into the hillside, producing steep and imposing rock faces - and elevated sections, where the road rises above the surrounding terrain, generating high acoustic and retaining walls that close the view and access from below. Each of these conditions produces a distinct barrier typology and a distinct spatial experience for the surrounding fabric.

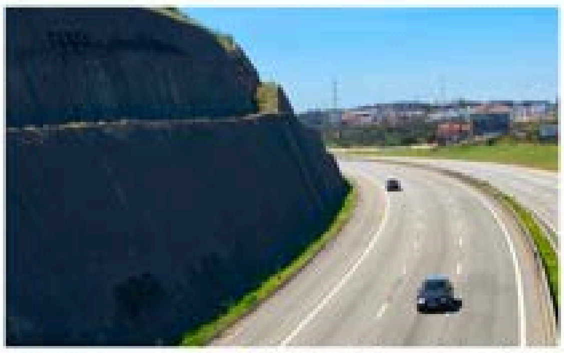


Figure 112. Terrain Deep Cuts - IC18.
Source: Original.

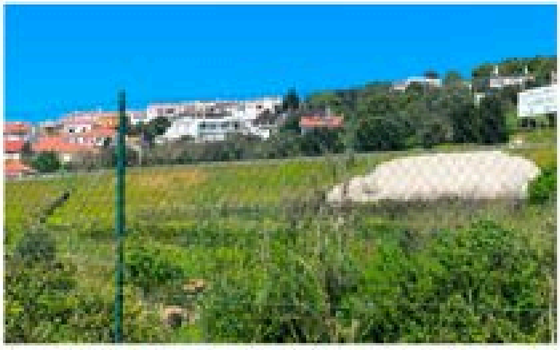


Figure 113. Barriers and Limitations of Access (a)- IC18
Source: Original.


Figure 114. Barriers and Limitations of Access (b) - IC18
Source: Original.

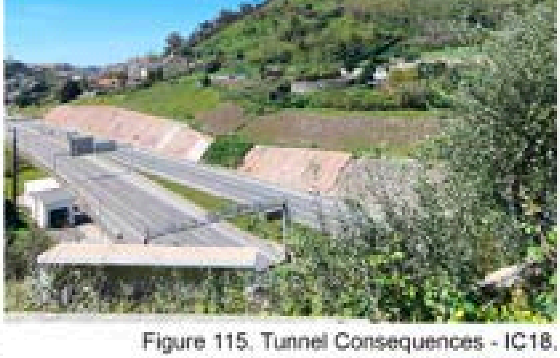


Figure 115. Tunnel Consequences - IC18.
Source: Original.

Acoustic barriers of significant height are installed along much of the corridor's extent, closing the visual and physical relationship between the road and the surrounding fabric. These barriers - while serving a noise mitigation function - add a further layer of physical and visual closure to a corridor that already imposes a severe barrier condition through its cross-section alone. Where the road is elevated above the terrain, walls and fences are imposed to the fabric, avoiding contact with the highway [Figures 113 & 114]. This condition creates a situation in which the immediate barriers along the IC18 form one of the most continuous and uninterrupted lines of the three corridors, with very few moments of openness or transition.

The tunnel, though located outside the Odivelas municipal boundary, has its entrance within the municipality, and this proximity produces some of the most severe infrastructural conditions along the entire corridor. The terrain cuts and retaining structures required to accommodate the tunnel approach occupy significant stretches of the municipal territory, generating large-scale excavations and slopes [Figure 112] that are among the most imposing physical presences along the CREL within Odivelas.

Figure 116. EN250 Underpass - IC22
Source: Original.

Figure 117. Historical Area Crossing - IC22
Source: Original.

The crossings [APPENDIX 41] at IC18 present the lowest number of crossings of the three corridors. Beyond the EN250 crossing [Figure 116], which constitutes the one of the only genuinely functional and legible points of negotiation along the corridor, the remaining crossings are predominantly vehicle-oriented and narrow [Figure 117]. There is, along the IC18, almost no meaningful pedestrian infrastructure, and no ecological provision whatsoever. The corridor does not merely separate, it excludes all and everything [Figure 118].

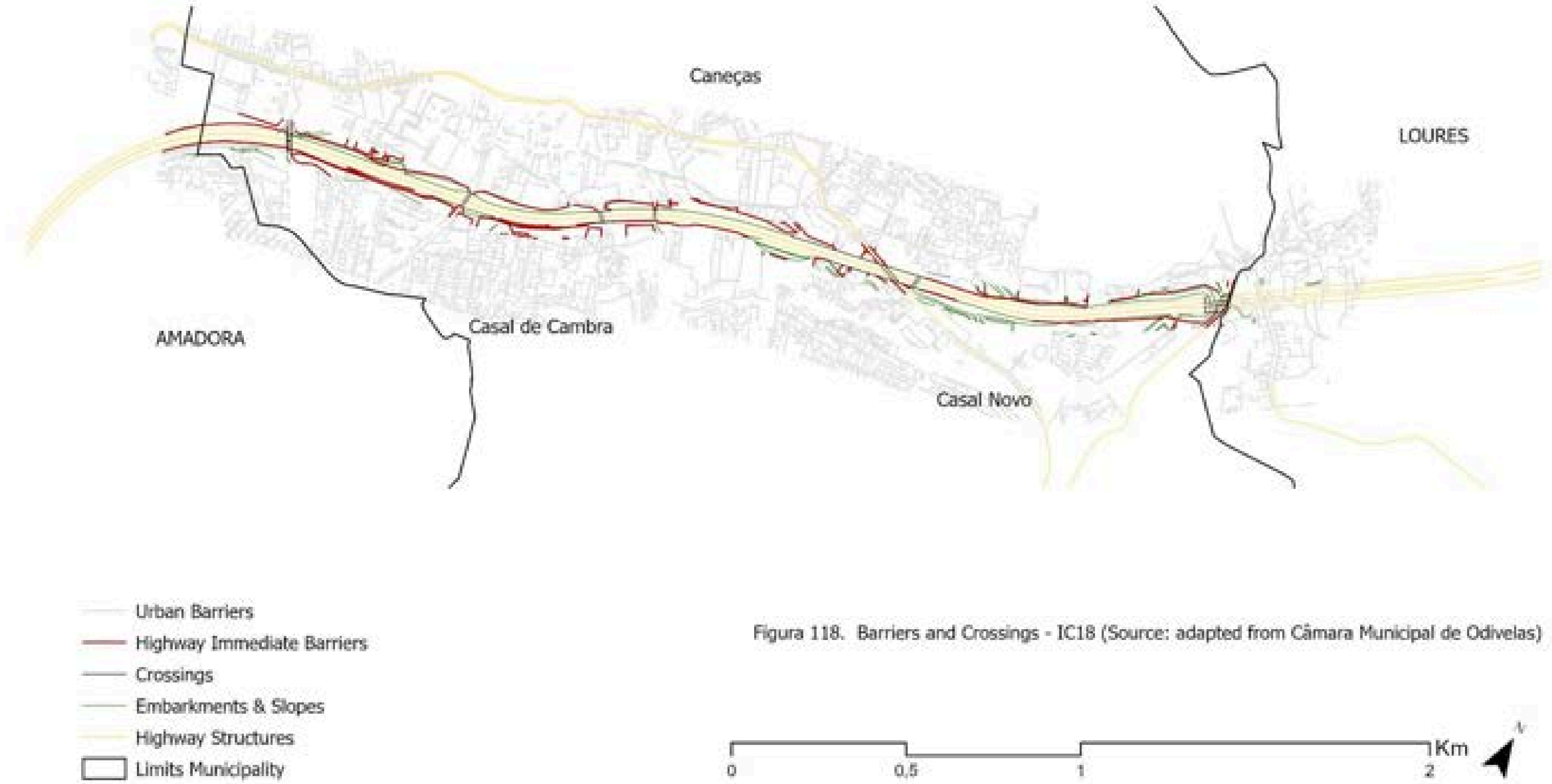


Figura 118. Barriers and Crossings - IC18 (Source: adapted from Câmara Municipal de Odivelas)

Urban Condition and Porosity.

The IC18 corridor runs through what is, of the three analysed highways, the most uniformly and peripheral urban condition. The PDM classifies much of the land adjacent to the IC18 as non-urban - rural or forest areas - which is paradoxically correct, given the absence of built occupation, but that also coexists with the barrier structures, terrain cuts, and embankments that dominate these spaces in practice. The spaces that planning calls rural are in reality infrastructural margins, not necessarily natural nor meaningfully urban, existing in a condition of formal and spatial indeterminacy that contributes to the disconnection between the corridor and the territory it traverses. The urban space itself is effectively offset by the highway's marginal condition [APPENDIX 42], pushed back and compressed by the infrastructural domain without being able to establish any relationship with it. This reality becomes even more significant if looking into a fundamental fact, as addressed before, the IC18 has no junctions, no nodes, no connections to Odivelas territory itself. Yet, despite this absence of connection, its presence has profoundly conditioned the spatial development of the territory on both sides, reducing opportunities for transversal movement and contributing to the physical separation of neighbouring urban areas. Therefore, it is an infrastructure that interrupts the local network, with no access, no centrality, no integration [APPENDIX 43]. The lack of qualified crossings, for pedestrians more than for cars, contributes even more to this reading.

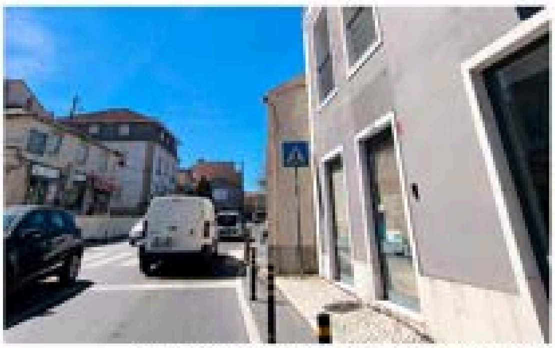


Figure 119. Caneças Center EN250 - IC18
Source: Original.

Figure 120. Casal de Cambra - IC18
Source: Original.

The corridor divides two distinct urban realities. To the north of the corridor - in the area of Caneças, one of the oldest areas of Odivelas - the fabric is comparatively more consolidated, with a modest degree of centrality associated with the EN250 development [Figure 119]. It is here that the few equipment and services present along the entire corridor are concentrated, including daily use facilities and health services, even if not much compared to the other corridors [APPENDIX 44]. To the south - in Casal Novo and Casal de Cambra - the condition is different. These areas developed later, especially due to not having a residual centrality to anchor itself such as the EN250, through informal processes, with a lot of the areas being classified as AUGIS [Figure 120] [APPENDIX 45].

By reading the fabric at a closer scale [Figures 121 - 124], it is possible to confirm that Caneças, Casal de Cambra and Casal Novo share a similar relationship with the corridor. Public space is generally limited to local streets, small squares and residual open spaces embedded within the urban fabric. As the highway is approached, this public realm progressively gives way to embankments, retaining structures, acoustic barriers and residual land. The corridor therefore does not operate as a destination, frontage or organising element of public life. Rather, it functions as a spatial edge where the urban fabric effectively stops, reducing opportunities for continuity, interaction and collective use.

Figure 121 - Public Space Porosity. Caneças.
Source: adapted from Câmara Munciipal de Odivelas

Figure 122 - Public Space Porosity. Caneças.
Source: adapted from Câmara Munciipal de Odivelas

Figure 123 - Public Space Porosity. Casal de Cambra.
Source: adapted from Câmara Munciipal de Odivelas

Figure 124 - Public Space Porosity. Casal Novo.
Source: adapted from Câmara Munciipal de Odivelas

Beyond the division between the two sides of the corridor, the urban fabric of the IC18 presents a further layer of internal fragmentation that is particularly evident in this context. Within each area - in Caneças as much as in Casal de Cambra and Casal Novo - the morphology is deeply discontinuous, as already seen in the Urban Atlas Land Use Map [see section 4.2.2.2], with vacant lots, unbuilt parcels, and abrupt transitions between different building typologies creating ruptures within the fabric itself, independently of the highway. This internal fragmentation, combined with the barrier condition of the infrastructure, produces a territory that is broken not only between its two sides but also within them, a condition that makes the activation of the marginal spaces of the corridor more consequential.

It is precisely in this context that the formal activation of the corridor's marginal spaces becomes most consequential, where the urban fabric is too dense, too informal, and too under resourced to generate qualified public or ecological space from within, the infrastructural margins represent the only available surface through which a different territorial logic can begin to be introduced [Figure 125].

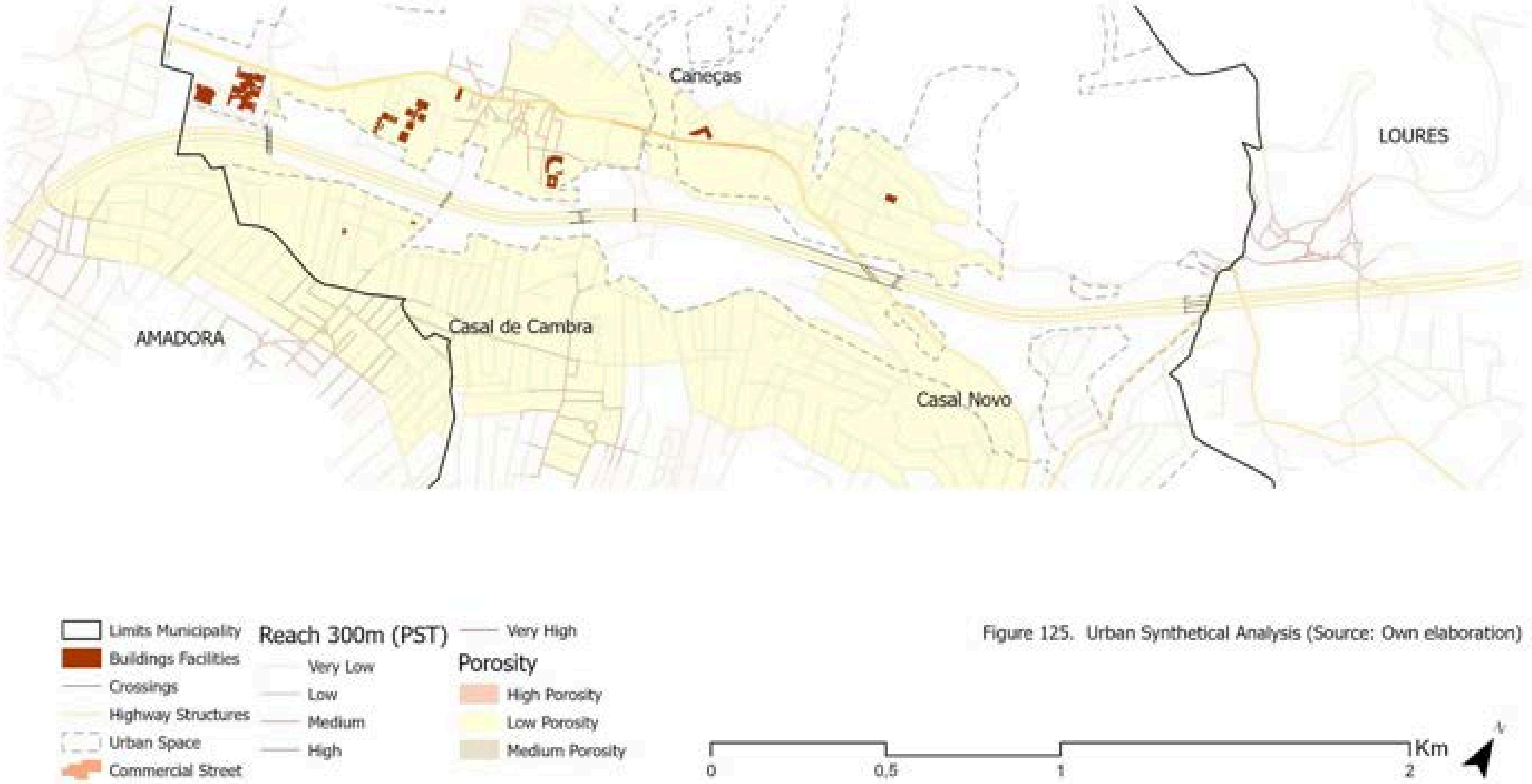


Figure 125. Urban Synthetical Analysis (Source: Own elaboration)

Green Infrastructure Condition

The green reading [Figure 128] of the IC18 reveals a condition that is, the most ecologically expressive of the three corridors, yet simultaneously the most structurally fragmented and least functionally coherent. The IC18 traverses a territory that the PDM formally classifies as non-urban, yet the Cohesion analysis [APPENDIX 46] confirms that this designation does not translate into ecological function.

Watercourse, which is the only meaningful hydrological element along the IC18 within Odivelas, one of, if not the only one of, the corridor's most significant ecological asset. Its presence could explain a little why certain areas adjacent to the via have remained unbuilt, the flooding zones mapped in the PDM [APPENDIX 47] confirm that the land immediately alongside the stream is periodically inundated, a condition that has inadvertently preserved some of the most ecologically valuable spaces along the corridor. The geotechnical risk map [APPENDIX 48] reinforces this, areas of greatest slope instability concentrate and the terrain cuts generated by the infrastructure, confirming that the IC18 was implanted in one of the most physically fragile zones of the municipality. Rather than a problem to be mitigated, this overlapping condition of flooding, geotechnical risk, and ecological potential represents a latent system that the infrastructure has interrupted but not entirely disrupted.

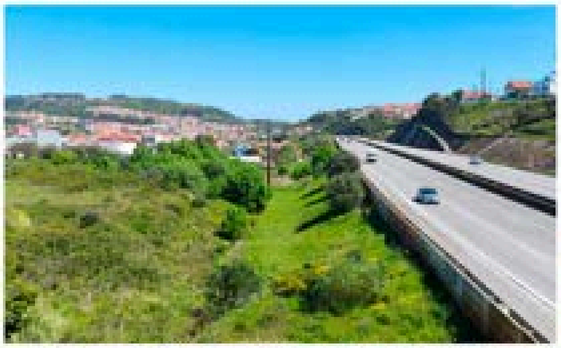


Figure 126. Caneças Stream along highway - IC18
Source: Original.

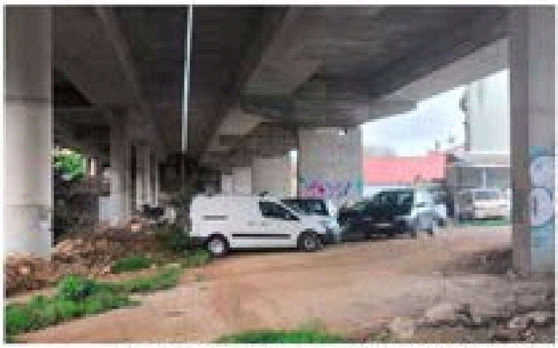


Figure 127.Underpass with EN250 and IC18
Source: Original.

The point where the watercourse passes beneath the corridor is particularly revealing. Through field work, it was observed that this crossing produces a condition similar to what was identified along the IC17, a dark, enclosed, and abandoned space beneath the infrastructural structure, where water, vegetation, and movement converge without any meaningful ecological or spatial activation. It is also here that the only crossing of any hydrological significance occurs along the entire IC18, and like other crossings, it does not have any ecological legibility. The natural patches that exist are fragmented and discontinuous, and what coherence they present follows almost exclusively the alignment of the small watercourse that runs alongside and beneath the corridor [Figures 126 & 127].

The NDVI and Tree Cover Density maps [APPENDIX 49] confirm this reality and clearly show that the vegetation that exists follows the hydrological logic, concentrating along the watercourse and in the less consolidated rural northern sector. Still, the thermal reading [APPENDIX 50] reinforces a bit of the “green” condition, with the IC18 presenting comparatively lower heat island intensities than the other two corridors, a direct consequence of its lower urban density and greater residual vegetation. However, a zone of higher thermal stress is identifiable precisely where the non-urban designation coexists with low NDVI values, suggesting that formal land use classification alone does not produce ecological cooling, and that vegetation cover is the determining factor.

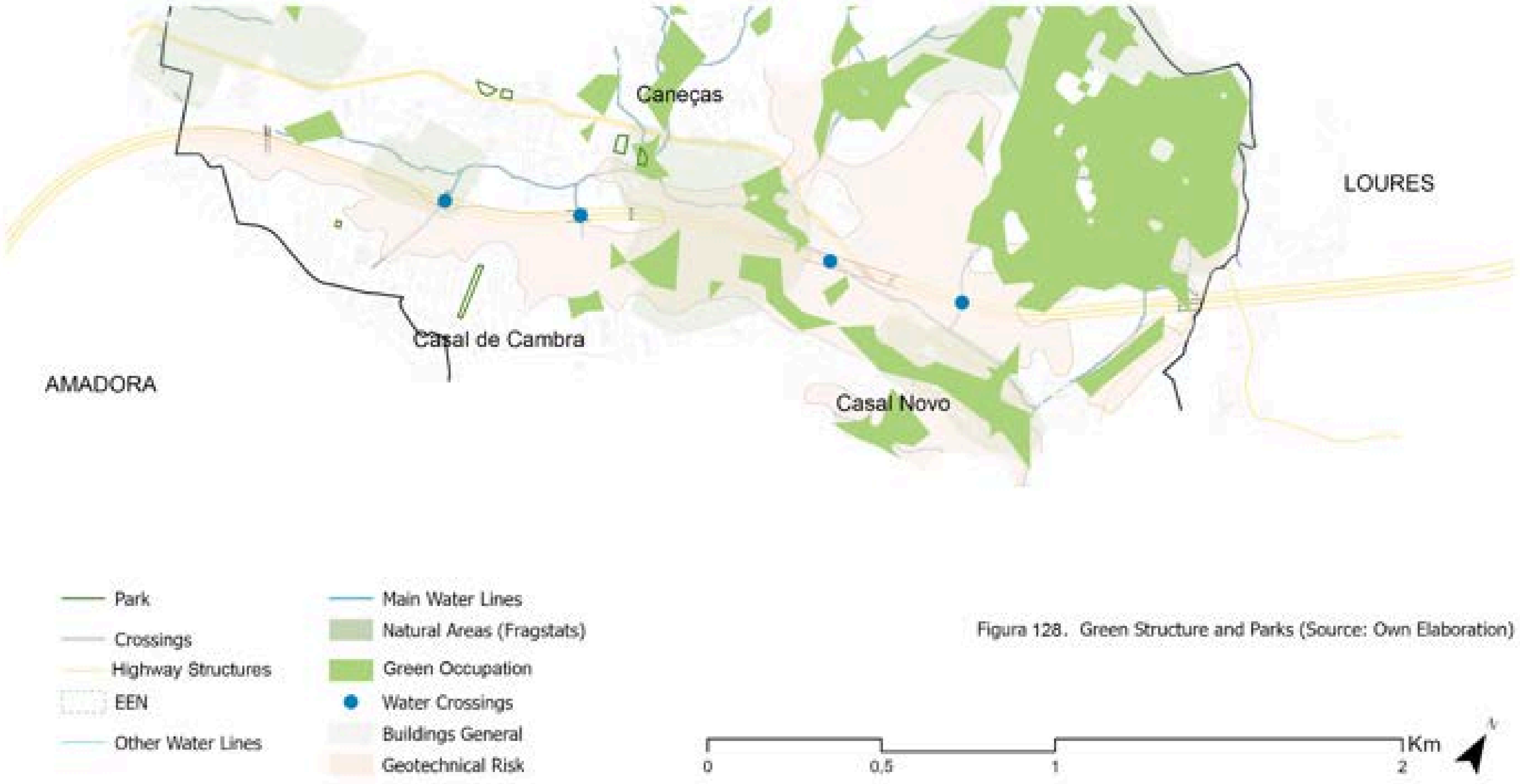


Figura 128. Green Structure and Parks (Source: Own Elaboration)

A synthetical view of IC18 and its latent potential condition.

More than any other corridor analysed, the IC18 is defined by detachment. Unlike the IC17, where multiple systems converge, or the IC22, where urban functions compete for limited space, the IC18 operates largely outside the territorial logic of Odivelas itself. Its junctions lie beyond the municipal boundary, its crossings are scarce and predominantly vehicle-oriented, and its surrounding spaces remain weakly connected to both the urban fabric and the ecological systems that traverse them. Rather than structuring urban life, the corridor functions primarily as a line of separation.

This condition of detachment is expressed not only in urban terms but also ecologically. Although the corridor traverses areas formally designated in the PDM as rural or forest land, ecological continuity remains fragmented and functionally weak. Natural patches persist, particularly around the watercourse that accompanies part of the corridor, yet they rarely operate as a coherent system. The corridor therefore separates territories that are themselves already fragmented, reinforcing existing discontinuities rather than generating new forms of connection.

The corridor's most significant ecological structure is the watercourse that accompanies part of its alignment. It concentrates flood-prone areas, residual riparian vegetation and some of the most ecologically relevant spaces along the corridor. At the point where it passes beneath the infrastructure, however, this potential is interrupted, producing one of the clearest examples of the corridor's condition of abandonment and disconnection.

The interstitial spaces identified along the IC18 emerge directly from this condition of detachment. Concentrated around terrain cuts, embankments, infrastructural margins and hydrological crossings, they occupy territories that belong neither fully to infrastructure nor to landscape. Following the criteria established in Chapter 2, these spaces were identified where the tensions between infrastructural occupation, hydrological systems and weak ecological performance become most evident [Figure 129]. Their significance lies less in urban pressure or ecological intensity than in their condition of abandonment and indeterminacy.

The transformative potential of the IC18 lies precisely within this condition of detachment. Because the corridor currently contributes little beyond movement and separation, interventions within its interstitial spaces have the capacity to introduce functions that are largely absent today: ecological continuity, public accessibility, spatial integration, and territorial identity. The corridor's greatest opportunity therefore emerges from the same condition that defines it: its distance from the territory it traverses. By transforming spaces that currently belong to neither infrastructure nor landscape, it becomes possible to establish relationships where none presently exist.

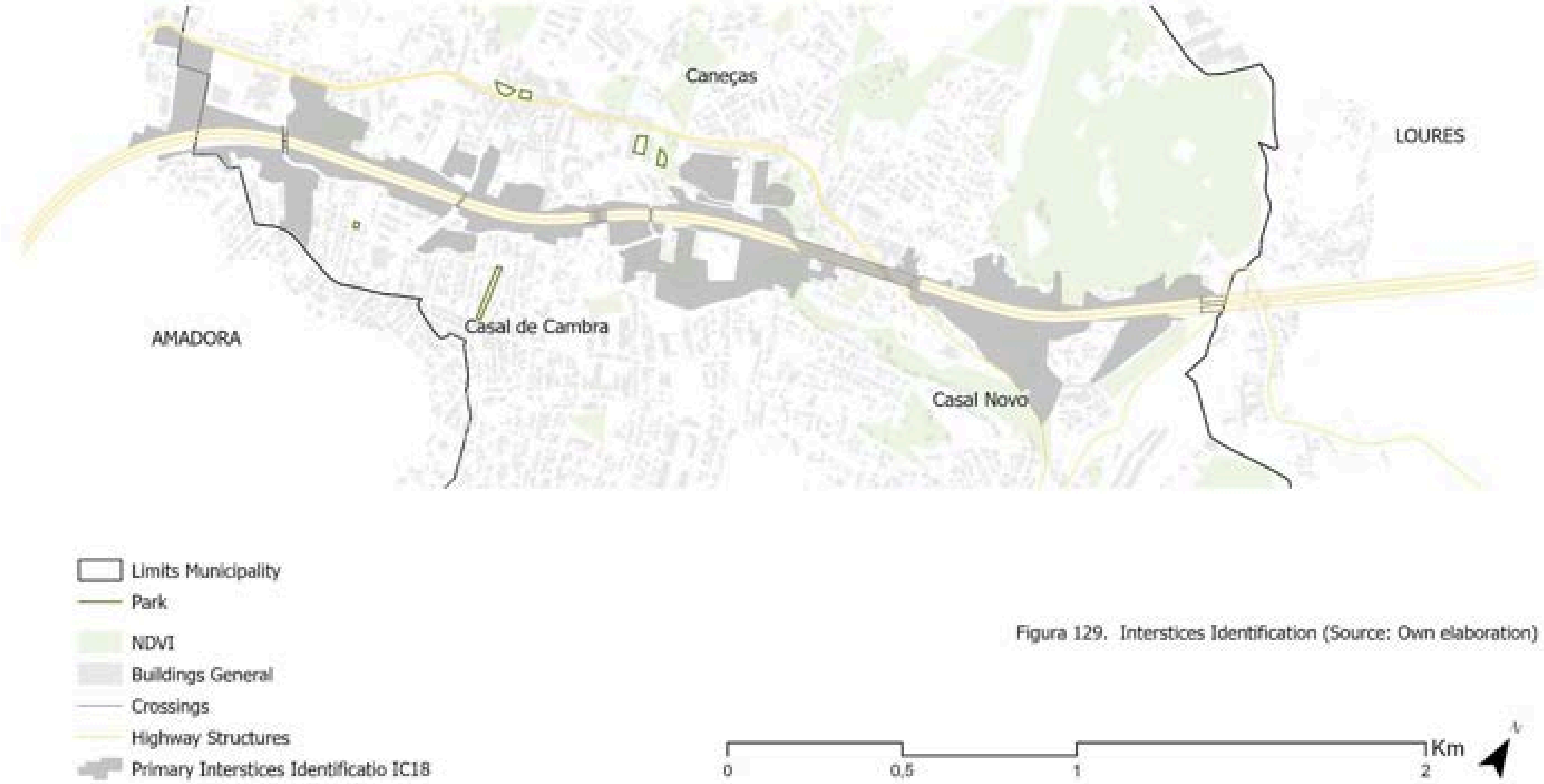


Figura 129. Interstices Identification (Source: Own elaboration)

4.3. Typological Synthesis of Infrastructural Interstices

The corridor-scale analysis developed throughout this chapter enables a municipal reading of the interstitial spaces produced by Odivelas' highway network. Rather than emerging from a single environmental, morphological, or infrastructural condition, these spaces result from the interaction between multiple territorial processes identified along the IC17, IC22 and IC18. The objective of the typology is not to classify ecological, urban or infrastructural conditions individually, but to classify recurring spatial situations generated by their overlap [Figure 130].

Based on the spaces mapped throughout the three corridors, this research proposes a typological classification of interstitial spaces specific to the Odivelas context. The objective is not to establish a universal framework applicable to all infrastructural landscapes, but rather to organise the recurrent spatial conditions observed within the study area into a coherent analytical structure. The proposed categories should therefore be understood as interpretative and operative tools that support the reading and discussion of these spaces in Odivelas, while remaining open to refinement through application in other territorial contexts.

The first type is Understructure and Node Spaces, corresponding to spaces generated by the geometry of interchanges, viaducts and elevated infrastructural structures. Concentrated primarily at major highway junctions, these spaces emerge where ramps, bridges and circulation systems overlap, producing spaces that remain subordinate to the operational requirements of the infrastructure itself. Unlike other interstitial conditions, their boundaries are often relatively legible, being defined by the infrastructural structures that enclose them. Their indeterminacy lies not in their physical limits but in their function and territorial role. Frequently designed to accommodate visibility requirements, maintenance access, drainage systems or safety clearances, these spaces rarely support ecological, social or urban functions beyond those required by the infrastructure. As a result, they often remain spatially present but functionally unresolved, occupying strategic locations within the territorial structure while lacking a clearly defined relationship with the surrounding urban fabric.

The second category, Embankments and Marginal Linear Spaces, refers to spaces generated directly through the physical construction of the corridor itself. These include embankments, terrain cuts, slopes and other linear spaces produced by the topographic adjustments required to accommodate the highway. Their form and extent are inseparable from the infrastructural geometry that generated them. While, at the same time, their linear configuration, steep topography and proximity to the highway often make them difficult to access and occupy. As a result, they frequently operate as spaces whose primary role remains infrastructural, while their ecological, social and public-space functions remain limited or underdeveloped. Positioned between the operational domain of the highway and adjacent urban or ecological systems, they often function as elongated transitional landscapes between infrastructure and the surrounding territory.

The third category, Primary Marginal Open Spaces, comprises the largest and most spatially extensive interstitial areas identified throughout the municipality. These spaces are generally located adjacent to highway corridors but are not directly generated by specific infrastructural elements such as embankments, interchanges or engineering structures. Instead, they emerge from the broader territorial influence of the infrastructure, occupying areas that remain undefined between urban development, ecological systems and infrastructural domains. Their defining characteristic is not a particular morphology but rather their condition of openness and availability. Unlike the more constrained infrastructural typologies, these spaces are not entirely determined by engineering requirements, allowing greater flexibility in how they may evolve over time. Although frequently perceived as vacant or residual land, they often concentrate multiple territorial processes simultaneously, including ecological remnants, informal uses, hydrological systems, and future urban pressures. This condition makes them some of the most adaptable and potentially transformative spaces within the interstitial landscape.

Beyond these primary categories, a complementary group of spaces was also identified. Named here as Secondary Open Spaces, these areas are not directly adjacent to the highway network but maintain spatial and functional relationships with the broader interstitial system. Frequently disconnected from surrounding urban structures and lacking a clear role within existing planning frameworks, they reinforce the fragmented character of the territory while simultaneously revealing opportunities for future articulation and transformation. The proposed typology should not be understood as a definitive classification system. Interstitial spaces frequently combine characteristics associated with multiple categories, reflecting the inherent ambiguity and indeterminacy that define infrastructural landscapes. The categories therefore operate less as fixed definitions than as analytical devices through which recurrent spatial conditions can be identified, discussed and compared. Their boundaries remain interpretative - as a reflection of the indeterminate character of interstitial spaces themselves - and the classification of individual spaces inevitably involves degrees of approximation and judgement.

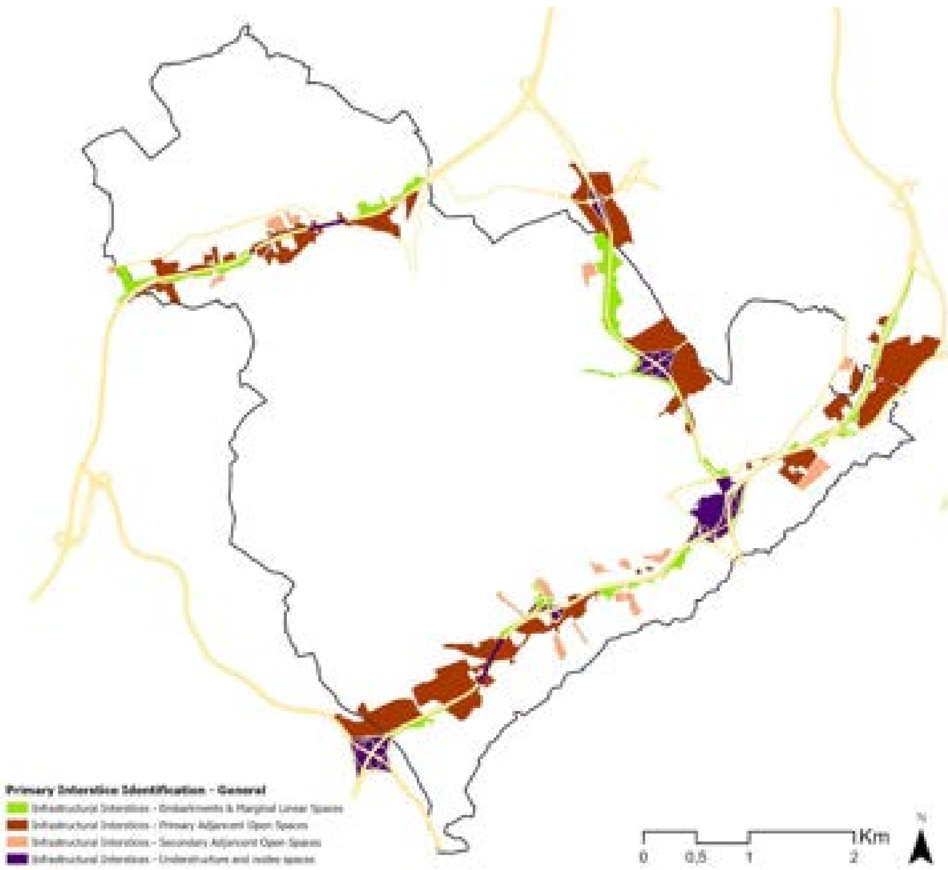


Figure 130. Municipal Interstices Typologies. Source: Own Elaboration

5. Natural-Based Solutions: Strategic Directions for the Activation of Odivelas Interstitial Spaces

5.1. Typological Synthesis of Infrastructural Interstices

The typological synthesis developed in Chapter 4 identified a series of recurrent interstitial conditions produced by Odivelas' highway network. While these spaces differ significantly in their morphology, location and territorial context, they share a common characteristic: the coexistence of spatial fragmentation, ecological disruption and functional indetermination. In many cases, these conditions are reinforced by the very infrastructural systems that generated them, producing spaces that remain disconnected from both urban and ecological networks.

Within this context, Nature-Based Solutions (NBS) are explored not as predefined design interventions for specific typologies, but as a means of responding to the specific challenges identified within each interstitial condition. Their relevance derives from their capacity to address multiple territorial issues simultaneously, be it ecologically or morphologically. Rather than introducing entirely new systems, NBS operate by revealing, restoring or strengthening ecological processes already present within the territory.

The proposals developed in this chapter therefore emerge directly from the analytical reading established in the previous chapters. The objective is not to identify a universal set of solutions applicable to all interstitial spaces, but to explore how different Nature-Based Solutions can respond to the distinct conditions observed within Odivelas. Particular attention is given to the role these spaces can play within a broader municipal green infrastructure network, supporting ecological connectivity, hydrological functioning and landscape continuity across the territory.

To operationalise this approach, five representative interstitial spaces were selected from the corridors analysed previously. The selection does not seek to cover every possible condition present within the municipality. Instead, the chosen cases represent different expressions of the typological categories identified and allow the relationship between spatial condition, territorial challenge and Nature-Based Solution to be examined in greater detail.

The evaluation of each case is presented through a structured analytical sheet. Each sheet combines two complementary dimensions. The first establishes a diagnostic reading of the site through Barrier Condition, Latent Connectivity and Functional Indetermination, identifying the principal challenges and latent opportunities present within the space. The second examines its operative capacity through Mediation, Requalification and Articulation, exploring how intervention may contribute not only to the transformation of the site itself but also to broader territorial systems. At the end, the Nature-Based Solutions proposed for each case are a direct response to this reading. They are not prescribed in advance and then applied, but derived from the conditions and capacities identified, calibrated to what each space demands and what each space makes possible, and how it can integrate into a green infrastructure plan. The NBS families and typologies suggested area draw directly from the international and national references analysed in chapter 2, is proposed.

The resulting Nature-Based Solution proposals are not intended as definitive projects, but as demonstrative applications of the analytical approach developed throughout this research, adapted to different interstitial conditions within Odivelas.

Sheet 1: Under-viaduct space - IC17 (CRIL)

Western viaduct, Serra da Luz - Western Paiã Rural Area

LOCATION & CHARACTERIZATION

Intervention Area | Surrounding Interstices | Equipments | Parks

Watercourses | Productive Area

0 0,05 0,1 0,2 Km

CORRIDOR	IC17 (CRIL) - Western sector
TYPOLOGY	Understructure and node space
CONTEXT	AUGI zones of Serra da Luz and rural area

KEY CHALLENGES

- Culverted watercourse, suppressed riparian function
- Ecological marginalisation
- Inaccessibility and poor safety conditions
- Disconnection from AUGI communities and rural landscape

Before NBS Interventions

After NBS Interventions

ANALYTICAL CONDITIONS - How the space is defined by its....

Barrier Condition

- Viaduct structure suppresses the spatial and ecological expression of the watercourse.
- Riparian vegetation is absent, and the hydrological system is reduced to a culverted condition.
- Viaduct structure physically encloses the space, limiting access and visibility, being dark, uninviting spatial character discourages pedestrian use.

Latent Connectivity

- Adjacent to AUGI communities with demand for public space and community equipment.
- Proximity to commercial uses and sports facilities.
- Hydrological sytems alignment of the buried watercourse.
- The site is embedded within a broader system of interstitial spaces with similar transformation potential.

Functional Indetermination

- The space performs no clearly defined ecological or social function
- Inside the Ecological Reserve and considered a Forestry space while abandoned, with informal waste disposal, waterlogged / flooded areas.
- Infrastructure dominate without generating complementary functions

OPERATIVE CAPACITY - What the space is capable of in...

Mediation

The site mediates between the lack of public space in nearby AUGI areas, the absence of vegetation and riparian ecological structure along the watercourse, and the underused residual space beneath the viaduct. It operates within a condition where social, ecological, and infrastructural deficits coexist in the same spatial field.

Requalification

The site is requalified through the transformation of the under-viaduct condition into a structured public and ecological landscape. This includes the reintroduction of vegetation, the reactivation of riparian systems along the watercourse, and the creation of accessible public space for surrounding AUGI communities. Shifting the abandoned are into and active enviromental and social space.

Articulation

The site articulates the hydrological system with surrounding urban and rural territories, establishing continuity between water infrastructure, urban fabric, and rural landscapes. Another point is that it contributes to the integration and activation of the surroundings interstitial spaces, aiming towards green infraestructure.

GREEN INFRASTRUCTUTRE ROLE

Plannig Positioning	Plannig Positioning	Obbligation	Aims	Role
Within a vital ecological corridor (PROT-AML, 2002), designated for landscape valorisation	Classified under REN and RAN (PDM Odivelas), formally constraining urbanisation	EU Nature Restoration Law: riparian restoration (Art. 9) and urban greening (Art. 8)	Primary ecological need: flood control and riparian system reactivation	Stepping stone between fragmented ecological patches along the IC17 valley corridor

Underneath Activation

The underside of the viaduct structure is transformed into a shaded public space with permeable paving and low-maintenance planting, new recreative uses, making the infrastructural presence legible as a covered public asset serving the adjacent AUGI communities and serving as a transition to the rural space.

Bioretention & Wetlands

The partially buried watercourse is revegetated with native riparian species (Aluvial & Calcarium), with bioretention planters filtering highway runoff before it reaches the restored hydrological system, while at the same time promoting **floodable parks**.

Urban Green Corridor

The existing crossing is reconfigured to support both pedestrian movement and ecological permeability, introducing native vegetation along its edges that follows the hydrological alignment of the watercourse. The focus here should be the ecological function it possesses, instead of human atraction.

Additionally, the flat margins adjacent to the space present potential for Urban Farming, formalising the informal agricultural occupation already present in the surrounding area; and a Spontaneous Occupation management strategy could reinforce native species along the embankments, following the logic of the Likoto métaprojet.

Sheet 2: IC17 - IC22 Interchange Node

Central Node of the IC17-IC22 Interchange, Odivelas Center / Senhor Roubado

T1 - Understructure & Node IC17 - IC22

LOCATION & CHARACTERIZATION

Intervention Area | Surrounding Interstices | Equipments | Parks

Watercourses | Productive Area

0 | 0,1 | 0,2 | 0,4 Km

CORRIDOR: IC17 (CRIL) - IC22

TYPOLOGY: Understructure and node space

CONTEXT: Major Node at Lisbon - Odivelas Entrance.

KEY CHALLENGES

- Inaccessibility and unsafe pedestrian crossings
- High thermal stress and flooding in the surroundings
- Large scale and complexity render the space illegible
- Urban and ecological disconnection.

Before NBS Interventions

After NBS Interventions

ANALYTICAL CONDITIONS - How the space is defined by its....

Barrier Condition

- The infrastructural system is composed of ramps, elevated sections, and diverging carriageways that fragment and enclose the central space.
- Access is discontinuous, producing multiple sub-zones that cannot be perceived or experienced as a unified spatial system.
- Enclosure is generated by infrastructural scale and complexity, along physical boundaries.

Latent Connectivity

- The site is positioned between key urban and ecological systems, including the Parque Urbano do Rio da Costa.
- At the north, a dense urban area (commercial streets and mobility nodes). At the south, AUGIS Communities.
- Inside the valley system and with hydrological system passing
- The site operates at the convergence of multiple mobility and landscape systems.

Functional Indetermination

- The space operates at a scale that exceeds conventional landscape or residual classifications.
- It contains fragmented and discontinuous uses, including informal plots, spontaneous invasive vegetation, and residual ground conditions.
- Informal agricultural areas occupies some areas.
- Structural scale reduces legibility of the valley space.

OPERATIVE CAPACITY - What the space is capable of in...

Mediation

The site mediates between high-intensity urban systems and peripheral AUGI territories. It sits between mobility infrastructure, ecological corridors, and fragmented urban fabric.

Requalification

The site is requalified through the includes the introduction of structured green areas, improved pedestrian conditions, and basic water management strategies that address flood vulnerability. At the same time, improvement of the existing crossing towards ecological and pedestrian movement. This process enhances environmental performance while creating accessible public space.

Articulation

This space has the capacity to extend and complete the Parque Urbano do Rio da Costa, transforming what is currently its end into a new node of a larger green system. It articulates with the Metro station and commercial street to the north, with the communities of Senhor Roubado to the south, and with the broader ecological corridor of the IC17 valley. Also connecting with Calçada do Carriche, being an new entrance to Odivelas.

GREEN INFRASTRUCTUTRE ROLE

Plannig Positioning	Plannig Positioning	Obbligation	Aims	Role
Near ecological bottleneck (PROT-AML), with potential to integrate the vital corridor network	Forest and recreational designation (PDM Odivelas), outside REN but adjacent to REN-integrated watercourse	EU Nature Restoration Law: riparian restoration (Art. 9) and urban greening (Art. 8)	Primary ecological need: heat island mitigation and stormwater management	Strategic green bridge between fragmented urban areas at metropolitan scale

NBS RESPONSES

Urban Green Corridor: The scale and central position of this space make it the most significant opportunity for a metropolitan green corridor in the municipality, a continuous ecological and public space system that connects the Parque Urbano do Rio da Costa and the center towards Calçada do Carriche.

Underneath Activation: The residual spaces beneath the elevated ramps and sections are activated as shaded public spaces, integrating community recreative uses, permeable surfaces, and native planting that transform the most enclosed and inaccessible sub-zones of the node into usable public infrastructure.

Floodable Park: The proximity to the Rio da Costa hydrological system and most surrounding floodable areas of the municipality should be integrated as a design feature rather than a constraint. A floodable landscape that manages water dynamics, reduces downstream flood risk.

Additionally, **Urban Farming** could formalise and expand the informal agricultural occupation already present; **Wildlife Crossings** could be introduced at key points establishing safe ecological traversal across the corridor, with new tunnels and enlarging of the structures, together with a safer pedestrian connection; and at a more radical point of view, the scale of this space opens the possibility of **Road Capping** interventions over specific ramp sections, creating continuous green surface above active infrastructure.

Sheet 3: Informal Parking Space - IC17 (CRIL)

Southern Edge of the Corridor, Vale do Forno / Senhor Roubado

T2 - Primary Marginal Open Space IC17

LOCATION & CHARACTERIZATION

Intervention Area | Surrounding Interstices | Equipments | Parks

Watercourses | Productive Area

Km 0 0.03 0.06 0.11

CORRIDOR IC17 (CRIL) - Southeast sector

TYPOLOGY Primary Marginal Open Space

CONTEXT Informal Parking Lot in the AUGI zones of Vale do Forno right next to the IC17

KEY CHALLENGES

- Suppressed riparian function
- Informal occupation
- Multiple overlapping barriers limits permeability.
- No connection to Parque Urbano do Rio da Costa,

Before NBS Interventions

After NBS Interventions

ANALYTICAL CONDITIONS - How the space is defined by its....

Barrier Condition

- IC17 carriageway defines a dominant infrastructural boundary.
- Physical separation is reinforced by metal grates that seal the corridor boundary and limit permeability.
- Watercourse is confined and ecologically suppressed.
- Crossings prioritize vehicular flow.

Latent Connectivity

- Urban agriculture in Vale do Forno and Senhor Roubado.
- Hydrological continuity along corridor.
- Parque Urbano do Rio da Costa, located on the opposite side of the IC17 with no patial connection between the park and the southern side of the corridor.
- The site sits within a broader system of marginal and underutilized landscapes

Functional Indetermination

- The site is occupied as an informal car park.
- Areas with low permeability and unmanaged.
- Flooding areas.
- Disjunction between use and spatial potential.

OPERATIVE CAPACITY - What the space is capable of in...

Mediation

The site mediates the absence of functional urban connections, public space with the presence of productive uses.It accommodates flood-prone conditions through its spatial configuration. It also reduces the legibility of surrounding infrastructure at pedestrian scale.

Requalification

Requalification occurs through the transformation of informal car occupation into a public and ecological space. The watercourse is reintroduced as a visible element, and vegetation is integrated to improve environmental quality, improved pedestrian conditions, and support the surrounding AUGIS. At the same time, supporting the current practices and the parking space is qualified.

Articulation

The site establishes connections between local conditions and wider territorial systems, including valley structures and green infrastructure networks. At a broader scale, it contributes to the continuity of ecological and urban systems across the municipality through the integration of multiple intervention points into a coherent landscape framework.

GREEN INFRASTRUCTUTRE ROLE

Plannig Positioning	Plannig Positioning	Obbligation	Aims	Role
High ecological value land (PDM Odivelas), adjacent to REN and RAN boundaries .	Forest and recreational designation (PDM Odivelas), outside REN but adjacent to REN-integrated watercourse	EU Nature Restoration Law: riparian restoration (Art. 9) and urban greening (Art. 8).	Primary ecological need: flood risk management and riparian system activation.	Support for AUGI communities, combining public green provision with risk mitigation.

NBS RESPONSES

Bioretention & Wetlands

The small watercourse running between the space and the highway is integrated as a visible and managed ecological feature, with bioretention areas that absorb and filter stormwater while creating a green edge that separates the activated space from the infrastructural domain.

Urban Farming

The space should formalise and expands the agricultural occupation already present in the Vale do Forno and Senhor Roubado, creating a productive landscape that connects the AUGI communities to the land they have already begun to appropriate, transforming an informal practice into a planned and supported public use.

Wildlife Crossing

The existing crossing is reconfigured to support both pedestrian movement and ecological permeability, introducing native vegetation along its edges that follows the hydrological alignment of the watercourse. The focus here should be the ecological function it possesses, instead of human atraction.

Additionally, an Urban Green Corridor logic could extend the activation beyond this space, connecting it longitudinally to other interstitial spaces along the IC17 valley; and a Spontaneous Occupation management strategy could consolidate the native vegetation already present along the margins.

Sheet 4: Embankment - IC22 (Radial de Odivelas)

Linear Embankment, Odivelas Center side / Póvoa de Santo Adrião.

T3 - Embankments and Marginal Linear Spaces IC22

LOCATION & CHARACTERIZATION

Intervention Area | Surrounding Interstices | Equipments | Parks

Watercourses | Productive Area

0 0.07 0.15 0.3 Km

CORRIDOR	IC22
TYPOLOGY	Embankments and Marginal Linear Spaces
CONTEXT	Between Odivelas Center and Póvoa de Santo Adrião, at Odivelas Center side.

KEY CHALLENGES

- Heat thermal stress and lack of vegetation.
- Presence of invasive species across the slope surface
- Absence of ecological function and public green space
- Disconnection between existing parks

Before NBS Interventions

After NBS Interventions

ANALYTICAL CONDITIONS - How the space is defined by its....

Barrier Condition

- IC22 and void surroundings makes an area with no need to "reach".
- Continuous embankment acts as a physical and visual barrier along its entire extent.
- Slope and geometry prevents access and continuity.
- Acoustic walls reinforce separation between infrastructure and residential areas.

Latent Connectivity

- Located between Póvoa de Santo Adrião and central Odivelas, making it Potential territorial continuity between two consolidated urban areas.
- Adjacent to interstitial open spaces along the embankment.
- Indirect connection to IC22 interchange and Parque Urbano da Quinta Nova.
- Existing crossings at the extremities and parks along the space.

Functional Indetermination

- Embankment operates as a purely infrastructural surface without public or ecological function.
- Physically present but not accessible or integrated into urban life.
- Spontaneous vegetation indicates informal ecological presence.

OPERATIVE CAPACITY - What the space is capable of in...

Mediation

The site mediates between the consolidated urban fabric of Odivelas and the less developed areas to the east, operating as a linear transition condition between contrasting urban territories. Its form allows a gradual spatial negotiation along the corridor, while also mediating the ecological loss resulting from urban development, where remaining natural systems are no longer continuous or clearly expressed.

Requalification

The embankment is requalified through its transformation from a technical infrastructural surface into a linear public and ecological space. This involves the introduction of native vegetation, the management of spontaneous growth, and the creation of continuous pedestrian accessibility along its full extent. The corridor is reactivated as a green infrastructure, improving environmental conditions and contributing to thermal regulation across the urban edge.

Articulation

The embankment articulates with the central park, adjacent open spaces, and existing interstitial areas along its length. Also articualtes consolidated urban areas, while it connects fragmented green elements into a continuous linear system and strengthens relationships between isolated urban fragments and ecological corridors.

GREEN INFRASTRUCTUTRE ROLE

Plannig Positioning	Plannig Positioning	Obbligation	Aims	Role
Partially within vital ecological corridor (PROT--AML) and high ecological value areas (PDM)	Forest recreational designation and classified within ecological structure outside REN	EU Nature Restoration Law: urban greening (Art. 8) and vegetation management.	Primary ecological need: urban heat mitigation in a dense consolidated urban context	Linear connector between fragmented green elements and to manage spontaneous vegetation

NBS RESPONSES

Urban Green Corridor

The linear extent of the embankment is activated as a continuous ecological corridor, introducing native vegetation along its slope and creating a longitudinal pedestrian path that connects the two existing crossings at its extremities into a coherent green spine alongside the corridor.

Spontaneous Occupation

The spontaneous vegetation already present on the slope is managed rather than cleared, reinforcing native species, suppressing invasive ones, and transforming an unintentional colonisation into a deliberate ecological asset, following the logic of the Likoto métaprojet.

Wildlife Crossing

Punctual ecological crossings are introduced at strategic points along the embankment's extent, creating transversal permeability for fauna movement between the two sides of the corridor, complementing the longitudinal green corridor logic with a transversal ecological dimension.

Additionally, at a more radical register, Road Capping interventions at specific points along the embankment could create moments of greater transversal continuity between the two sides, transforming the slope from a barrier into a surface of connection, and opening the possibility of a more complete morphological integration between Odivelas Centro and Póvoa de Santo Adrião.

Sheet 5: Floodplain - IC18 (CREL)

Central Sector, between Caneças and Casal Novo, Adjacent to the EN250.

T2 - Primary Marginal Open Spaces IC18

Intervention Area | Surrounding Interstices | Equipments | Parks

Watercourses | Productive Area

0 0,05 0,1 0,2 Km

CORRIDOR	IC18
TYPOLOGY	Primary Marginal Open Spaces
CONTEXT	Central sector, between Caneças and Casal Novo. Adjacent to the EN250

KEY CHALLENGES

- Morphological urban disconnection
- Absence of public green space
- Periodic flooding limiting accessibility and public activation
- Informal AUGI occupation progressively encroaching

Before NBS Interventions

After NBS Interventions

ANALYTICAL CONDITIONS - How the space is defined by its....

Barrier Condition

- The space is enclosed by metal grates along its edges.
- IC18 carriageway and EN250 define strong northern and eastern boundaries.
- Slightl slope reinforces separation from surroundings.
- Watercourse and periodic flooding increase physical inaccessibility.

Latent Connectivity

- Embedded within fragmented AUGI urban fabric.
- Watercourse and dam indicate a wider hydrological system beyond the site.
- Dense vegetation at the south suggests ecological continuity potential
- A single underpass provides limited transversal connection.

Functional Indetermination

- Site maintained in an unbuilt condition due to flooding and geotechnical constraints.
- Functions primarily as an infrastructural buffer between major roads and urban fabric.
- Low vegetation coverage and high thermal exposure.
- Informal residential uses reaching into the area.

OPERATIVE CAPACITY - What the space is capable of in...

Mediation

The site mediates between AUGI communities located on multiple sides of the corridor and the ecological and hydrological systems embedded within the site. It operates between socially vulnerable urban areas with limited access to green space and a landscape shaped by flooding dynamics and infrastructural constraints generated by the IC18 and EN250.

Requalification

The site is requalified through the activation of its ecological and hydrological processes as structuring elements of the landscape. Flooding is integrated as an operative condition rather than a constraint, supporting the introduction of native vegetation. Also enables the provision of accessible public green space and safer transversal connections for surrounding AUGI communities.

Articulation

It connects the dense southern vegetation with surrounding urban and rural territories, reinforcing the green continuity across the IC18 corridor. At a wider scale, it contributes to the legibility and functional coherence of the ecological network by linking isolated green systems into a more continuous territorial structure

GREEN INFRASTRUCTUTRE ROLE

Plannig Positioning	Plannig Positioning	Obbligation	Aims	Role
Classified within REN, formally constraining urbanisation and signalling ecological concern.	Forest and recreational designation (PDM Odivelas), outside REN but adjacent to REN-integrated watercourse	EU Nature Restoration Law: riparian restoration (Art. 9), urban greening and forest restoraition (Art. 8 *12)	Primary ecological need: flood risk management, geotechnical stabilisation and riparian restoration	Support for AUGI communities, combining public green provision with risk mitigation.

NBS RESPONSES

Floodable Park

The periodic flooding condition should be integrated as a defining feature of the space, designing a landscape that accommodates water fluctuation, creates wetland habitats during flood periods, and functions as an accessible public space during dry (and even flooded) periods.

Bioretention & Wetlands

The existing hydrological system - the watercourse, the dam, and the flooding dynamics - could be managed and revegetated with native species, restoring ecological function to a system that is present but currently suppressed, and improving water quality before it continues downstream.

Urban Green Corridor

The space is connected to the dense vegetation to the south through a longitudinal green system and also follows the hydrological alignment, making it possible to establisj a continuous ecological corridor that links the space begins to give spatial expression to a designation that currently exists only on paper.

Additionally, a Wildlife Crossing - in addition to the underneath activation of the viaduct on the western edge intervention - over the IC18 could serve a dual function, establishing ecological permeability across the corridor while simultaneously creating a safe pedestrian connection between the AUGI communities to the south and Caneças to the north.

5.2. Typological Synthesis of Infrastructural Interstices

The five Nature-Based Solution proposals developed throughout this chapter aim to demonstrate how interstitial spaces can be transformed through site-specific interventions. However, their significance extends beyond the boundaries of the spaces themselves. While each proposal responds to a particular set of local conditions, the ecological, spatial and social benefits they seek to generate depend largely on the relationships established with the surrounding territory.

The interstitial spaces identified throughout Odivelas do not function as isolated fragments. They are - by the very nature of interstitial spaces as understood in this research - embedded within wider ecological systems - embedded within wider ecological systems, urban fabrics, hydrological structures and mobility networks that extend beyond the limits of individual sites. In many cases, these spaces occupy strategic positions between ecological corridors, watercourses, urban parks, agricultural areas and infrastructural systems that currently remain weakly connected. Their transformation therefore presents an opportunity to reinforce ecological continuity while simultaneously improving accessibility, public space provision and landscape quality.

Consequently, the effects of Nature-Based Solutions cannot be understood solely through the transformation of the interstitial space itself, but also through their capacity to reinforce broader territorial relationships. Several of the ecological functions activated through the proposed interventions correspond directly to objectives promoted by contemporary environmental frameworks such as the EU Nature Restoration Law, including the restoration of riparian margins (Article 9), the reinforcement of ecological connectivity, the expansion of urban green infrastructure (Article 8) and the restoration of degraded landscapes (Article 4). At the same time, also reinforcing ecological structures already recognised through instruments such as the National Ecological Reserve and Municipal plans. From this perspective, the proposals contribute not only to the requalification of residual spaces, but also to the strengthening of municipal green infrastructure

Also, the transformation of interstitial spaces cannot rely exclusively on interventions within their boundaries. The surrounding urban fabric plays a decisive role in determining whether ecological and social functions can emerge and persist over time. Streets, sidewalks, public spaces and infrastructural edges therefore constitute a complementary field of action capable of extending the benefits generated within the interstitial spaces themselves. Through measures such as street tree planting, permeable surfaces, bioswales, pedestrian improvements and small-scale green corridors, adjacent urban areas can operate as transition zones between the highway landscape and the consolidated city. Acting around interstitial spaces therefore becomes another way of acting upon them, creating the conditions necessary for ecological processes, public use and territorial integration to develop beyond the limits of individual interventions.

Rather than functioning as isolated projects, the proposed interventions should therefore be understood as components of a broader territorial strategy. Their collective value lies not only in the transformation of specific sites, but in their capacity to contribute to a more connected, resilient and ecologically integrated municipal landscape.

6. Discussions and Conclusions

This research set out to understand how the regeneration of infrastructural spaces and their interstices, through Green Infrastructure and Nature-Based Solutions, can contribute to ecological and morphological connectivity, using Odivelas as a case study. To address this objective, the research combined ecological systems, urban morphology and infrastructural analysis within a multi-scale methodology, demonstrating how road infrastructures can be understood not only as mobility systems, but also as territorial structures capable of supporting broader ecological and spatial regeneration processes.

At the same time, the research demonstrates that these spaces are not uniform. They express distinct conditions - Barrier Condition, Latent Connectivity, and Functional Indetermination - that require differentiated readings and responses. Through a multi-scale diagnostic approach operating across metropolitan, municipal, and corridor scales, it was possible to read systematically how the IC17, IC22, and IC18 have conditioned the territory of Odivelas in distinct ways. The most significant interstitial spaces consistently emerge where ecological systems, urban discontinuities, and infrastructural barriers overlap, revealing them not as peripheral anomalies but as strategic territories for intervention.

The operative dimensions of mediation, requalification, and articulation demonstrated that this reading can be translated into concrete spatial and ecological proposals. Three typological categories - Understructure and Node Spaces, Primary Marginal Open Spaces, and Embankments and Marginal Linear Spaces - each reflect distinct spatial logics and transformative capacities. The analytical sheets developed for five representative cases showed that Nature-Based Solutions, when grounded in the specific conditions of individual spaces, can operate as territorial instruments capable of reconnecting fragmented infrastructural landscapes. The sheet format itself constitutes a methodological contribution, translating the analytical approach into an operative instrument applicable to otherl territories. The answer to the research question is affirmative: transport infrastructure can be reimagined as a carrier of ecological and spatial quality.

While ecological restoration, green infrastructure and public space projects are increasingly common, integrated approaches specifically focused on the transformation of highway landscapes remain relatively scarce. This research also contributes to the broader discussion on the integration of green and grey infrastructures, demonstrating an approach specifically focused on highway landscapes. The methodology developed here offers a replicable basis for application in different metropolitan highway corridors - within Lisbon, other Portuguese cities, or comparable European peripheries - allowing its categories to be tested, refined, and adapted to different territorial realities.

The research focuses on the extent of the three corridors within the Odivelas municipal boundary, but the infrastructural logic that produced these spaces does not respect administrative borders. The IC17, IC22, and IC18 continue beyond Odivelas into Loures, Amadora, and Lisbon, generating analogous interstitial conditions in adjacent municipalities that this research could not address. This territorial delimitation was a deliberate methodological decision that enabled a detailed reading of interstitial spaces within a specific planning context. Odivelas provides a particularly relevant case due to its ongoing efforts to structure a municipal green infrastructure system and its position within wider ecological and infrastructural networks of the Lisbon Metropolitan Area. Its application in Odivelas also contributes directly to ongoing planning processes, offering guidance aligned with the upcoming National Ecological Strategy and EU Nature Restoration Law obligations, where the mechanisms of infrastructural fragmentation remain spatially legible and directly addressable.

The absence of participatory processes constitutes one of the main limitations of the research. While the analytical sheets aim to demonstrate the operational capacity of the method, they cannot substitute the engagement of local communities and users in defining priorities and programmes for each space. Incorporating the knowledge and practices of residents, AUGI communities and informal farmers would likely strengthen and potentially alter some of the proposals developed.

A second limitation concerns governance at the institutional scale. The transformation of infrastructural interstices cannot be achieved exclusively through municipal planning instruments, as many of the spaces analysed remain under the jurisdiction of infrastructure agencies whose priorities are primarily oriented towards mobility, safety and maintenance. In the case of Odivelas, this is particularly evident in the relationship between Infraestruturas de Portugal, which manages the highway corridors, and the Câmara Municipal, which holds planning authority over the surrounding territory, two institutional logics operating largely independently over the same spatial reality. This institutional disconnect points to one of the most urgent directions for future research and practice: the development of coordination frameworks capable of bringing infrastructure managers and planning authorities into a shared operative logic. Until the spaces produced by highway infrastructure are recognised not only as technical margins by those who manage them, but also as ecological and urban assets by those who plan around them, their transformative potential will remain structurally inaccessible, regardless of the quality of the spatial proposals developed for them. The regeneration of interstitial spaces therefore depends not only on spatial design, but on long-term governance structures, institutional coordination, and cross-scalar planning frameworks that remain largely absent in most metropolitan peripheries, including Odivelas.

A third limitation concerns the definition of interstitial space itself. Although the research proposes a consistent reading method across the three corridors, the boundaries between interstitial typologies remain inherently ambiguous. Many spaces simultaneously exhibit multiple spatial conditions, making classification necessarily interpretive rather than absolute. This also means that another researcher, operating from a different methodological or analytical position, might classify individual spaces differently. This is not a failure of the method but a condition of the object: interstitial spaces resist clean categorisation precisely because they are produced by the same indeterminacy that defines them.

Finally, Nature-Based Solutions were engaged here as an operative tool rather than a fixed category, used to guide spatial and territorial strategies. The proposals presented should be understood as strategic territorial propositions whose ecological performance would require further development through implementation and monitoring.

These limitations define more precisely the territory that remains open for future investigation and reveal several directions through which the method proposed in this research can be further developed. The governance dimension remains perhaps the most urgent challenge: how interstitial activation can be coordinated, financed, implemented, and maintained across multiple municipal boundaries and institutional jurisdictions. Equally important is the development of participatory methodologies capable of integrating the situated knowledge, informal appropriations, and spatial practices already present within these spaces into the programming of Nature-Based Solutions, particularly within AUGI territories and peripheral metropolitan contexts.

A third direction, as already mentioned, concerns the transferability of the methodology itself. Extending the diagnostic and typological methodology developed for Odivelas to other metropolitan highway corridors, whether within Lisbon, other Portuguese cities, and comparable European peripheries, would allow both the proposed typological structure and the analytical sheets to be tested, refined and adapted to different territorial realities. At the same time, the establishment of long-term ecological monitoring protocols for infrastructural NBS interventions would help develop more robust links between spatial proposals and measurable ecological performance. Finally, the question of how existing informal uses, urban agriculture, spontaneous vegetation, and community appropriation can be recognised and integrated into formal planning instruments without displacing the communities that produced them remains one of the most politically and spatially complex challenges for future research and practice in this field.

The question should no longer be whether these spaces can be transformed. The question is whether the institutional will, the governance frameworks, and the collective commitment to doing so systematically and equitably can be built before the remaining opportunities are consumed by the same developmental pressures that produced the problem in the first place. The interstitial spaces of Odivelas demonstrate that infrastructural landscapes contain significant but often overlooked opportunities for ecological and spatial regeneration. Their transformation depends not only on design interventions, but on the capacity to integrate ecological objectives, planning instruments and infrastructural management within a shared territorial vision. In this sense, the spaces examined throughout this research should be understood not as residual margins of the city, but as potential components of its future green infrastructure network.

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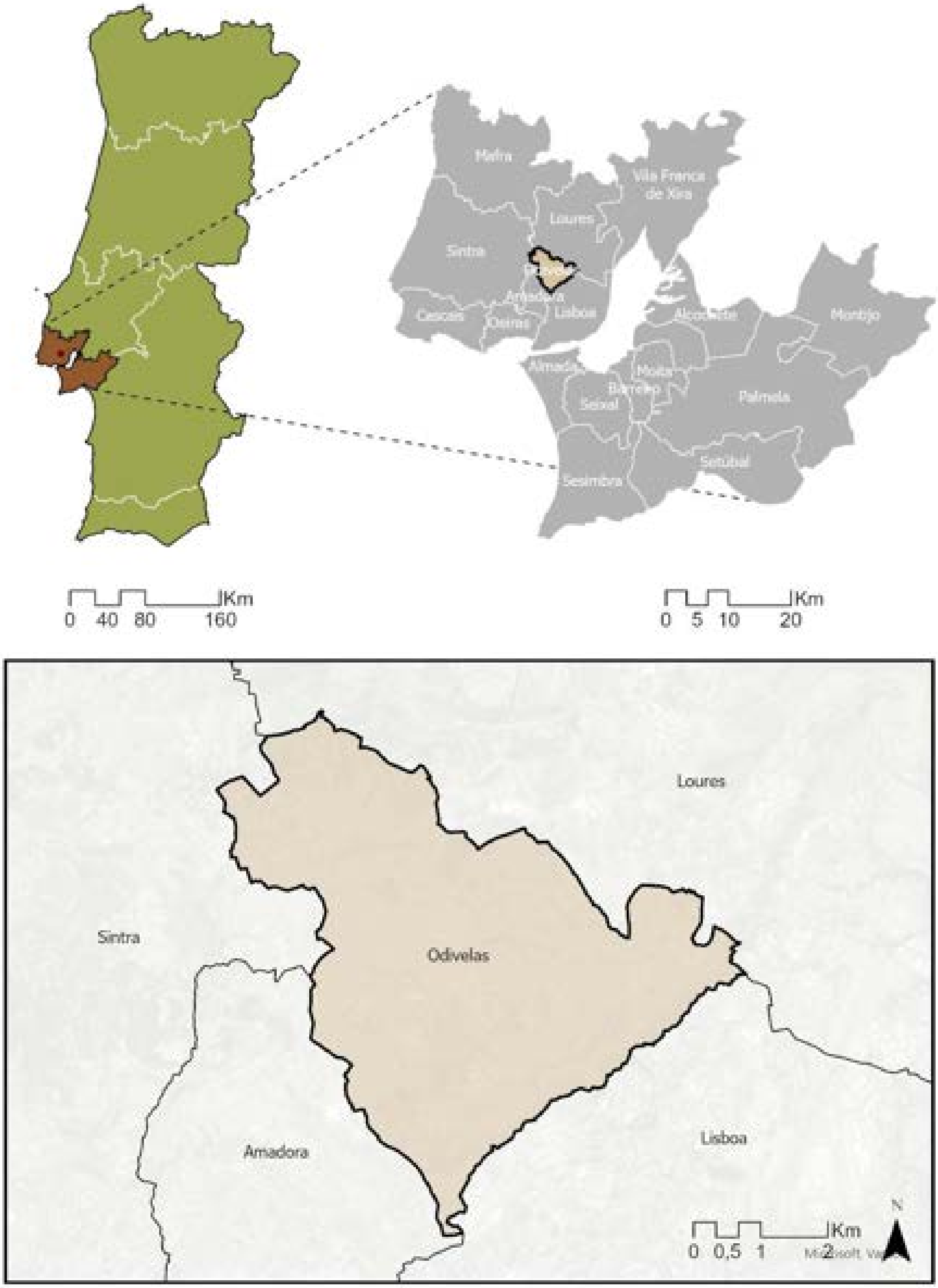
8. APPENDIX

Appendix 01 - NBS Typologies Table Analysis (Source: Own Elaboration)

Category / Row	Cheonggyecheon KOR	High Line USA	Goods Line AUS	Bonaventure CAN	A35 Highway. FRA	Likoto FRA/BEL	A16 Rotterdam NLD	Groene Singel NLD	Buffalo Bayou USA	Nus Trinitat ESP	Meguro Sky JPN	Ricardo Lara USA	Terrabank USA	Bicentenario COL	Bentway CAN	Natuurbrug NLD	Crab Bridge AUS	Loop Bridge USA	A7 Highway DEU	Alto Lumiar POR	Bela Vista POR	Ponte Verde POR	Urban Farm Amadora POR
TYPOLOGIES																							
Removal & Renaturation	●																						
Structural Reconversion		●	●																				
Functional Downgrading				●	●																		
Urban Green Corridor						●	●	●	●											●	●	●	
Road Capping / Deck Park																			●				
Underneath Activation															●								
Floodable Park									●														
Urban Farming										●		●	●						●			●	●
Wildlife Crossing																●	●	●					
Bioretention & Wetlands									●			●			●						●		
Spontaneous Occupation						●																	●
ECOSYSTEM SERVICES (MEA)																							
Provisioning										●	●	●	●		●							●	●
Regulating	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Cultural	●	●	●	●	●		●	●	●	●	●	●		●	●				●	●	●	●	●
Supporting	●				●	●	●	●	●	●		●	●			●	●	●	●	●	●	●	
NBS FAMILY (WORLD BANK 2021)																							
Urban Forests		●	●		●	●	●	●	●	●	●								●	●	●	●	
Green Corridors		●	●		●	●	●	●	●										●	●	●	●	
River & Stream Renaturation	●								●														
Building Solutions					●						●			●	●				●				
Open Green Spaces	●	●	●	●	●		●	●	●	●	●	●		●	●				●	●	●	●	
Urban Farming										●		●	●						●			●	●
Bioretention Areas					●				●			●			●					●	●		
Natural Inland Wetlands	●								●														
Constructed Inland Wetlands									●			●											
River Floodplains	●				●				●														
Terraces and Slopes					●	●	●	●		●										●		●	
SCALE																							
Metropolitan	●					●	●	●	●										●				
Municipal		●	●	●				●	●	●		●		●	●				●	●	●	●	
Local										●	●	●	●	●	●	●	●	●					●
PROJECT STAGE																							
Completed	●	●	●	●					●	●	●	●		●	●	●	●	●		●	●		●
In progress								●											●				
Planning																						●	
Proposal						●	●						●										

● Green headers = Portuguese projects | Source: World Bank (2021); Smith et al. (2015); Delbaere (2020); project sources as cited in text

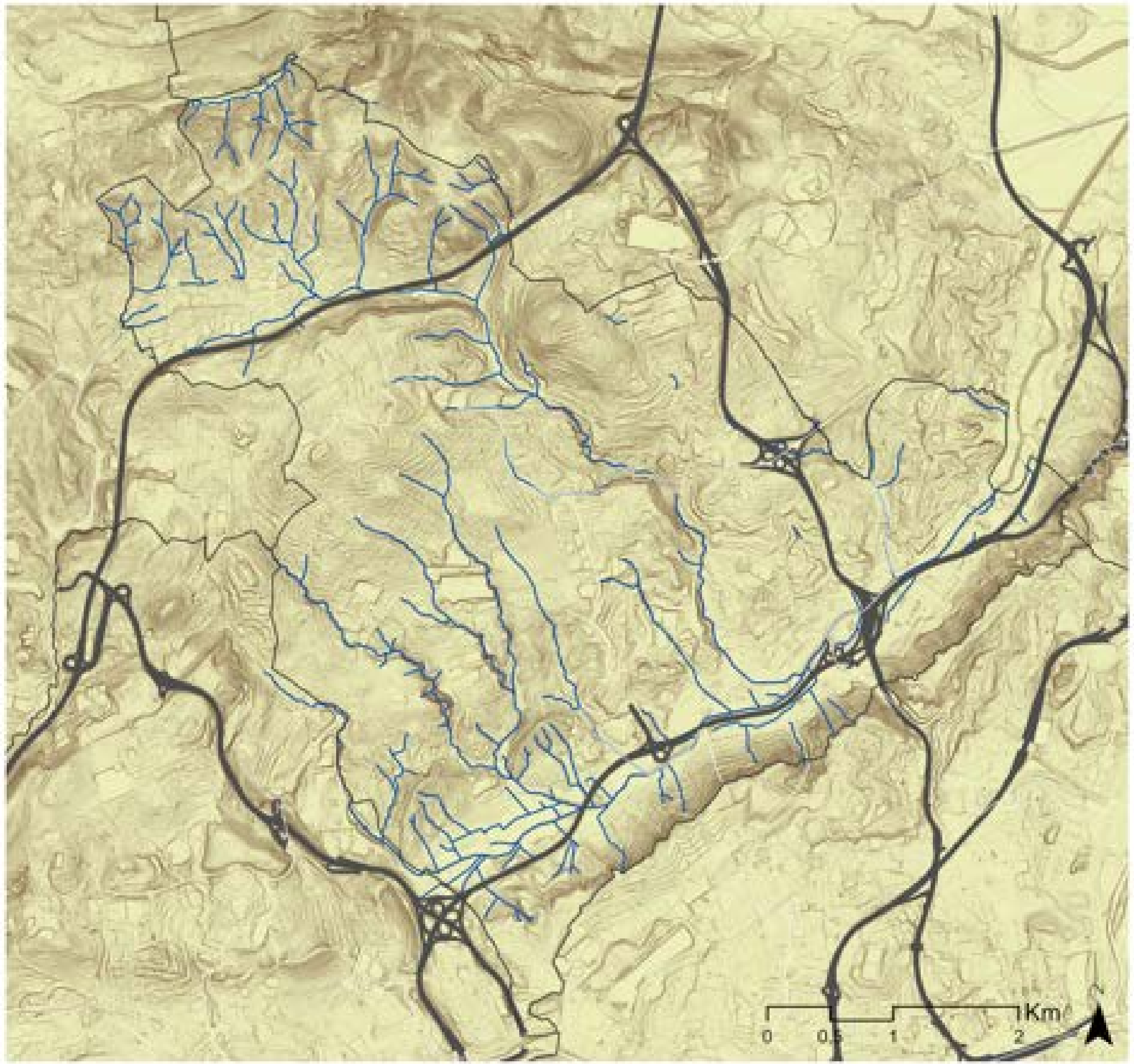
Appendix 02 - Odivelas Location (Source: Adapted from CAOP)



APPENDIX 03 - Slope Map

The slope map was done using DGT-MDT data and the tool of Slope in Arcgis, while the hidrgophic map was obtanied from the Municipality of Odivelas; it reveals how topography can be a primary conditioning force in Odivelas, shaping not only its ecological structure but also its urban form and risk distribution. The territory is marked by a clear altimetric contrast between the steeper slopes and ridges concentrated to the north and southeast, which have historically limited urban occupation due to their physical constraints, and the intermediate plateaus where the city predominantly disperse, while also occupying the topographic middle ground between the steep slopes and the valley floors. The valleys themselves, following the hydrographic network, constitute the lowest and most dynamic areas of the territory, as is to be shown in future sections.

Also, the major highway corridors reveal a particularly significant relationship with this topographic structure. Rather than adapting to the terrain, they cut transversally across valleys and slopes through embankments, cuttings, and viaducts - a capacity to overcome topographic constraints that enables regional connectivity but simultaneously generates some of the most severe barrier conditions in the territory, interrupting the natural continuity of valleys that functioned as corridors of water, air, and fauna.



APPENDIX 04 Ecological Structure

The National Ecological Structure (EEN) can be understood as a strategic reference for identifying ecological continuities and territorial support systems. But also, it is a framework that guides their articulation within the territory, supporting functions such as water regulation, soil protection, biodiversity maintenance, and the mitigation of natural risks (Magalhães, 2013).

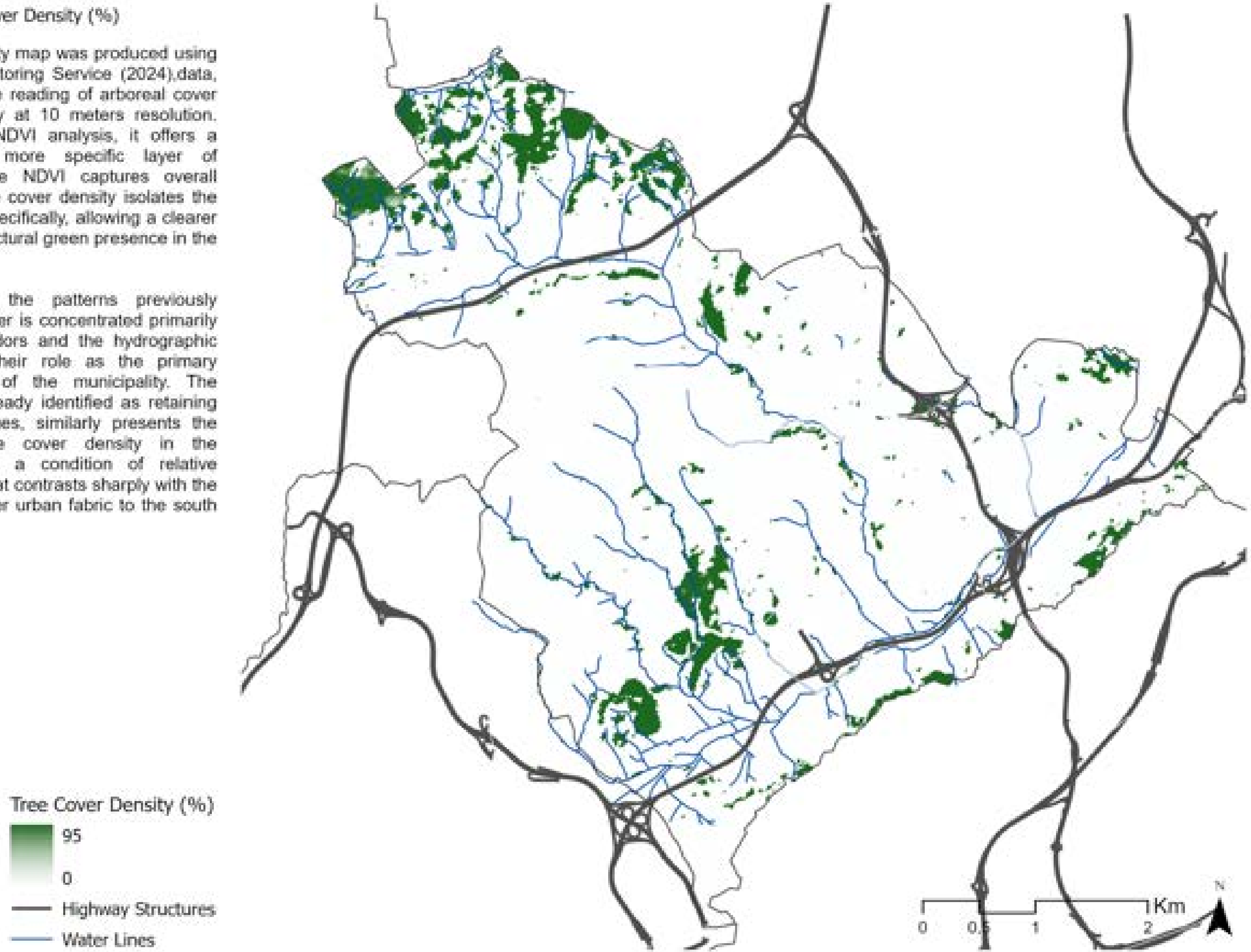
This ecological structure is expressed through two distinct levels - EEN 1 and EEN 2 - which reflect different degrees of sensitivity and ecological function. This data was obtained in EPIC WebGIS Portugal (Magalhães, 2013).

The ecological relationships within the municipality are strongly associated with the hydrographic structure, which acts as the main organizing element, concentrating areas of higher ecological value and supporting key natural processes. At the same time, part of this structure develops alongside road infrastructures and urban areas, revealing an overlap between ecological systems and technical systems. This coexistence highlights a significant potential for transformation: spaces currently shaped by infrastructural logic - often fragmented or underused - can be reinterpreted as ecological corridors or areas that reinforce environmental connectivity. In this sense, the EEN in Odivelas is not limited to isolated "natural" areas, but also emerges in hybrid contexts where infrastructure and ecology coexist.

APPENDIX 05 Tree Cover Density (%)

The Tree Cover Density map was produced using Copernicus Land Monitoring Service (2024),data, providing a quantitative reading of arboreal cover across the municipality at 10 meters resolution. Read alongside the NDVI analysis, it offers a complementary and more specific layer of information that, while NDVI captures overall vegetation activity, tree cover density isolates the arboreal component specifically, allowing a clearer assessment of the structural green presence in the territory.

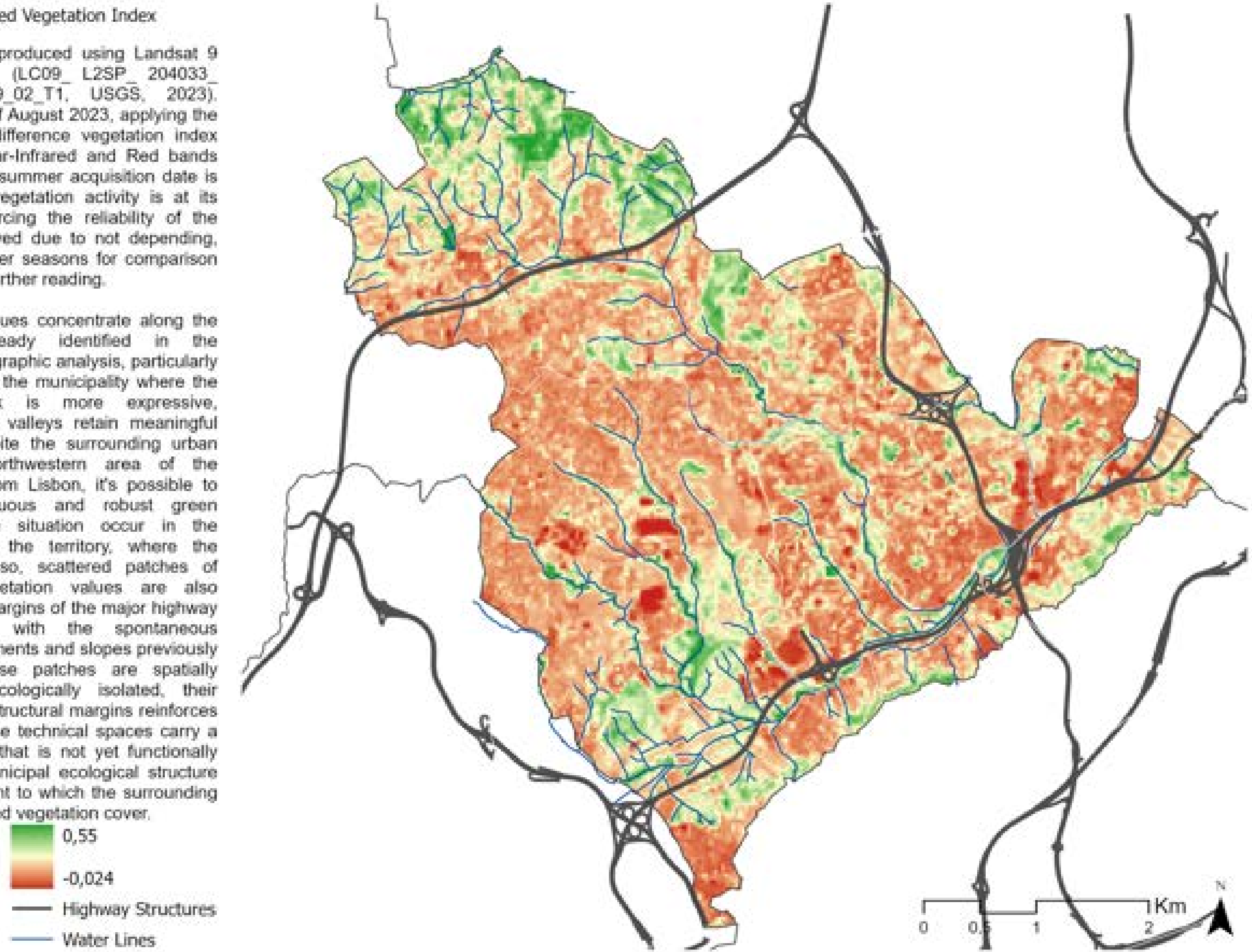
The map confirms the patterns previously identified: arboreal cover is concentrated primarily along the valley corridors and the hydrographic network, reinforcing their role as the primary ecological backbone of the municipality. The northwestern area, already identified as retaining higher vegetation values, similarly presents the most expressive tree cover density in the municipality, reflecting a condition of relative ecological continuity that contrasts sharply with the predominantly low-cover urban fabric to the south and east.



APPENDIX 06 Normalized Vegetation Index

The NDVI map was produced using Landsat 9 Collection 2 imagery (LC09_ L2SP_ 204033_ 20230827_ 20230829_02_T1, USGS, 2023). Acquired on the 27th of August 2023, applying the standard normalized difference vegetation index formula using the Near-Infrared and Red bands (Bands 5 and 4). The summer acquisition date is analytically relevant: vegetation activity is at its seasonal peak, reinforcing the reliability of the spatial patterns observed due to not depending, even though using other seasons for comparison could contribute for a further reading.

The highest NDVI values concentrate along the valley corridors already identified in the topographic and hydrographic analysis, particularly in the western area of the municipality where the hydrographic network is more expressive, suggesting that these valleys retain meaningful vegetation cover despite the surrounding urban pressure. In the northwestern area of the municipality, further from Lisbon, it's possible to see a more continuous and robust green presence. The same situation occur in the southeastern part of the territory, where the pronounced slope. Also, scattered patches of relatively higher vegetation values are also detectable along the margins of the major highway corridors, consistent with the spontaneous vegetation on embankments and slopes previously discussed. While these patches are spatially discontinuous and ecologically isolated, their recurrence along infrastructural margins reinforces the argument that these technical spaces carry a latent green potential that is not yet functionally integrated into the municipal ecological structure and revealing the extent to which the surrounding urban fabric has reduced vegetation cover.



APPENDIX 07 - National Ecological Reserve (REN)

The Reserva Ecológica Nacional (REN) map, under Decree Law 166/2008, and adapted from the AML Atlas, shows the areas of highest ecological sensitivity in Portugal, restricting urbanisation and certain land uses in order to protect natural resources, ensure ecological continuity, and mitigate natural risks. Unlike the EEN, which is a planning concept that organises ecological spaces into a functional network, the REN is a legal protection instrument, it defines what cannot occur in designated areas, rather than prescribing how ecological systems should connect and function. In Odivelas, the REN operates as a dominant structuring element of the territory, and its spatial distribution is directly revealing of the ecological and infrastructural tensions that characterise the municipality.

To the north, the REN manifests as larger and more continuous patches, where ecological presence is stronger and less contested by urban development, a condition consistent with the higher vegetation cover already identified by other biophysical readings. To the south, however, it organises itself not as large preservation areas but as an interconnected system of valleys that penetrates deeply into the urban fabric, following the hydrographic network that constitutes one of its primary ecological backbones. These valley corridors, againm represent the potential spine of ecological continuity in Odivelas, the spaces where natural processes, water flow, and biodiversity movement could be sustained and reinforced.
The overlapping of the REN with the main road network implies, together with the historical analysis, that Odivelas' development is the product of a sustained tension between natural constraints and infrastructural impositions - a tension that the REN has been unable to fully resolve, and that any strategy for ecological transformation in the municipality must directly address.

REN - Ecological Reserve
Highway Structures

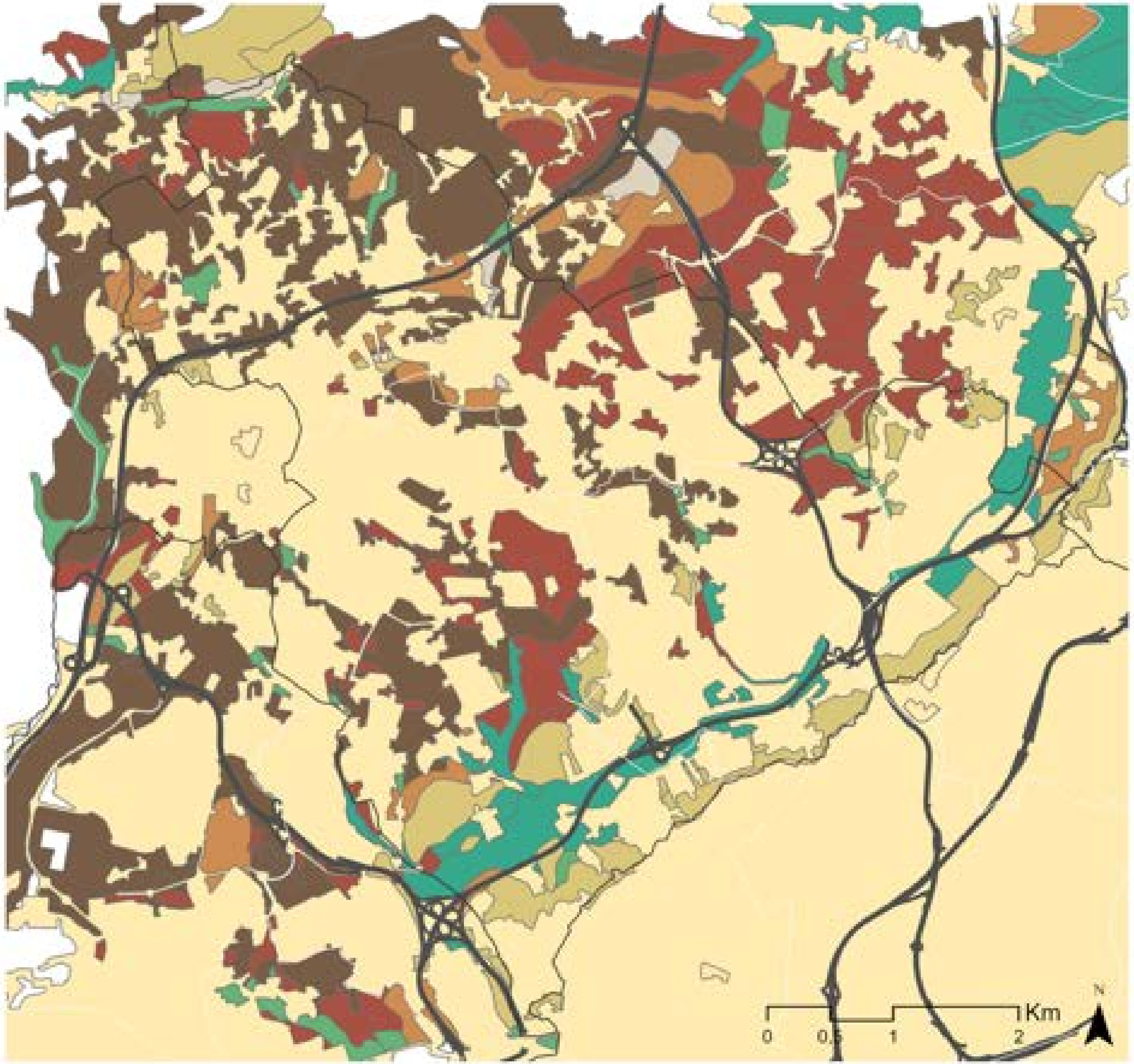
APPENDIX 08 Soil Typology

The soil map was produced using data from the DGADR (Direção-Geral de Agricultura e Desenvolvimento Rural), from which, based on the classifications and definitions provided by the same institution, a simplified typology of 9 generic soil types was elaborated to represent the principal categories present in the municipal territory. This generalisation was done to allow a more legible and analyticalreading at the municipal scale, without losing the essential distinctions that condition ecological and urban potential across the territory.

The result of this map reveals a clear spatial logic in the distribution of soil types. A significant portion of the territory is classified as non-soil - reflecting the extent of urban impermeabilisation and surface sealing. This alone could be a significant indicator of the degree to which natural soil functions have been suppressed by urbanisation.

Among the remaining soil types, the distribution is spatially structured in Alluvial soils, which concentrate along the valley floors and lower areas associated with the hydrographic network. Their presence in these locations reinforces the ecological value of the valley system already identified, while also signalling a tension: these are precisely the areas most exposed to urban pressure and infrastructural encroachment.
Around the CREL, litholic soils are more prevalent, being less fertile and with reduced infiltration capacity, limiting both ecological and productive potential. Another condition emerges along several highway margins, where barro soils -characterised by high water retention but poor drainage - concentrate, creating conditions that require active water management and that amplify flood risk in adjacent low-lying areas.

Aluviais
Argiluviados
Barros
Calcários
Coluviais
Litólicos
Outros
Urbano
Highway Structures



APPENDIX 09 - MACAT

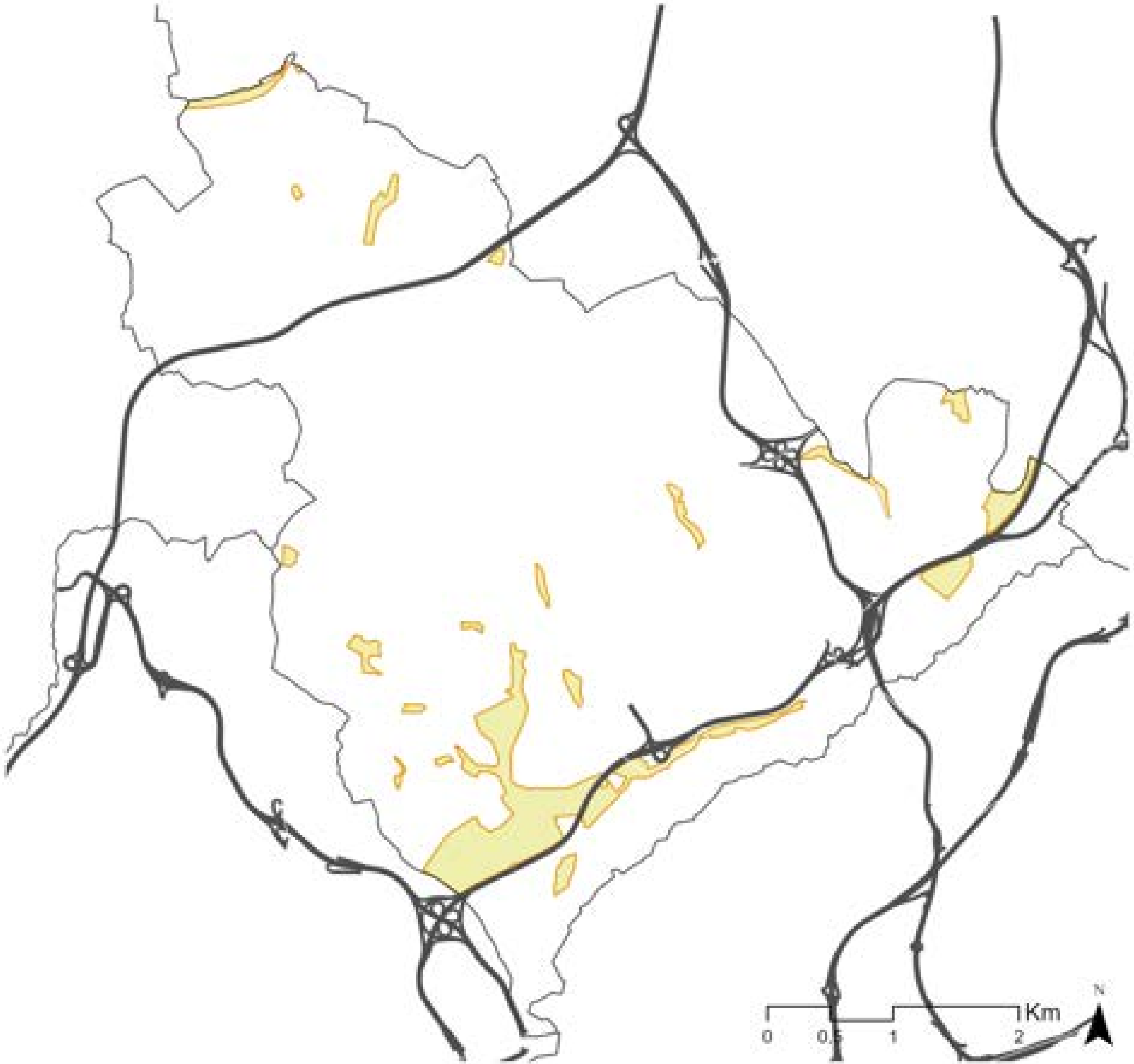
The MACAT (Mapa Anual de Culturas Agrícolas Temporárias) was obtained from the DGT SMOS platform and represents the annual distribution of temporary agricultural activity across the territory.

In Odivelas, the temporary crop activity, is scarce and dispersed. Where it does appear, however, it tends to concentrate in relatively larger patches, suggesting that where agricultural activity persists, it does so with some spatial consistency, even if isolated from broader productive areas. Read alongside the soil analysis, these patches coincide in several cases with the alluvial soils previously identified as having the highest productive and ecological potential, reinforcing the argument that the valley corridors and infrastructural margins retain a latent productive capacity that current planning instruments have neither adequately protected nor strategically activated.

APPENDIX 10 - National Agricultural Reserve

The Reserva Agrícola Nacional (RAN), established under Decree Law 73/2009, was obtained from the Municipality of Odivelas defines the areas of highest agricultural potential in Portugal, aiming to protect productive soils from urban encroachment and ensure their long-term availability for food production. It is important to clarify that the RAN does not imply active agricultural use, it is a protective designation that restricts urban expansion over soils of recognised productive value, without necessarily determining or requiring their cultivation. In Odivelas, the RAN presents itself as a markedly residual presence, scattered in small patches or concentrated mainly in the central and southern areas of the municipality, surrounded by consolidated urban expansion that has progressively reduced its spatial expression.
Despite their reduced extent, these designated areas signal where soils of higher agricultural potential persist within an otherwise predominantly urban territory.

The spatial relationship between the RAN and the major highway corridors is equally revealing. The RAN boundaries tend to follow these corridors, appearing as linear designations running alongside the major infrastructures, which could suggest that the residual soils of agricultural potential have been progressively shaped and contained by the infrastructural logic, occupying the spaces that the road network left behind rather than reflecting an independent agricultural structure.



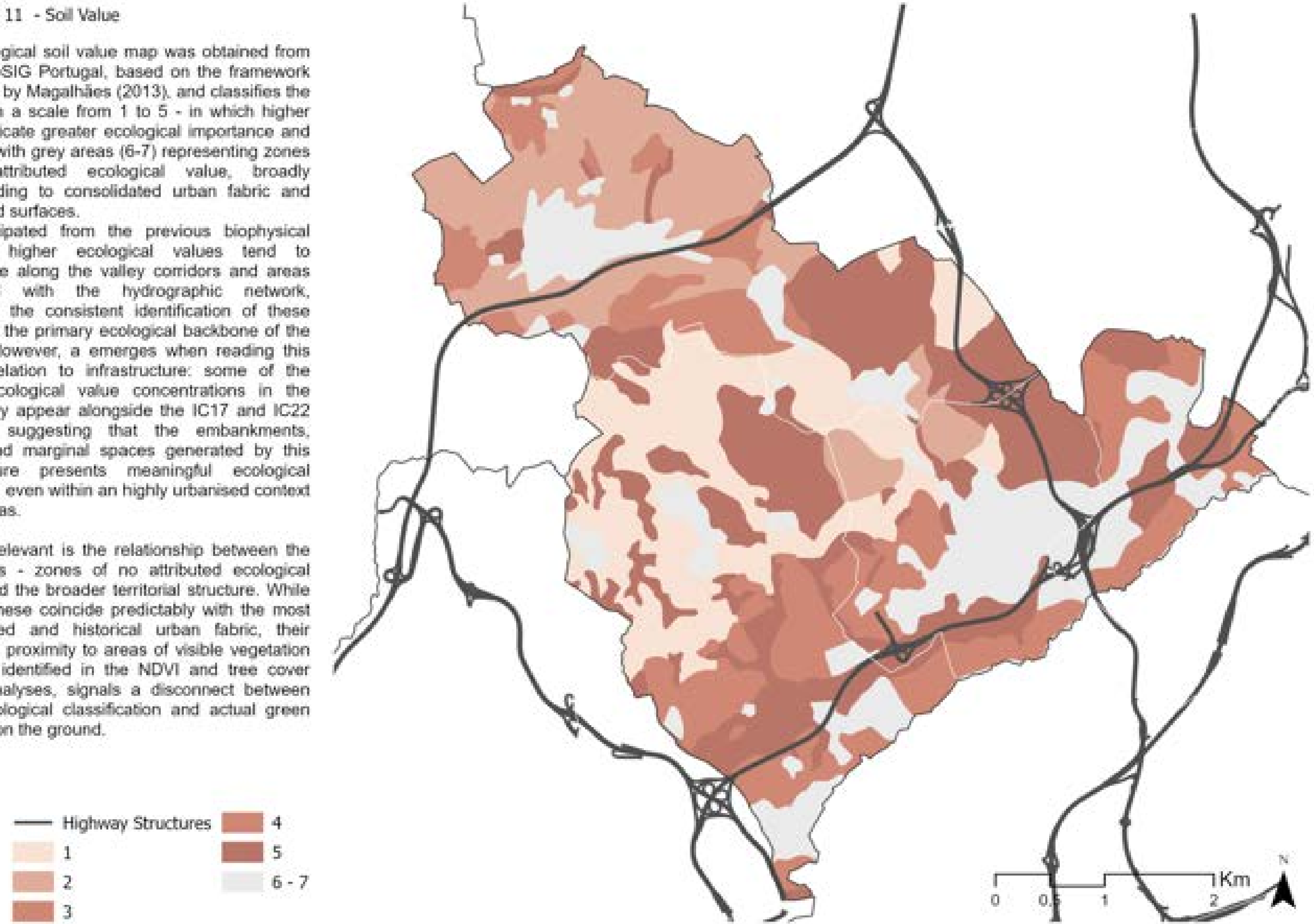
RAN - Agricultural Reserve
Highway Structures

APPENDIX 11 - Soil Value

The ecological soil value map was obtained from EPIC WebSIG Portugal, based on the framework developed by Magalhães (2013), and classifies the territory on a scale from 1 to 5 - in which higher values indicate greater ecological importance and potential- with grey areas (6-7) representing zones without attributed ecological value, broadly corresponding to consolidated urban fabric and fully sealed surfaces.

As anticipated from the previous biophysical readings, higher ecological values tend to concentrate along the valley corridors and areas associated with the hydrographic network, reinforcing the consistent identification of these spaces as the primary ecological backbone of the territory. However, a emerges when reading this map in relation to infrastructure: some of the highest ecological value concentrations in the municipality appear alongside the IC17 and IC22 corridors, suggesting that the embankments, slopes, and marginal spaces generated by this infrastructure presents meaningful ecological conditions, even within an highly urbanised context like Odivelas.

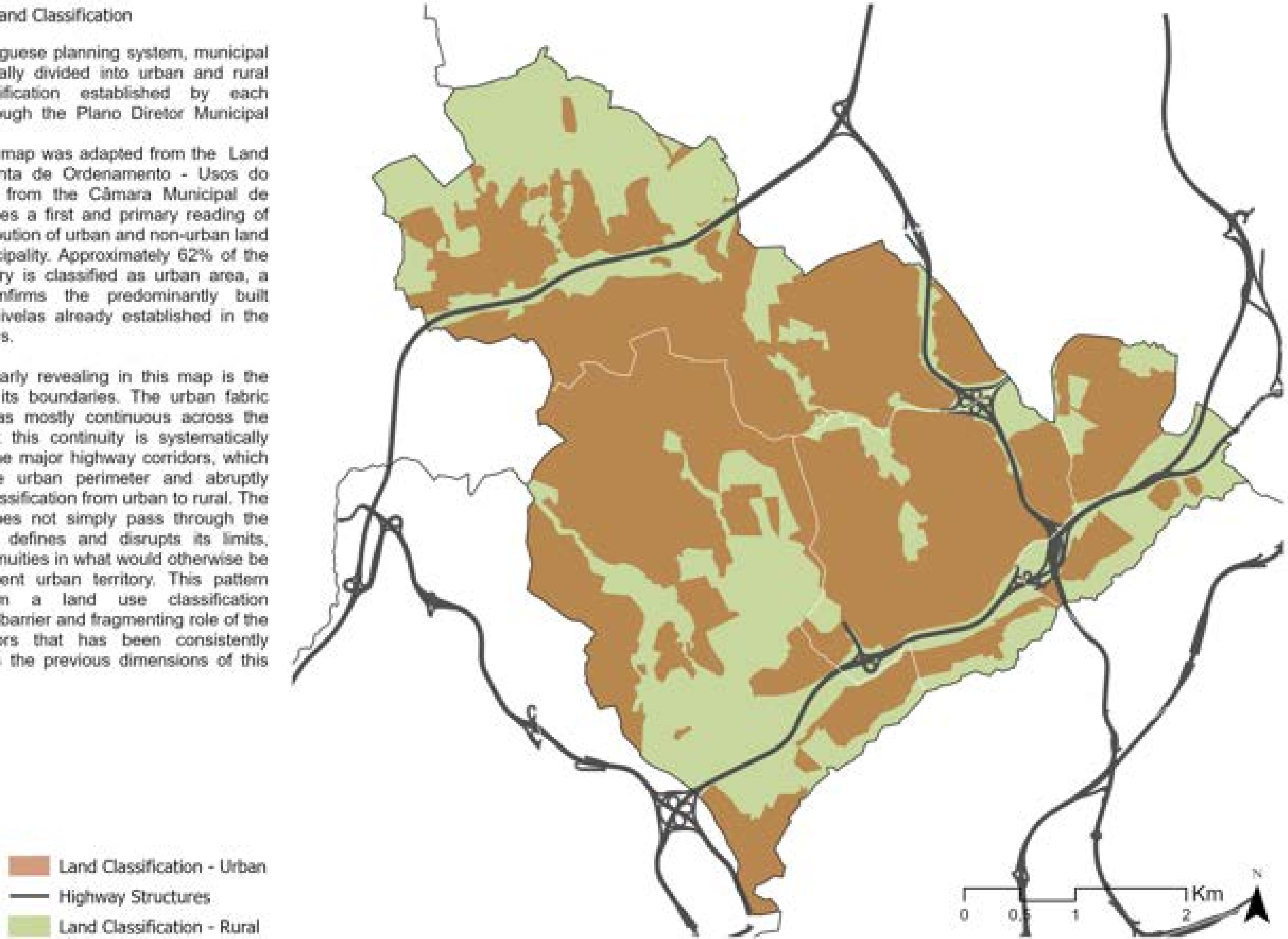
Equally relevant is the relationship between the grey areas - zones of no attributed ecological value - and the broader territorial structure. While many of these coincide predictably with the most consolidated and historical urban fabric, their occasional proximity to areas of visible vegetation cover, as identified in the NDVI and tree cover density analyses, signals a disconnect between formal ecological classification and actual green presence on the ground.



APPENDIX 12 Land Classification

Within the Portuguese planning system, municipal territory is formally divided into urban and rural land, a classification established by each municipality through the Plano Diretor Municipal (PDM).
The urban area map was adapted from the Land Use Chart (Planta de Ordenamento - Usos do Solo), obtained from the Câmara Municipal de Odivelas, provides a first and primary reading of the spatial distribution of urban and non-urban land within the municipality. Approximately 62% of the municipal territory is classified as urban area, a figure that confirms the predominantly built character of Odivelas already established in the previous readings.

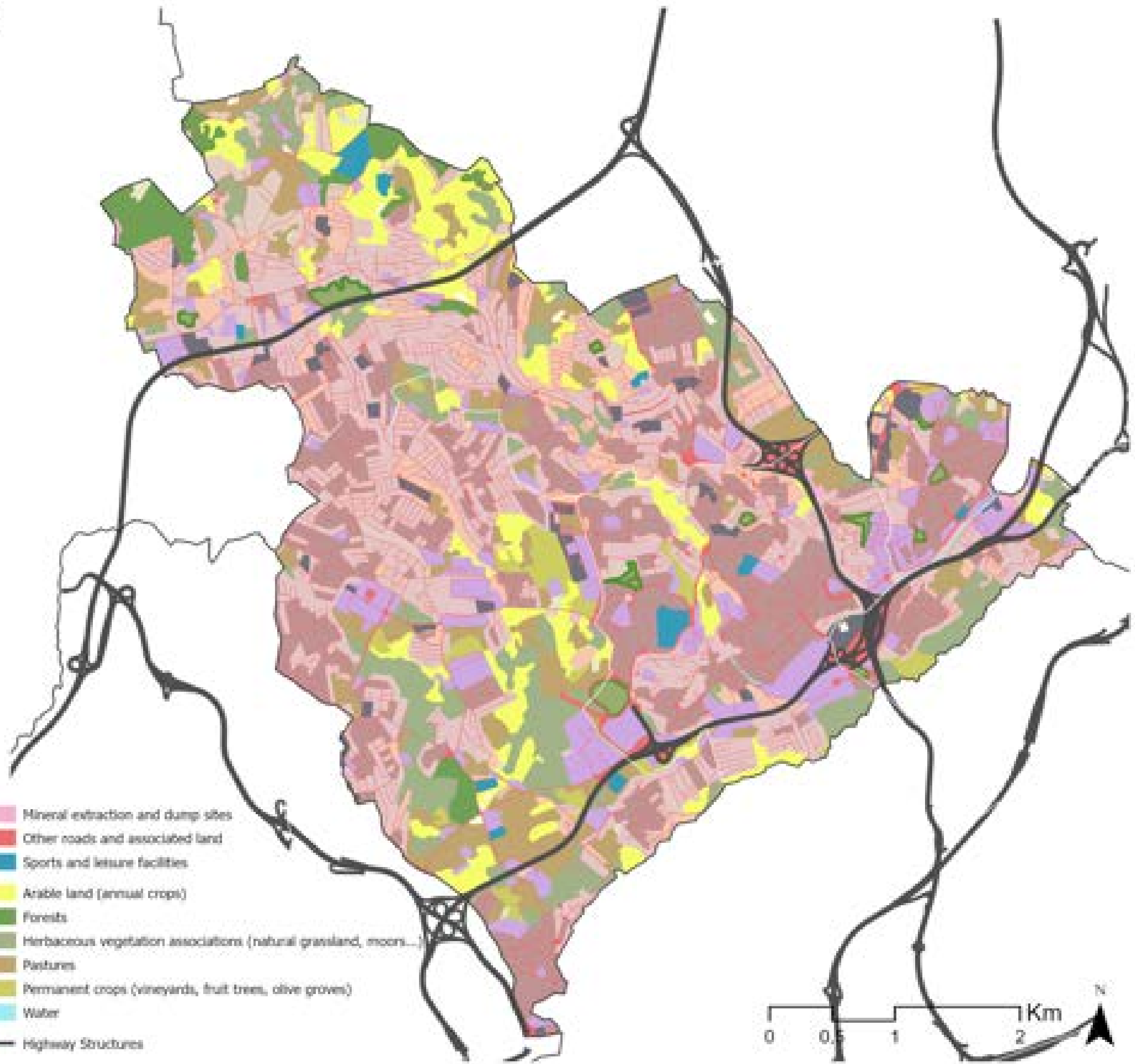
What is particularly revealing in this map is the spatial logic of its boundaries. The urban fabric presents itself as mostly continuous across the municipality, but this continuity is systematically interrupted by the major highway corridors, which cut through the urban perimeter and abruptly transition the classification from urban to rural. The infrastructure does not simply pass through the built fabric - it defines and disrupts its limits, creating discontinuities in what would otherwise be a largely coherent urban territory. This pattern reinforces, from a land use classification perspective, the barrier and fragmenting role of the highway corridors that has been consistently identified across the previous dimensions of this analysis.



APPENDIX 13 - Land Use Classification: Urban Atlas

The land use map was produced using Urban Atlas data, providing a more granular and categorised reading of land occupation across the municipality.
The map reinforces the reading of Odivelas as a largely continuous urban territory, with residential uses dominating the majority of the municipal area. However, this continuity is not homogeneous, the Urban Atlas classification reveals a fabric marked by significant variation in density and typology, from more consolidated and higher-density residential areas concentrated along the major arterial corridors, to lower-density and more fragmented occupation in transitional and peripheral zones. This morphological diversity reflects the successive waves of road-centric urbanisation described in the historical reading, where different periods of growth produced different urban forms in direct response to the infrastructural conditions of each moment.

The non-urban categories - green urban areas, agricultural land, and natural or semi-natural spaces - appear as residual and dispersed presences within this predominantly built context, largely confined to the valley corridors and northern areas already identified as retaining higher ecological value. Their limited extent and spatial discontinuity reinforce the argument that the green structure of Odivelas is fragmented and insufficient relative to the scale and density of its urban fabric.

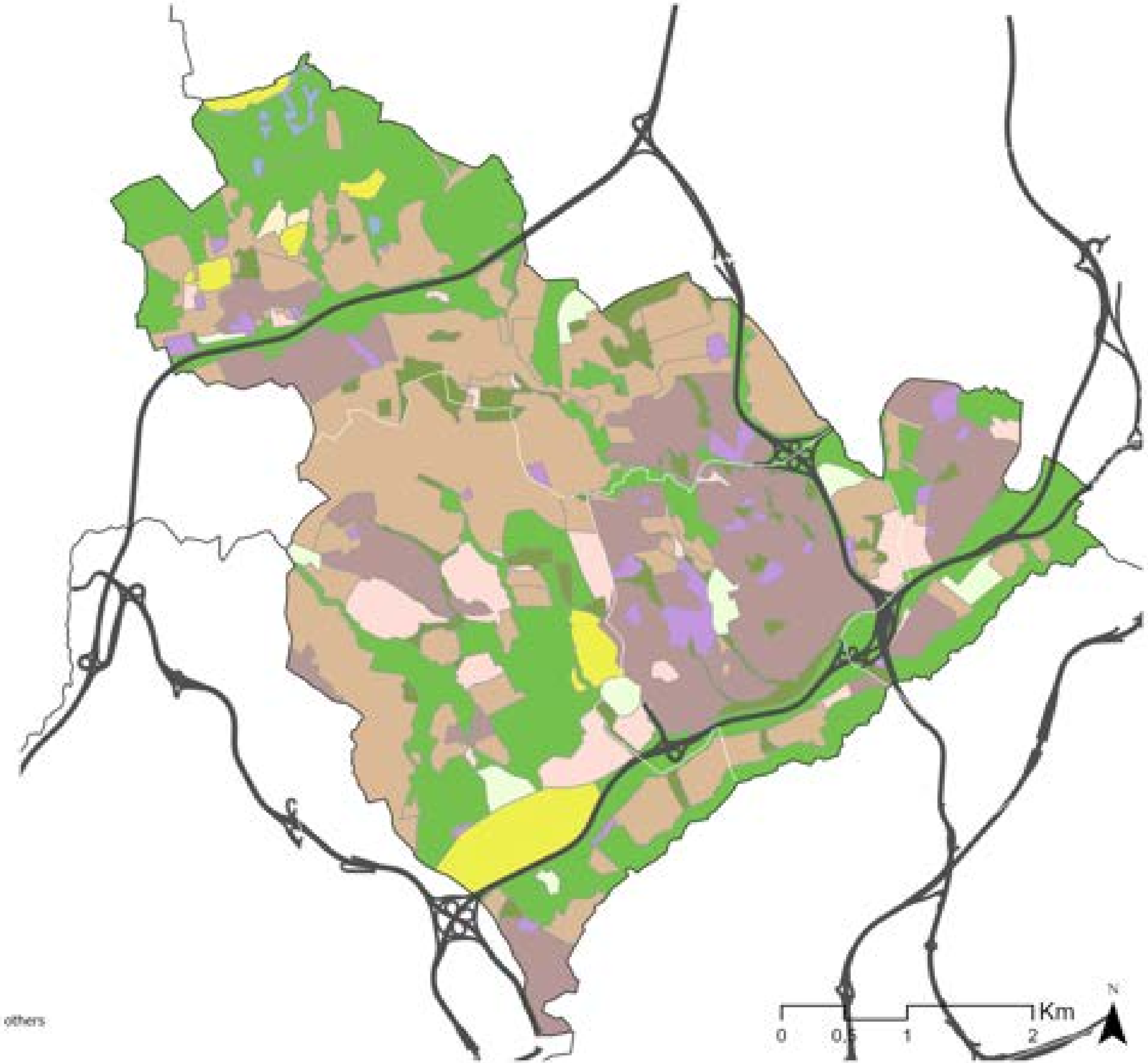


APPENDIX 14 - Land Use Classification: PDM

The Municipal Master Plan (PDM) of Odivelas provides the formal planning framework through which land use is regulated and projected across the municipality. Read alongside the Urban Atlas and the urban area map, it offers a normative counterpart to the empirical land use readings already conducted, showing not only how the territory is currently occupied, but how it is formally intended to develop and be protected. From it, it's possible to see that most of the urban area is to be kept similar, with the focus being on consolidation rather than expansion.

One of the most analytically significant aspects of the PDM in the context of this research is the classification of the areas surrounding the major highway corridors. These marginal and interstitial spaces - the embankments, slopes, and residual land generated by the infrastructural network - are predominantly designated as forest areas within the PDM. This classification simultaneously acknowledges the non-urban character of these spaces and assigns them a passive, largely infrastructural designation, stopping sright beforeof integrating them into a coherent ecological or public green structure. The forest designation, in this sense, reflects the planning logic that has historically treated infrastructural margins as leftover sp aces rather than as strategic opportunities. The map was elaborated by adapting the Land Use Chart (Planta de Ordenamento - Uso dos Solos) provided by the Câmara Municipal de Odivelas.

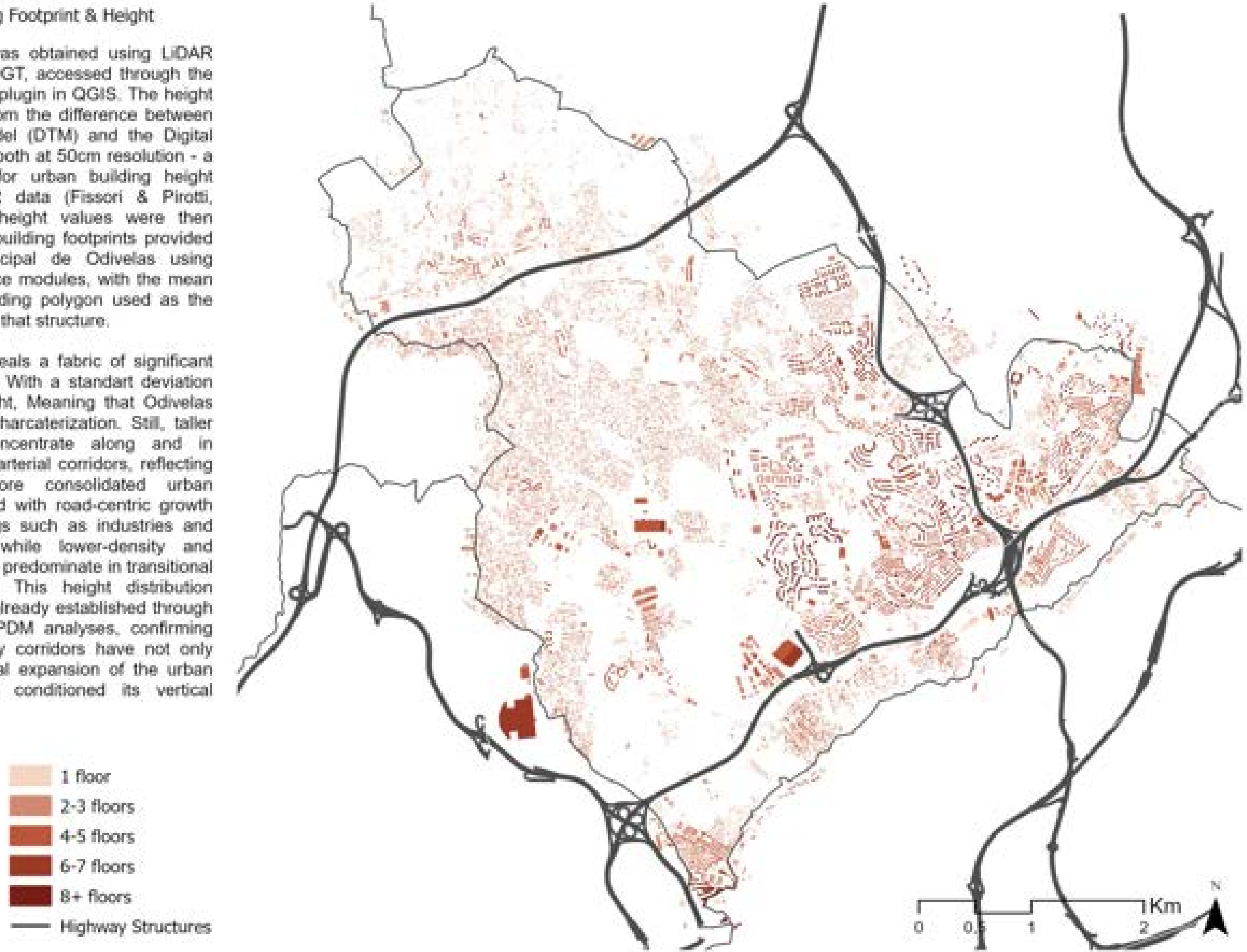
- Agricultural Spaces
- Central Spaces
- Economic Activities Spaces
- Special Use Spaces
- Espaços florestais
- Residential Spaces
- Low Density Urban Spaces
- Green Spaces
- Other Rustic Categories - Rural Aglomerates
- Other Rustic Categories - Spaces for facilities, infrastructures and others
- Highway Structures



APPENDIX 15 - Building Footprint & Height

Building height data was obtained using LiDAR data provided by the DGT, accessed through the DGT CDD Downloader plugin in QGIS. The height values were derived from the difference between the Digital Terrain Model (DTM) and the Digital Surface Model (DSM), both at 50cm resolution - a method widely used for urban building height extraction from LiDAR data (Fissori & Pirotti, 2019). The resulting height values were then spatially joined to the building footprints provided by the Câmara Municipal de Odivelas using GeoParquet open-source modules, with the mean height within each building polygon used as the representative value for that structure.

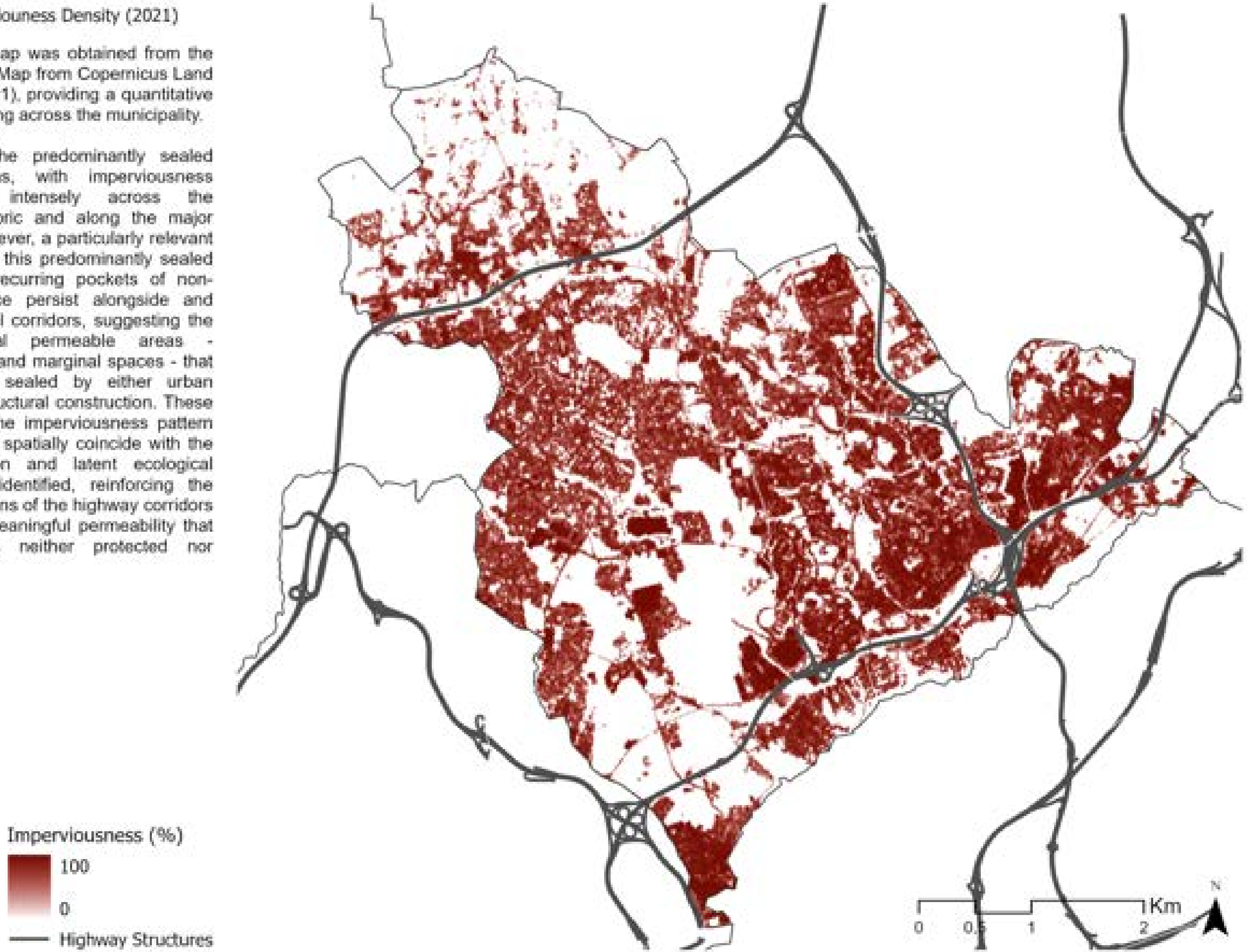
The resulting map reveals a fabric of significant morphological diversity. With a standart deviation of 3.74 meters in height, Meaning that Odivelas does not have a tall charcaterization. Still, taller buildings tend to concentrate along and in proximity to the major arterial corridors, reflecting the denser and more consolidated urban development associated with road-centric growth and/or specific buildings such as industries and commercial centers, while lower-density and smaller-scale structures predominate in transitional and peripheral areas. This height distribution reinforces the reading already established through the Urban Atlas and PDM analyses, confirming that the major highway corridors have not only structured the horizontal expansion of the urban fabric but have also conditioned its vertical dimension.



APPENDIX 16 - Imperviouness Density (2021)

The imperviousness map was obtained from the Imperviouness Density Map from Copernicus Land Monitoring Service (2021), providing a quantitative reading of surface sealing across the municipality.

The map confirms the predominantly sealed character of Odivelas, with imperviousness concentrating most intensely across the consolidated urban fabric and along the major highway corridors. However, a particularly relevant pattern emerges within this predominantly sealed condition: small and recurring pockets of non-impermeabilised surface persist alongside and within the infrastructural corridors, suggesting the presence of residual permeable areas - embankments, slopes, and marginal spaces - that have not been fully sealed by either urban development or infrastructural construction. These small interruptions in the imperviousness pattern are not incidental; they spatially coincide with the spontaneous vegetation and latent ecological conditions previously identified, reinforcing the argument that the margins of the highway corridors retain a residual but meaningful permeability that current planning has neither protected nor strategically activated.



APPENDIX 17 - AUGI & Execution Units

The map identifying AUGI (Áreas Urbanas de Génese Ilegal) boundaries and active execution units was elaborated from two sources: the execution unit maps made available by the Câmara Municipal de Odivelas, and the DGT Report on AUGI reconversion processes (2020). Together, these layers provide a reading of the informal urbanisation legacy and its current planning response across the municipality.

The spatial distribution of AUGIs and execution units reveals a pattern that is both historically and territorially coherent. The majority of these areas concentrate in the more central zone of the municipal territory - not the historic urban core, but the intermediate areas that developed most intensively during the periods of unplanned periurban growth. Their presence alongside several highway corridors is particularly significant: it suggests that the implantation of road infrastructure created conditions for informal urban occupation in its margins and surroundings, attracting settlement that followed infrastructural accessibility rather than planned urban logic. This dynamic is consistent with the historical reading already established, infrastructure arrived, and informal urbanisation followed, producing a fabric whose reconversion remains an ongoing planning challenge.

The active execution units, representing the formal planning response to this informal legacy, signal where the municipality is currently attempting to regularise and qualify these areas. Their distribution alongside the highway corridors reinforces the argument that the infrastructural margins of Odivelas are not only ecologically and environmentally significant, but also socially and urbanistically complex - spaces where planning debt, ecological potential, and infrastructural pressure converge simultaneously.

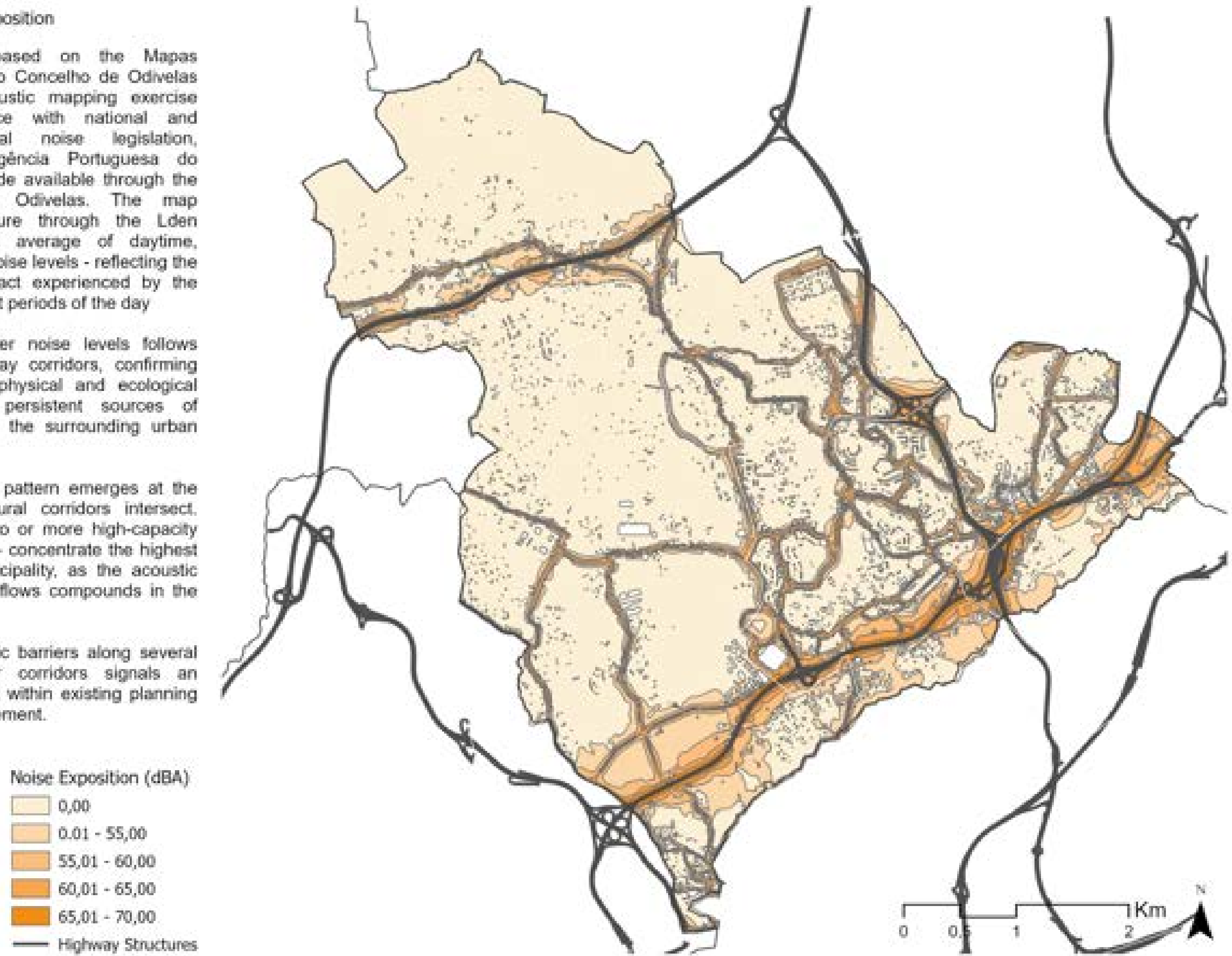
APPENDIX 18 - Noise Exposition

The noise map is based on the Mapas Estratégicos de Ruído do Concelho de Odivelas (2022), a strategic acoustic mapping exercise produced in compliance with national and European environmental noise legislation, coordinated by the Agência Portuguesa do Ambiente (APA) and made available through the Câmara Municipal de Odivelas. The map measures noise exposure through the Lden indicator - a weighted average of daytime, evening, and night-time noise levels - reflecting the cumulative acoustic impact experienced by the population across different periods of the day

The distribution of higher noise levels follows closely the major highway corridors, confirming their role not only as physical and ecological barriers but also as persistent sources of environmental stress for the surrounding urban fabric.

A particularly significant pattern emerges at the points where infrastructural corridors intersect. These nodes - where two or more high-capacity roads converge or cross - concentrate the highest noise levels in the municipality, as the acoustic impact of multiple traffic flows compounds in the same spatial unit.

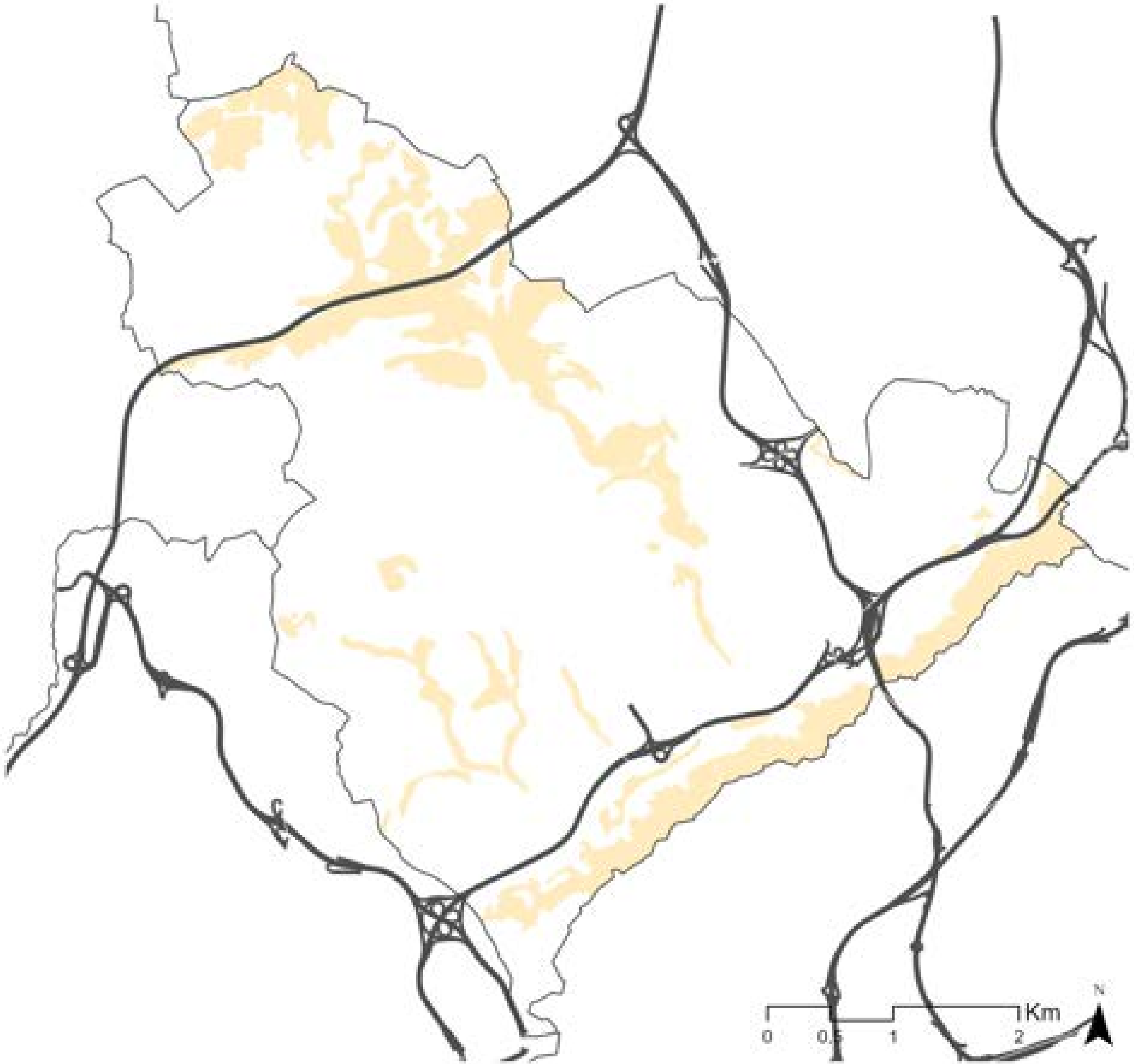
The presence of acoustic barriers along several stretches of the major corridors signals an awareness of this impact within existing planning and infrastructure management.



APPENDIX 19 - Geotechnical Risks

The geotechnical risk map is based on the Risks Prevention Areas Chart (Odivelas PDM - Planta de Ordenamento - Áreas sujeitas a Prevenção de Riscos) and was obtained from the Câmara Municipal de Odivelas and reveals the spatial distribution of slope instability and geotechnical vulnerability across the municipality. The pattern is concentrated and spatially coherent, risk areas are predominantly associated with the A9 (CREL) corridor and with the pronounced slope that marks the southeastern boundary of the municipality, where the significant altimetric difference in relation to Lisbon, previously identified in the topographic reading, generates the steepest and most physically unstable conditions in the territory.

This distribution carries a dual significance. On one hand, it signals a direct risk to the infrastructural structures themselves, he embankments, cuttings, and slopes that sustain the highway corridors in these areas are exposed to instability conditions that generate ongoing maintenance demands and potential structural vulnerability. On the other hand, the residential and urban fabric that has developed in proximity to these slopes is equally exposed, placing populations in direct contact with geotechnical risk conditions that are both spatially persistent and difficult to mitigate through conventional urban management.



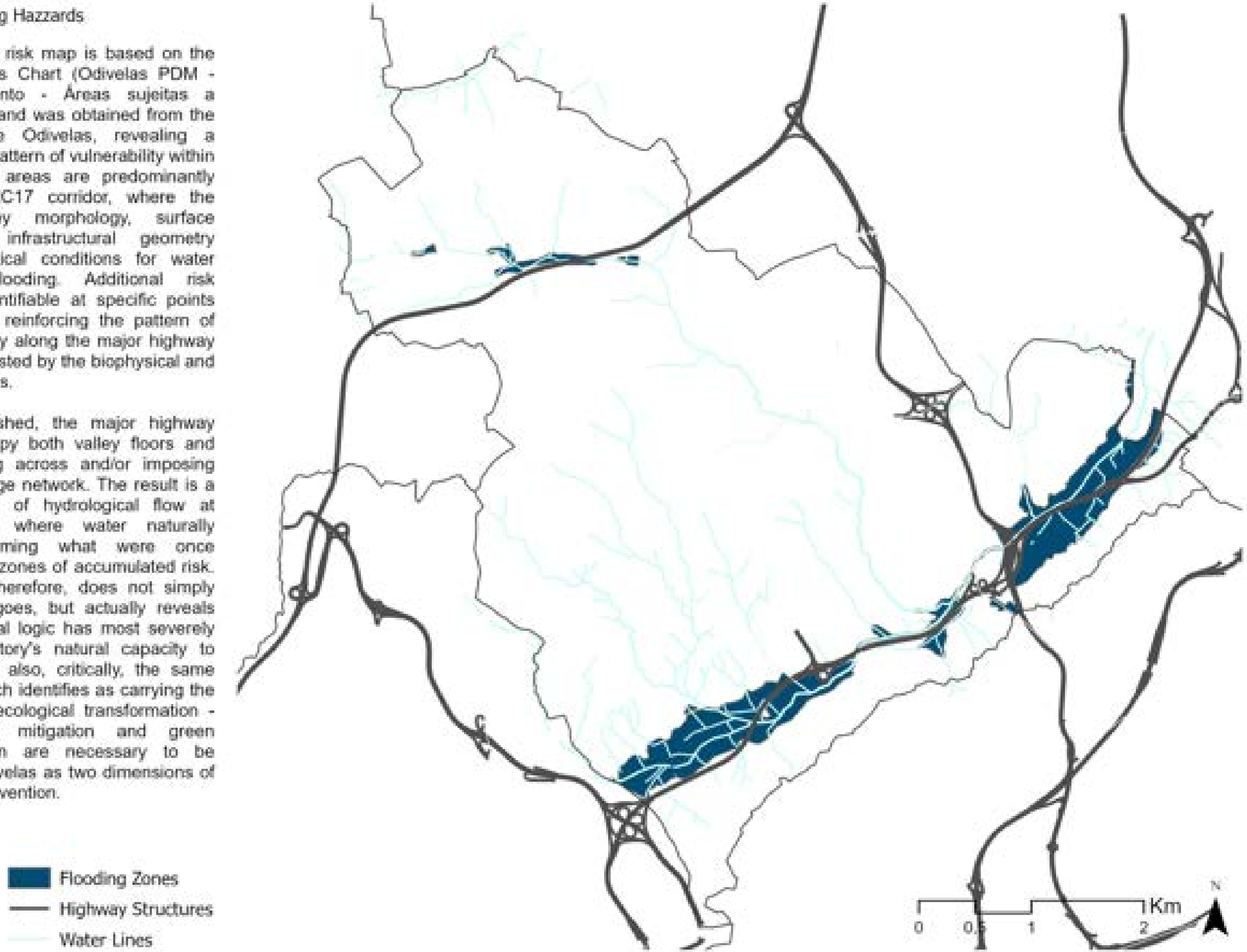
Highway Structures

Risco Geotecnico

APPENDIX 20 - Flooding Hazzards

The flodding hazzards risk map is based on the Risks Prevention Areas Chart (Odivelas PDM - Planta de Ordenamento - Áreas sujeitas a Prevenção de Riscos) and was obtained from the Câmara Municipal de Odivelas, revealing a spatially concentrated pattern of vulnerability within the municipality. Risk areas are predominantly associated with the IC17 corridor, where the intersection of valley morphology, surface impermeability, and infrastructural geometry creates the most critical conditions for water accumulation and flooding. Additional risk concentrations are identifiable at specific points along the A9 (CREL), reinforcing the pattern of hydrological vulnerability along the major highway corridors already suggested by the biophysical and imperviousness readings.

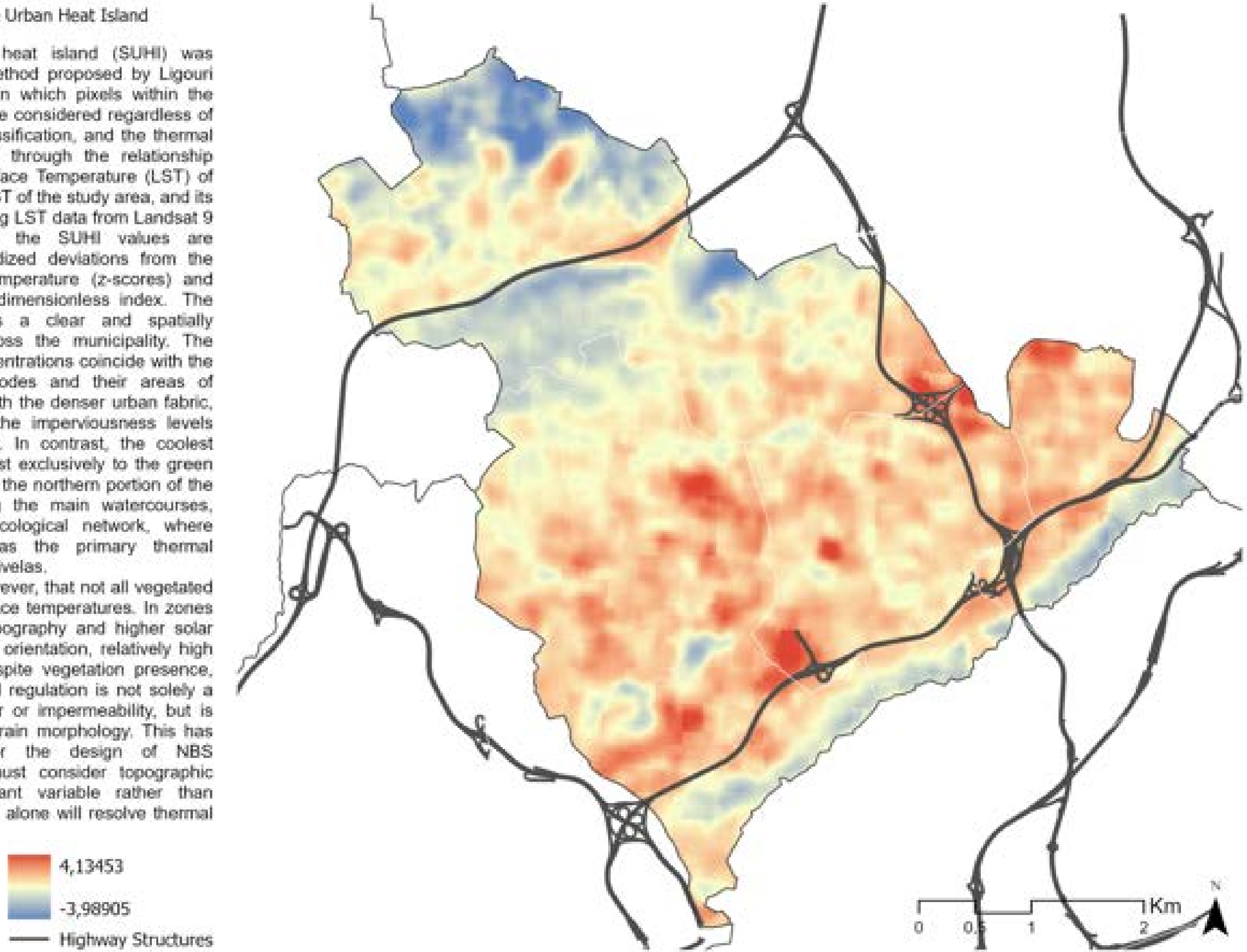
As previously established, the major highway corridors tend to occupy both valley floors and ridge positions, cutting across and/or imposing over the natural drainage network. The result is a systematic interruption of hydrological flow at precisely the points where water naturally concentrates, transforming what were once drainage corridors into zones of accumulated risk. The flood risk map, therefore, does not simply identify where water goes, but actually reveals where the infrastructural logic has most severely compromised the territory's natural capacity to manage it. These are also, critically, the same spaces that this research identifies as carrying the greatest potential for ecological transformation - confirming that risk mitigation and green infrastructure activation are necessary to be worked together in Odivelas as two dimensions of the same territorial intervention.



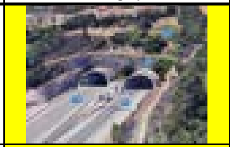
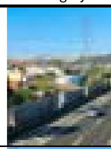
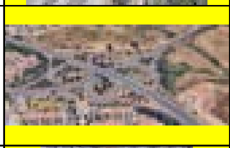
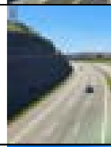
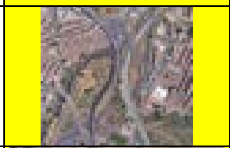
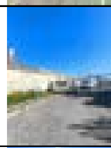
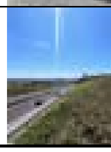
APPENDIX 21 - Surface Urban Heat Island

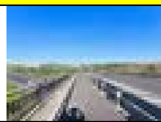
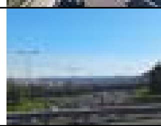
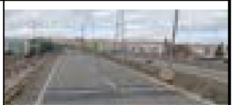
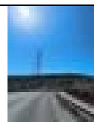
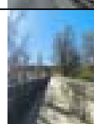
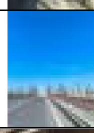
The surface urban heat island (SUHI) was assessed using the method proposed by Ligouri and Monteiro (2024), in which pixels within the analysed urban area are considered regardless of their urban or rural classification, and the thermal balance is determined through the relationship between the Land Surface Temperature (LST) of each pixel, the mean LST of the study area, and its standard deviation, using LST data from Landsat 9 (USGS). This means the SUHI values are expressed as standardized deviations from the mean land surface temperature (z-scores) and therefore constitute a dimensionless index. The resulting map reveals a clear and spatially consistent pattern across the municipality. The most intense heat concentrations coincide with the major infrastructural nodes and their areas of influence, as well as with the denser urban fabric, a direct reflection of the imperviousness levels documented previously. In contrast, the coolest areas correspond almost exclusively to the green corridors, particularly in the northern portion of the municipality and along the main watercourses, confirming that the ecological network, where preserved, operates as the primary thermal regulation system in Odivelas.

It is worth noting, however, that not all vegetated areas present low surface temperatures. In zones with more exposed topography and higher solar incidence due to slope orientation, relatively high LST values persist despite vegetation presence, suggesting that thermal regulation is not solely a function of green cover or impermeability, but is also conditioned by terrain morphology. This has direct implications for the design of NBS interventions, which must consider topographic exposure as a relevant variable rather than assuming that greening alone will resolve thermal stress conditions.

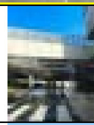
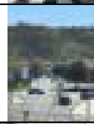
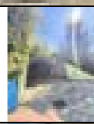
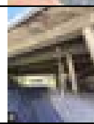
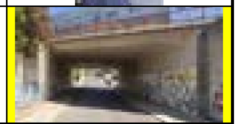
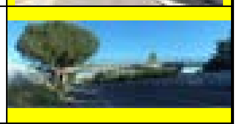
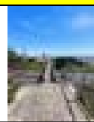
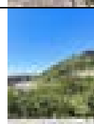
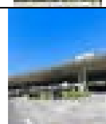
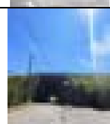
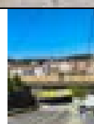
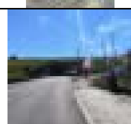
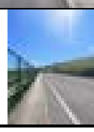


Appendix 22 (a) - Odivelas Highway Structures (Source: Own Elaboration)

Main Structures			Physical Barriers		
Type of Structure	Associated Highway	Imagery	Type of Structure	Associated Highway	Imagery
Tunnel	IC18 (CREL)		Accoustic Wall	All	
Interchange	IC22		Terrain Cuts	IC17 (CRIL) / IC18 (CREL)	
Interchange	ICC22/IC17 (CRIL)		Retaining Wall	IC17 (CRIL) / IC18 (CREL)	
Interchange	IC16/IC17 (CRIL)		Slopes & Embankments	IC22 / IC17 (CRIL) / IC18 (CREL)	

	Type of Structure	Associated Highway	Crossing Condition (Above x Below)	Mobility Type	Pedonal Condition	Imagery
1	Bridge	IC17 (CRIL)	Above	Mixed	Small	
2	Bridge	IC17 (CRIL)	Above	Car	-	
3	Bridge	IC17 (CRIL)	Above	Mixed	Private	
4	Underpass	IC17 (CRIL)	Below	Mixed	Small	
5	Underpass	IC17 (CRIL)	Below	Pedonal	Off Road	-
6	Tunnel	IC17 (CRIL)	Below	Mixed	Small	
7	Connection Bridge	IC17 (CRIL)	Above	Car	-	
8	Bridge	IC17 (CRIL)	Above	Mixed	Medium	
9	Walkway	IC17 (CRIL)	Above	Pedonal	Medium	
10	Bridge	IC17 (CRIL)	Above	Mixed	Medium	
						

Appendix 22 (b) - Odivelas Highway Structures (Source: Own Elaboration)

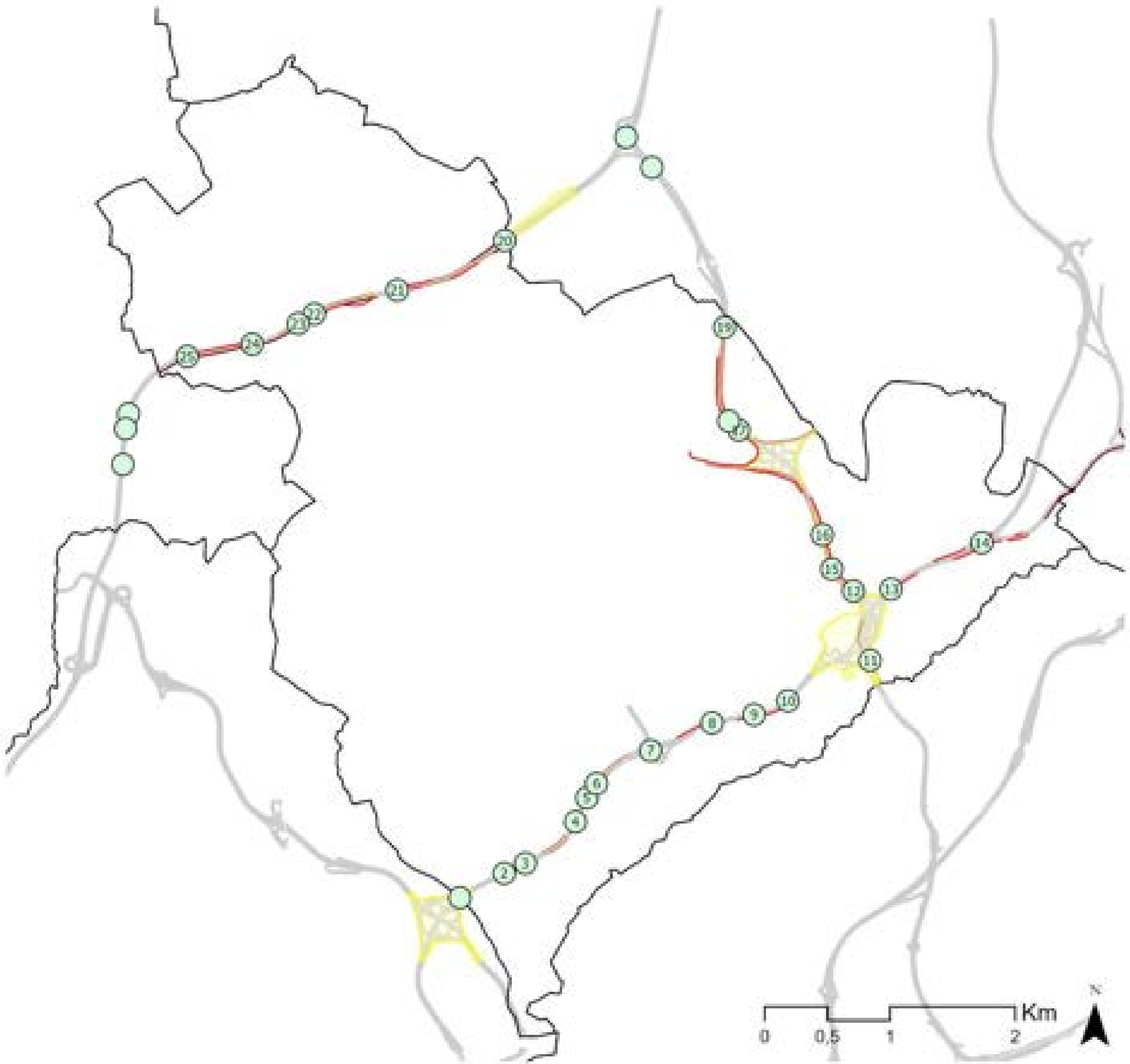
11	Tunnel	IC17 (CRIL) / IC22	Above	Mixed	Medium	
12	Underpass	IC17 (CRIL) / IC22	Below	Mixed	Medium	
13	Bridge	IC17 (CRIL)	Above	Mixed	Medium	
14	Tunnel	IC17 (CRIL)	Below	Mixed	Small	
15	Tunnel	IC22	Below	Pedonal	Large	
16	Underpass	IC22	Below	Mixed	Large	
17	Tunnel	IC22	Below	Mixed	Medium	
18	Walkway	IC22	Above	Pedonal	Large	
19	Walkway	IC22	Above	Pedonal	Large	
20	Road	IC18 (CREL)	Above	Car	-	
21	Underpass	IC18 (CREL)	Below	Mixed	Small	
22	Tunnel	IC18 (CREL)	Below	Car	-	
23	Underpass	IC18 (CREL)	Below	Mixed	Small	
24	Tunnel	IC18 (CREL)	Below	Mixed	Small	
25	Bridge	IC18 (CREL)	Above	Mixed	Medium	
	Source: Original Pictures. Yellow marked ones are obtained from Google Street View & Earth.					

APPENDIX 23 - Highway Structures

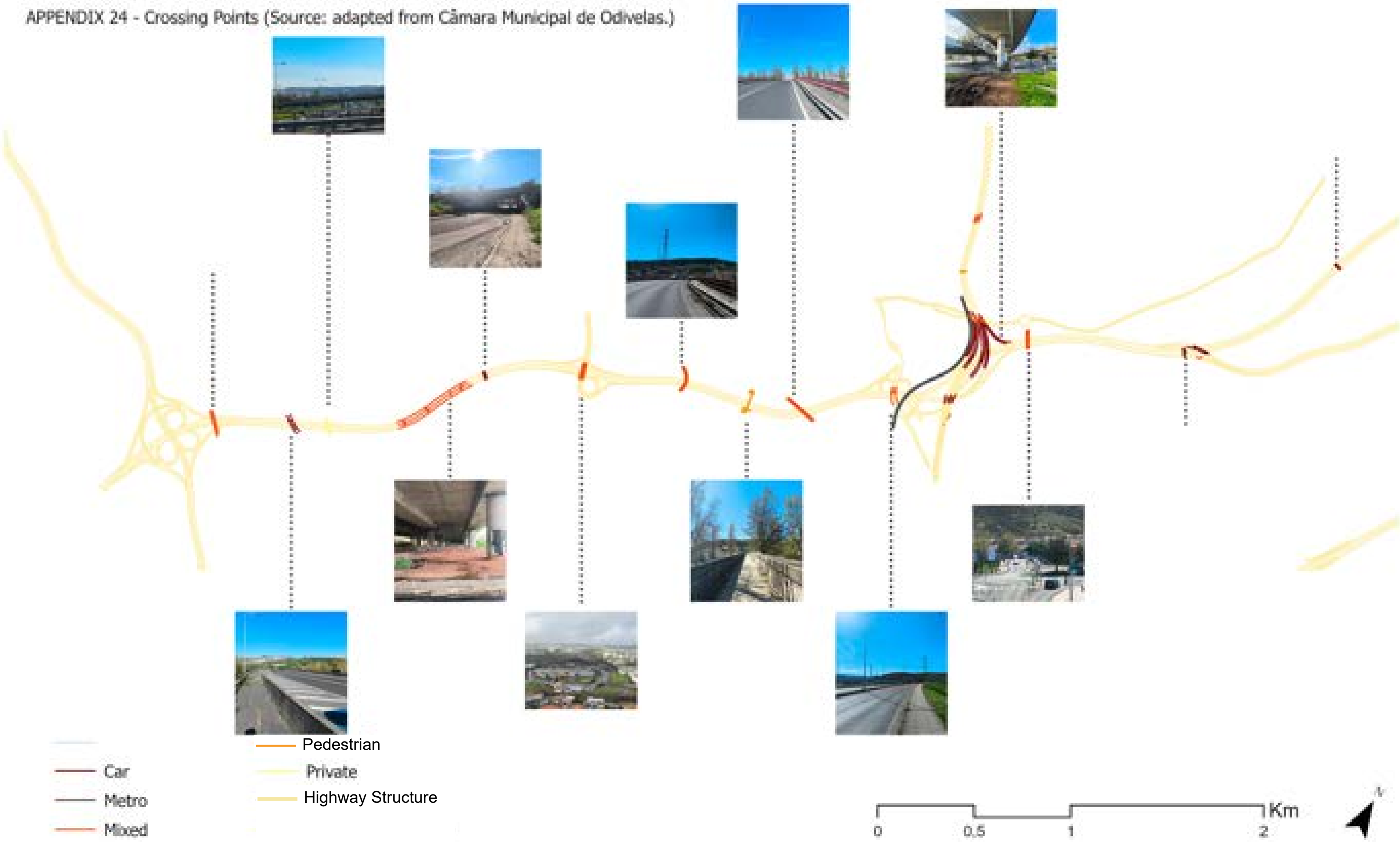
The physical barrier and crossing inventory was elaborated through a systematic field-based and cartographic reading of the highway network across the municipality, mapping the spatial elements that together compose the infrastructural impact of each corridor. The inventory organises these elements into three categories: Main Structures; Physical Barriers; and Crossings.

The IC17 (CRIL) concentrates the highest number of mixed-use crossings - shared by vehicles and pedestrians - reflecting its deeper insertion into the consolidated urban fabric and the density of local connections it must accommodate, though frequently with reduced pedestrian infrastructure that signals a contradiction between crossing provision and pedestrian priority. The IC22 presents the greatest proportion of pedestrian-oriented crossings, suggesting either a higher demand for foot movement or, conversely, the absence of vehicular crossing alternatives in certain stretches. The IC18 (CREL), operates predominantly through underpasses and submerged crossings, generating spatial conditions that are "darker," more enclosed, a qualitative dimension of barrier that also affects the experiential relationship between the corridor and its surroundings.

Taken together, this inventory confirms that the crossings are not merely technical responses to a connectivity problem, but spatial conditions in their own right, with each typology producing a distinct quality of transition that directly shapes whether and how the surrounding urban fabric can relate across the infrastructural barrier.



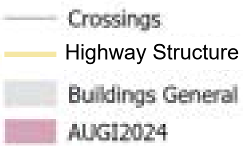
APPENDIX 24 - Crossing Points (Source: adapted from Câmara Municipal de Odivelas.)



APPENDIX 25 - Urban Space & Building Height (Source: adapted from Câmara Municipal de Odivelas)

The building height distribution confirms the spatial asymmetry of the IC17 corridor. The northern side - Odivelas Center and Póvoa de Santo Adrião - concentrates the tallest buildings, reflecting its historical consolidation. The southern AUGI areas of Serra da Luz and Vale do Forno are predominantly 1 to 2-floor dwellings. The transition between typologies is abrupt, confirming the corridor as a hard boundary between two distinct urban logics.

APPENDIX 26 - AUGIS & Execution Units (Source: DGT Exprience Builder)



The AUGI distribution along the IC17 confirms the southern margin as the primary zone of informal occupation, with Serra da Luz and Vale do Forno pressing directly against the infrastructural edge. The contrast with the northern side is spatially legible, where consolidated urban fabric gives way abruptly to informal settlement at precisely the line the corridor imposes. The execution units signal ongoing regularisation processes, but fragmented.

APPENDIX 27 - Space Syntax: Reach 300m (Source: Own Elaboration - PST)

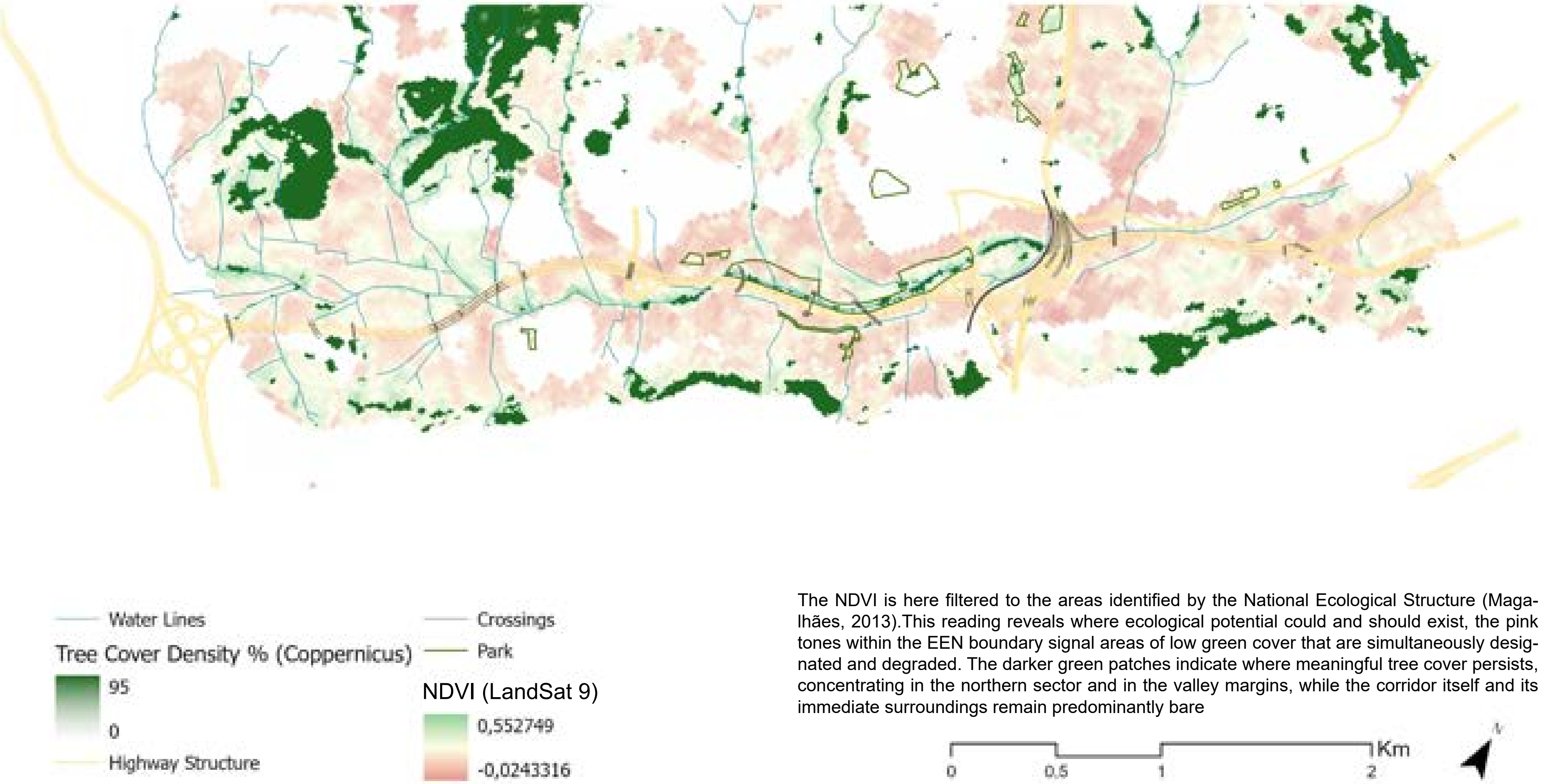
- Very Low
- Low
- Medium
- High
- Very High
- Crossings
- Highway Structure

The Reach analysis at corridor scale reinforces what the municipal reading already suggested. Higher values concentrate in the northern consolidated fabric, while the southern margins - particularly in Serra da Luz and Vale do Forno - present predominantly low to very low values. The corridor itself is legible as a discontinuity in the reach pattern, with the few crossings failing to generate meaningful local accessibility on the southern side.

APPENDIX 28 - Facilities, Commerces & Services (Source: adapted from Câmara Municipal de Odivelas)

Facilities are organised into four clusters reflecting patterns of daily use: markets and schools as essential daily services, health and safety, sports and culture, and civic uses. Commercial centres and streets - concentrated along the northern side and at infrastructural nodes - reflect the car-oriented retail logic characteristic of highway-adjacent landscapes. The southern margin remains largely unserved, with facilities almost entirely absent from Serra da Luz and Vale do Forno.”

APPENDIX 29 - Green Structure (Source: Own Elaboration)



The NDVI is here filtered to the areas identified by the National Ecological Structure (Magalhães, 2013).This reading reveals where ecological potential could and should exist, the pink tones within the EEN boundary signal areas of low green cover that are simultaneously designated and degraded. The darker green patches indicate where meaningful tree cover persists, concentrating in the northern sector and in the valley margins, while the corridor itself and its immediate surroundings remain predominantly bare

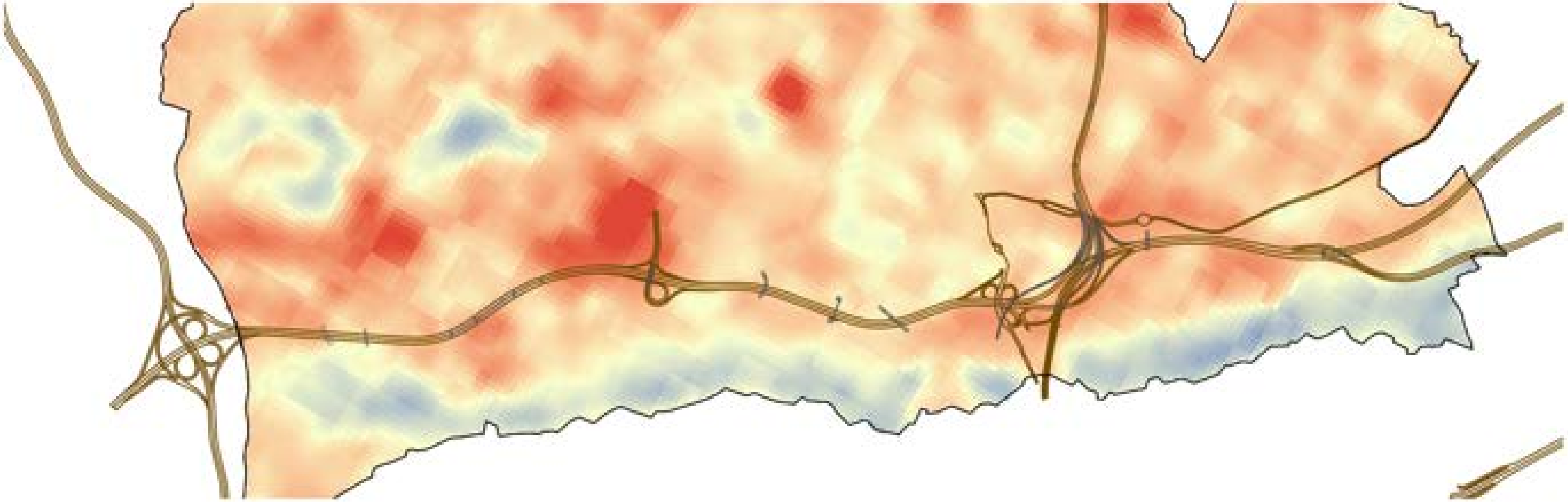
APPENDIX 30 - Hydrography & Flooding Hazzards (Source: adapted from Câmara Municipal de Odivelas)

The flooding zones follow the hydrographic network, concentrating along the watercourses that predate the infrastructure. The highway was implanted across these valley systems, intersecting them at the water crossing points marked, moments where the infrastructural logic was imposed over an existing hydrological one. The flooding condition is therefore also consequence of the highway burying watercourses and sealing the valley floor that would otherwise absorb and regulate water flow.

APPENDIX 31 - Fragstats: Natural Cohesion (Source: Own Elaboration)

The Natural Cohesion index measures the physical connectedness of natural patches. the higher the value, the more continuous and spatially coherent the natural areas are. Along the IC17, the map confirms what field observation suggested: significant natural patches exist on both sides of the corridor but are systematically interrupted at the valley floor in which the highway was imposed, precisely where ecological continuity would be most needed. The corridor operates as a hard break in what could otherwise form a connected green system.

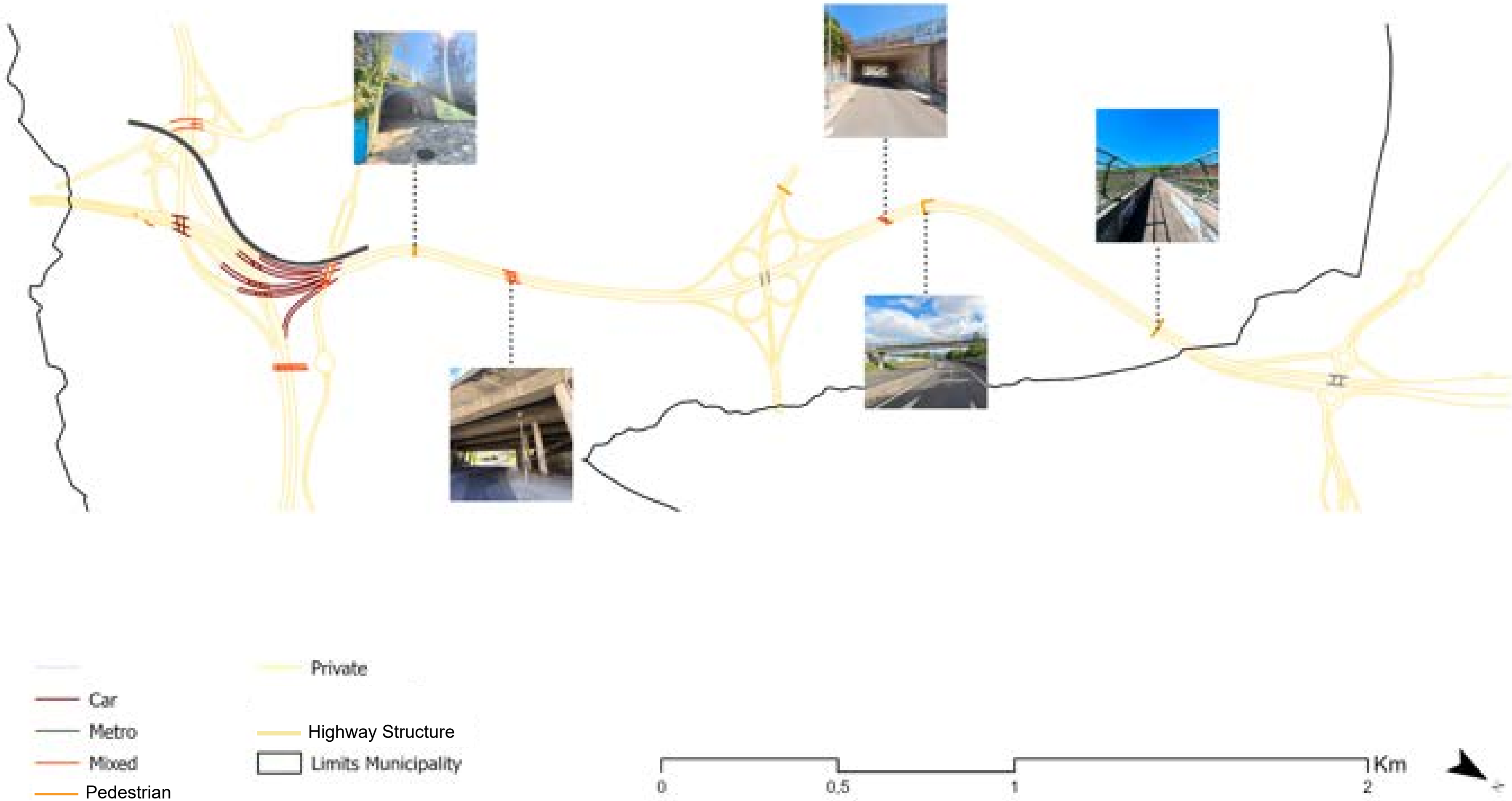
APPENDIX 32 - Urban Heat Islands (Source: Own Elaboration from Landsat 9)



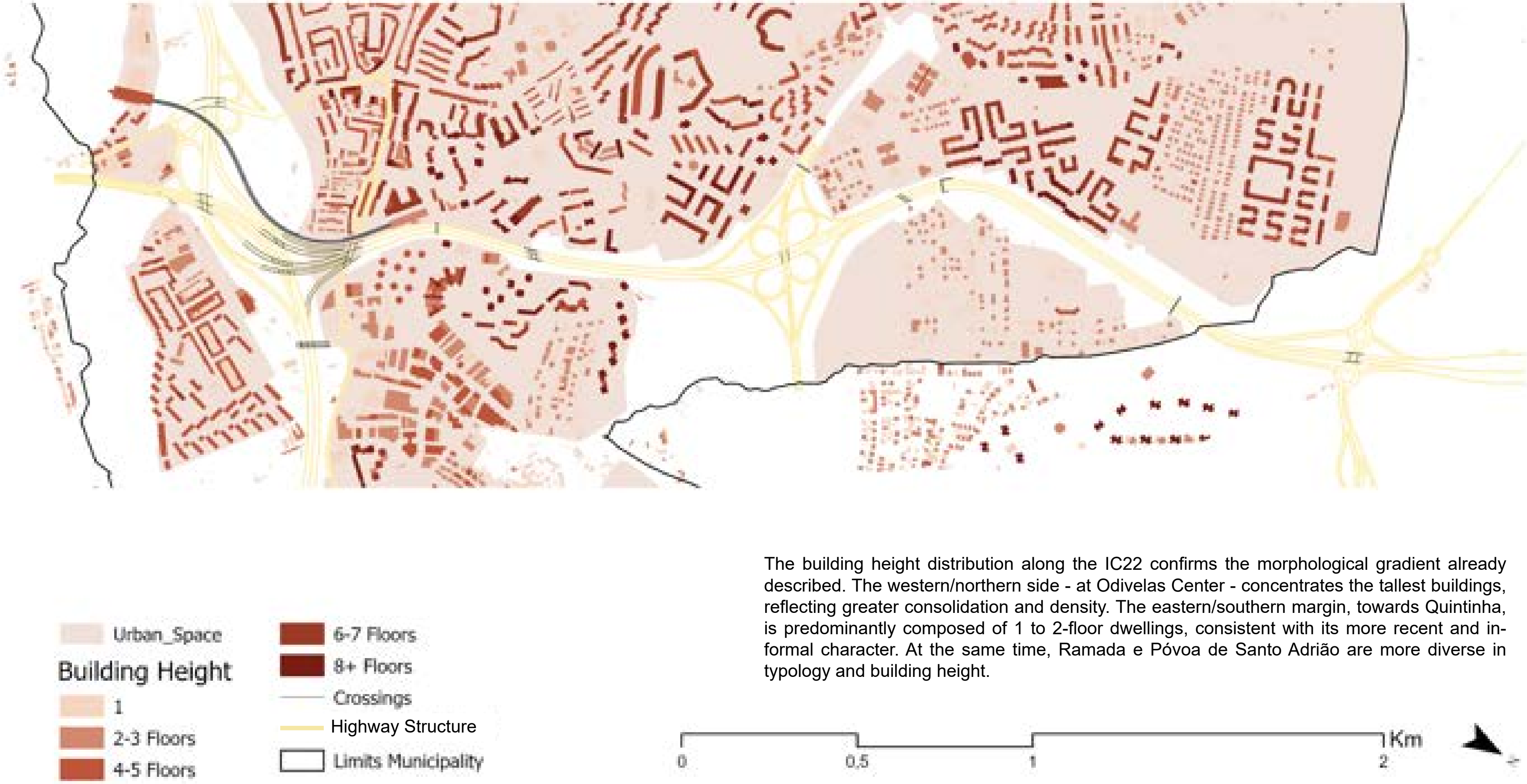
Crossings
Highway Structure
Limits Municipality
LST Urban Heat Islands
Value
4,13453
-3,98905

The LST map reveals higher thermal stress concentrating in the southern margin and along the infrastructural domain itself, where impervious surfaces and the absence of vegetation amplify heat retention. This condition is further intensified by the topographic position of the corridor, occupying one of the lowest points of the municipality, where heat could be prone to accumlate due to lack of air cooling. The cooler areas correspond to greater tree cover from previous analysis, but also the higher slope positioning on the southern border of Odivelas.

APPENDIX 33 - Crossings Points (Source: adapted from Câmara Municipal de Odivelas)



APPENDIX 34 - Urban Space & Building Height - IC22 (Source: adapted from Câmara Municipal de Odivelas.)



The building height distribution along the IC22 confirms the morphological gradient already described. The western/northern side - at Odivelas Center - concentrates the tallest buildings, reflecting greater consolidation and density. The eastern/southern margin, towards Quintinha, is predominantly composed of 1 to 2-floor dwellings, consistent with its more recent and informal character. At the same time, Ramada e Póvoa de Santo Adrião are more diverse in typology and building height.

APPENDIX 35 - AUGIS & Execution Units - IC22 (Source: DGT Experience Builder)

The AUGI zones concentrate on the eastern side of the corridor - Quintinha to the south and partially in Ramada to the north - pressing directly against the infrastructural edge. The execution units signal ongoing regularisation processes within these areas, but their fragmented distribution reflects the absence of a coordinated planning response to the informal condition."

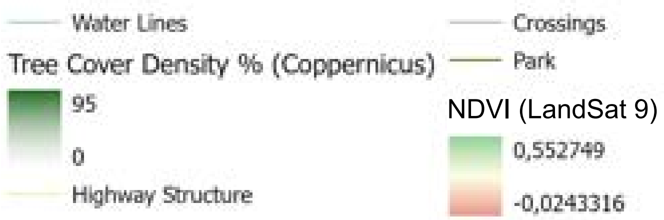
APPENDIX 36 - Space Syntax: Reach 300m - IC22 (Source: Own Elaboration - PST)

The Reach analysis confirms the accessibility gradient along the IC22. Higher values concentrate on the western side - Center- where the denser street network supports greater local movement. While Póvoa e Ramada are also diverse in reachness. Quintinha on the eastern side present predominantly low to very low values.

APPENDIX 37 - Facilities, Commerces & Services: IC22 (Source: adapted from Câmara Municipal de Odivelas)

Equipment and services concentrate predominantly on the western side - Centro and Ramada - with daily use facilities, commercial streets and parks reflecting the greater consolidation of this side. The eastern side - Póvoa de Santo Adrião and Quintinha, specially this last one - remains largely underserved, with almost no daily use facilities or green spaces, confirming the service asymmetry that the corridor reinforces. The facilities in Póvoa, mainly exists arondthe crossings towards Odivelas Center.

APPENDIX 38 - Green Structure: IC22 (Source: Own Elaboration)

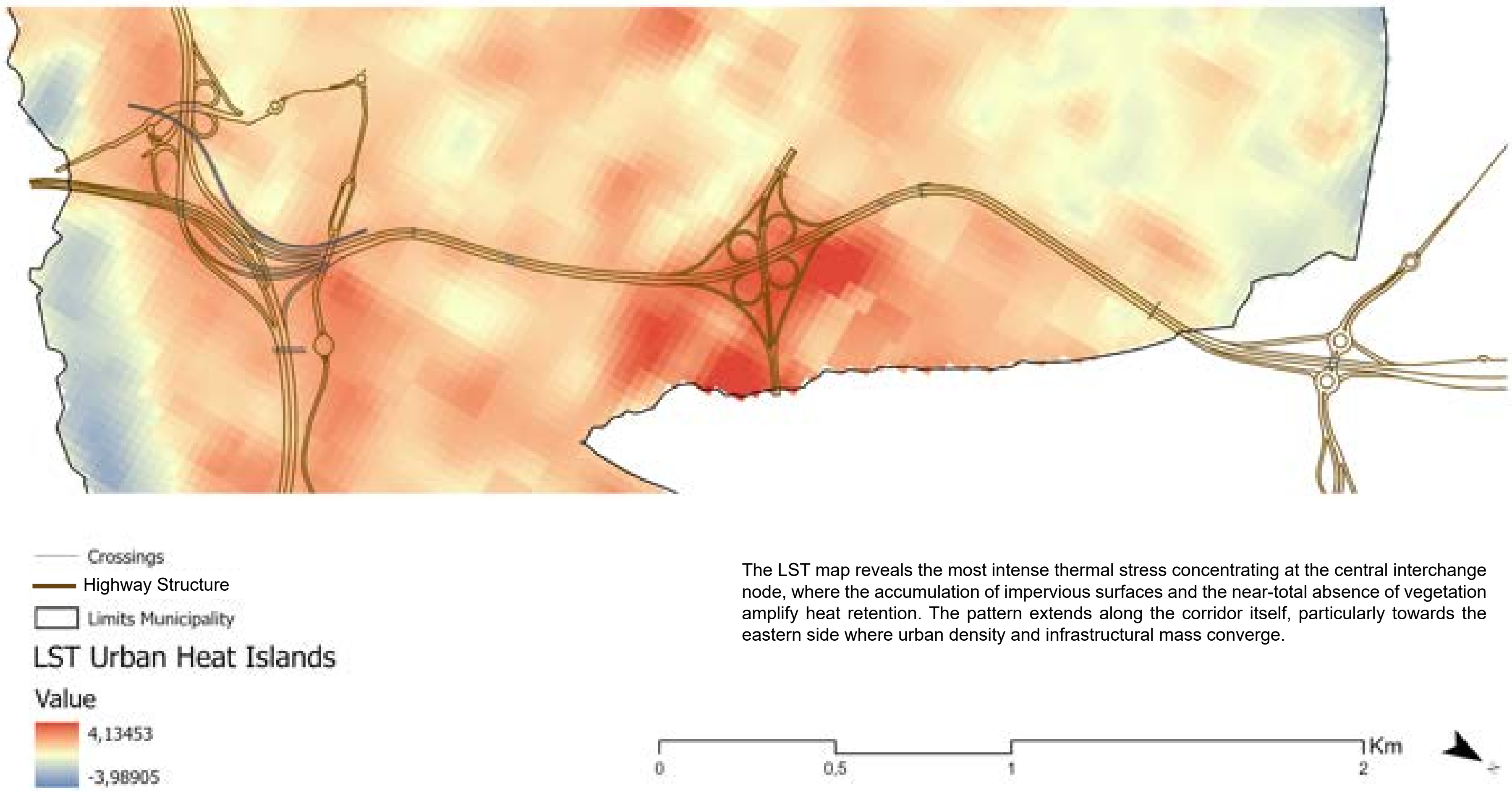


The NDVI filtered to the National Ecological Structure confirms the near-total absence of active green mass along the IC22 corridor. The few green patches concentrate at the central interchange node and on the eastern margin towards Póvoa de Santo Adrião, where natural areas press towards the corridor before being interrupted by it. The predominantly pink tones across the buffer signal that the potential ecological areas have been largely urbanised, leaving almost no functional vegetation cover within the corridor's influence zone.

APPENDIX 39 - Fragstats: Natural Cohesion- IC22 (Source: Own Elaboration)

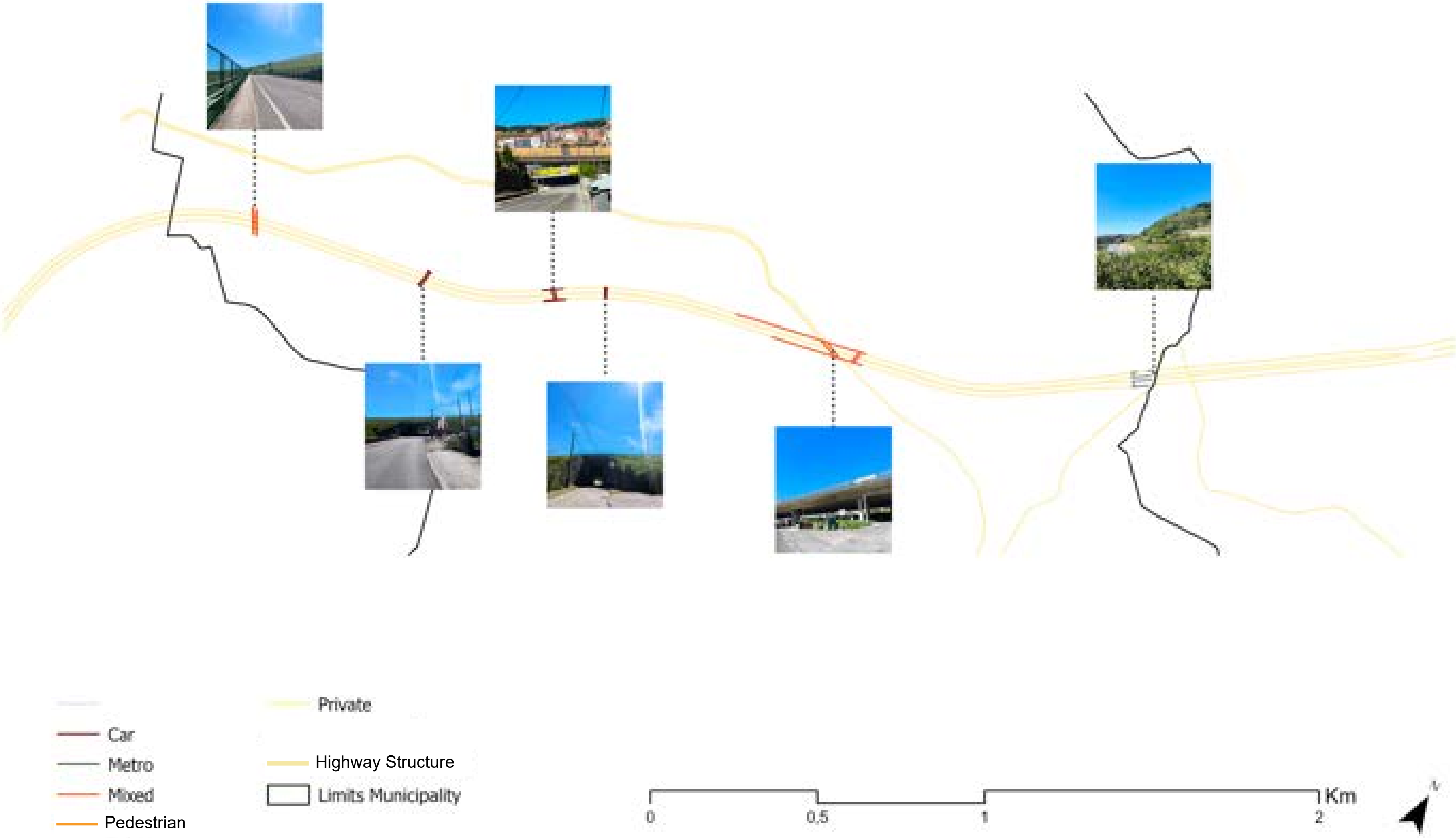
The Natural Cohesion index reveals larger and more continuous natural patches on the eastern side of the corridor - towards Póvoa de Santo Adrião and Quintinha - reflecting its less urbanised character. These patches press towards the corridor from both sides without ever connecting across it, confirming the IC22 as a hard interruption of what could otherwise form a more coherent ecological system.

APPENDIX 40 - Urban Heat Islands - IC22 (Source: Own Elaboration with Landsat 9)



The LST map reveals the most intense thermal stress concentrating at the central interchange node, where the accumulation of impervious surfaces and the near-total absence of vegetation amplify heat retention. The pattern extends along the corridor itself, particularly towards the eastern side where urban density and infrastructural mass converge.

APPENDIX 41 - Crossing Points (Source: adapted from Câmara Municipal de Odivelas.)



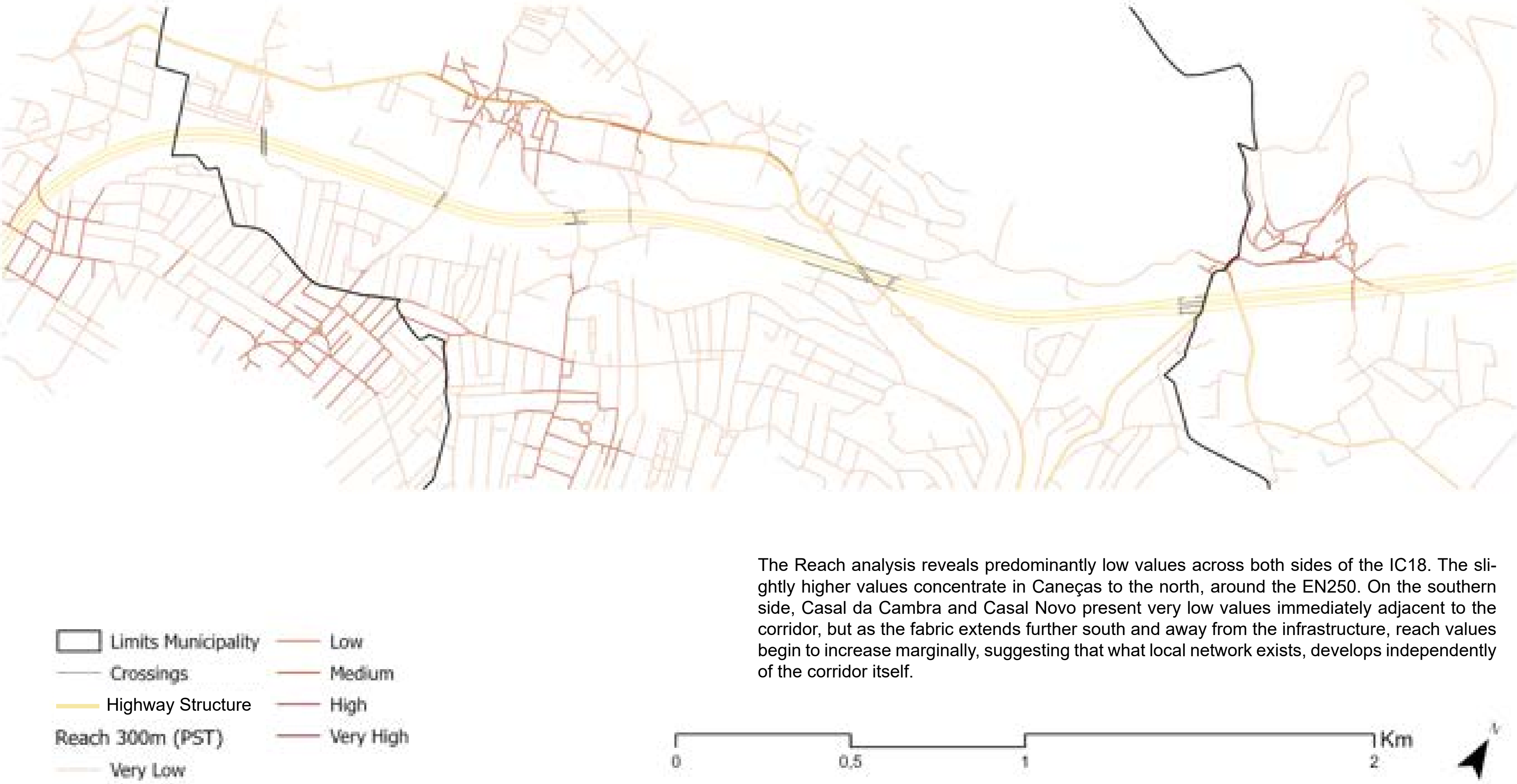
APPENDIX 42 - Urban Space & Building Height (Source: adapted from Câmara Municipal de Odivelas.)

Urban_Space
Building Height
1
2-3 Floors
4-5 Floors
6-7 Floors
8+ Floors
Crossings
Highway Structure
Limits Municipality

The building height distribution confirms the low-rise character of the IC18 corridor on both sides. Caneças to the north presents comparivelly taller buildings, but not tall ones, reaching aroun 3-4 floors, reflecting its modest historical consolidation along the EN250. Casal da Cambra and Casal Novo to the south are composed of 1 to 2-floor dwellings, consistent with their informal and more recent development. The corridor itself can be read as a boundary between two similarly low-rise but distinctly different urban conditions.

0 0,5 1 2 Km

APPENDIX 43 - Space Syntax: Reach 300 - IC18 (Source: Own Elaboratio - PST)



The Reach analysis reveals predominantly low values across both sides of the IC18. The slightly higher values concentrate in Caneças to the north, around the EN250. On the southern side, Casal da Cambra and Casal Novo present very low values immediately adjacent to the corridor, but as the fabric extends further south and away from the infrastructure, reach values begin to increase marginally, suggesting that what local network exists, develops independently of the corridor itself.

APPENDIX 44 - Facilities, Commerces & Services (Source: adapted from Câmara Municipal de Odivelas)

Facilities and services concentrate almost entirely in Caneças to the north, along the EN250, confirming its role as the only meaningful service centre along the corridor. The southern side - Casal de Cambra and Casal Novo - is virtually unserved, with only isolated occurrences scattered across a largely underequipped territory. The asymmetry is the most pronounced of the three corridors, reflecting the IC18's condition as an infrastructure that passes without generating any local centrality.

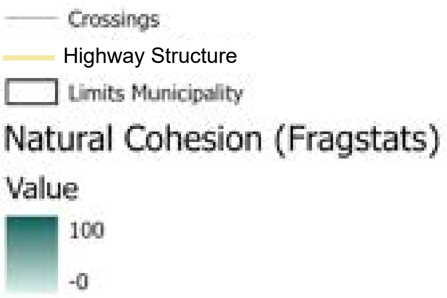
APPENDIX 45 - AUGIS & Execution Units (Source: DGT Expeience Builder)

Crossings
Highway Structure
Buildings General
AUGI2024
Execution Units
Limits Municipality

The AUGI zones dominate the southern side of the corridor, with Casal de Cambra and Casal Novo forming an extensive and largely continuous informal fabric that presses directly against the infrastructural edge. The northern side - Caneças - presents a comparatively more regularised condition. The execution units signal ongoing regularisation processes, but their scale relative to the extent of informal occupation suggests that the southern margin remains largely unresolved.

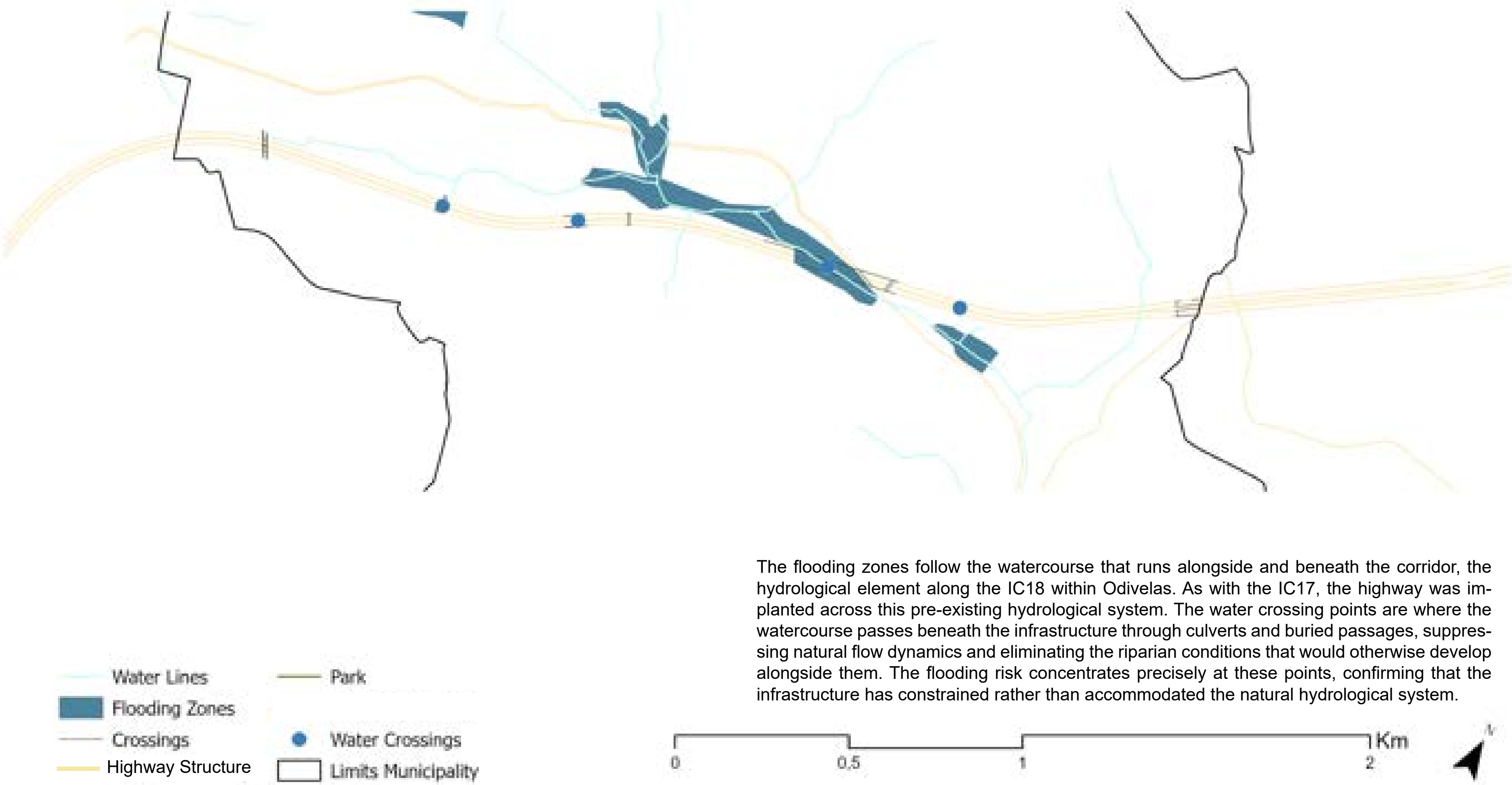
0 0,5 1 2 Km

APPENDIX 46 - Fragstats: Natural Cohesion - IC18 (Source: Own Elaboration)



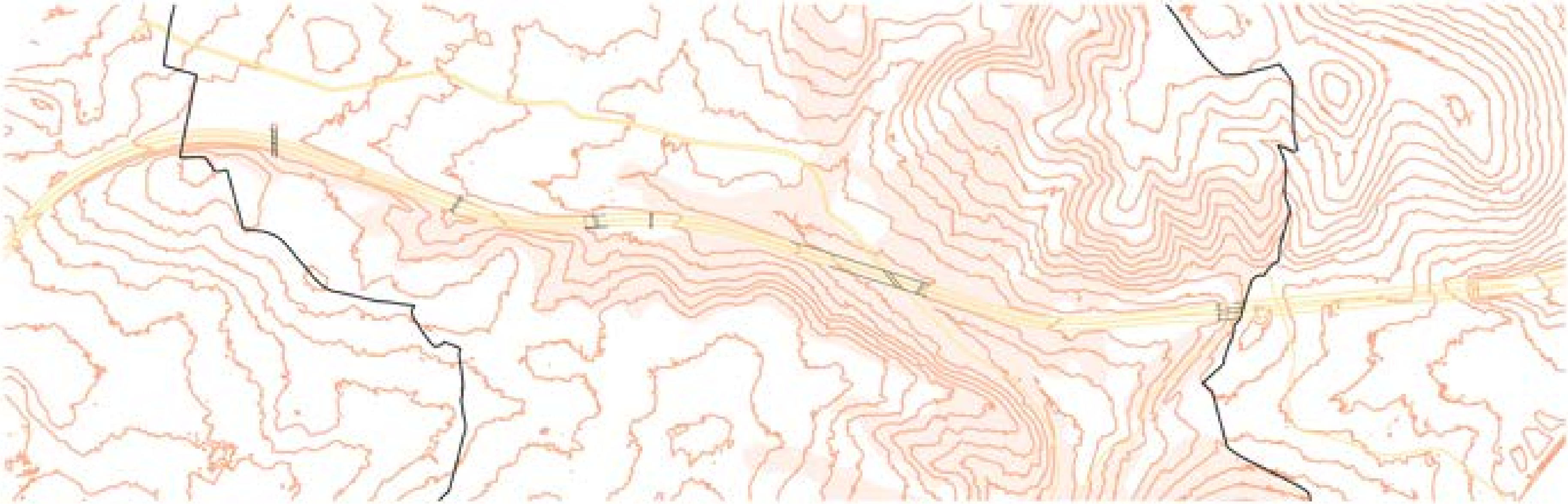
The Natural Cohesion index reveals larger natural patches along the IC18 corridor, a condition likely related to the lower urbanisation density and the PDM's non-urban designation of much of the surrounding territory. However, as the NDVI confirms, these designations do not translate into ecological function, the patches remain fragmented and interrupted by the corridor, never forming a connected system as formally designated in the PDM.

APPENDIX 47 - Floodings Risks - IC18 (Source: adapted from Câmara Municipal de Odivelas)



The flooding zones follow the watercourse that runs alongside and beneath the corridor, the hydrological element along the IC18 within Odivelas. As with the IC17, the highway was implanted across this pre-existing hydrological system. The water crossing points are where the watercourse passes beneath the infrastructure through culverts and buried passages, suppressing natural flow dynamics and eliminating the riparian conditions that would otherwise develop alongside them. The flooding risk concentrates precisely at these points, confirming that the infrastructure has constrained rather than accommodated the natural hydrological system.

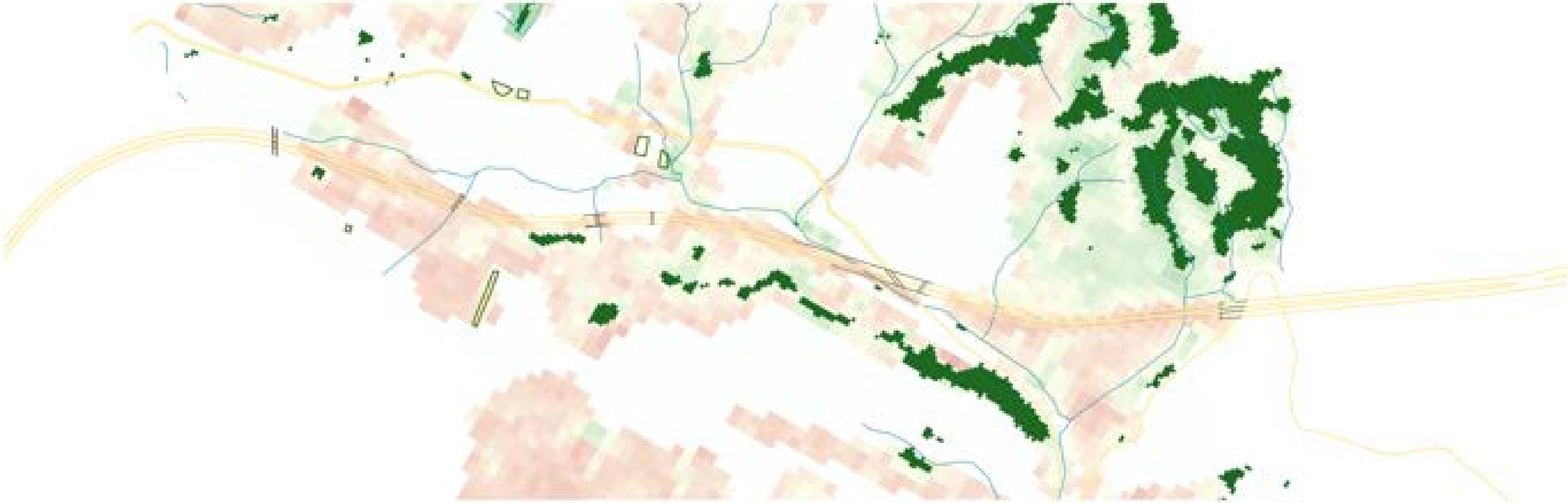
APPENDIX 48 - Geotechnical Risks & Countours- IC18 (Source: adapted from Câmara Municipal de Odivelas and DGT Lidar)



Limits Municipality
10m Countour Lines (DGT - MDT)
Geotechnical Risk
Crossings
Highway Structure

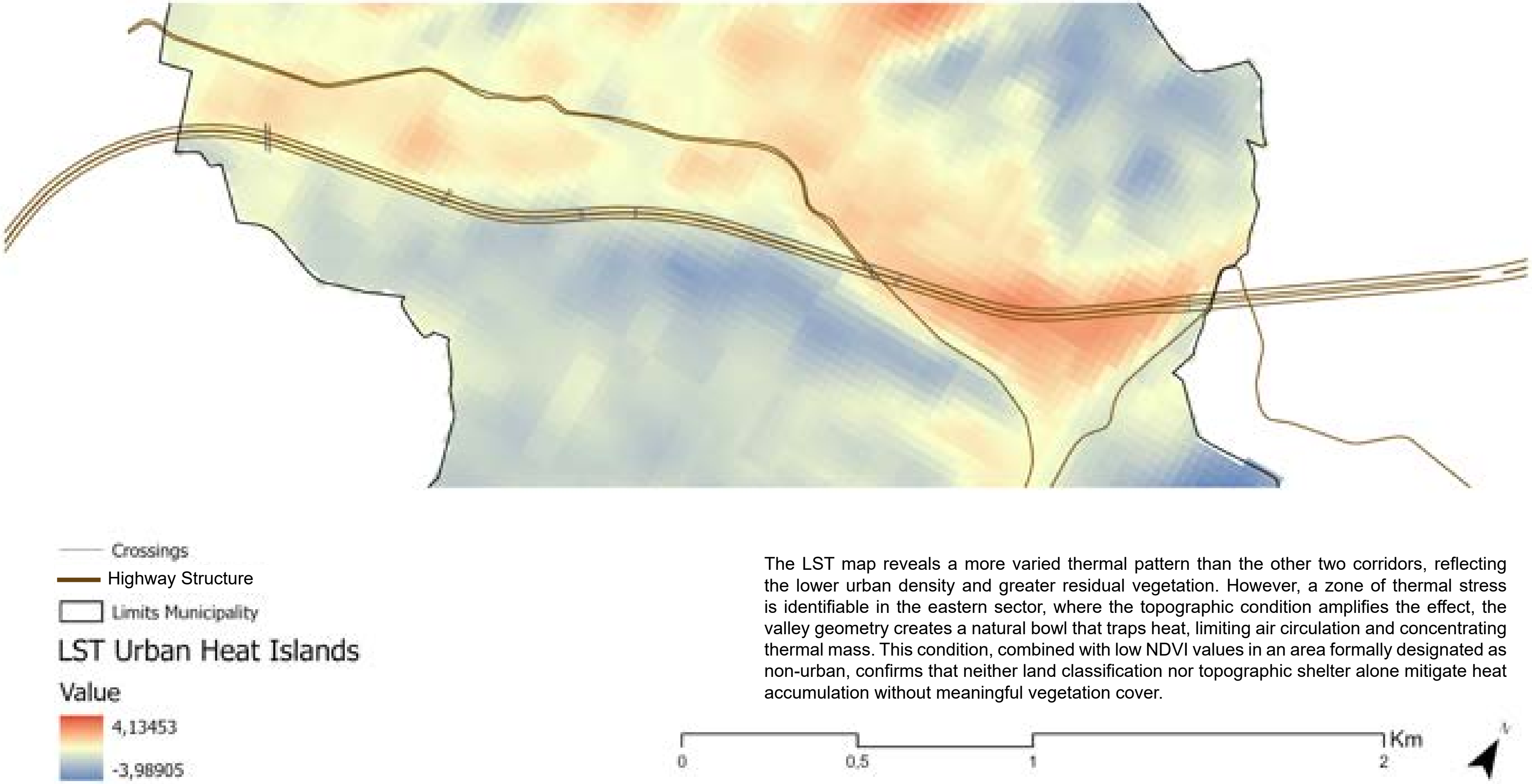
The contour lines, derived from DGT LiDAR data and processed in ArcGIS through the Contour tool, reveal the topographic complexity of the IC18 corridor. The valley structure is clearly legible, with the tightly spaced contours on the southern side confirming the steep slopes that generate the geotechnical instability risks identified by the Câmara Municipal. The corridor runs precisely along this topographically fragile zone, where terrain cuts and embankments required by the infrastructural alignment coincide with the areas of greatest slope instability.

APPENDIX 49 Green Structure: IC18 (Source: Own Elaboration)



The NDVI filtered to the National Ecological Structure reveals a more expressive vegetation presence than the other two corridors, concentrating particularly on the eastern sector where tree cover density is also higher. However, this vegetation follows the hydrological logic of the watercourse rather than forming a coherent ecological system, the pink tones immediately alongside the corridor confirm that the infrastructural domain itself remains largely bare, with the green presence retreating away from the highway rather than engaging with it.”

APPENDIX 50 - Urban Heat Islands - IC18 (Source: Own Elaboration with Landsat 9)



The LST map reveals a more varied thermal pattern than the other two corridors, reflecting the lower urban density and greater residual vegetation. However, a zone of thermal stress is identifiable in the eastern sector, where the topographic condition amplifies the effect, the valley geometry creates a natural bowl that traps heat, limiting air circulation and concentrating thermal mass. This condition, combined with low NDVI values in an area formally designated as non-urban, confirms that neither land classification nor topographic shelter alone mitigate heat accumulation without meaningful vegetation cover.